Lectures in Discrete Mathematics

CS 2800

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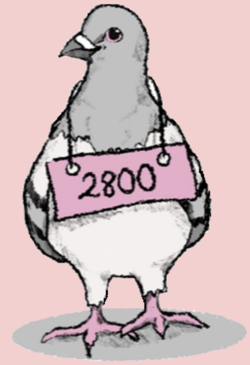


Preface

Discrete mathematics serves as the theoretical backbone of computer science. From the design of logic gates in low-level hardware to the development of verifiable correct code in large software systems, computer scientists rely on tools from discrete mathematics to design and reason about computational tools. Beyond this, developing mathematical maturity – being able to read and understand new definitions, trace through and develop multi-step procedures and arguments, and verify or disprove claims – is crucial for programming, research, and scientific communication. There are many great references on Discrete Mathematics, books that we have read and learned from and which have served as inspiration in the preparation of these notes. In particular, we mention

- *Discrete Mathematics and its Applications*, by Kenneth Rosen.
- *Essential Discrete Mathematics for Computer Science*, by Harry Lewis and Rachel Zak.
- *Mathematics for Computer Science*, by Eric Lehmann, Albert Meyer, and Tom Leighton.
- *Discrete Mathematics*, zyBook by Sandy Irani (this is an interactive online text).

In developing these notes, we sought to create a reference that was tailor-made for *CS 2800: Mathematical Foundations of Computing* at Cornell University. The notes are self-contained, as they include every definition and main result that we expect to present in the course; no more, no less. The notes are separated into lectures, with the expectation that most material from each chapter can be covered in a single 50-minute period. Each lecture has roughly 10 included exercises that range from easier questions that recall and apply that lecture's definitions to harder exercises that require more thought and the synthesis of material from multiple modules in the course. Harder exercises are marked with a "$\star$". Exercises that require knowledge of material outside of this course are marked with a "$\diamond$". The completion of all of the exercises should coincide with mastery of the material.

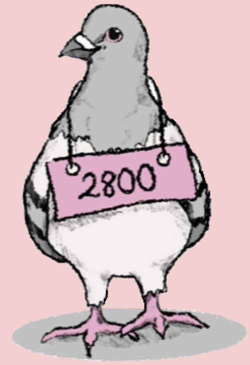
The preparation of such an expansive set of notes was by no means a one-person endeavor. An earlier version of these notes was prepared in collaboration with Alexandra Silva, with whom I had the pleasure of teaching CS 2800 in Fall 2022. A lot of the feedback that inspired the reorganization of the notes into the current version was provided by our TAs and students from that semester (along with corrections to well over 100 typos!). Joe Halpern, Anke van Zuylen, Noah Stephens-Davidowitz, and Frans Schalekamp have also taught recent versions of CS 2800 and its honors version (CS 2802). Their course materials and insightful feedback have been invaluable throughout the development of various drafts of these notes.

The course is organized into seven modules.

1. Foundations (Logic and Sets)
2. Induction
3. Graphs and Relations
4. Functions and Cardinality
5. Languages and Automata
6. Combinatorics
7. Probability and Statistics

Beyond the material presented in these notes, we expect students to come away from the course with a level of mathematical sophistication and appreciation. They should be able to read, understand, write, and prove precise mathematical statements about important objects in computer science.

December 2024, Ithaca, NY
Matt



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I Foundations

Formalizing Mathematics

We begin our study of mathematics at a very young age. In the early grades, we learn to count, and we study our math facts. We talk about how to sort, categorize, and describe objects. Later on, we are introduced to more advanced topics such as geometry, algebra, trigonometry, and calculus. Along the way, we learn many facts (or *theorems*) that help us to solve problems more easily. These facts build on each other as we advance in math, and they allow us to reason about more and more complicated settings.

This course is likely very different from previous math courses that you have taken. Rather than looking at harder problems or trying to solve more intricate equations, we are going to look backward toward the foundations of math itself. We will explore *why* and *how* math works. We will come to appreciate math as a language. Natural languages combine words (nouns, verbs, adjectives, etc.) to form sentences. These sentences in turn can be used to express nuanced information, ideas, and thoughts. Similarly, in math, we can build up to more complicated theories using simple building blocks such as sets, functions, and relations. We will study each of these in the course. Extending our language analogy, words alone are not enough to convey meaning; they must be carefully arranged according to various grammatical rules in order to make sense. Math is also governed by rules and structure, known as *logic*.

Logic is the study of reasoning. In mathematical logic, we seek to understand truth — what mathematical statements can we deem to be “correct” — and proof — how can we verify the correctness of these statements? To answer both of these questions, we will need to introduce a more formal basis for mathematical reasoning. In this first module of the course, we will do just this. We will discuss both the syntax (notation) and semantics (meaning) of expressions in propositional and predicate logic. As we proceed in the course, you will come to appreciate how all of the mathematical statements that we make will ultimately boil down to this formal logical language.

However, just as the structure of a language on its own is uninteresting without things to talk about, we will need to develop a class of mathematically relevant objects. The foundation for these objects is the theory of *sets*. We’ll define sets (albeit not at a level deep enough to be fully rigorous) and some common set operations. Armed with sets, we’ll be at a point where we can start to really do math. We will be able to make interesting statements about sets and develop logically-sound strategies for verifying that they are correct. In all, this first module should be appreciated as a foundation for what is to come. While logic and sets have intrinsic beauty and interest, their true power is as tools to model the more advanced topics in mathematics and computer science that we will explore later in the course.

1. Propositional Logic

1.1 Propositions

In our first lecture, we study propositions, the basic units of truth in mathematics. Propositions serve as the basis for formal reasoning; developing a careful understanding of propositions and their operations is critical for writing clear and precise arguments.

> **Definition 1.1 — Proposition.**
> A *proposition* is a statement that can be assigned a truth value (true or false).

Example 1.1 The following are all examples of propositions:

(a) "CS 2800 meets in Statler Hall."

(b) "$3 \cdot 7 + 4 = 25$"

(c) "$250 - 179 = 81$"

(d) "Cornell was founded in 2011."

(e) "It will rain tomorrow."

(f) "CS 2800 has 3 lectures and 1 discussion per week."

Here, (c) and (d) illustrate that propositions can be false statements. (e) shows that the truth value of a proposition may be indeterminate (with our current knowledge).

Example 1.2 The following are non-examples of propositions:

(a) "4": The number 4 is an object (a noun) that isn't inherently true or false. A proposition would assert a property about this number (e.g. it is even).

(b) "Are there infinitely many prime numbers?": A question (even a yes/no question) is not true or false.

(c) "$(n+1)^2 > 3n+2$": Here, there is a *free* variable, n, in the statement. We cannot determine whether the statement is true without knowing the value of n. This is an example of a predicate, which we will study soon.

The proposition (f) in Example 1.1 differs from the others. It can be broken down into two smaller propositions, "CS 2800 has 3 lectures per week," and "CS 2800 has 1 discussion per week," whereas the other examples cannot be broken down. For this reason, (f) is an example of a *compound* proposition, while the others are examples of *atomic* propositions.

Definition 1.2 — Atomic/Compound Propositions.

An *atomic* proposition cannot be broken into smaller propositions. Its truth value is intrinsic.

A *compound* proposition is formed from multiple atomic propositions. Its truth value is determined by the truth values of these atomic pieces.

We'll use lowercase letter variables (p, q, r) to refer to propositions. Note that these propositions may be atomic or compound. We form compound propositions from atomic propositions using *logical operations*, which we will describe soon.

1.1.1 Syntax vs. Semantics

Before we give a list of the logical operations, it is important to distinguish between the notions of *syntax* and *semantics*.

Definition 1.3 — Syntax, Semantics.

The *syntax* of a formal system describes the symbols that it uses and the orders/ways that these symbols can be combined to form valid expressions.

The *semantics* of a formal system describes the meaning of the expressions.

For the "formal system" of the English language, syntax includes grammatical rules about parts of speech and sentence structure. Semantics tells us what information a collection of words conveys. In the system of *propositional logic*, the syntax will assign different symbols to various logical operations and tell us which arrangements of symbols correspond to valid (compound) propositions. The semantics will give us rules for determining the truth value (true or false) of a compound proposition based on the truth value of its arguments.

1.2 Logical Operations

An *operation* is a rule that takes in one or more *arguments* and returns a *value*. In the case of logical operations, both the arguments and value are propositions, so operations describe a way to build a more complicated (compound) proposition out of simpler propositions. We will introduce five logical operations. Each is notated using a certain symbol, its *syntax*. The semantics of these operations can be summarized with truth tables.

Definition 1.4 — Truth Table.

Given a compound proposition c expressed in terms of argument propositions $p_1, \ldots, p_n$, the *truth table* for c is a chart with the following form:

$$\begin{array}{c|c|c||c|c} p_1 & \cdots & p_n & \cdots & c \\ \hline \mathrm{T} & \cdots & \mathrm{T} & \cdots & \mathrm{F} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \mathrm{F} & \cdots & \mathrm{F} & \cdots & \mathrm{T} \end{array}$$

To the left of the double vertical lines, there are n columns, one for each argument proposition. There is one row for each possible combination of truth assignments to the argument variables. The rightmost column gives the truth value of c for the corresponding truth assignments of its argument propositions. Intermediary columns give the values of smaller compound propositions used to build c.

As described above, the truth table summarizes *all possible* situations that we could encounter, depending on the truth values of the argument propositions. This makes truth tables a convenient tool for reasoning about *how* we can argue (or *prove*) that a proposition is true. We'll summarize these key proof insights in pink boxes below.

1.2.1 Negation ($\neg$)

Negation is a *unary* (one argument) operation. When we negate a proposition, it switches its truth value. For example, if the atomic proposition $p =$ "It is sunny today.", then its negation, written $\neg p$, is the proposition "It is not sunny today." The semantics of negation are described by the truth table:

p	$\neg p$
T	F
F	T

The first row tells us that when p is true $\neg p$ is false. The second row tells us that when p is false $\neg p$ is true. These are the only possibilities, so the truth table completely describes the semantics of $\neg$.

1.2.2 Conjunction ($\wedge$)

Conjunction (along with all of the other operations we'll introduce) is a *binary* (two argument) operation. When we conjoin two propositions, we link them with "and." For example, if $p =$"Today is Wednesday." and $q =$ "Today it is sunny." then their conjunction, written $p \wedge q$, is "Today is Wednesday, and it is sunny." Semantically, this statement is true only when *both* of its arguments are true. We summarize this with the truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Suppose we wish to argue that $p \wedge q$ is true. Looking at the table, we must show we are in the scenario described by the first row, meaning both p and q are true.

> Given propositions p and q, to prove that $p \wedge q$ is true, you must both argue that p is true and also that q is true. Conversely, if you know that $p \wedge q$ is true, you can use this to conclude that p is true and also that q is true.

1.2.3 Disjunction ($\vee$)

Disjunction joins two propositions with an "or." Using our previous example, the disjunction of p and q, written $p \vee q$, is "Either today is Wednesday, or it is sunny (or both)." Semantically, this statement is true as long as *at least one* of its arguments is true. We summarize this with the truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If we want to argue that $p \vee q$ is true, we only need to eliminate the possibility of being in the last row. To do this, it is enough to either show that p is true, or that q is true.

> Given propositions p and q, to prove that $p \vee q$ is true, you must either argue that p is true or argue that q is true. Conversely, if you know that $p \vee q$ is true, this is not enough to conclude that p (specifically) is true or q (specifically) is true.

1.2.4 Biconditional ($\Leftrightarrow$)

The biconditional is our formalization of "if and only if." In our example, the biconditional of p and q, written $p \Leftrightarrow q$, is "Today is Wednesday if and only if it is sunny." Semantically, this statement should be declared true either if it is both Wednesday and sunny, or if it is neither Wednesday nor sunny. If only one of the arguments is true, we declare the biconditional to be false. We summarize this with the truth table:

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

> Given propositions p and q, to prove that $p \Leftrightarrow q$, we must argue that if p is true then q is also true, and separately argue that if q is true then p is also true.

1.2.5 Implication ($\Rightarrow$)

Finally, implication is used to represent statements of the form "if ... then ..." Syntactically, we write $p \Rightarrow q$ to mean "if p then q," or "p implies q." In our example, $p \Rightarrow q =$ "If today is Wednesday, then it is sunny." Semantically, we'd assert that this statement was false if it happened to be Wednesday, but was not sunny. In any other case, we have no evidence that the implication is false, so we declare it to be true. We summarize this with the truth table:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Study this truth table carefully! If the proposition on the left of the implication (the *antecedent*) is false, then the implication is automatically true.

> Given propositions p and q, to prove that $p \Rightarrow q$ is true, it is enough to either argue that p is false or to argue that q is true.

We will talk more specifically about strategies for proving implications in upcoming lectures.

1.3 More Complicated Propositions

Despite appearing relatively simple, the operations we just introduced are powerful because of their *inductive* structure. We defined more complicated propositions in terms of simpler argument propositions; however, nothing requires these arguments to be atomic; they could be other compound propositions. Using this idea, we can compose logical operations to build up more and more complicated propositions that depend on many variables.

Example 1.3 Consider the expression $(p \vee q) \Rightarrow (r \wedge p)$, where p, q, r are atomic. We argue that this is a proposition. The left side of the implication, $p \vee q$ is the disjunction of two atomic propositions, so it is a proposition that we can call s. Similarly, the right side of the implication, $r \wedge p$, is the conjunction of two atomic propositions, so it is a proposition that we call t. Then, our original expression is $s \Rightarrow t$, an implication of two propositions, so it is a proposition.

$$\underbrace{(p \vee q)}_{s} \Rightarrow \underbrace{(r \wedge p)}_{t}$$

We can describe the semantics of this proposition with a truth table. To the left of the double vertical lines are three columns, one for each atomic proposition p, q, r. We'll need 8 rows to account for the 8 possible truth assignments of the atomic propositions. On the right, we'll use columns to build up our proposition piece by piece. We'll add one column for each logical operation that we apply.

p	q	r	$p \vee q$	$r \wedge p$	$(p \vee q) \Rightarrow (r \wedge p)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	F	F	T
F	F	F	F	F	T

To fill out the columns on the right side of the table, we identify the operation that column is performing, find the two columns to its left that represent its arguments, and then consult with the truth tables from Section 1.2 to determine the truth value to write in each row.

R In an earlier math class, you should have learned about an "order of operations" for arithmetic. When you see an expression like "$3+4\cdot 5$", you know that multiplication should take precedence over addition, so this expression is simplified "$3+4\cdot 5=3+20=23$." If we want the operations evaluated in the opposite order, we use parentheses: "$(3+4)\cdot 5 = 7\cdot 5 = 35$." There is also a precedence order for logical operations: $\neg,\wedge,\vee,\Rightarrow,\Leftrightarrow$, so we'd interpret "$\neg p\wedge q \Rightarrow r\vee s$" as $((\neg p)\wedge q)\Rightarrow(r\vee s)$. Even where not strictly necessary, it is useful to clarify complicated compound propositions with parentheses.

1.4 Equivalent Propositions

When writing down logical arguments (in English) different people will articulate the same argument in different ways. It does not mean one of them is wrong. The notion of *logical equivalence* is used to formalize that these alternatives are equally valid.

Definition 1.5 — Logical Equivalence.

Two propositions p and q are equivalent (which we express using the notation $p\equiv q$) if they have the same truth value regardless of the truth values of their atomic propositions. In other words, their columns in a truth table have the same entries.

Example 1.4 Consider the propositions $\neg(p\wedge q)$ and $\neg p\vee\neg q$. The truth tables for these propositions are given by:

p	q	$p\wedge q$	$\neg(p\wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

p	q	$\neg p$	$\neg q$	$\neg p\vee\neg q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Since the columns of these propositions are the same, $\neg(p\wedge q)\equiv\neg p\vee\neg q$.

This logical equivalence is one of De Morgan's Laws.

Lemma 1.1 — De Morgan's Laws. Given propositions p and q,

1. $\neg(p\wedge q)\equiv\neg p\vee\neg q$
2. $\neg(p\vee q)\equiv\neg p\wedge\neg q$

Example 1.5 Given an implication $p\Rightarrow q$, we can obtain its *contrapositive* by negating both propositions and reversing the implication: $\neg q\Rightarrow\neg p$. The contrapositive is logically equivalent to the original implication, as demonstrated by the following truth table:

p	q	$p\Rightarrow q$	$\neg q$	$\neg p$	$\neg q\Rightarrow\neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

We will use this observation soon when we introduce *contrapositive proofs* for implications.

Main Takeaways:

- *Propositions* are statements that can be deemed true or false.
- We can use logical operations to build more complicated *compound* propositions out of simpler *atomic* propositions.
- Truth tables show the truth value of a compound proposition in *all possible* scenarios (truth assignments of its arguments).
- Propositions are *logically equivalent* if they have the same truth value regardless of the truth assignments of their arguments.

Exercises

1.1 Suppose that p,q,r,s are atomic propositions such that p and s are true and q and r are false. Determine the truth value of each of the following compound propositions. Justify your answer by showing the steps of your calculation.

(a) $s \vee \neg q$

(b) $p \vee (\neg q \wedge r)$

(c) $(s \Rightarrow r) \Rightarrow (p \Rightarrow q)$

(d) $\neg(p \wedge \neg r) \Leftrightarrow (q \vee p)$

1.2 Write out a truth table for each of the following compound propositions. Include a column for *each* logical operation.

(a) $p \wedge (q \vee \neg p) \Rightarrow \neg q$

(b) $(p \wedge q) \Rightarrow (r \vee \neg q)$

(c) $(p \Rightarrow q) \wedge (q \Rightarrow r)$

(d) $(p \vee q) \wedge (r \vee s)$

1.3 Consider the following three atomic propositions:

$$p = \text{“Sam makes the lacrosse team.”}$$
$$q = \text{“Sam practices on Saturday.”}$$
$$r = \text{“Sam goes out for ice cream.”}$$

Express the following compound propositions symbolically.

(a) “Sam did not make the lacrosse team.”

(b) “Sam practices on Saturday and goes out for ice cream.”

(c) “Sam makes the lacrosse team or she does not go out for ice cream.”

(d) “If Sam does not practice on Saturday, then she will not make the lacrosse team.”

(e) “It isn’t true that Sam both went out for ice cream and did not make the lacrosse team.”

1.4 Consider the following four atomic propositions.

$$\begin{aligned} p &= \text{“The Buffalo Bills win their game.”} \\ q &= \text{“There is a snow storm.”} \\ r &= \text{“School is canceled on Monday.”} \\ s &= \text{“The children are happy.”} \end{aligned}$$

Write an English interpretation of the following compound propositions.

(a) $q \Rightarrow r$ (b) $p \wedge r$ (c) $\neg r \wedge \neg s$

(d) $(p \vee r) \Rightarrow s$ (e) $s \Leftrightarrow p$ (f) $\neg s \Rightarrow (\neg p \vee (q \wedge \neg r))$

1.5 A binary logical operation $\star$ is *commutative* if $p \star q \equiv q \star p$. Use a truth table to determine whether each of the following operations is commutative.

(a) $\wedge$ (b) $\vee$ (c) $\Rightarrow$ (d) $\Leftrightarrow$

1.6 A binary logical operation $\star$ is *associative* if $(p \star q) \star r \equiv p \star (q \star r)$. Use a truth table to determine whether each of the following operations is associative.

(a) $\wedge$ (b) $\vee$ (c) $\Rightarrow$ (d) $\Leftrightarrow$

1.7 A binary logical operation $\star$ *(left) distributes* over another binary logical operation $\circ$ if $p \star (q \circ r) \equiv (p \star q) \circ (p \star r)$. Use a truth table to determine whether each of the following statements is true.

(a) $\wedge$ distributes over $\vee$. (b) $\vee$ distributes over $\wedge$.

(c) $\Rightarrow$ distributes over $\wedge$. (d) $\vee$ distributes over $\Rightarrow$.

(e) $\wedge$ distributes over $\Leftrightarrow$. (f) $\Rightarrow$ distributes over $\Leftrightarrow$.

1.8 For each of the following sub-exercises, suppose that the given compound proposition is true. Based on this information, select the statement that accurately describes a conclusion we can draw about proposition p.

- p must be true.
- p must be false.
- It is possible that p is true or that p is false.

(a) $p \wedge q$ (b) $p \vee q$ (c) $\neg p \wedge \neg q$ (d) $\neg(p \wedge q)$

(e) $p \Rightarrow q$ (f) $q \Rightarrow p$ (g) $\neg(p \Rightarrow q)$ (h) $\neg(q \Rightarrow p)$

(i) $p \Rightarrow \neg p$ (j) $\neg p \Rightarrow p$ (k) $q \wedge (\neg q \vee p)$ (l) $(p \wedge q) \vee (p \wedge \neg q)$

1.9 *Exclusive-or* ($\oplus$) is another binary logical operation. Semantically, $p \oplus q$ is true when exactly one of its arguments is true.

(a) Write out the truth table for $\oplus$.

(b) Find two propositions that are logically equivalent to $p \oplus q$ but only use $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$.

(c) Is $\oplus$ commutative? (See Exercise 1.5)

(d) Is $\oplus$ associative? (See Exercise 1.6)

(e) Does $\oplus$ distribute over $\wedge$? (See Exercise 1.7)

(f) Does $\wedge$ distribute over $\oplus$?

(g) Does $\oplus$ distribute over $\vee$?

(h) Does $\vee$ distribute over $\oplus$?

1.10 The boolean operations are realized in computer hardware by logical *gates*, which take in two input voltages (F = low voltage, T = high voltage) and output the voltage corresponding to their logical operation (so an AND gate puts out a high voltage exactly when both of its input voltages are high).

In many cases, these logic gates are sold on chips that contain many copies of the same gate. Therefore, to save money, it can often be desirable to rewrite a boolean expression to use fewer *types* of logical operations, even if this increases the *number* of operations.

(a) Explain how you can transform a formula including $\wedge, \vee$, and $\neg$ into a formula involving $\wedge$ and $\neg$ only (Hint: Use DeMorgan's Laws).

(b) Use your answer from part (a) to rewrite the following expressions using only $\wedge$ and $\neg$. For the latter expressions, it may be helpful to first express them in terms of $\wedge, \vee$, and $\neg$.

i. $\neg p \vee \neg q$

ii. $p \vee q \vee r$

iii. $p \Rightarrow q$

iv. $p \Leftrightarrow q$

$\star$ (c) The logic gate NAND, sometimes represented by the symbol $\uparrow$, is defined to be the negation of $\wedge$; that is, $p \uparrow q = \neg(p \wedge q)$. Express the following three logical expressions using only $\uparrow$.

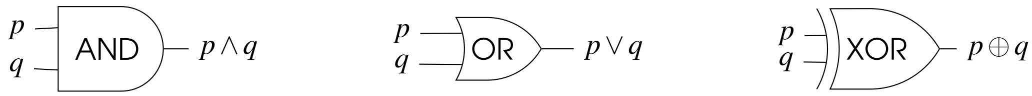
i. $\neg p$

ii. $p \wedge q$

iii. $p \vee q$

Together, parts (b) and (c) show that every logical operation from Section 1.2 can be expressed using only NAND gates.

◇ **1.11** Continuing in the spirit of the previous question, we can interpret all of the calculations in a computer through the lens of logical operations on propositions by interpreting the binary digits as boolean values (so T=1 and F=0). We refer to this view as *boolean algebra*. We can think about the input wires to a computer circuit as atomic propositions and each output wire after passing through a series of logical gates as a compound proposition. We'll focus on three types of gates (see Exercise 1.9).



To design more interesting circuits, we can *compose* these gates, feeding the output of one gate in as an input to another gate.

(a) Draw the circuit with input wires p, q, r that computes the expression $(p \wedge q) \vee (q \oplus r)$.

For the rest of this exercise, we'll consider designing a circuit that can perform addition.

To start, we'll consider the simpler problem of adding together two bits b_0 and b_1, each of which is either 0 or 1. The possible results are $0 = (00)_2$, $1 = (01)_2$, and $2 = (10)_2$, where we use the $(\cdot)_2$ notation to mean the representation as a binary number. Since these outputs can have two binary digits, we will need two output bits. We'll call these s_1 (the sum, or the 1s place of the answer) and c (the carry, or the 2s place of the answer).

(b) Write out a table with rows representing the possible values of the input wires a and b, and there are columns representing the values of s_1 and c corresponding to these inputs. This will look like a truth table but filled with 0s and 1s.

(c) Express s_1 and c as compound propositions in terms of a and b.

(d) Draw the diagram of a circuit with inputs a and b and outputs s_1 and c (along with any necessary gates and intermediary wires). This circuit is called a *half adder*.

To design a more practical addition circuit, we will need to be able to deal with these carry bits. That is, we should consider adding together a and b, along with a carry-in bit c_i. The answer will be between 0 and 3, so it still can be represented with two output bits, s_1 and c_o.

(e) Similar to part (b), write out a table that expresses s_1 and c_o for all possible values of a, b and c_i.

(f) Similar to part (c), express s_1 and c_o as compound propositions in terms of a, b, and c_i.

(g) Draw the diagram of a circuit with inputs a, b, and c_i and outputs s_1 and c_o (along with any necessary gates and intermediary wires). This circuit is called a *full adder*.

(h) Now, suppose that our inputs are larger (say 4-bit) binary numbers $a = (a_4a_3a_2a_1)_2$ and $(b_4b_3b_2b_1)_2$. Explain how we can compute their sum $s = (s_5s_4s_3s_2s_1)_2$ using 4 full adders.

What you have just described in part (h) is a *ripple-carry adder*. This approach gives a valid way to add binary numbers of any size in hardware, but it is not particularly efficient (the input values need to flow through a lot of gates before reaching the output). More clever tricks can be used to design a more efficient logical circuit (look up *carry-lookahead adders* if you are interested).

⋆ **1.12** (Based on the puzzles of Raymond Smullyan) In the magical swamp, statements made by toads are always true and statements made by frogs are always false. Unfortunately, since you have devoted your studies to computer science rather than biology, you cannot tell the species apart. Rather, you must use their statements to logically deduce whether they are frogs or toads.

(a) Suppose that the swamp creature A states proposition p. If we let q represent the proposition "A is a toad.", explain why the proposition $p \Leftrightarrow q$ must be true.

(b) On your walk in the swamp, you encounter two creatures, A and B, who make the following statements:

A : "B is a frog."
B : "A and I are from the same species."

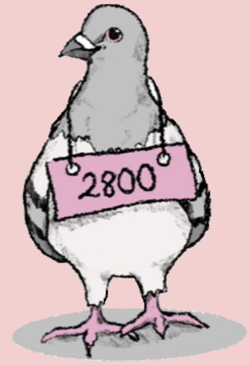
Based on this exchange, can you determine whether A is a frog or a toad? How about B?

(c) Repeat part (b) with the following dialog:

A : "B and I are from different species."
B : "A is a toad."

(d) Same questions, but now there are three creatures.

A : "B and C are from the same species."
B : "A and C are from the same species."
C : "B is a frog."



2. Sets and Predicates

In the first lecture, we introduced propositions as statements with an inherent truth value. We discussed the syntax and semantics of various logical operations that form new, more complicated propositions from simpler ones, and some strategies for verifying that these propositions are true.

On its own, propositional logic has limited use. Atomic propositions express facts about *one specific* object, whereas we are often interested in establishing facts about entire *families* of objects. For example, suppose that we wish to prove the following claim:

Every odd (natural) number can be written as the difference of two perfect squares.

If we tried to write this as a proposition, we'd end up with something like

$$\big(1 \text{ is the difference of perfect squares}\big) \wedge \big(3 \text{ is the difference of perfect squares}\big) \wedge \ldots,$$

a conjunction of infinitely many atomic propositions. We cannot reason about this statement with a truth table. Instead, we must develop a new way to represent collections of objects and their properties. We will do this with *sets* and *predicates*. In the next lecture, we'll see how another operation, *quantification*, can be used to construct propositions that express facts about these collections, such as our above claim.

2.1 Sets

Sets are the fundamental building blocks of mathematics. In the language of mathematics, sets are nouns. They allow us to organize, categorize, count, and classify. Everything we'll learn in the course can be traced back to sets. The intuition behind a set is rather straightforward; *mathematical objects* are anything that we are interested in studying (numbers, words, a deck of cards, a *function*), and a set collects together these objects.

Definition 2.1 — Set, Elements, Inclusion.

A *set* is an unordered collection of unique mathematical objects. The objects contained within the set are its *elements*.

When an object s is contained in S, we write $s \in S$, which we read "s is an element of S" or "s in S."

When a set S is finite, we denote the number of elements in S by $|S|$.

We typically use uppercase letter variables to represent sets. Lowercase letter variables are used to talk about generic elements within sets, and we match the letters whenever possible (e.g., an element $s \in S$ or $a \in A$). When a set S is small, we can express its contents as a comma-separated list in curly brackets.

Example 2.1

- The set A of integers between 1 and 9 (inclusive) is written $A = \{1,2,3,4,5,6,7,8,9\}$. Since sets are unordered, we could also write, for example, $A = \{2,7,6,3,8,5,4,1,9\}$.
- The set T of CS 2800 Fall 2023 head TAs is $T = \{$Alkim, Ambrose, Annabel, Jiho, Sean, Tuni $\}$. Note that $|T| = 6$.
- We may view a deck of cards as a set $C = \{2\spadesuit, 3\spadesuit, \dots, \mathrm{A}\diamondsuit\}$, with $|C| = 52$.
- The elements in a set can be more complicated objects, such as other sets. For example, the set $\{\{1,3\},\{1\}\}$ has two elements: the 2-element set containing 1 and 3 and the 1-element (singleton) set containing 1.

There are two key features in the definition of a set: it is unordered, and its elements are unique. These features distinguish sets from two other mathematical structures.

Definition 2.2 — Multiset.

A *multiset* is an unordered collection of possibly repeated elements.

We often denote a multiset using "decorated" brackets, such as $\{\!\!\{1,1,2,3,3,3\}\!\!\}$.

Definition 2.3 — Sequence.

A *sequence* is an ordered collection of (possibly repeating) elements.

Sequences can have finite or infinite length (finite sequences are sometimes called *tuples*) and are often written surrounded by parentheses (alternatively, sometimes square brackets are used), such as $(2,4,5,2,3)$. Since sequences are ordered, $(2,4,5,2,3) \neq (2,2,3,4,5)$.

Before moving on, we list special notations for some common sets.

Example 2.2 — Notations for Common Sets.

- The *empty set* contains no elements and is denoted by the symbol $\varnothing$.

- The (infinite) set of all natural numbers $\{0,1,2,\ldots\}$ is denoted $\mathbb{N}$. When we wish to exclude 0, we write $\mathbb{N}^{+}$.
- We use similar "blackboard" notation for the infinite sets of integers ($\mathbb{Z}$), rational numbers ($\mathbb{Q}$), and real numbers ($\mathbb{R}$).
- Given $n \in \mathbb{N}$, $[n]$ refers to the set $\{1,2,\ldots,n\}$. Note that $\big|[n]\big| = n$. As a special case, $[0] = \varnothing$.

In this course, we always adopt the convention to include 0 in the natural numbers!

2.2 Predicates

Let's return to the example that we gave at the start of the lecture. We can imagine writing atomic propositions such as

$$\begin{aligned} p &= \text{“1 can be written as the difference of two perfect squares.”} \\ q &= \text{“3 can be written as the difference of two perfect squares.”} \\ &\vdots \end{aligned}$$

Each proposition has the same structure but has a different number at the beginning. We can imagine using a variable x as a placeholder for this number, giving the statement

$$P(x) = \text{“}x \text{ can be written as the difference of two perfect squares.”}$$

This statement $P(x)$ doesn't have a truth value (that is, it is not a proposition); it can be true or false depending on which integer we plug in for x. We call such a statement a *predicate*.

Definition 2.4 — Predicate, Variable, Domain.

A *predicate* $P(x_1,\ldots,x_n)$ is a statement that is parameterized by *variables* $x_1,\ldots,x_n$. When we assign (or *bind*) a specific value to each variable, the statement evaluates to either true or false. The *domain* D_i of each variable x_i is the set of its possible values.

We say that the variables in a predicate are *free* since they can represent many different values. They become *bound* once they are assigned to a particular value. In this terminology, a predicate becomes a proposition once *all of* its free variables become bound. We typically represent predicates with uppercase letters (P,Q,R) and use lowercase letters for their parameter variables. We will always assume that all the variables in the predicate's parameter list appear in its statement. When you define a predicate, you need to clarify the domain of each of its variables. This will be especially important when we discuss quantifiers in the next lecture.

Example 2.3 $O(x) =$ "x is an odd number." is a predicate with one free variable, x, with domain $\mathbb{Z}$. When we bind the variable x to a particular integer (say 4), we obtain a proposition $O(4) =$ "4 is an odd number," which is false.

Things are more nuanced when a predicate has more than one variable.

Example 2.4 Consider the statement $G(x,y) =$"$x > y$". This statement has two (free) variables, x and y, which we can interpret as having the domain $\mathbb{R}$. If we fix the values of both variables (say $x = 1.2$ and $y = 0.87$), we obtain an inequality of two numbers, $G(1.2, 0.87) =$ "$1.2 > 0.87$", which is either true or false (that is, a proposition), so $G(x,y)$ is a predicate.

Now, suppose that we have fixed the value of $y = 0.87$, but allow x to remain free. This gives $G(x, 0.87) =$ "$x > 0.87$". The truth value of this statement depends on the value that we assign to x. Thus, $G(x, 0.87)$ is a predicate in the variable x (only), which we can notate using a new letter, say $H(x)$. We can similarly fix $x = 1.2$ but allow y to remain free, giving a predicate $J(y)$ in only variable y.

This example illustrates the following facts about predicates.

- When we bind all variables in a predicate, we fix its truth value; it becomes a proposition.
- If we bind only some variables in a predicate, it remains a predicate, but only in the variables that remain free.

In the next lecture, we'll see that *quantifiers* provide another way to bind variables. For the rest of this lecture, we will see how to use predicates to define new sets.

2.3 Restriction and Set Builder Notation

Definition 2.5 — Set Builder Notation.

Given a set S and a predicate $P(s)$, the set $R = \{s \in S \mid P(s)\}$ contains exactly the elements of S that make P true.

In set-builder notation, we must include the set S to the left of the vertical bar, as this specifies the domain of the predicate. To the right of the bar is a comma-separated list of the properties (predicates) that must *all* be satisfied.

Example 2.5

- The odd numbers form a set, $O = \{z \in \mathbb{Z} \mid z \text{ is odd }\} = \{z \in \mathbb{Z} \mid z = 2y + 1 \text{ for some } y \in \mathbb{Z}\}$.
- The irrational numbers form a set, $\mathbb{I} = \{r \in \mathbb{R} \mid r \notin \mathbb{Q}\}$. We draw a slash through the symbol $\notin$ to mean "not in".
- $P = \{n \in \mathbb{N} \mid n < 20, n \text{ prime}\} = \{2, 3, 5, 7, 11, 13, 17, 19\}$ is a set.

In this last example, we interpret the comma in the condition "$n < 20$, n prime" as an "and"; we require that both conditions are true for n to belong to set P. In general, we see that a restriction R of S contains some elements of S (unless no $s \in S$ satisfies the conditions), but likely not all of S. We say that R is a *subset* of S.

Definition 2.6 — Subset, Superset.

A set R is a *subset* of a set S, notated $R \subseteq S$ if every element of R is an element of S. In this case, we also say that S is a *superset* of R.

Example 2.6 Consider the set $S = [9] = \{1,2,3,4,5,6,7,8,9\}$. The following are all subsets of S.

- The one-digit prime numbers, $\{2,3,5,7\}$.
- The one-digit even numbers, $\{2,4,6,8\}$.
- The singleton set containing 5, $\{5\}$.
- The set S, itself.
- The empty set, $\varnothing$.

The last two examples are the edge cases. Let's confirm that they satisfy the definition. First, $S \subseteq S$ since (tautologically) every element of S is an element of S. Next, for $\varnothing$, the definition holds *vacuously*. Since $\varnothing$ contains no elements, it is true that every element in $\varnothing$ is also in S.

Main Takeaways:

- *Sets* are unordered collections of unique objects.
- We list the elements of small sets in curly braces. We can use *set builder notation* to write down large (even infinite) sets.
- *Predicates* are statements parameterized by *variables* that can assume any value in their *domain*. When all variables in a predicate are *bound*, it becomes a proposition.
- A subset $R \subseteq S$ contains only elements that belong to S.

Exercises

2.1 Use set builder notation to give a formal description of the following sets.

(a) Negative integers

(b) Multiples of 9

(c) Even integers that are not multiples of 4

(d) Positive integers with exactly two digits

(e) The set $[n]$ for some $n \in \mathbb{N}$

2.2 Give a complete description (i.e. write out all elements) of the following sets.

(a) Multiples of 13 between 1-100.

(b) Subsets of $\{a,b,c\}$

(c) $\{n \in \mathbb{N} \mid n = 10 - m^2 \text{ for some } m \in \mathbb{N}\}$

(d) Sequences of length 3 with entries from $\{H,T\}$.

(e) Subsets of $\{\spadesuit,\clubsuit,\heartsuit,\diamondsuit\}$ that contain at most 2 elements.

(f) $\{q \in \mathbb{Q} \mid 0 \leq q \leq 1, 7q \in \mathbb{Z}\}$

2.3 Give an example of three sets A, B, C with the following properties:

- $A \subseteq C$
- $B \subseteq C$
- $A \not\subseteq B$
- $B \not\subseteq A$

2.4 Consider the following 3 sets:

$$A = \{\varnothing\} \qquad B = \{\varnothing, \{\varnothing\}\} \qquad C = \{\{\varnothing, \{\varnothing\}\}\}$$

Determine whether each of the following mathematical statements is true or false.

(a) $A \in B$
(b) $A \subseteq B$
(c) $A \in C$
(d) $A \subseteq C$
(e) $B \in C$
(f) $B \subseteq C$
(g) $\varnothing \in A$
(h) $\varnothing \subseteq A$
(i) $\varnothing \in C$
(j) $\varnothing \subseteq C$

2.5 In this exercise, you'll think about all possible subsets of some small sets.

(a) Write down all subsets of $[2] = \{1,2\}$. How many subsets are there?

(b) Write down all subsets of $[3] = \{1,2,3\}$. How many subsets are there?

(c) In general, how many subsets will $[n]$ have? Does your formula work for $n = 0$ and $n = 1$? Can you explain why your formula is correct?

2.6 Given any $n \in \mathbb{N}$, define the set

$$M_n = \left\{ z \in \mathbb{Z} \mid z = yn \text{ for some } y \in \mathbb{Z} \right\}.$$

(a) Explain in words the numbers that are contained in M_n.

(b) Find a predicate $P(k,n)$ such that $M_k \subseteq M_n \Leftrightarrow P(k,n)$ is true.

2.7 For each of the following predicates, give two examples of how the variables can be assigned to make the predicate true and two examples of how they can be assigned to make the predicate false.

(a) $P(x) =$ "$x > 3$, x is odd" $\qquad x \in \mathbb{Z}$

(b) $Q(x,y) =$ "$2x + 3y = 12$" $\qquad x, y \in \mathbb{N}$

(c) $R(m,n) =$ "$(m = 12) \Rightarrow (n = 4)$" $\qquad m, n \in \mathbb{Q}$

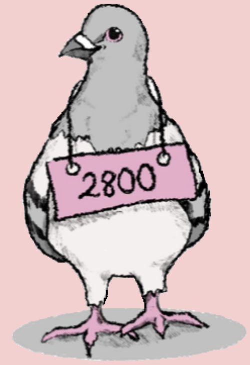
⋆ **2.8** In the previous lecture, we defined the logical operations $\{\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow\}$ over propositions. That is, each of these operations took in one or two propositions as arguments and returned a new, more complex proposition. We can extend these operations to predicates. Syntactically, they behave the same, but we must define the semantics. For example:

> Given a predicate $P(x)$ where x has domain D, we interpret its negation $(\neg P)(x)$ to be the predicate defined such that for each $d \in D$, the proposition $(\neg P)(d) = \neg(P(d))$. That is, every $d \in D$ that makes $P(d)$ true makes $\neg P(d)$ false, and vice versa.

(a) Describe the semantics of the remaining binary logical operations on predicates $P(x)$ and $Q(x)$ with $x \in D$.

(b) Describe how you can extend the semantics from part (a) to work for predicates $P(x_1, \ldots, x_n)$ and $Q(x_1, \ldots, x_n)$ with *any number* of parameters (but where P and Q have the *same* parameters). You should introduce notation to describe the domains of these parameters.

(c) What happens if P and Q have different parameters? As an example, think about how you would define the conjunction $P \wedge Q$ of two predicates $P(x)$ and $Q(y)$ where x has domain D_1 and y has domain D_2.



3. Quantifiers

In the last lecture, we introduced predicates, logical statements parameterized by one or more (*free*) variables. By assigning (or *binding*) each variable to a specific value in its domain, we can turn a predicate into a proposition and ascertain its truth value. In today's lecture, we'll see another way to bind variables in predicates using *quantifiers*. Almost every statement we will prove throughout the course includes one or more quantifiers; being able to write, parse, and verify quantified statements will be an essential skill.

3.1 The Existential Quantifier

A predicate $P(x)$ can be either true or false for each possible value of x in its domain. The *existential* quantifier asserts that at least one assignment of x makes $P(x)$ true.

Definition 3.1 — Existential Quantifier.
To *existentially quantify* a predicate $P(x)$ where variable x has domain D, we write

$$\exists\, x \in D.\; P(x),$$

which asserts that there is some element $d \in D$ such that assigning $x = d$ makes $P(x)$ true (i.e., $P(d)$ is a true proposition). We read this "There exists $x \in D$ such that $P(x)$ (is true)."

R In this definition, we chose the letters carefully to distinguish two things: the *variable* x and the domain element d. Just as in coding, variables are named entities that reference some underlying value, so we think of x as being a placeholder that can be assigned any value from the domain set D. The letter d is referring directly to some element in D. Usually, though, this distinction is swept aside and we write things like "$\exists\, x \in X$" with matching letters and think of x as both the variable and the element from set X.

When we quantify a variable in a predicate, it binds that variable. Just as before, once all of the variables in a predicate have been bound, we are left with a proposition.

Example 3.1

- The expression "$\exists\, x \in \mathbb{N}.\, x^2 = 4$" is a proposition that is true. This is because setting $x = 2$ makes $x^2 = 2^2 = 4$.
- The expression "$\exists\, x \in \mathbb{N}.\, x^2 = 3$" is a proposition that is false. No natural number squares to the number 3.

Similar to our discussion last lecture, we can also consider predicates with more than one free variable. Each quantifier binds one variable, so there are cases where after quantification, we may still be left with a predicate (in one fewer variable).

Example 3.2 The statement "$\exists\, x \in \mathbb{N}.\, x^2 = y$" is a predicate in one variable, $y \in \mathbb{N}$. The innermost expression "$x^2 = y$" is a predicate in two variables. Although we have bound the variable x by the existential quantifier, the variable y remains free. This expression is equivalent to the predicate $P(y) =$ "y is a perfect square."

We make the following observations about proving existentially quantified statements:

- To argue that an existential statement $\exists\, x \in D.\, P(x)$ is true, it suffices to find an *example*, one value $d \in D$ for which $P(d)$ is true.
- To argue that an existential statement $\exists\, x \in D.\, P(x)$ is false, we must show that there is no $d \in D$ for which $P(d)$ is true. In other words, we must show that for any *arbitrary* $d \in D$, $P(d)$ is false.

Example 3.3

- Suppose that we wish to prove that the statement "$\exists\, x \in \mathbb{N}.\, \big(x \geq 4 \,\wedge\, x^2 - 5 \text{ is prime}\big)$" is true. Doing this is very straightforward. If we let $x = 4$, we can verify that $4 \geq 4$ and that $4^2 - 5 = 16 - 5 = 11$ is prime.
- Suppose that we wish to argue that the statement "$\exists\, x \in \mathbb{N}.\, \big(x \geq 4 \,\wedge\, x^2 - 4 \text{ is prime}\big)$" is false. Supplying a single piece of evidence (for example, noting that $7^2 - 4 = 45$, which is not prime) is *not enough*. All this tells us is that our particular choice of x does not satisfy the predicate. How would we know that a different choice also will not? As an alternate approach, note that we can factor $x^2 - 4 = (x-2)(x+2)$. Given *any* natural number $x \geq 4$, this shows us that $x^2 - 4$ is the product of two natural numbers ≥ 2, so x is not prime. In other words, *no matter which* x we consider, we know that it will not make the predicate true. This allows us to conclude that our existential statement is false.

3.2 The Universal Quantifier

The *universal* quantifier asserts that *every* possible assignment of x to an element of D makes $P(x)$ true; that is, the predicate holds *universally*.

Definition 3.2 — Universal Quantifier.
To *universally quantify* a predicate $P(x)$ where variable x has domain D, we write

$$\forall\, x \in D.\ P(x),$$

which asserts that for every element $d \in D$, assigning $x = d$ makes $P(x)$ true. We read this "For all $x \in D$, $P(x)$ (is true)."

Example 3.4

- The expression "$\forall\, x \in \mathbb{N}.\ x^2 \in \mathbb{N}$" is a proposition that is true. Squaring a natural number always results in a natural number.
- The expression "$\forall\, x \in \mathbb{N}.\ x \geq 1$" is a proposition that is false. Remember that 0 is a natural number, so taking $x = 0$ makes the predicate false.
- The expression "$\forall\, x \in \mathbb{N}.\ x^2 = y$" is a predicate in one variable, $y \in \mathbb{N}$. The innermost expression "$x^2 = y$" is a predicate in two variables. Although we have bound the variable x by the universal quantifier, the variable y remains free. This expression is equivalent to the predicate $P(y) =$ "y is equal to the square of every natural number," which is certainly false for all $y \in \mathbb{N}$.

We make the following observations about proving universally quantified statements:

- To argue that a universal statement $\forall\, x \in D.\ P(x)$ is true, we must show that every way to assign $x = d$ makes $P(x)$ true. To do this, we argue that $P(d)$ is true for an *arbitrary* $d \in D$.
- To argue that a universal statement $\forall\, x \in D.\ P(x)$ is false, it suffices to find a single *counterexample*, one value $d \in D$ for which $P(d)$ is false.

In general, it is more difficult to show that a universal statement is true than false. To show it is true, we need to consider every element in the domain, whereas to show it is false, we only need to find a single piece of evidence which we call a *counterexample*.

R It may seem at first glance that $\forall$ statements are much stronger than $\exists$ statements. Formally, this would mean that $\big(\forall\, x \in D.\ P(x)\big) \Rightarrow \big(\exists\, x \in D.\ P(x)\big)$, with the intuition that if $P(x)$ holds for every possible value of x, then it definitely holds for one particular value of x. This is almost always true, except in one corner case. What happens if the domain $D = \varnothing$? In this case, an existential statement over domain D is false. Since the domain is empty, there is no way to choose $x \in D$ to make $P(x)$ true. However, *any* universal statement quantified over an empty domain is true. Since there are no x in the domain to refute $P(x)$, it is *vacuously* true that every $x \in D$ satisfies $P(x)$.

3.3 Statements with Multiple Quantifiers

Just as in propositional logic, our definition of quantifiers permits multiple quantifiers to appear in the same expression. In some earlier examples, we saw that if we quantify a predicate but leave at least one free variable, we are left with another predicate (which we can in turn quantify). When a statement has multiple quantifiers, we interpret them from right to left. Then, to reason about their truth value, we work from left to right. This is best illustrated with some examples.

Example 3.5 We consider the following statement with two quantifiers:

$$\forall\, x \in \mathbb{N},\ \exists\, y \in \mathbb{N}.\ y > x.$$

Let's interpret this piece by piece, working from right to left. "$y > x$" is a predicate in two variables. When we existentially quantify this predicate, "$\exists\, y \in \mathbb{N}.\ y > x$", we end up with a predicate in variable x only. This predicate asserts that there is some natural number (y) greater than x. Finally, we universally quantify x in this predicate. In all, we read this statement "For each natural number x, there is some natural number y that is greater than x."

Now, we'd like to give a proof of this statement. This is the first time in the course that we are asking for this type of formal reasoning, which can be a bit jarring. Nevertheless, we can do this by breaking down this statement and reasoning about the quantifiers from left to right using the strategies that we outlined in the pink boxes above. As we go on in the course, we will develop a large toolbox of tips and techniques for proving mathematical statements, but really the best way to get comfortable understanding and writing proofs is to jump in.

Proof. $\underbrace{\text{Given any (arbitrary) } x \in \mathbb{N}}_{\forall\, x\in\mathbb{N}},\underbrace{\text{we take } y = x+1 \in \mathbb{N}}_{\exists\, y\in\mathbb{N}}$, so that $y = x+1 > x$. ■

These proofs are all about "gives and takes." If we're proving a statement is true, then when we see a universal quantifier, we imagine being given any element from that variable's domain. When we see an existential quantifier, we are free to choose (take) any value that we'd like for that variable which will help us satisfy the predicate.

Example 3.6 We'll contrast the previous example with a similar-looking statement:

$$\exists\, y \in \mathbb{N},\ \forall\, x \in \mathbb{N}.\ y > x.$$

Again, let's interpret this statement from the inside out. "$y > x$" is the same predicate in two variables. When we universally quantify this predicate, "$\forall\, x \in \mathbb{N}.\ y > x$", we end up with a predicate in variable y only. This predicate asserts that y is greater than every natural number. Finally, we existentially quantify y in this predicate. In all, we read this statement "There is some natural number y such that for each natural number x, y that is greater than x." Said more succinctly, "There is a natural number that is greater than every natural number." We argue that this statement is false, again reasoning from left to right.

Proof. $\underbrace{\text{Given any (arbitrary) } y \in \mathbb{N}}_{\exists\, y\in\mathbb{N}},\underbrace{\text{we take } x = y \in \mathbb{N}}_{\forall\, x\in\mathbb{N}}$, so that $y = x \not> x$. ■

Notice how all of the strategies are "flipped" when we wish to show a statement is false. Now, the existential quantifiers become the "gives" and the universal quantifiers become the "takes". This is related to De Morgan's Laws for Quantifiers, that we'll see in a little while.

R The previous two examples show that the order of the quantifiers matters. Swapping quantifiers can change the truth value of a statement. Often times, definitions or theorems will contain many quantifiers, so an important skill is being able to track quantifiers and understand the order that they are applied.

3.4 Predicate Calculus

In our first lecture, we saw that since logical operations transform propositions into other propositions, we can repeatedly apply these operations to build up arbitrarily complicated propositions. We also saw that there were often many different formulas that were *logically equivalent*, and we used this idea to formulate properties and identities such as De Morgan's Laws, which we could verify using truth tables.

The same phenomenon holds for logical statements involving predicates and quantifiers. The formal system that encapsulates all of this is referred to as *predicate calculus* or *first order logic*. This system includes the definitions of all of the operations, both their syntax and their semantics (the semantics of the logical operations acting on predicates is explored in Exercise 2.8). Within the system, we can also derive many identities that allow us to transform logical expressions into equivalent logical expressions. These identities include the properties of propositions (associativity, distributivity, De Morgan's Laws, etc.) from two lectures ago that we can prove with truth tables. When logical equivalences include predicates and quantifiers, truth tables are no longer sufficient; we cannot enumerate all possible assignments of the variables (which may have infinite domains) in different rows. Instead, we will need to use the proof strategies that we listed above in pink, along with some more proof techniques that we will learn about soon. For example, proving De Morgan's Laws for Quantifiers requires a technique known as a *proof by contradiction*.

Lemma 3.1 — De Morgan's Laws for Quantifiers. Given a predicate $P(x)$ with $x \in X$:

- $\neg\left(\exists\, x \in D.\ P(x)\right) \quad \equiv \quad \forall\, x \in D.\ \neg P(x).$
- $\neg\left(\forall\, x \in D.\ P(x)\right) \quad \equiv \quad \exists\, x \in D.\ \neg P(x).$

For now, we will focus on explaining informally why the equivalences hold. Try to return to these notes and develop a more rigorous proof once we have explored this new proof technique. At a high level, the first identity says that if there is no $x \in D$ that makes $P(x)$ true, then it must be the case that every $x \in D$ makes $P(x)$ false. The second identity says that if it isn't true that every $x \in D$ makes $P(x)$ true, then there must be some $x \in D$ that makes $P(x)$ false.

We would hope to someday fill our proof toolbox and have a comprehensive list of proof strategies that could be used to reason about any proposition. Unfortunately, such a toolbox is unattainable. The celebrated results in logic known as *Gödel Incompleteness Theorems* tell us that any logical system that we develop will be unable to prove or disprove every statement. While this can be unsettling, it means that mathematics will always remain a fruitful area where we can pose new questions and search for novel ways to answer them.

Main Takeaways:

- When we quantify a predicate, we remove its dependence on that variable, resulting in either a proposition or a predicate in fewer variables.
- The existential quantifier ($\exists$) asserts that there is some way to fix the variable to make the predicate true.
- The universal quantifier ($\forall$) asserts that the predicate is true no matter how we fix the variable.
- Given a statement with multiple quantifiers, we reason about it by working from right to left and argue its truth value by reasoning from left to right.

Exercises

3.1 Determine whether each of the following expressions is a predicate or a proposition. Identify which variables are free and which are bound.

(a) $P(x) \wedge Q$

(b) $P(x,y) \Rightarrow Q(z)$

(c) $\forall\, x \in X.\ (P(x) \vee Q(x))$

(d) $\forall\, x \in X.\ (P(x,y) \vee Q(y))$

(e) $\forall\, x \in X,\ \exists\, y \in Y.\ P(x,y)$

(f) $\forall\, x \in X,\ \exists\, y \in Y.\ P(x,y,z)$

3.2 Use De Morgan's Laws to negate the following logical expressions, moving the negations as far inward as possible.

(a) $\exists\, x \in X.\ \big(P(x) \vee Q(x)\big)$

(b) $\forall\, x \in X.\ \big(P(x) \wedge (Q(x) \vee R(x))\big)$

(c) $\forall\, x \in X,\ \exists\, y \in Y.\ \big(P(x,y) \wedge Q(x,y)\big)$

(d) $\exists\, x \in X.\ \Big(\big(\forall\, y \in Y.\ P(x,y) \vee Q(y)\big) \wedge \big(\exists\, z \in Z.\ R(x,z) \vee S(z)\big)\Big)$

3.3 Write out the property $S \subseteq A$ formally, using only variables and logical symbols.

3.4 For each of the following subparts, let us define the sets $X = \{1,2,3,5,8\}$ and $Y = \{3,4,5,6,10\}$. Determine whether each of the following propositions is true or false, justifying your answer with a proof or (counter)example.

(a) $\exists\, x \in X.\ x \in Y$

(b) $\exists\, x \in X.\ x \notin Y$

(c) $\forall\, x \in X,\ \exists\, y \in Y.\ y > x.$

(d) $\exists\, y \in Y,\ \forall\, x \in X.\ y > x.$

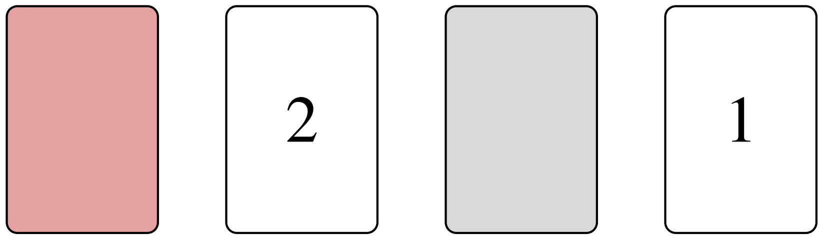
(e) $\forall\, x \in X,\ \exists\, y \in Y.\ y < x$

(f) $\exists\, y \in Y,\ \forall\, x \in X.\ y < x$

(g) $\exists\, x \in X,\ \exists\, y \in Y.\ x = y + 3$

(h) $\forall\, x_1 \in X,\ \exists\, x_2 \in X,\ \exists\, y \in Y.\ x_1 + x_2 = y$

3.5 In a deck of cards, one side of each card is either red or gray, and the other side has either the number 1 or 2. Alice selects four (not necessarily unique) cards from this collection and places them on the table in front of Bob. He sees:



Alice makes the following statement about these four cards: "Every card with a two on it is red on the other side."

(a) If Alice wishes to prove that her statement is correct, what is the fewest number of cards that she will need to flip over? Which cards?

(b) If Bob wishes to show that Alice's statement is incorrect, which cards should he flip over?

(c) Formalize this scenario in symbols by introducing any predicates, variables, and domain sets that you need to model Alice's statement as a proposition. Then explain how you would go about proving or disproving this proposition.

3.6 Translate the following logical expressions into English. Then, determine whether they are true or false.

(a) $\exists\, x \in \mathbb{N}.\ 3x = 6$

(b) $\forall\, x \in \mathbb{N}.\ x > 13$

(c) $\exists\, x \in \mathbb{N},\ \exists\, y \in \mathbb{N}.\ x + y = 71$

(d) $\forall\, x \in \mathbb{N},\ \exists\, y \in \mathbb{N}.\ x + y = 1{,}003$

(e) $\exists\, x \in \mathbb{N},\ \forall\, y \in \mathbb{N}.\ x \leq y$

(f) $\forall\, y \in \mathbb{N},\ \exists\, x \in \mathbb{N}.\ x \leq y$

(g) $\forall\, x \in \mathbb{N},\ \forall\, y \in \mathbb{N},\ \exists\, z \in \mathbb{N}.\ x + y = z$

(h) $\exists\, x \in \mathbb{N},\ \forall\, y \in \mathbb{N},\ \exists\, z \in \mathbb{N}.\ x + y = z$

(i) $\exists\, x \in \mathbb{N},\ \forall\, y \in \mathbb{N},\ \forall\, z \in \mathbb{N}.\ x + y = z$

(j) $\forall\, x \in \mathbb{N},\ \exists\, y \in \mathbb{N},\ \forall\, z \in \mathbb{N}.\ x + y = z$

3.7 Determine whether each of the following is a valid logical identity. If it is, justify with a proof. If it is not, provide a counterexample. Is one "direction" of each identity true? In other words, if $p \not\equiv q$, is it still always true that $p \Rightarrow q$ or $q \Rightarrow p$?

Note: When providing a counterexample, you need to specify all of the involved objects, which means choosing a specific domain set for each quantified variable and a specific predicate or predicates.

(a) $\exists\, x \in X,\ \exists\, y \in Y.\ P(x,y) \quad \equiv \quad \exists\, y \in Y,\ \exists\, x \in X.\ P(x,y)$

(b) $\forall\, x \in X,\ \forall\, y \in Y.\ P(x,y) \quad \equiv \quad \forall\, y \in Y,\ \forall\, x \in X.\ P(x,y)$

(c) $\forall\, x \in X,\ \exists\, y \in Y.\ P(x,y) \quad \equiv \quad \exists\, y \in Y,\ \forall\, x \in X.\ P(x,y)$

(d) $\forall\, x \in X,\ \exists\, y \in Y.\ P(x,y) \quad \equiv \quad \forall\, y \in Y,\ \exists\, x \in X.\ P(x,y)$

(e) $\exists\, x \in X,\ \forall\, y \in Y.\ P(x,y) \quad \equiv \quad \exists\, y \in Y,\ \forall\, x \in X.\ P(x,y)$

(f) $\exists\, x \in X.\ \big(P(x) \wedge Q(x)\big) \quad \equiv \quad \big(\exists\, x \in X.\ P(x)\big) \wedge \big(\exists\, x \in X.\ Q(x)\big)$

(g) $\forall\, x \in X.\ \big(P(x) \wedge Q(x)\big) \quad \equiv \quad \big(\forall\, x \in X.\ P(x)\big) \wedge \big(\forall\, x \in X.\ Q(x)\big)$

(h) $\exists\, x \in X.\ \big(P(x) \vee Q(x)\big) \quad \equiv \quad \big(\exists\, x \in X.\ P(x)\big) \vee \big(\exists\, x \in X.\ Q(x)\big)$

(i) $\forall\, x \in X.\ \big(P(x) \vee Q(x)\big) \quad \equiv \quad \big(\forall\, x \in X.\ P(x)\big) \vee \big(\forall\, x \in X.\ Q(x)\big)$

3.8 Let us define the predicate

$$M(x,y) = \text{“}\exists\, k \in \mathbb{N},\ x = k \cdot y,\text{”}$$

where both x and y have domain $\mathbb{N}$.

(a) Describe in words the predicate M.

(b) We say that a natural number $n > 1$ is *prime* if its only natural number factors are 1 and n.

Define the predicate $P(n) =$ “n is prime” using only mathematical symbols (no words). Your answer should incorporate the predicate M.

(c) We say that two natural numbers m and n are *relatively prime* if their only common natural number factor is 1.

Define the predicate $R(m,n) =$ “m and n are relatively prime” using only mathematical symbols (no words). Your answer should incorporate the predicate M.

⋆ (d) We say that a number $g \in \mathbb{N}$ is the *greatest common divisor* (gcd) of $m,n \in \mathbb{N}$ if it is the largest number that is a factor of both m and n.

Define the predicate $G(g,m,n) =$ “$g = \gcd(m,n)$” using only mathematical symbols (no words).

4. Operations on Sets

Now that we have finished our discussion of logic — propositions, predicates, and quantifiers — we are ready to put these ideas into practice, using logical reasoning to formally describe mathematical objects and operations and prove assertions about their properties. As a first example of a setting where we can practice this reasoning, we will return to our discussion of sets. In this lecture, we'll introduce many operations we can perform on sets. Next lecture, we'll use these definitions to formulate additional properties of sets that we will formally verify with a collection of proof techniques.

4.1 Restriction Operations

A set *operation* is a rule for constructing a new set from one or more other sets. We have already seen one example of such an operation two lectures ago, *set restriction*. Recall that when we restrict a set A on a predicate P (with domain A), we filter out some of its elements, leaving only those that make P true. We are left with a new set S that is a subset of A. By choosing to restrict A on particular predicates involving a new set B, we can define two new *binary* set operations: *set intersection* and *set difference*.

4.1.1 Set Intersection

The intersection of two sets A and B contains all of the elements that belong to both sets.

Definition 4.1 — Set Intersection.

Given two sets A and B, their *intersection*, denoted $A \cap B$, consists of all elements that belong to both A and B. In set builder notation, we have

$$A \cap B = \{a \in A \mid a \in B\} = \{b \in B \mid b \in A\}.$$

Example 4.1 Suppose that $A = \{1,3,5\}$ and $B = \{1,2,3,4\}$. Then, $A \cap B = \{1,3\}$ since 1 and 3 are the only elements that belong to both A and B.

An immediate consequence of this definition is that $(A \cap B) \subseteq A$ and $(A \cap B) \subseteq B$. This is because given any element $x \in (A \cap B)$, the definition tells us that $x \in A$ and that $x \in B$, and these are exactly the implications that appear in the subset definition.

When the intersection of two sets is the empty set, the sets have no elements in common, and we say that the sets are *disjoint*.

Example 4.2 The sets $S = \{1,4,5,7\}$ and $T = \{2,3,6,8\}$ are disjoint since $S \cap T = \varnothing$.

As an edge case, we note that the empty set is disjoint from every set, since $S \cap \varnothing = \varnothing$.

4.1.2 Set Difference

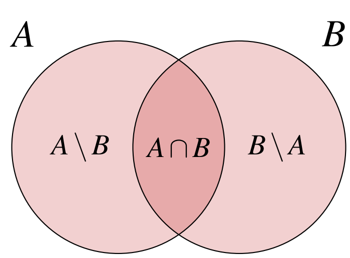
While the set intersection collects elements that belong to both of its argument sets, set differences collect elements that belong to exactly one of these sets.

Definition 4.2 — Set Difference.
Given two sets A and B, their *difference* $A \setminus B$ contains all elements of A that are not in B,

$$A \setminus B = \left\{ a \in A \mid a \notin B \right\}.$$

Example 4.3 We return to our example with $A = \{1,3,5\}$ and $B = \{1,2,3,4\}$. The set difference $A \setminus B = \{5\}$, as this is the only element of A that does not belong to B. The set difference $B \setminus A = \{2,4\}$, as these elements of B do not belong to A.

This example demonstrates that set difference is not a *commutative* operation; the order of its two argument sets A and B matters. We can visualize the set operations we have just introduced using a *Venn diagram*.



The left circle contains all of the elements of A, and the right circle contains all of the elements of B. All of the elements of $A \cap B$ will be in the overlapping section of the circles, and the elements in the set differences are contained in the non-overlapping sections of the circles.

We see from this diagram that the three sets $A \setminus B$, $A \cap B$, and $B \setminus A$ are *pairwise disjoint*, meaning any two of these sets do not have any elements in common. When we use Venn diagrams, it is also important to consider the possibility that one (or both) of the sets is empty. For example, if $B = \varnothing$ in the above picture, we can think that the B circle shrinks down to a point. In this case, the overlapping region will also disappear, which tells us that all of the elements in A sit in the crescent-shaped $A \setminus B$ segment. In other words, we find that $A \setminus \varnothing = A$.

4.2 Set Union and De Morgan's Laws

The remaining operations we will define cannot be expressed as restrictions. They all give us ways to build *bigger* sets from larger parts. The ability to carry out these operations is taken as an *axiom*, or assumed property, of sets. The simplest of these operations is the set *union*.

Definition 4.3

Given two sets A and B, their union $A \cup B$ consists of all elements that belong to either A or B (or both).

Example 4.4 We return to our example with $A = \{1,3,5\}$ and $B = \{1,2,3,4\}$. Their union $A \cup B = \{1,2,3,4,5\}$, all elements from either of the sets.

In our Venn diagram, $A \cup B$ comprises all the space within the two circles. We can write this symbolically as

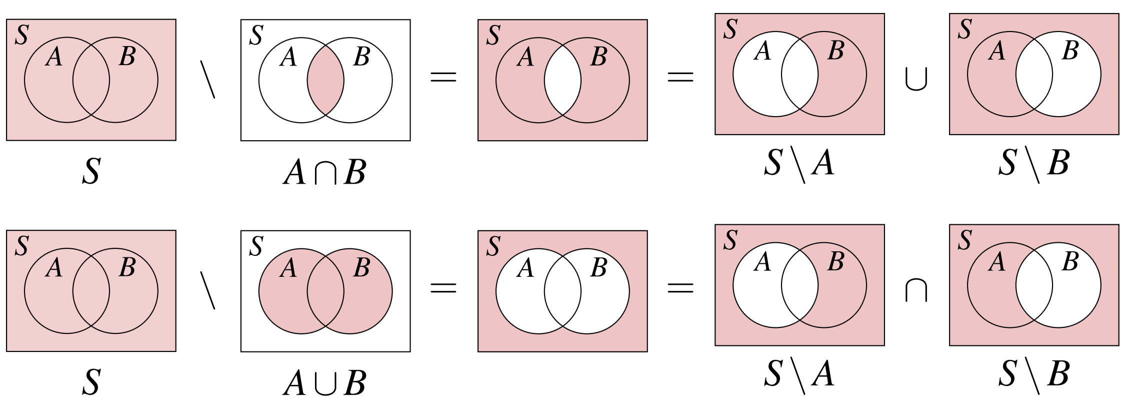
$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

This is a set identity, an equality of sets that involves various set operations. We can construct many more identities with these three set operations ($\cap, \setminus, \cup$), including the so-called *De Morgan's Laws for Sets*.

Lemma 4.1 — De Morgan's Laws for Sets. Suppose that A, B are subsets of a set S. Then,

1. $S \setminus (A \cap B) = (S \setminus A) \cup (S \setminus B)$
2. $S \setminus (A \cup B) = (S \setminus A) \cap (S \setminus B)$

We can give an intuitive illustration of these identities by shading regions of Venn diagrams that represent the various sets that they include. In the next lecture, we'll discuss more principled and rigorous techniques that we can use to verify such identities.



You may recognize a connection between De Morgan's Laws for Sets and for Propositional (and perhaps Predicate) Logic. More generally, you may see a connection between the set operations we just introduced and the logical operations we saw a few lectures ago. You can explore this connection more in Exercise 4.5.

4.3 Building Sets with More Complicated Elements

So far, the operations we've introduced all construct sets packed with the same "types" of elements we started with. If A and B are sets of numbers, then $A \cap B$, $A \setminus B$, and $A \cup B$ will also contain numbers. The last two operations we will see, *Cartesian products* and *power sets* will produce sets whose elements are more complicated mathematical objects.

4.3.1 Cartesian Products

One of the basic properties of a set is that its elements are unordered. However, many objects in math rely on ordered collections (the coordinates of points, vectors, sequences, etc.). We formalize the notion of order using a construct called a pair.

Definition 4.4 — (Ordered) Pair.
A *pair* is an ordered collection of two (possibly identical) elements. We write the ordered pair with first element a and second element b as (a,b).

We describe how we can represent ordered pairs as sets in Exercise 4.7. While such a construction is important for formalizing ordered pairs, it is cumbersome in practice. For this reason, we *abstract*; we introduce new definitions and notations that allow us to argue at a higher level. The intuitive definition of an ordered pair above is sufficiently rigorous so we will stick with that from now on.

The Cartesian product is an operation that allows us to construct sets of ordered pairs.

Definition 4.5 — Cartesian Product.
Given two sets A and B, their *Cartesian product*, denoted $A \times B$, consists of all ordered pairs whose first element comes from A and whose second element comes from B.

$$A \times B = \{(a,b) : a \in A, b \in B\}.$$

Note that we can take the Cartesian product of a set with itself. In this case, $A \times A$ contains all ordered pairs with both elements from A. We often abbreviate $A \times A$ as A^2. More generally, we use the notation A^n to refer to the Cartesian product of A with itself n-times. In other words, A^n consists of all *n-tuples* $(a_1, a_2, \ldots, a_n)$ with each $a_i \in A$.

Example 4.5 We return to our example with $A = \{1,3,5\}$ and $B = \{1,2,3,4\}$. Their Cartesian product $A \times B$ is a 12-element set of ordered pairs. We can visualize the elements of this Cartesian product as cells in a 2-dimensional grid, where the rows are labeled with the elements of A and the columns are labeled with the elements of B.

		B: 1	2	3	4
	1	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$
A	3	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$
	5	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$

We give a few more examples of Cartesian products below.

Example 4.6

- We can formalize a deck of cards as the Cartesian product of the set of suits $S = \{\spadesuit, \clubsuit, \heartsuit, \diamondsuit\}$ and the set of values $V = \{2,3,4,5,6,7,8,9,10,\mathrm{J},\mathrm{Q},\mathrm{K},\mathrm{A}\}$.
- The *Cartesian plane* $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ consists of all ordered pairs of real values. This is the usual 2-dimensional grid on which we graph functions.
- Given any set A, $\varnothing \times A = A \times \varnothing = \varnothing$. We cannot construct any ordered pairs whose first (or second) coordinate is an element from $\varnothing$, as there are no such elements.

4.3.2 Power Sets

The *power set* operation is a unary operation that constructs a set whose elements are subsets.

Definition 4.6 — Power Set.
The *power set* of a set S, denoted $\mathcal{P}(S)$, is the set of all subsets of S.

Example 4.7

- The power set of $\{x,y\}$ contains $2^2 = 4$ elements,

$$\mathcal{P}(\{x,y\}) = \big\{\varnothing, \{x\}, \{y\}, \{x,y\}\big\}.$$

- The power set of $A = \{1,3,5\}$ contains $2^3 = 8$ elements,

$$\mathcal{P}(A) = \big\{\varnothing, \{1\}, \{3\}, \{5\}, \{1,3\}, \{1,5\}, \{3,5\}, \{1,3,5\}\big\}.$$

- The power set of $\mathbb{N}$ contains all subsets of natural numbers. The sets of even numbers, prime numbers, perfect squares, etc. are all elements of $\mathcal{P}(\mathbb{N})$.
- $\mathcal{P}(\varnothing) = \big\{\varnothing\big\}$. $\varnothing$ is vacuously a subset of every set (including $\varnothing$), but this is the only subset of $\varnothing$.

Main Takeaways:

- Set operations give us a way to describe new sets in terms of old sets.
- The set *intersection* ($\cap$), *difference* ($\setminus$), and *union* ($\cup$) operations are related to the logical operations $\{\neg, \wedge, \vee\}$.
- Set *identities* can be used to describe how the set operations interact. We can visualize these relationships with *Venn diagrams*.
- The *Cartesian product* and *power set* operations can be used to construct sets with more complicated elements (pairs and subsets).

Exercises

4.1 Let us define sets,

$$A = \{1,2,3,4,5,6,7\}$$
$$B = \{1,3,5,7,9\}$$
$$C = \{2,3,6,8\}$$
$$D = \{2,4,7\}$$

Write down the following sets.

(a) $A \cap B$
(b) $A \cup C$
(c) $B \setminus C$
(d) $C \setminus B$
(e) $B \cap (C \cup D)$
(f) $(B \cap C) \cup D$
(g) $A \cap D$
(h) $D \setminus A$
(i) $D \setminus (D \setminus A)$
(j) $(A \cap B) \setminus (C \cup D)$
(k) $C \times D$
(l) $(C \setminus B)^2$
(m) $A^2 \cap A$
(n) $(A \cap C) \times (B \cap D)$
(o) $\mathcal{P}(C)$
(p) $\mathcal{P}(D) \times D$
(q) $\mathcal{P}(A \setminus B)$
(r) $\mathcal{P}(B) \cap \mathcal{P}(D)$

4.2 Shade in a (3-circle) Venn diagram to indicate each of the following sets.

(a) $A \cup B \cup C$
(b) $A \cap B \cap C$
(c) $A \cup (B \cap C)$
(d) $A \cap (B \cup C)$
(e) $A \setminus (B \cup C)$
(f) $(A \setminus B) \cup C$

4.3 Given any $n \in \mathbb{N}$, define the set

$$M_n = \{z \in \mathbb{Z} : \exists y \in \mathbb{Z}.\ z = y \cdot n\}.$$

(a) Express each of the following as M_n for some n:

i. $M_2 \cap M_3$

ii. $M_4 \cap M_7$

iii. $M_2 \cap M_6$

iv. $M_6 \cap M_{10}$

◇ (b) Can you come up with a general formula for $M_a \cap M_b$?

◇ (c) Try to do the same thing for $M_a \cup M_b$. What is different in this case?

4.4 Give an example of two sets A and B such that:

$$|A| = 7, \qquad |A \cap B| = 4, \qquad |A \cup B| = 9.$$

4.5 In this exercise, we explore some connections between the logical operations that we discussed last lecture and the set operations that we discussed in this lecture. Given two sets A and B and some element x, we'll consider the following two atomic propositions:

$$p = \text{“}x \in A\text{''} \qquad q = \text{“}x \in B\text{''}$$

We can connect the logical and operation ($\wedge$) and the set intersection operation ($\cap$) by noticing:

$$(x \in A) \wedge (x \in B) \quad \Leftrightarrow \quad x \in (A \cap B).$$

(a) Write down a similar expression that connects the logical or ($\vee$) and set union ($\cup$) operations.

(b) What is the set analog to the logical negation ($\neg$) operation? (Hint: It may help to think about a *superset* $S \supseteq A$.)

(c) What is the logical analog to the subset ($\subseteq$) operation?

(d) What is the logical analog to the set equality ($=$) operation?

4.6 This exercise builds on the ideas of Exercise 4.5. Observe how De Morgan's Laws for Sets are just a "reframing" of the De Morgan's Laws for logic using the connections in Exercise 4.5. Similarly, provide a "set version" of the following properties from logic.

(a) Distributivity of $\vee$: $\big(p \wedge (q \vee r)\big) \Leftrightarrow \big((p \wedge q) \vee (p \wedge r)\big)$

(b) Distributivity of $\wedge$: $\big(p \vee (q \wedge r)\big) \Leftrightarrow \big((p \vee q) \wedge (p \vee r)\big)$

(c) Absorption Law: $p \Leftrightarrow \big(p \wedge (p \vee q)\big)$

(d) Involution Law: $p \Leftrightarrow \big(\neg(\neg p)\big)$

(e) Contrapositive Law: $\big(p \Rightarrow q\big) \Leftrightarrow \big(\neg q \Rightarrow \neg p\big)$

4.7 Here, we explore one way to formally construct ordered pairs using sets (due to Kazimierz Kuratowski in 1921). To do this, we'll represent the ordered pair (a,b) by the set $\{a, \{a,b\}\}$.

(a) Show that this construction does enforce an ordering of the elements. In other words, argue that the set representations of (a,b) and (b,a) are different if $a \neq b$.

(b) We declare two ordered pairs (a_1,b_1) and (a_2,b_2) to be equal if and only if they are equal *coordinate-wise*; that is if $a_1 = a_2$ and $b_1 = b_2$. Using this definition, argue that if $(a_1,b_1) = (a_2,b_2)$, then their set representations are the same.

⋆ (c) As the converse (reverse implication) to part (b), argue that if (a_1,b_1) and (a_2,b_2) have the same set representations, then $(a_1,b_1) = (a_2,b_2)$. (Be careful that you don't take anything for granted. It may help to look ahead at some proof strategies from the next lecture.)

4.8 Give a complete description (write down all of the elements) of each of the following sets.

(a) $\mathcal{P}(\{1\})$

(b) $\mathcal{P}(\mathcal{P}(\{1\}))$

(c) $\mathcal{P}(\mathcal{P}(\varnothing))$

(d) $\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing)))$

4.9 In this connection, we explore the connection between power sets, set inclusion, and subsets.

(a) Given any set S, explain why it is always true that $S \in \mathcal{P}(S)$.

(b) Given two examples of sets S such that $S \not\subseteq \mathcal{P}(S)$.

⋆ (c) Given two examples of sets T such that $T \subseteq \mathcal{P}(T)$.

4.10 Suppose that A and B are finite sets with $|A| = a$, $|B| = b$ and $|A \cap B| = c$. Express the cardinalities (i.e., the number of elements) of the following sets in terms of a, b, c.

(a) $A \cup B$ (b) $A \setminus B$ (c) $B \setminus A$ (d) $A \times B$

(e) $A \times A$ (f) $(A \times A) \setminus A$ (g) $(A \setminus B) \times (B \setminus A)$

4.11 Suppose that A and B are finite sets with $|A| = 14$ and $|B| = 8$.

(a) What is the maximum possible value of $|A \cap B|$? When does this occur?

(b) What is the minimum possible value of $|A \cap B|$? When does this occur?

(c) What is the maximum possible value of $|A \cup B|$? When does this occur?

(d) What is the minimum possible value of $|A \cup B|$? When does this occur?

4.12 Let A, B, C be three sets, and suppose that $|A| = 8$, $|B| = 10$, $|C| = 11$, $|A \cap B| = 7$, and $|A \cup C| = 13$. List all possible values of

(a) $|A \cup B|$ (b) $|A \cap C|$ (c) $|A \cup B \cup C|$

(d) $|A \cap B \cap C|$ (e) $|B \cap C|$ (f) $|B \cup C|$

⋆ **4.13** Recall that a multiset is a collection that may contain duplicated elements. We can represent a multiset M as a set of ordered pairs (m, n) with $m \in M$ and $n \in \mathbb{N}^+$ which encodes that the element m appears in the multiset n times. For example, the multiset $\{\!\{a, a, b, c, c, c\}\!\}$ would be encoded $\{(a, 2), (b, 1), (c, 3)\}$.

(a) We say that two multisets M and M' are equal ($M = M'$) if they contain the same elements, each with the same *multiplicity* (that is, the elements appear the same number of times). Briefly explain how this property is enforced by the ordered-pairs construction.

(b) How should we define the intersection of two multisets so that it behaves similarly to the set intersection? In particular, what should be the intersection of

$$M_1 = \{\!\{a, a, b, c, c, c\}\!\} \quad \text{and} \quad M_2 = \{\!\{a, b, b, b, c, d\}\!\}?$$

(c) How should we define the union of two multisets so that it behaves similarly to the set intersection? In particular, what should be the union of

$$M_1 = \{\!\{a, a, b, c, c, c\}\!\} \quad \text{and} \quad M_2 = \{\!\{a, b, b, b, c, d\}\!\}?$$

5. Proof Techniques

The fairly basic definitions that we gave for sets and their operations last lecture can be used to establish many interesting properties. In this lecture, we'll explore some of these properties; more importantly, we'll talk about how to prove them. *Proofs* serve as the backbone for higher-level mathematics and are a central focus of this course.

Definition 5.1 — Proof.

A *proof* is a formal argument that a proposition is true. Generally, a proof consists of a series of assertions. Each assertion must either already be known (or assumed) to be true, or logically follows from previous assertions.

You can compare writing a proof to solving a jigsaw puzzle. The pieces of the puzzle are the facts that we already know. These may be definitions, assumptions that are given as part of the proposition, or previous propositions that we have already proven. The definition says that each step in the proof should "logically follow" from the previous assertions. This means that we can use a line of logical reasoning to verify that the next statement is a consequence of the previous statements. Just as there are strategies that we can use to help solve a jigsaw puzzle (build the edges first, sort pieces by color, consult the picture on the box), there are also some common techniques that we use to write proofs. We'll briefly survey many of these techniques today, and they will all come up many times over the course of the semester.

5.1 Direct Proofs

Most propositions that we wish to prove are (biconditional) implications. Their statements begin by introducing objects and asserting some properties of these objects. Then, under these assumptions, a consequence (or *consequent*) is stated. Consider the following lemma.

Lemma 5.1 Let A, B, C be sets with $A \subseteq B$. Then, $(A \cap C) \subseteq (B \cap C)$.

The mathematical objects that the lemma references are three sets A, B, and C. The assumption (*antecedent*) of the lemma is $A \subseteq B$, and the consequent is $(A \cap C) \subseteq (B \cap C)$. We can symbolically write the lemma as the statement

$$(A \subseteq B) \Rightarrow \big((A \cap C) \subseteq (B \cap C)\big)$$

for any sets A, B, C. For a particular choice of sets, both sides of this implication are propositions; it is either true or false that one set is a subset of another. In a *direct proof* of an implication, we assume the antecedent is true and use this to show that the consequent must also be true.

Definition 5.2 — Direct Proof.

A *direct proof* of an implication $p \Rightarrow q$ starts by assuming that p is true and then uses this to deduce that q is true.

In this proof, we'll assume that $A \subseteq B$ and try to show that $(A \cap C) \subseteq (B \cap C)$. Notice that we don't have any information about the elements in any of these sets, so our proof will need to rely only on the definitions of $\cap$ and $\subseteq$. What does it mean that $(A \cap C) \subseteq (B \cap C)$? Looking back at the definition, this means that every element of $A \cap C$ is also an element of $B \cap C$. This means to argue that $(A \cap C) \subseteq (B \cap C)$, we can consider an *arbitrary* element a in $A \cap C$ and show that a is also in $B \cap C$.

Proving $\subseteq$:

To prove that $R \subseteq S$, consider an *arbitrary* element $r \in R$ and prove that $r \in S$.

The key word in this is *arbitrary*. If $r \in R$ is arbitrary, then we cannot assume anything about r other than its membership in R. Think of r as a variable that stands in for any element of R in the proof. Since we can (one by one) plug in each element of R for r in the proof and still have correct statements, we can conclude that *every* element of R belongs to S, so $R \subseteq S$.

Proof of Lemma 5.1. Let a be an arbitrary element of $A \cap C$. Since $a \in A \cap C$, we can use the definition of set intersection to conclude that $a \in A$ and that $a \in C$. But then since $a \in A$ and $A \subseteq B$, we know that $a \in B$. Now, we use the definition of $\cap$ again. Since $a \in B$ and $a \in C$, we conclude that $a \in B \cap C$. As a was chosen arbitrarily, we conclude that $(A \cap C) \subseteq (B \cap C)$. ■

Let's stop and make a few observations about the proof.

- The proof is written in complete sentences. The argument is explained in words, not just shown in symbols.
- Each step of the argument clearly lists the facts and definitions that it is applying.
- The variable a is defined before it is used.
- The end of the proof is delineated by a "■" symbol.

Proofs are meant to be read, and all of these conventions help ensure that proofs can be easily followed and understood.

5.1.1 Proofs by Case Analysis

Sometimes, a proof does not have such a clear, linear argument as the previous example. Rather, the argument needs to branch out to separately handle different cases. An example of this phenomenon arises in the proof of the following lemma.

Lemma 5.2 Let A, B, C be sets with $A \subseteq B$. Then, $(A \cup C) \subseteq (B \cup C)$.

Similar to the previous proof, we will start by letting a be an arbitrary element of $A \cup C$. However, at this point, it's not clear how to proceed. We can't assume that $a \in A$, since this isn't necessarily true of *every* element of $A \cup C$ (a may just belong to C). To deal with this uncertainty, our proof will branch into two cases based on whether $a \in A$ or $a \notin A$. We know that one of these two cases will be true, so if we can show that $a \in (B \cup C)$ in both cases, we will be finished.

Definition 5.3 — Proof by Case Analysis.

In a *proof by case analysis*, the argument separates into multiple cases that *together account for all possibilities*. If the claim is proven in each case, we may conclude that it is true.

Proof of Lemma 5.2. Let a be an arbitrary element of $A \cup C$. We consider two cases.

Case 1: $a \in A$ Since $A \subseteq B$, we must have $a \in B$. Then, by the definition of set union, we conclude that $a \in (B \cup C)$.

Case 2: $a \notin A$ Since $a \in A \cup C$ but $a \notin A$, we must have $a \in C$. Then, by the definition of set union, we conclude that $a \in (B \cup C)$.

As $a \in (B \cup C)$ in both cases and since the cases exhaust all possible $a \in (A \cup C)$, we conclude that $(A \cup C) \subseteq (B \cup C)$. ■

5.1.2 Proofs in Two Directions

Many times, a proof will require us to establish a biconditional. As an example, consider the following lemma.

Lemma 5.3 Given any sets A and B, $A = (A \cap B) \cup (A \setminus B)$.

We are trying to prove an equality of sets; that is, we must argue that the sets on both sides of the "=" contain exactly the same elements. To do this, we can argue that $A \subseteq (A \cap B) \cup (A \setminus B)$ and also argue that $(A \cap B) \cup (A \setminus B) \subseteq A$.

Proving Set Equality:

To prove that $S = T$, separately argue that $S \subseteq T$ and $T \subseteq S$.

An equivalent statement in propositional logic is that we can prove $p \Leftrightarrow q$ by separately arguing $p \Rightarrow q$ and $q \Rightarrow p$, i.e., proving implication in "both directions." This is justified by the fact that the two highlighted columns in the following truth table are identical.

p	q	$p \Leftrightarrow q$	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Proof of Lemma 5.3. We separately argue both directions of containment. Both arguments will require a short case analysis.

$A \subseteq (A \cap B) \cup (A \setminus B)$: Let $a \in A$ be an arbitrary element. We separate into two cases based on whether or not $a \in B$. If $a \in B$, then $a \in A \cap B$. Otherwise, if $a \notin B$, then $a \in A \setminus B$. In either case, $a \in (A \cap B) \cup (A \setminus B)$. Since a was chosen arbitrarily, $A \subseteq (A \cap B) \cup (A \setminus B)$.

$(A \cap B) \cup (A \setminus B) \subseteq A$: Let $x \in (A \cap B) \cup (A \setminus B)$ be an arbitrary element. We separate into two cases based on whether or not $x \in (A \cap B)$. If so, then we also know that $x \in A$. If not, then since $x \in (A \cap B) \cup (A \setminus B)$, we conclude that $x \in (A \setminus B)$. But then again, in this case, $x \in A$. Since x was chosen arbitrarily $(A \cap B) \cup (A \setminus B) \subseteq A$.

Since $A \subseteq (A \cap B) \cup (A \setminus B)$ and $(A \cap B) \cup (A \setminus B) \subseteq A$, we conclude that $A = (A \cap B) \cup (A \setminus B)$. ■

5.2 Indirect Proofs

Sometimes, it is not easy (or even possible) to argue a claim directly, following a chain (or chains) of logical conclusions starting from the given assumptions. In this case, we must use different proof techniques. Two of the most common indirect proof techniques are *contrapositive proofs* and *proofs by contradiction*.

5.2.1 Contrapositive Proofs

When we discussed propositions, we saw that the implication $p \Rightarrow q$ is equivalent to its contrapositive $\neg q \Rightarrow \neg p$. Thus, if we can show that the contrapositive is true, we can conclude that the original implication is also true.

Definition 5.4 — Contrapositive proof.

A *contrapositive proof* of an implication $p \Rightarrow q$ starts by assuming that q is false and then uses this to deduce that p is false.

Consider the following lemma.

Lemma 5.4 Given sets A, B, suppose that $A \setminus B \subseteq B$. Then, $A \subseteq B$.

It is not clear how we could prove this implication directly. This would require us to take an arbitrary element $a \in A$ and use the fact that $A \setminus B \subseteq B$ to conclude that $a \in B$. However, a contrapositive proof of this lemma is quite natural.

Proof. Suppose that $A \not\subseteq B$. By definition, there is some element $a \in A$ that does not belong to B. By the definition of set difference, $a \in A \setminus B$. Since we have located an element a that belongs to $A \setminus B$ but does not belong to B, we conclude that $A \setminus B \not\subseteq B$. Applying the contrapositive, we find that if $A \setminus B \subseteq B$, then $A \subseteq B$. ■

5.2.2 Proofs by Contradiction

Definition 5.5 — Contradiction, Proof by Contradiction.

We have reached a *contradiction* if we have shown that a proposition q and its negation $\neg q$ are simultaneously true.

To prove a proposition p by contradiction, we assume that $\neg p$ is true and show that under this assumption, we reach a contradiction.

The intuition behind a proof by contradiction is the so-called *Law of the Excluded Middle*. If assuming that $\neg p$ is true causes a contradiction, then our assumption must be wrong, meaning $\neg p$ must be false, i.e. p is true. This framework is more difficult to understand at first and is probably best illustrated by an example. We'll prove the following lemma by contradiction.

Lemma 5.5 Given sets A, B, $(A \cap B) \cap (A \setminus B) = \varnothing$.

Here, a direct proof doesn't seem possible. Using our earlier ideas, part of the proof would require us to take an arbitrary $x \in (A \cap B) \cap (A \setminus B)$ and argue that $x \in \varnothing$, but what does it mean for an element to belong to the empty set!? From the perspective of a proof by contradiction, we should instead assume that the lemma is false and use this to derive a contradiction.

Proof of Lemma 5.5. For sake of contradiction, suppose that $(A \cap B) \cap (A \setminus B) \neq \varnothing$. In other words, there is some element x belonging to the set $(A \cap B) \cap (A \setminus B)$. By the definition of set intersection, this means that $x \in (A \cap B)$ and $x \in (A \setminus B)$. Since $x \in (A \cap B)$, we may conclude that $x \in B$. Moreover, since $x \in (A \setminus B)$, we may conclude that $x \notin B$. These two statements are in contradiction. Hence, our earlier supposition must be incorrect, meaning $(A \cap B) \cap (A \setminus B) = \varnothing$. ■

As proofs by contradiction require more roundabout thinking, it is a good practice to alert the reader that this technique is being used at the start of the proof (for example, by using the phrase "*for sake of contradiction*").

R People often confuse contrapositive proofs and proofs by contradiction for implications since both styles of proof start by assuming that some of the given information is false. If we wish to prove $p \Rightarrow q$, a contrapositive proof assumes that the *consequent* q is false and concludes that the antecedent p must be false. On the other hand, a proof by contradiction would assume that the *entire implication* $p \Rightarrow q$ is false (in other words, it would assume that p is true and q is false) and use this to derive some other contradiction. If your contradiction is that p is false, your proof by contradiction is most likely a contrapositive proof in disguise.

5.3 Erroneous Proofs

Reading proofs carefully and spotting mistakes is an important skill. These can be mathematical errors or logical errors (asserting statements that do not follow from previous statements). For example, consider the "proofs" of the following propositions.

Proposition 5.5: Given any sets A,B,C, if $A \subseteq B$, then $(C \setminus B) \subseteq (C \setminus A)$.

Proof. To conclude that $C \setminus B$ is a subset of $C \setminus A$, we can argue that there are no elements in $C \setminus B$ that are not in $C \setminus A$ (or, said differently, that $x \notin C \setminus A \Rightarrow x \notin C \setminus B$). To this end, suppose that $x \notin C \setminus A$. This tells us that $x \notin C$, since every element in $C \setminus A$ belongs to C. Now, if $x \notin C$, we also know that $x \notin C \setminus B$, which is a subset of C. Since the element x was chosen arbitrarily, we conclude that $(C \setminus B) \subseteq (C \setminus A)$. ■

Here, an error is made in the third sentence ("This tells us..."). There can be elements that don't belong to $C \setminus A$ but belong to C (namely, those in $A \cap C$). It is also alarming that this "proof" never uses the assumption that $A \subseteq B$. Typically, we will only include assumptions in the proposition statement that are necessary for its proof.

Proposition 5.6: Given any sets A,B,C, if $(C \setminus B) \subseteq (C \setminus A)$, then $A \subseteq B$.

Proof. Consider an arbitrary $a \in A$. By the definition of set difference, we know that $a \notin C \setminus A$. Under the assumption that $(C \setminus B) \subseteq (C \setminus A)$, we know that every element of $C \setminus B$ must belong to $C \setminus A$. Hence, $a \notin C \setminus B$. Again applying the definition of set difference, we conclude that $a \in B$. Since $a \in A$ was chosen arbitrarily, $A \subseteq B$. ■

A mistake is made in the second last sentence of the proof. Just because a does not belong to $C \setminus B$ doesn't mean we can conclude that $a \in B$; it could also be the case that $a \notin C$ and $a \notin B$. Carefully considering all possibilities can help avoid mistakes like this in your proofs.

It turns out that Proposition 5.5 is true (see Exercise 5.1), even though the sequence of assertions given in its proof is flawed. Contrast this with Proposition 5.6, which is simply false. Any purported proof of this statement would necessarily contain an error. To justify that this proposition is false, we can provide evidence in the form of a *counterexample*.

5.4 Counterexamples

In general, the statements we wish to prove are universally quantified; we want to show that *all* objects that satisfy the assumptions of the proposition statement also satisfy its conclusion. For example, Proposition 5.6 is a statement quantified over the choice of three sets (A,B,C). To establish that this proposition is false, it suffices to supply a particular choice of these objects that refutes the proposition statement. In other words, we must define *all* of the objects appearing in the proposition statement so that:

- All the assumptions that the proposition makes about these objects are satisfied (true).
- The conclusion of the proposition is not true.

In the case of Proposition 5.6, suppose that we consider sets $A = \{1,2\}, B = \{1\}$, and $C = \{1,3\}$. Then, $C \setminus A = C \setminus B = \{3\}$, meaning $(C \setminus B) \subseteq (C \setminus A)$. However, $A \not\subseteq B$ since $2 \in A \setminus B$. Therefore, this choice of A,B,C is a valid counterexample and demonstrates that the proposition is false.

Main Takeaways:

- A proof is a formal mathematical argument (written in complete sentences) that begins with some premises and, using logical rules, proceeds to establish a conclusion.
- There are many useful proof techniques. *Direct proofs* establish an implication $p \Rightarrow q$ by assuming that p is true and using this to show that q is true. *Proofs by case analysis* separately consider scenarios that together account for all possibilities. *Contrapositive proofs* establish an implication $p \Rightarrow q$ by assuming that q is false and using this to show that p is false. In a *proof by contradiction*, we assume the proposition is false and use this to derive a contradiction.
- To prove a biconditional statement, separately argue both directions of implication.
- A counterexample can be used to demonstrate that a proposition is false.

Exercises

5.1 Give a correct proof of Proposition 5.5.

5.2 Suppose that $A \subseteq B$ and $C \subseteq D$. Argue each of the following inclusions.

(a) $(A \cap C) \subseteq (B \cap D)$

(b) $(A \cup C) \subseteq (B \cup D)$

(c) $(A \cap C) \subseteq (B \cup D)$

(d) $(A \times C) \subseteq (B \times D)$

Now, disprove each of the following inclusions by providing a counterexample.

(e) $(A \cup C) \subseteq (B \cap D)$

(f) $(A \setminus C) \subseteq (B \setminus D)$

5.3 Use the techniques introduced in this lecture to prove the two De Morgan's Law for Sets: Let A, B, S be any sets such that $A \subseteq S$ and $B \subseteq S$. Then,

(a) $S \setminus (A \cap B) = (S \setminus A) \cup (S \setminus B)$.

(b) $S \setminus (A \cup B) = (S \setminus A) \cap (S \setminus B)$.

(c) In your proofs, was it necessary that $A \subseteq S$ and $B \subseteq S$. That is, if we removed these assumptions, are you still able to prove the identities in parts (a) and (b)?

5.4 Recall that a binary operation $\star$ is *commutative* if $A \star B = B \star A$. Determine whether each of the following set operations is commutative. If so, explain why. If not, provide a counterexample.

(a) $\cap$ (b) $\setminus$ (c) $\cup$ (d) $\times$

5.5 Recall that a binary operation $\star$ is *associative* if $(A \star B) \star C = A \star (B \star C)$. Determine whether each of the following set operations is associative. If so, explain why. If not, provide a counterexample.

(a) $\cap$ (b) $\setminus$ (c) $\cup$ (d) $\times$

5.6 Recall that a binary operation $\star$ *distributes* over another binary operation $\circ$ if $A \star (B \circ C) = (A \star B) \circ (A \star C)$. Determine whether each of the following statements is true, providing either a proof or a counterexample.

(a) $\cap$ distributes over $\cup$.

(b) $\cup$ distributes over $\cap$.

(c) $\cap$ distributes over $\setminus$.

(d) $\setminus$ distributes over $\cap$.

(e) $\cup$ distributes over $\setminus$.

(f) $\setminus$ distributes over $\cup$.

(g) $\times$ distributes over $\cap$.

(h) $\cup$ distributes over $\times$.

5.7 Suppose that $A \subseteq B$ and $B \subseteq C$. Argue that $A \subseteq C$. As we will learn when we study relations, this shows that "$\subseteq$" is *transitive*.

5.8 Given any sets A, B, argue that $(A \setminus B) \cap (B \setminus A) = \varnothing$.

5.9 Given any sets A, B, argue that $(A \cap B) = (A \cup B) \Leftrightarrow A = B$.

5.10 Given any sets A, B, argue that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cup \mathcal{P}(B)$, then either $A \subseteq B$ or $B \subseteq A$.

5.11 Given any two sets A, B, we may define their *symmetric difference* (written $A \triangle B$) by

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

(a) Compute the symmetric difference of $A = \{0, 1, 2, 3, 4, 5, 6\}$ and $B = \{1, 3, 5, 7, 9, 11\}$.

(b) Argue that $A \triangle B = (A \cup B) \setminus (A \cap B)$.

(c) Is $\triangle$ commutative? (See Exercise 5.4)

(d) Is $\triangle$ associative? (See Exercise 5.5)

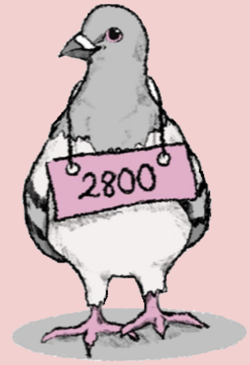
(e) Does $\cap$ distribute over $\triangle$? (See Exercise 5.6)

(f) Does $\cup$ distribute over $\triangle$?

$\star$ (g) Given any two sets A, B, argue that $A \cup B = (A \triangle B) \triangle (A \cap B)$.

$\star$ (h) Given any three sets A, B, C, argue that $(A \triangle B) \triangle (B \triangle C) = A \triangle C$.

II Induction



6. Deductive and Inductive Reasoning

In the previous lecture, we introduced a collection of *proof techniques*, methods that we can use to rigorously demonstrate that a mathematical statement is true. We used these techniques to prove some properties of sets and their operations, which were based on the definitions of these operations we introduced two lectures ago. In today's lecture, we will continue our discussion of proofs. First, we will show how the general techniques from the last lecture can be used to reason about things beyond sets. Later, we'll introduce another proof technique, *induction*, that plays a central role in computer science.

6.1 Beyond Set Proofs

A few lectures ago, we motivated predicates by considering how to formalize the statement

> *Every odd (natural) number can be written as the difference of two perfect squares.*

Now, we have the tools to do this, along with the techniques to prove that the statement is true. First, we can note that this statement is an implication about a natural number: *if* a natural number is odd, *then* it can be written as the difference of two perfect squares. We can represent this logically as

$$\forall\, n \in \mathbb{N}.\ \Bigg[\underbrace{\Big(\exists\, k \in \mathbb{N}.\ n = 2k-1\Big)}_{n \text{ is odd}} \Rightarrow \underbrace{\Big(\exists\, j \in \mathbb{N},\ \exists\, h \in \mathbb{N}.\ n = j^2 - h^2\Big)}_{n \text{ is the difference of two perfect squares}}\Bigg].$$

We know that the implication will be true whenever the antecedent is false, meaning the bracketed predicate is true for all even n. We are left to show that it is true for an *arbitrary* odd natural number. For this, we can use a direct proof. Since the consequent consists of two existential quantifiers, we simply need to find particular choices for j and h that satisfy the equality.

Proof. Given any odd natural number n, we can represent $n = 2k-1$ for some $k \in \mathbb{N}$. Letting $j = k$ and $h = k-1$, we find that $j^2 - h^2 = k^2 - (k-1)^2 = k^2 - (k^2 - 2k + 1) = 2k - 1 = n$, so n is the difference of two perfect squares. ■

Taking a step back, we see that writing this proof required us to put together the pieces we have been collecting over our first few lectures. We used logic, predicates, and quantifiers to formalize the statement. Then, we used the various proof strategies techniques (collected in the pink boxes in earlier lecture notes) to reason about the resulting logical expression. Finally, we had to use a definition (in this case, the definition of an odd number) to get the final piece (the introduction of the variable k) we needed to make the proof work.

6.2 Inductive Reasoning

The proof we just wrote (along with all of the proofs we have seen over the past few lectures) is *deductive*. It starts with a big claim we want to prove and breaks it down into simpler pieces which are assembled using logical inferences. While deductive proofs are important, they are not the only approach; sometimes, we need to use an alternate approach known as *induction*. Consider the following example.

Example 6.1 Suppose that we wished to find a formula for the sum of the first n odd numbers (for all $n \in \mathbb{N}$). Trying out the first few small cases, we can see that 0 is the sum of the first 0 odd numbers, 1 is the sum of the first 1 odd number, then $1+3=4$, $1+3+5=9$, $1+3+5+7=16$. All of these are perfect squares, so we can conjecture that

> For each $n \in \mathbb{N}$, the sum of the first n odd natural numbers is n^2.

This statement is a predicate $P(n)$ that has been universally quantified over the natural numbers. Using summation notation, we can represent this as

$$\forall\, n \in \mathbb{N}.\ \underbrace{\sum_{i=1}^{n}(2i-1) = n^2}_{P(n)}.$$

Your first instinct might be to show $P(n)$ is true for an *arbitrary* $n \in \mathbb{N}$, the strategy we discussed previously. While this is good intuition, it doesn't seem very practical here. We would need a way to reason about summations with an arbitrary number of terms, which seems tricky. *Proofs by induction* provide us another strategy for proving statements of the form "$\forall\, n \in \mathbb{N}.\ P(n)$." We call the predicate $P(n)$ an *inductive statement*.

Definition 6.1 — Inductive Statement.
In a proof by induction, the *inductive statement* is the predicate $P(n)$ with domain $\mathbb{N}$ that we wish to prove holds for all $n \in \mathbb{N}$.

After identifying the inductive statement, the rest of the proof by induction is split into two steps: proving the base case(s), and establishing the inductive step.

6.2.1 Base Case(s)

In the base cases, we directly reason about the smallest possible value or values of n. Plugging $n = 0$ into $P(n)$ in our example, we must show that

$$\sum_{i=1}^{0}(2i-1) = 0^2 = 0.$$

Notice that the summation on the left side has a lower bound $i = 1$ that is greater than its upper bound $i = 0$. Since there are no values of i between these bounds, we call this an *empty sum*, which by convention has value 0 (when we add zero numbers together, we get the additive identity 0). Thus, we conclude that $P(0)$ is true.

6.2.2 Inductive Step

The inductive step is the heart of a proof by induction. In its most basic form, the inductive step requires us to prove the implication $P(k) \Rightarrow P(k+1)$ for an arbitrary $k \in \mathbb{N}$. We typically prove this implication directly, making the assumption that $P(k)$ is true. This is called the *inductive hypothesis*.

Definition 6.2 — Inductive Hypothesis.

When we make an *inductive hypothesis* in a proof by induction, we assume that the inductive statement $P(n)$ is true for an arbitrary value we represent by a variable k.

In our example, we assume that

$$\sum_{i=1}^{k}(2i-1) = k^2.$$

We must use this fact to show that $P(k+1)$ is also true, or that

$$\sum_{i=1}^{k+1}(2i-1) = (k+1)^2.$$

When we start the inductive step, we should ask ourselves the question:

> *How can I use the inductive hypothesis?*

In this example, this question becomes "*How does knowing the value of the sum of the first k odd numbers help me calculate the sum of the first $k+1$ odd numbers?*" We can observe that the latter sum is just the former sum with one more term added at the end. Thus, we can obtain our desired result by "pulling" this extra term out of the sum and applying the inductive hypothesis:

$$\sum_{i=1}^{k+1}(2i-1) = (2(k+1)-1) + \sum_{i=1}^{k}(2i-1) \overset{\text{(IH)}}{=} (2(k+1)-1) + k^2 = k^2 + 2k + 1 = (k+1)^2.$$

6.2.3 The Principle of Induction

We can finish the proof by appealing to the following *Principle of Induction*.

Definition 6.3 — Principle of Induction.

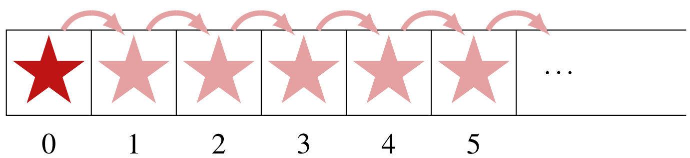
Given a predicate $P(n)$ with domain $\mathbb{N}$, and some $k_0 \in \mathbb{N}$ if we can argue that

Base Case: $P(k_0)$ is true,

Inductive Step: $P(k) \Rightarrow P(k+1)$ for all $k \geq k_0$,

then we may conclude that $P(n)$ is true for all $n \geq k_0$.

In this lecture, we'll only consider the case $k_0 = 0$, which allows us to prove that $P(n)$ holds for all $n \in \mathbb{N}$. We will consider different values of k_0 in later lectures. Let's think about why the Principle of Induction works. We can imagine a row of boxes representing the natural numbers, and we will place a star in box n once we know that $P(n)$ is true. The base case tells us that $P(0)$ is true, so we may place a star in box 0. Now, let's plug $k = 0$ into the implication in the inductive step. This tells us that $P(0) \Rightarrow P(1)$. Since we know $P(0)$ is true, we can conclude that $P(1)$ is also true, so we can draw a star in box 1. Next, if we plug $k = 1$ into the inductive step, we know that $P(1) \Rightarrow P(2)$. Since we know $P(1)$ is true, we conclude that $P(2)$ is true and draw a star in box 2.



By continuing this chain of reasoning, we will eventually be able to draw a star in any *arbitrary* box. Thus $P(n)$ is true for an arbitrary n, so must be true for all $n \in \mathbb{N}$.

To summarize, the following steps are necessary for any proof by induction that we will write in this course (as you move on to higher math and computer science courses, some of the steps may be abridged or omitted):

Steps in a Proof by Induction:

1. Write out the inductive statement.
2. Write out and prove the base case $P(0)$.
3. Fully write out the inductive hypothesis $P(k)$.
4. Prove the implication $P(k) \Rightarrow P(k+1)$ in the inductive step.
5. Cite the Principle of Induction to establish the claim.

Applying these steps to our running example we obtain the following self-contained proof.

Proof. We proceed by induction on n, with the inductive statement $P(n) =$ "$\sum_{i=1}^{n}(2i-1) = n^2$".

Base Case: When $n = 0$, we obtain the equality $\sum_{i=1}^{0}(2i-1) = 0$, which is true since the summation on the left side is empty.

Inductive Hypothesis: Given some arbitrary $k \in \mathbb{N}$, let us assume that $P(k)$ is true. That is, $\sum_{i=1}^{k}(2i-1) = k^2$.

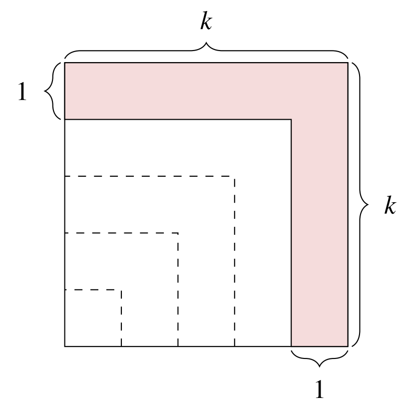
Inductive Step: We may simplify,

$$\sum_{i=1}^{k+1}(2i-1) = (2(k+1)-1) + \sum_{i=1}^{k}(2i-1) \overset{\text{(IH)}}{=} (2(k+1)-1) + k^2 = k^2 + 2k + 1 = (k+1)^2.$$

Here, the first equality pulls out the $i = k+1$ term from the summation, the second equality applies the inductive hypothesis, and the remaining steps are algebraic. Thus, $P(k+1)$ is true.

By the Principle of Induction, we conclude that $\sum_{i=1}^{n}(2i-1) = n^2$ for all $n \in \mathbb{N}$. ∎

R The two proofs that we've seen in the lecture are related, and we can visualize this connection with the following diagram:



The area of the shaded region is the difference in the area of two squares, the larger $k \times k$ square and the inner white $(k-1) \times (k-1)$ square. We can compute the pink area as $2k-1$, which can be any odd number by choosing different $k \in \mathbb{N}^+$. Thus, every odd number is the difference of two (consecutive) perfect squares. On the other hand, note that we can split apart the $k \times k$ square by repeatedly "peeling off" these L-shaped layers, starting with the pink-shaded one. In total, we will end up with k of these layers, whose areas are the first k odd numbers which sum to the total k^2 area.

6.3 Another Proof by Induction

Throughout the course, we will see many more examples of proofs by induction. To conclude today's lecture, we will prove an important formula that will come back at the end of the course in our discussion of probability and statistics.

Lemma 6.1 — Geometric Series Formula. Given any $x \in \mathbb{R}$ with $x \neq 1$ and any $n \in \mathbb{N}$,

$$\sum_{i=0}^{n} x^i = \frac{1-x^{n+1}}{1-x}.$$

In this lemma, we are universally quantifying over two variables: the real number x and the natural number n. We will use different strategies to handle each of these quantification. We will let x remain arbitrary throughout the proof; that is, we will not make any assumptions about the value of x other than $x \neq 1$ throughout the proof, meaning it will be valid for every $x \neq 1$. To handle the quantification over n, we will use induction.

Proof. We will establish the identity by induction on n, with inductive statement

$$P(n) = \text{“}\forall\, x \in \mathbb{R} \setminus \{1\}.\ \sum_{i=0}^{n} x^i = \frac{1-x^{n+1}}{1-x}.\text{”}$$

Base Case: Given any $x \neq 1$, substituting $n = 0$ into this equation gives $\sum_{i=0}^{0} x^i = \frac{1-x}{1-x}$. Using our assumption that $x \neq 1$, we can simplify the right side of this equation to 1. On the left side, the summation has a single term corresponding to $i = 0$ which is $x^0 = 1$ (here, we use the convention that $0^0 = 1$ to handle the possibility that $x = 0$). Thus, the identity holds, meaning $P(0)$ is true.

Inductive Hypothesis: Given an arbitrary $k \in \mathbb{N}$, we assume that $P(k)$ is true: for all $x \neq 1$,

$$\sum_{i=0}^{k} x^i = \frac{1-x^{k+1}}{1-x}.$$

Inductive Step: Given an arbitrary $x \neq 1$, we may simplify

$$\begin{aligned}
\sum_{i=0}^{k+1} x^i &= x^{k+1} + \sum_{i=0}^{k} x^i && \text{(pull out } i = k+1 \text{ term from summation)} \\
&= x^{k+1} + \frac{1-x^{k+1}}{1-x} && \text{(Inductive Hypothesis)} \\
&= \frac{x^{k+1} - x^{(k+1)+1}}{1-x} + \frac{1-x^{k+1}}{1-x} && \text{(common denominator)} \\
&= \frac{1-x^{(k+1)+1}}{1-x}.
\end{aligned}$$

Thus, $P(k+1)$ is true under the inductive hypothesis.

By the Principle of Induction, we have proven the geometric series formula. ∎

By using properties of limits, one can use this lemma to establish the infinite geometric series formula

$$\sum_{i=0}^{\infty} x^i = \lim_{n\to\infty} \sum_{i=0}^{n} x^i = \lim_{n\to\infty} \left(\frac{1-x^{n+1}}{1-x} \right) = \frac{1}{1-x}$$

for all $-1 < x < 1$.

Main Takeaways:

- We can use the proof techniques that we have learned to argue about claims from different areas of math. The first step is always to carefully translate these claims into formal statements and understand their logical structure.
- *Induction* is a proof technique to prove that a predicate $P(n)$ over $\mathbb{N}$, called an *inductive statement*, is true for all $n \in \mathbb{N}$.
- In the *base case*, we must show that $P(0)$ is true.
- In the *inductive step*, we make an *inductive hypothesis* by assuming that $P(k)$ is true for an arbitrary $k \in \mathbb{N}$ and uses this to show that $P(k+1)$ is true.
- The Principle of Induction says that once we establish the base case and the inductive step, we may conclude that $P(n)$ is true for all $n \in \mathbb{N}$.

Exercises

6.1 Write each of the following propositions using only mathematical symbols. Then, give a proof of the proposition, Identify which proof techniques you are using.

(a) The square of any odd integer is odd.

(b) An integer is even if and only if its perfect square is even.

(c) Given any integer x, if $x^2 - 2x$ is odd, then x is odd.

(d) The sum of two odd integers is even.

(e) If the sum of two integers is even, then their difference is also even.

(f) $x^2 + 3x$ is even for any integer x. (Hint: use a case analysis.)

(g) Given any three integers, two of them sum to an even number.

6.2 In the lecture, we proved that every odd natural number can be written as the difference between two perfect squares.

(a) Prove that every natural number that is a multiple of 4 can be written as the difference of two perfect squares.

(b) Prove that if the difference of two perfect squares is even, it must be a multiple of 4. In other words, $j^2 - h^2$ can never be equal to a number *congruent to* 2 *modulo* 4 $(2, 6, 10, \ldots)$. (Hint: use a case analysis based on whether j and h are even/odd.)

(c) Argue that every integer can be represented as $a^2 - b^2 - c^2$ for some choice of $a, b, c \in \mathbb{N}$.

6.3 Find a formula for the sum of the first n positive even integers $(2, 4, 6, \ldots)$. Verify that your formula is correct using induction.

6.4 Use induction to prove the following identities.

(a) $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

(b) $\displaystyle\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

(c) $\displaystyle\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4} = \left(\sum_{i=1}^{n} i\right)^2$

$\star$ (d) $\displaystyle\sum_{i=1}^{n} i^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$

(e) $\displaystyle\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$

(f) $\displaystyle\sum_{i=1}^{n} \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1}$

6.5 An integer n is divisible by another integer d if we can represent $n = k \cdot d$ for some integer k. Prove each of the following statements of divisibility in two ways:

1. By a direct algebraic proof (factoring the expression and performing a case analysis)
2. Using induction

Which approach did you find to be easier?

(a) n^2+5n+6 is even (i.e., divisible by 2) for all $n \in \mathbb{N}$.

(b) n^3+3n^2-4n is divisible by 3 for all $n \in \mathbb{N}$.

(c) n^3+5n is divisible by 6 for all $n \in \mathbb{N}$.

(d) 9^n-1 is divisible by 8 for all $n \in \mathbb{N}$.

6.6 Prove that for all $n \in \mathbb{N}$ and for all $m \in \mathbb{N}^+$, m^n-1 is divisible by $m-1$. This is a generalization of Exercise 6.5(d). (Hint: let m be arbitrary and perform induction on n)

6.7 Given sets A and $B_1,\ldots,B_n$ for some $n \in \mathbb{N}$, prove the following properties. We have seen the special cases with $n=2$ in earlier lectures. Here, we use the "big operator notation" in a similar way to summations, so

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \ldots \cap A_n, \qquad \bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \ldots \cup A_n.$$

Hints:

- In the inductive steps, how can we represent the intersection/union of $k+1$ sets in terms of the intersection/union of k sets?
- How should we define the "empty union" and "empty intersection"?

(a) Distributivity of $\cap$: $\quad A \cap \left(\bigcup_{i=1}^{n} B_i\right) = \bigcup_{i=1}^{n} (A \cap B_i)$

(b) Distributivity of $\cup$: $\quad A \cup \left(\bigcap_{i=1}^{n} B_i\right) = \bigcap_{i=1}^{n} (A \cup B_i)$

(c) DeMorgan's Law 1: $\quad A \setminus \left(\bigcap_{i=1}^{n} B_i\right) = \bigcup_{i=1}^{n} (A \setminus B_i)$

(d) DeMorgan's Law 2: $\quad A \setminus \left(\bigcup_{i=1}^{n} B_i\right) = \bigcap_{i=1}^{n} (A \setminus B_i)$

$\star$ **6.8** Prove that for all $n \in \mathbb{N}$ and all $k \in \mathbb{N}^+$,

$$\sum_{i=1}^{n} \prod_{j=0}^{k-1} (i+j) = \frac{1}{k+1} \prod_{j=0}^{k} (n+j).$$

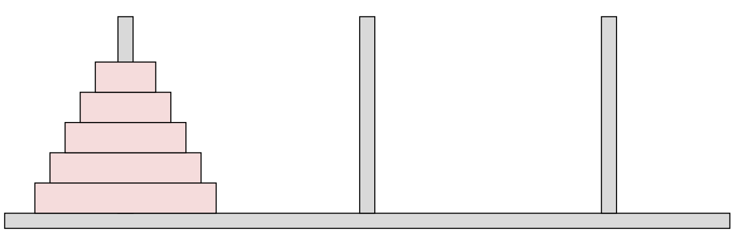
Here, the "pi" notation $\prod$ denotes the product of its arguments as its variable assumes all values within its bounds, similar to the summation notation $\sum$. For example $\prod_{i=1}^{3} i = 1 \cdot 2 \cdot 3 = 6$.

7. Recursion and Recurrences

In the previous lecture, we introduced a new proof technique, *induction*, to prove statements of the form $\forall\, n \in \mathbb{N}.\ P(n)$ for some predicate $P(n)$. Induction works by directly proving the smallest case (i.e., showing that $P(0)$ is true) and then showing how we can use smaller cases to build the solutions to bigger cases (i.e., using the assumption that $P(k)$ is true to argue that $P(k+1)$ must also true). If you have experience with programming, this structure feels very similar to writing a recursive function, which directly returns the answer to its simplest inputs and computes the answer to more complicated inputs by calling itself on simpler inputs. This similarity is no coincidence, as induction is the theoretical tool we use to verify the correctness of recursive algorithms (giving induction a central importance in computer science). In this lecture, we'll use a simple puzzle, the *Towers of Hanoi* to explore this connection. Along the way, we'll see a connection to another mathematical tool, *recurrence relations*, which are important to analyzing recursive algorithms.

7.1 The Towers of Hanoi

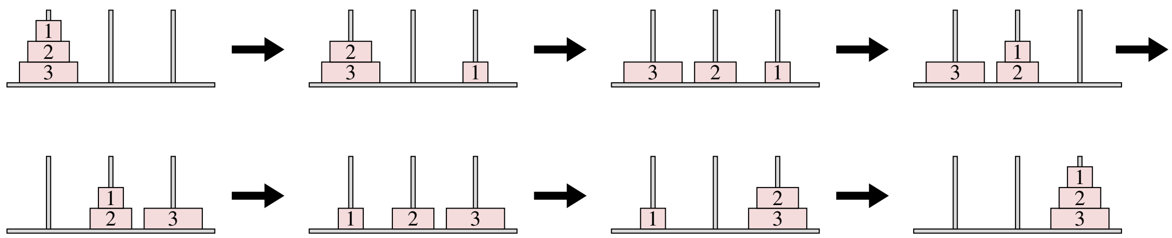
The *Towers of Hanoi* is a puzzle devised by the mathematician Édouard Lucas in 1883. Its name pays homage to the architectural style of Vietnamese pagoda temples. In the puzzle, there is a board with three pegs arranged in a line. On the leftmost peg, there is a stack of n washers ($n \in \mathbb{N}$) that decrease in diameter as they go up. Here, we visualize the game when $n = 5$.



The goal of the puzzle is to move the entire stack of washers to the rightmost peg while obeying the following two rules:

- In any move, the topmost washer can be removed from a peg and placed on the top of the stack on another peg.
- A larger washer can never sit on top of a smaller washer.

We would like to understand for which values of n, this puzzle has a solution. When there are very few washers, the puzzle is not too difficult. For example, here is a solution for $n = 3$ washers.



As more washers are added, finding such a solution becomes more difficult. Is there some "critical" number N of washers, such that once there are at least N washers, the puzzle becomes too "crowded" to be solved? We can formalize this question by defining the following predicate,

> $P(n) =$ "There's a sequence of legal moves to transfer a stack of n washers from the leftmost peg to the rightmost peg."

The domain of this predicate is $\mathbb{N}$. Our question asks whether there is some n for which $P(n)$ is false. The perhaps surprising answer is *no*. The puzzle can be solved for any number of washers, so the proposition "$\forall\, n \in \mathbb{N}.\ P(n)$" is true, but how can we prove this formally? Similar to what we saw in the previous lecture, our instinct to "show $P(n)$ is true for an *arbitrary* $n \in \mathbb{N}$," does not seem practical. It would require us to be able to describe a sequence of moves for any number of washers. However, these solutions become complex for even modestly sized instances; over 1000 moves are needed when $n = 10$. As an alternative, we can attempt a proof by induction.

7.1.1 A First Attempt

Let's follow the structure of an inductive proof we outlined in the previous lecture. We defined our inductive statement $P(n)$ above.

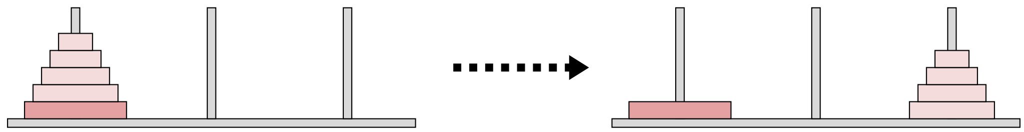
Base Case: Since we want to show that $P(n)$ is true for all natural numbers, we must start with the smallest natural number, 0. Substituting $n = 0$ into our inductive statements gives,

> $P(0) =$ "There's a sequence of legal moves to transfer a stack of 0 washers from the leftmost peg to the rightmost peg."

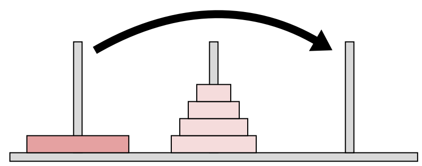
Is this true? Well, if there are no washers on the leftmost peg to begin with, then we can "move" the entire stack of washers without doing anything. This is a legal sequence of moves (we haven't broken either of the rules), and it has gotten us from the starting state of the puzzle to the desired ending state. Thus, $P(0)$ is true, albeit for some strange, seemingly trivial, reasoning. This will often be the case in the base cases; however, it is nonetheless important that we think through the base case carefully, as it serves as the foundation for the entire inductive proof.

Inductive Hypothesis: Given some arbitrary k, we assume that $P(k)$ is true: there is a sequence of legal moves to transfer a stack of k washers from the leftmost peg to the rightmost peg.

Inductive Step: As usual, we should start our inductive step by thinking about how we might use the inductive hypothesis. We should ask "How is moving k pegs from left to right going to help us move $k+1$ pegs from left to right?" At this point, we get somewhat stuck. By using our inductive hypothesis, we know that it is possible to move the k smallest washers to the right.



What about the largest washer? At some point, it needs to move to the rightmost peg, but now the smaller washers are in the way. The *only* way we can move this largest washer to the rightmost peg is if all of the other washers are stacked on the middle peg:



To do this, we need a way to move the smallest k washers to the middle peg. Our inductive statement doesn't allow this; it only allows us to move the stack from the left peg to the right peg. If we consider a stronger inductive statement, however, we can gain the flexibility we need to complete the proof.

7.1.2 Attempt 2: Strengthened Inductive Statement

Proof. We'll proceed by induction on n, with the inductive statement

> $P'(n)$ = "There is a sequence of legal moves to transfer a stack of n washers from any starting peg s to any final peg f."

Letting s be the leftmost peg and f be the rightmost peg gives our original statement, so proving this stronger claim will suffice.

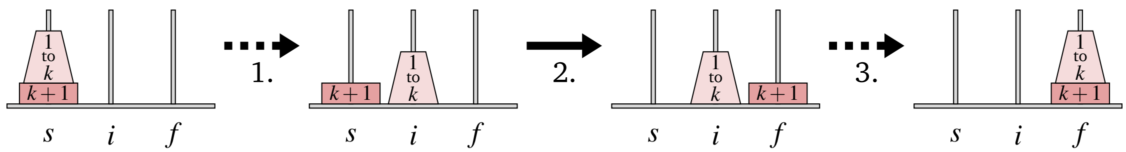
Base Case: Just as before, we require no moves to transfer a stack of 0 washers between any two pegs.

Inductive Hypothesis: Given an arbitrary $k \in \mathbb{N}$, we assume that $P'(k)$ is true: that is, there is a sequence of legal moves to transfer k washers from *any* peg to *any other* peg.

Inductive Step: Let us suppose that there are $k+1$ washers stacked on peg s which we wish to move to peg f. We'll use the letter i to denote the other (*intermediary*) peg. We can use the following three-step procedure to transfer the washers:

1. Transfer the top k washers from peg s to peg i.
2. Move the largest washer from peg s to peg f.
3. Transfer the top k washers from peg i to peg f.

We illustrate this procedure in the following diagram, where we use a trapezoid to represent the smallest k washers.



Our inductive hypothesis ensures that steps 1 and 3 can be completed using a sequence of legal moves; the $(k+1)$st washer can be safely ignored as it is below the k washers that are moved in this step. In addition, step 2 constitutes a single legal move; we are moving the topmost washer from peg s and placing it on top of peg f, where it is not atop another washer. In all, our procedure gives a legal way to transfer $k+1$ washers to a new peg, which shows that $P(k+1)$ is true under our assumption that $P(k)$ is true.

By the Principal of Induction, we conclude that $P'(n)$ is true for all $n \in \mathbb{N}$. In particular, the *Towers of Hanoi* puzzle can be solved for any number of washers. ∎

R The three-step procedure that we developed in our proof is a *recursive algorithm* since steps 1 and 3 carry out the same procedure on fewer washers. We could implement the procedure by writing code such as:

```
1. hanoi(n,s,f,i):
2.     if n == 0: return
3.     hanoi(n-1,s,i,f)
4.     % move washer n from s to f
5.     hanoi(n-1,i,f,s)
```

Our inductive argument establishes the correctness of this recursive algorithm. The base case of the recursion (line 2), which directly handles the “simplest” inputs to the function, is verified by the base case(s) of the inductive proof. In the inductive step, we assume that the recursive calls (lines 3 and 5) have the correct behavior (our inductive hypothesis) and show that this leads the outer function call to have the correct behavior. Many people will refer to the verification of the correctness of recursive algorithms as *structural induction* since we are inducting over the recursive call structure of the algorithm.

7.2 Recurrence Relations and Closed Formulas

How many moves will our Hanoi procedure require? Let us define h_n to be the number of moves required for an instance n washers. We know that $h_0 = 0$; there are no washers to move. Also, the solution we saw earlier for $n = 3$ washers is a special case of this procedure and requires 7 moves, so $h_3 = 7$. Can we find a formula for any n? Let’s look back at the 3-step procedure that we gave in our proof. To relocate $n+1$ washers, we first relocate n washers (h_n moves), then we move the largest washer to a new peg (1 move), and then we relocate n washers again (h_n moves). Adding the move counts from these three steps together, we find that $h_{n+1} = 2h_n + 1$. These facts,

$$h_0 = 0, \qquad h_{n+1} = 2h_n + 1 \quad \text{for all } n \in \mathbb{N}$$

describe a *recurrence relation* for h_n, since later terms are expressed using earlier terms. We would like to instead find a *closed formula*, which expresses h_n only in terms of n.

Definition 7.1 — Recurrence Relation, Closed Formula.
A *recurrence relation* for a sequence $(a_n)_{n\in\mathbb{N}}$ expresses later terms in the sequence using earlier terms.

In contrast, a *closed formula* for the sequence expresses each a_n only in terms of n.

The following lemma gives this closed form, which we can prove by induction.

Lemma 7.1 Let h_n represent the number of moves that our Towers of Hanoi solution requires to move n washers, so that $h_0 = 0$ and $h_{n+1} = 2h_n + 1$ for all $n \in \mathbb{N}$. Then, $h_n = 2^n - 1$.

Proof. We perform a proof by induction with inductive statement $P(n) =$ "$h_n = 2^n - 1$."

Base Case: When $n = 0$, our recurrence relation tells us that $h_0 = 0 = 2^0 - 1$.

Inductive Hypothesis: We assume that $P(k)$ is true for some $k \in \mathbb{N}$. That is, $h_k = 2^k - 1$.

Inductive Step: We wish to argue that $P(k+1)$ is true, meaning $h_{k+1} = 2^{k+1} - 1$. By using our recurrence relation, we have

$$h_{k+1} = 2h_k + 1 \overset{\text{IH}}{=} 2\cdot(2^k - 1) + 1 = 2\cdot 2^k - 2 + 1 = 2^{k+1} - 1.$$

The labeled equality is where we have applied the inductive hypothesis.

Conclusion: Since we have established the base and inductive steps, we conclude by the principle of induction that $h_n = 2^n - 1$ for all $n \in \mathbb{N}$. ■

A recurrence relation gave us a natural way to express the complexity (number of moves) of our recursive algorithm, and a proof by induction served as a natural way to verify a closed formula for this recurrence relation.

7.2.1 A Proof of Optimality

As a final example, we can use induction to argue that the three-step procedure that we described to solve the *Towers of Hanoi* puzzle is *optimal*; it solves the puzzle in the smallest possible number of moves.

Lemma 7.2 Any procedure for the *Towers of Hanoi* puzzle with n washers requires at least $2^n - 1$ moves.

Proof. We perform induction on n, where the inductive statement $P(n)$ is given in the lemma.

Base Case: When there are $n = 0$ washers, it is not possible to complete the puzzle in fewer than $2^0 - 1 = 0$ moves.

Inductive Hypothesis: We assume that at least $2^k - 1$ moves are required to move a stack of k washers.

Inductive Step: This argument is similar to our earlier reasoning. We know that at some point in the procedure, we must move the largest washer $(k+1)$. To do this, the other k washers must all be stacked on a different peg. Our inductive hypothesis says this can only occur after at least $2^k - 1$ moves. Then, we must use a move to relocate the largest washer. Finally, when the largest washer is first moved to the rightmost peg, the k smaller washers must again all be stacked on a different peg. To move them onto the rightmost peg will require at least $2^k - 1$ moves. In all, we have argued that any procedure requires at least $2^k - 1 + 1 + 2^k - 1 = 2^{k+1} - 1$ moves.

Conclusion: By induction, we conclude that any procedure for the Towers of Hanoi puzzle with n washers requires at least $2^n - 1$ moves. As our procedure uses exactly this many moves, it is optimal. ■

Main Takeaways:

- A *recursive* algorithm carries out a computation on a larger input by using results of the same computation carried out on smaller inputs. Induction provides a natural framework for reasoning about recursive algorithms.
- Sometimes, we need to strengthen an inductive statement beyond what we are trying to prove to give us more tools to work with in the inductive step.
- Recurrence relations express the terms of a sequence as a function of earlier terms, while a closed formula expresses the n'th term of the sequence directly as a function of n.

Exercises

7.1 The *No-Jump Hanoi* puzzle has the same setup as the Towers of Hanoi but with one extra rule: a washer cannot be directly moved between the leftmost and rightmost pegs (in other words, moves must be to adjacent pegs). In this exercise, you'll use induction to establish results similar to those we obtained for the original puzzle.

(a) Explain why the *Towers of Hanoi* solution for $n = 3$ given in the lecture is not a legal solution to the *No-Jump Hanoi* puzzle.

(b) Find a solution to the *No-Jump Hanoi* puzzle when $n = 3$. How many moves does your solution require?

(c) Argue that the No-Jump Hanoi puzzle has a solution for each possible number of washers. You should be careful about how you define the inductive statement, as this will be critical for the rest of the exercise.

(d) Let h'_n be the number of moves required to solve the No-Jump Hanoi puzzle using your procedure from part (a). Write a recurrence relation for h'_n. Then, use induction to establish the closed formula $h'_n = 3^n - 1$.

⋆ (e) Use induction to argue the optimality of this solution.

7.2 In the *Double Hanoi* puzzle there are $2n$ washers, with two washers with each of n distinct sizes (also arranged on the leftmost peg with larger washers below smaller washers). The goal is again to move all washers to the rightmost peg obeying the same rules as before. It does *not* matter which of the same-sized washers sits atop the other; they are indistinguishable. Alice and Bob each believe they have found a solution to this puzzle.

<u>Alice's Solution:</u> Use the Towers of Hanoi procedure for n washers, but replace each single move with two moves, moving both washers of that size.

<u>Bob's Solution:</u> Use the Towers of Hanoi procedure for $2n$ washers (effectively viewing the lower of the same-sized washers as "slightly larger" than the upper washer).

(a) Convince yourself that both of these solutions work. (There's nothing to write here; just make sure that you understand why these approaches are valid.)

(b) Before you move on to the next part, which solution to you expect to be more efficient? Why do you think this?

(c) Find closed formulas for A_n, the number of moves needed for Alice's solution with $2n$ washers, and B_n, the number of moves for Bob's solution. Which solution requires fewer moves? (No induction is needed here.)

(d) Use induction to prove that better solution is optimal for this puzzle.

When we posed this problem, we assumed that the order of the two same-sized washers does not matter. However, it is interesting to consider how these procedures affect the order. Suppose that in the initial configuration, we labeled the upper washer of each size with a "u" and the lower washer with an "ℓ" and then carried out each procedure.

(e) Explain why the "u" washers will all sit atop the same-sized "ℓ" washers after carrying out Bob's procedure.

(f) Explain why this will not be the case for Alice's procedure. Can you figure out which size(s) of washers will have the "u" washer below the "ℓ" washer? (Hint: Exercise 7.4 will be helpful.)

$\star$ (g) Find (with proof) the optimal solution that ensures the washers end up in the same order they started in (i.e., the "u" washer always sits on top of the "ℓ" washer of the same size). How many moves are required?

$\star$ **7.3** Argue that in a Towers of Hanoi instance with n washers it is possible to get from any legal position (that is, a position where no larger washer sits atop a smaller washer) to any other legal position. This property is often referred to as *transitivity*. (Hint: Think about how you can break up this task into smaller sub-problems and use this to come up with an inductive statement.)

7.4 Given an instance of size n, label the washers $0,\dots n-1$, where 0 is assigned to the smallest washer and n is assigned to the largest washer (this is slightly different from the numbering in the lecture figures). For each $n \in \mathbb{N}$ and each size $s < n$, let $m(n,s)$ be the number of times that washer s changes pegs in our n-washer Towers of Hanoi procedure. We wish to find a closed formula for $m(n,s)$.

(a) By tracing through some small examples, calculate some values of $m(n,s)$ and use these to guess a closed formula.

(b) Verifying this closed formula is different from the others you've seen because there are two variables we need to quantify over. One way to do this is as follows:

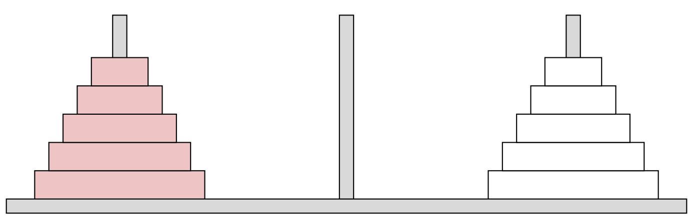
- To handle the quantification over s, we will first select an *arbitrary* s.
- Given this value of s, perform induction on n starting from $n = s+1$. (The next lecture gives more detail about induction with a different base case.)

Draw out a "stars and boxes" diagram similar to the one from the previous lecture which has a box for each (n,s) for which we must verify the closed formula. This picture will be part of a 2D grid. Identify which boxes are verified by the base cases, and draw arrows that show which boxes are verified by the inductive steps. You should find that every box in the grid will be verified eventually.

(c) Carry out the induction as described in part (b). In the inductive step, you should refer back to the 3-step procedure from the lecture to derive a recurrence relation for the number of times a particular washer is moved.

(d) Lemma 7.1 tells us that the Towers of Hanoi procedure with n washers requires $2^n - 1$ moves. Verify that your closed formula is correct by showing that $\sum_{s=0}^{n-1} m(n,s) = 2^n - 1$ for each $n \in \mathbb{N}$. Does this formula look familiar?

⋆ **7.5** In the *Swapping Towers* puzzle, there are two stacks of n washers: a pink stack on the leftmost peg and a white stack on the rightmost peg. The corresponding washers of each color have the same size. The instance with $n = 5$ is pictured below.



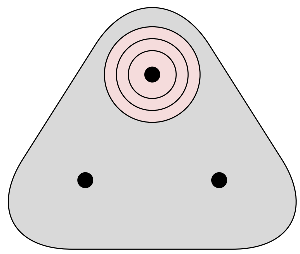
The goal of the puzzle is to switch the positions of the towers, so the pink is on the rightmost peg and the white is on the leftmost peg. Similar to *Towers of Hanoi*, only one washer (of either color) can be moved at a time, and each washer can only be placed on top of an equal-sized or larger washer. The colors of the washers do not factor into the stacking rule.

Argue that for each $n \geq 1$ the *Swapping Towers* puzzle has a solution that requires at most

$$S_n = \tfrac{1}{6}\left(7 \cdot 2^{n+2} + (-1)^{n+1} - 18n - 21\right)$$

moves. (Hint: Break down the puzzle into many smaller tasks and reason separately about each tasks. It may help to separately consider the even/odd values of n to eliminate the $(-1)^n$ term from the above expression.)

7.6 In the *Cyclic Hanoi* puzzle, imagine that the three pegs on the board are arranged in a triangle rather than a line. An overhead view of the puzzle for $n = 3$ looks like:



The goal of the puzzle is to move all of the washers from the top peg to the lower-right peg. The moves must obey the original *Hanoi* rules (i.e., one washer is moved at a time, and a larger washer cannot sit atop a smaller washer). Additionally, washers can only be moved clockwise (from the top peg to lower-right, from lower-right to lower-left, and from lower-left to top).

(a) Find a solution to the *Cyclic Hanoi* puzzle when $n = 3$. How many moves did your solution require?

(b) Use induction to argue that the *Cyclic Hanoi* puzzle has a solution for any number of washers. (Hint: To develop your algorithm, it may help to write two procedures, one that moves a stack of n washers clockwise, and one that moves a stack of n washers counterclockwise.)

Let R_n be the number of moves required for the optimal solution of the *Cyclic Hanoi* puzzle (in other words, R_n is the minimum number of moves required to move a stack of n washers one peg clockwise). Similarly, let L_n be minimum number of moves required to move a stack of n washers one peg counterclockwise.

(c) Write a *mutual recurrence relation* for R_n and L_n. This means you should express R_n in terms of earlier terms of the R and L sequences. Similar to what we did in lecture, you can derive these recurrences by reasoning about the recursive procedure that you developed in part (b).

⋆ (d) Use induction to *simultaneously* verify that R_n and L_n satisfy the closed formulas:

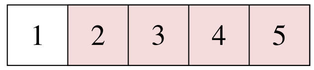
$$R_n = \frac{(1+\sqrt{3})^{n+1} - (1-\sqrt{3})^{n+1}}{2\sqrt{3}} - 1, \qquad L_n = \frac{(1+\sqrt{3})^{n+2} - (1-\sqrt{3})^{n+2}}{4\sqrt{3}} - 1.$$

7.7 Consider a row of n pink boxes. You would like to re-color all these boxes to be gray while adhering to the following rules:

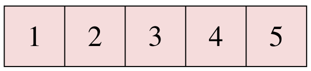
- At any point, each box must be pink, white, or gray.
- Boxes must be re-colored one at a time.

- Each re-coloring may only change the leftmost box of a color. In other words, a box may only be re-colored if it was the leftmost box of its old color, *and* it becomes the leftmost box of its new color.

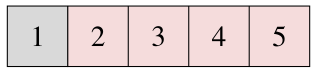
To help understand these rules, consider the following state after one move is made.



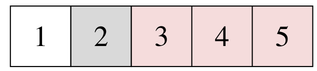
From this position, there are three possible moves:



We can re-color Box 1 pink. Here, Box 1 was the leftmost white box, and now is the leftmost pink box.



We can re-color Box 1 gray. Here, Box 1 was the leftmost white box, and now is the leftmost gray box.



We can re-color Box 2 gray. Here, Box 2 was the leftmost pink box, and now is the leftmost gray box.

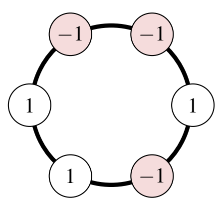
Note that we *cannot* re-color Box 2 white, as it would *not* become the leftmost white box.

(a) Find a valid procedure to re-color a row of $n = 3$ boxes.

(b) Use induction to prove that it is always possible to legally re-color all boxes to gray.

(c) Let b_n be the number of moves required to re-color a row of n boxes. Find a recurrence relation and a closed formula for b_n. Verify that your closed formula is correct using induction.

7.8 On a circle, there are $2n$ distinct labeled points. n of these points have been labeled with "1" and n of these points have been labeled with a "-1". For example, if $n = 3$, one possible labeling is:

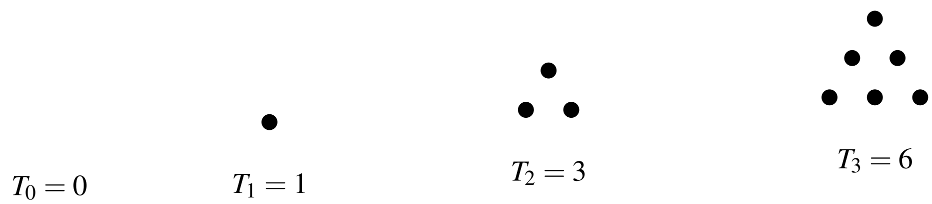


Consider the following procedure carried out starting at some point p (not one of the labeled points) on the circle:

> Start at point p with value $v = 0$. Then, travel around the circle in a clockwise direction. Each time you encounter a point labeled 1, increase v by 1. Each time you encounter a point labeled -1, decrease v by 1. Stop when you return to p.

Argue that for each n, and no matter how the labels are assigned, there is some choice of p that ensures that v is never negative during this procedure. For the pictured example, choosing p to be the bottommost point on the circle will work.

7.9 The *triangular number* T_n ($n \in \mathbb{N}$) is the number of dots in a triangular array with n rows. The first few triangular numbers are:



(a) Find a recurrence relation for the triangular numbers.

(b) Use your recurrence relation to write each triangular number as a summation.

(c) Use induction to prove that $T_n = \frac{n(n+1)}{2}$.

7.10 The *tetrahedral number* Te_n ($n \in \mathbb{N}$) is the number of dots in a tetrahedral (triangular pyramid) arrangement of dots with n layers. They are described by the formula

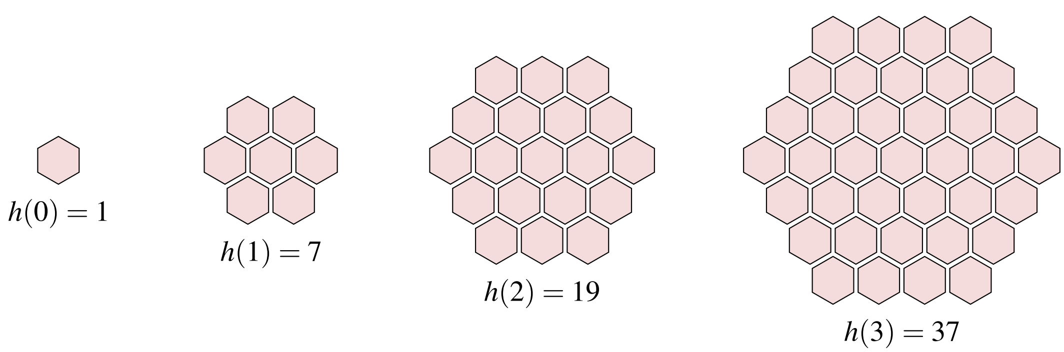
$$Te_n = \sum_{i=1}^{n} T_i,$$

where T_i is the i'th triangular number (See Exercise 7.9).

(a) Write down a recurrence relation for Te_n. (Hint: first, substitute in the answer from Exercise 7.9(c).

(b) Use induction to argue that $Te_n = \frac{n(n+1)(n+2)}{6}$.

7.11 When honeybees construct their hives, they arrange the cells in a hexagonal pattern. Let's suppose that the overall structure of the hive takes the shape of a larger hexagon, consisting of a central cell and ℓ concentric layers of cells around it. We'll let h_ℓ denote the number of cells in a hive with ℓ layers for each $\ell \in \mathbb{N}$. We visualize the first few values of h_ℓ below:



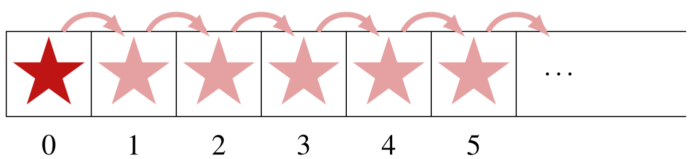
(a) Write a recurrence relation for h_ℓ. Explain why your recurrence is, in fact, satisfied by the hexagonal numbers.

(b) Find a closed formula for the hexagonal numbers h_ℓ (Hint: Relate them to the triangular numbers from Exercise 7.9.)

(c) Use induction to verify the correctness of your closed formula from part (b).

8. Strong Induction

In the previous lectures, we introduced induction as a technique for verifying that a predicate $P(n)$ is true for all $n \in \mathbb{N}$. We used the following figure to help visualize the method.



The first step of an inductive proof was to directly prove that the predicate is true for $n = 0$ (after which we drew a dark star in the 0 box). Next, by assuming that $P(k)$ was true for an arbitrary $k \in \mathbb{N}$ and using this to show that $P(k+1)$ was also true, we proved that $P(k) \Rightarrow P(k+1)$ for all $k \in \mathbb{N}$, which we visualized by drawing arrows from each box to the next box on its right. Then, whenever there is a star in the box at the start of an arrow, we can draw a star in the box at the end of the arrow; by repeating this process, we can place a star in any box, meaning we can show that $P(n)$ is true for all $n \in \mathbb{N}$.

This inductive framework is very useful. As we saw last lecture, it has many applications in computer science and mathematics. However, the outline that we gave is very rigid. We always started with the single base case, and we always established the implication $P(k) \Rightarrow P(k+1)$. Today we'll see some examples where this framework does not lead to the most natural argument. Instead, we'll see how being able to start with a different base case (or base cases) and working with different inductive steps can be useful in our proofs.

8.1 The Cupcake Problem

As our first example, we will consider the following question:

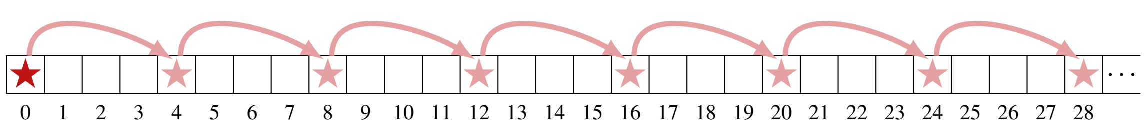
> *A bakery packages their cupcakes into square containers, selling them in boxes of 4 (small) and 9 (large). Which numbers of cupcakes is it possible to order?*

We can start trying to answer this question by playing around with some small examples. There are some numbers that it is easy to see how to order. For example, we can order 4 cupcakes by purchasing one small box, we can order 9 cupcakes by purchasing a large box, and we can order 13 cupcakes by purchasing one of each. There are other numbers that we cannot order. For example, we cannot order 5 cupcakes. Ordering one small box will not be enough cupcakes. However, if we add another small box or swap to a large box, this will be too many cupcakes. From these examples, we see that the question is really asking for us to decide how many small boxes and how many large boxes we should order to get a particular number of cupcakes. By this observation, we see that the predicate

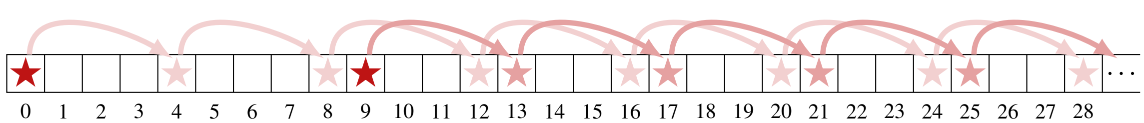
$$P(n) = \text{“}\exists\, x \in \mathbb{N},\ \exists y \in \mathbb{N}.\ 4x + 9y = n\text{”}$$

will be true if and only if it is possible to order exactly n cupcakes. Our task is to figure out which n makes $P(n)$ true. Our basic induction proof, which establishes that $P(n)$ is true for all $n \in \mathbb{N}$, will not work; this conclusion is false since $P(5)$ is false.

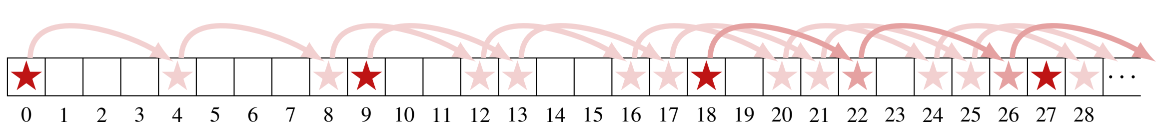
The issue here is with our inductive hypothesis. The implication $P(k) \Rightarrow P(k+1)$ cannot be true for all $k \in \mathbb{N}$ since $P(4) \not\Rightarrow P(5)$. Let's think about what would be a better choice of an inductive hypothesis by asking ourselves the following question: *If we know that $P(k)$ is true for some $k \in \mathbb{N}$, which other numbers do we know will make P true?*. In the language of the problem, if we can order k cupcakes, which other numbers of cupcakes can we order? Well, by adding one more small box, we could order $k+4$ cupcakes. This means that we have the implication $P(k) \Rightarrow P(k+4)$. We definitely can order 0 cupcakes, and this implication tells us that we can also order 4 cupcakes, or 8 cupcakes, or 12 cupcakes, etc. We can visualize this in our diagram by drawing arrows that point four boxes to the right.



This diagram tells us that $P(n)$ is true for all multiples of 4. Is it possible to fill in the gaps? We still haven't considered the box of 9 cupcakes. $n = 9$ as a second base case, we can use the implication $P(k) \implies P(k+4)$ to argue that we can also order 13, 17, 21, etc. cupcakes.



Similarly, we could order two large boxes and any number of small boxes, allowing us to order $18, 22, 26, \ldots$ cupcakes. If we purchase three large boxes, we could order 27 cupcakes.



Let's stop and examine this last diagram. Notice how at the very right, we see four consecutive boxes with stars, $n = 24, 25, 26, 27$. Once we see this, we know that it is possible to order any larger number of cupcakes. This is because the four arrows leaving these boxes will place stars in the next four boxes $n = 28, 29, 30, 31$, and the four arrows leaving these will in turn place stars in the next four boxes $n = 32, 33, 34, 35$. By iterating this idea, we could eventually place a star in any box $n \geq 24$. Thus, we have an answer to our original question.

It is possible to order $0, 4, 8, 9, 12, 13, 16, 17, 18, 20, 21, 22$*, or any* $n \geq 24$ *cupcakes.*

8.2 Strong Induction

To give a formal proof of this claim, we will need to introduce a more general framework called *strong induction*.

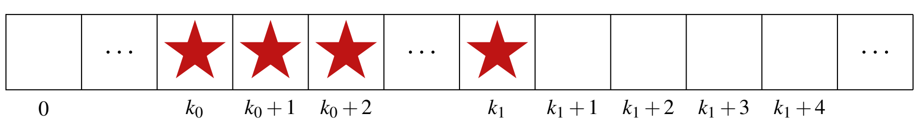
Strong Induction:

Given a predicate $P(n)$ over the natural numbers, suppose that we can argue that:

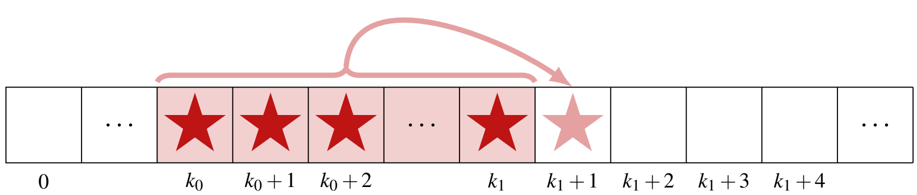
- **Base Step(s)**: $P(k_0), P(k_0+1), \ldots, P(k_1)$ are true.
- **Inductive Step**: For any $k \geq k_1$, if $P(j)$ is true for *all* $k_0 \leq j \leq k$, then $P(k+1)$ is also true.

Then, we may conclude that $P(n)$ is true for all $n \geq k_0$.

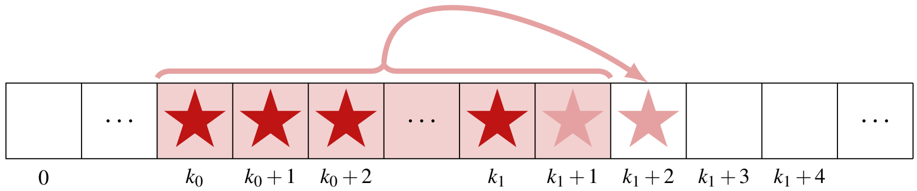
k_0 is the starting point of our induction; notice that the conclusion depends on k_0. Just as with ordinary induction, we start by directly reasoning about some small cases. In our diagram, we visualize this as drawing red stars in the range of boxes from k_0 to k_1.



Now, given a particular value of k, our inductive hypothesis is that $P(j)$ is true for all j between k_0 and k (inclusive), and the inductive step says that this is enough to conclude that $P(k+1)$ is true. We can think about this implication as scanning to check that all of the cells from k_0 to k have been filled with stars, and then using this as justification to place a star in cell $k+1$. When $k = k_1$, we check the highlighted stars and draw a pink star in cell k_1+1.



Similarly, when $k = k_1+1$, we check the highlighted stars (which now include cell k_1+1) and draw a pink star in cell k_1+2.



By iterating this process, we can eventually draw a star in any cell to the right of cell k_0, which is exactly the conclusion of strong induction.

Let's use this strong induction framework to formalize our result from the cupcake problem.

Lemma 8.1 Every natural number $n \geq 24$ can be represented as $n = 4x + 9y$ for some $x, y \in \mathbb{N}$.

In our proof, we will use starting point $k_0 = 24$ and last base case $k_1 = 27$.

Proof. We perform a proof by strong induction using the inductive statement

$$P(n) = \text{“}\exists\, x \in \mathbb{N},\ \exists y \in \mathbb{N}.\ 4x + 9y = n.\text{”}$$

Base Step: We directly establish four base cases.

$$\begin{aligned} n = 24: &\qquad 4 \cdot 6 + 9 \cdot 0 = 24 \\ n = 25: &\qquad 4 \cdot 4 + 9 \cdot 1 = 25 \\ n = 26: &\qquad 4 \cdot 2 + 9 \cdot 2 = 26 \\ n = 27: &\qquad 4 \cdot 0 + 9 \cdot 3 = 27 \end{aligned}$$

Inductive Hypothesis: Given $k \geq 27$, we assume that $P(j)$ is true for all $24 \leq j \leq k$. In particular, this tells us that $P(k-3)$ is true, so $k - 3 = 4x + 9y$ for some $x, y \in \mathbb{N}$.

Inductive Step: We must argue that $P(k+1)$ is true. That is, we must find a way to represent $k+1$ as a sum of 4s and 9s. Applying the inductive hypothesis, we have

$$k + 1 = (k-3) + 4 \overset{\text{(IH)}}{=} 4x + 9y + 4 = 4(x+1) + 9y.$$

Since x is an integer, so is $x+1$. Thus, we have found a way to represent $k+1$ as 4 times an integer plus 9 times an integer, meaning $P(k+1)$ is true.

Conclusion: By strong induction, we conclude that $P(n)$ is true for all $n \geq 24$. ■

This lemma gives a formal justification for our answer for all $n \geq 24$. It doesn't say anything about $n < 24$. However, there are only finitely many more cases to check, which we can do by brute force. For each $n < 24$, one can either describe how to order n cupcakes or show that it is impossible. Exercise 8.1 asks you to do this for $n = 23$, which will show that our choice of $k_0 = 24$ in our inductive proof was the smallest possible.

8.3 Induction and Games

As a second application, we will use strong induction to analyze a game.

Definition 8.1 — Game, State, Action, N-position, P-position.

In a two-player *game*, players take turns choosing *actions* with the goal of winning. On each turn, the player observes the game *state* and selects from an available set of actions. Each action either causes them to win the game, lose the game, or transition the game to a new state, at which point the turn passes to the other player.

A game state is an *N-position* if it affords a winning strategy to the *next* player to move, meaning they can take actions that will eventually guarantee them a win regardless of the actions taken by the other player. Otherwise, a game state is a *P-position*, since the *previous* player to move has a winning strategy.

R For our purposes, we will only consider games that are guaranteed to end after a finite number of moves, which ensures that each state can be labeled as an N-position or P-position. More generally, a game can get caught in a “stalemate”, where the players can cycle around a fixed sequence of states to prevent the game from ever concluding unfavorably.

The *terminal* states of a game are those in which a winner has been determined, meaning no subsequent actions will be taken. The game rules dictate whether these terminal positions are N-positions or P-positions. Any non-terminal state in the game will require an additional move by the current player, and we can evaluate the position according to the following two rules (which we may recall from our analysis of *Chomp*).

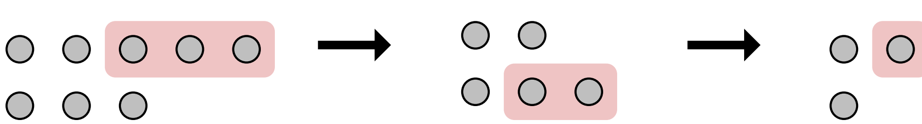
- A non-terminal state is an N-position if the current player has at least one action to transition the game to a P-position.
- A non-terminal state is a P-position if every possible action by the current player transitions the game to an N-position.

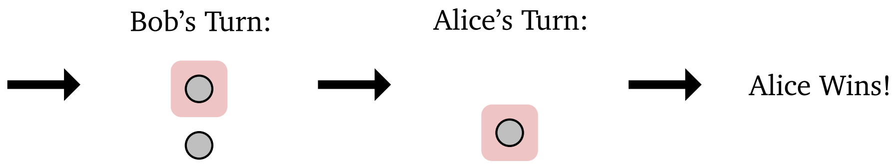
Now, let’s use these definitions to analyze a game called *2-Pile Nim*:

> *The game starts with two rows of stones. The top row has r_1 stones and the bottom row has r_2 stones. On their turn, a player must remove some number (≥ 1) of stones from one of the rows. A player loses if there are no stones left for them to take.*

As a convention, we’ll call the players Alice and Bob and assume that Alice goes first. If the rows have sizes $r_1 = 5$ and $r_2 = 3$, one possible run of the game is:

Alice’s Turn: Bob’s Turn: Alice’s Turn:





In this case, Alice won the game. Was this guaranteed, or did she get lucky? In other words, is game state with one row of 5 stones and one row of 3 stones an N-position or a P-position? One way to answer this question would be to draw out a *graph* of all the possible game states and apply the rules above to label the states from the terminal positions back to the starting position. However, this procedure grows unwieldy even for moderate row sizes.

Instead, we can look at the game from the perspective of strong induction. When a player takes a turn, they remove some of the stones from the board and then send the other player to a new, smaller game state. If we could classify all smaller states as N-positions or P-positions, we would know how to classify the current state. In the case of 2-Pile Nim, there is a rather simple characterization of the P-position given by the following lemma.

Lemma 8.2 In 2-Pile Nim, the state with row sizes r_1 and r_2 is a P-position if and only if $r_1 = r_2$.

Proof. We'll do a proof by strong induction with the inductive statement

$P(n) =$ "Any 2-Pile Nim state with rows sizes r_1 and r_2 with $r_1 + r_2 = n$ is a P-position if and only if $r_1 = r_2$."

Base Step: When $n = 0$, the only game position with $r_1 + r_2 = 0$ has $r_1 = r_2 = 0$. This state is reached when the previous player takes the last stone and wins the game, so is a P-position.

Inductive Hypothesis: Given $k \in \mathbb{N}$ we assume that $P(j)$ holds for all $0 \leq j \leq k$: Any game state with at most k stones is a P-position if and only if the rows are even.

Inductive Step: We consider an arbitrary game position with $r_1 + r_2 = k + 1$ stones. We separate the analysis into two cases.

<u>Case 1:</u> $r_1 = r_2$

No matter what the current player does, since they must remove at least one stone from one row but no stone from the other, they will make the rows uneven and leave the game in a position with strictly fewer stones. By the inductive hypothesis, the game is left an N-position. Thus, the current state is a P-position.

<u>Case 2:</u> $r_1 \neq r_2$

The current player should remove $|r_1 - r_2|$ stones from the longer row to make the rows even. By the inductive hypothesis, the game is left in a P-position. Thus, the current state is an N-position.

Together, these cases show that $P(k+1)$ is true.

Conclusion: By strong induction, we conclude that $P(n)$ is true for all $n \in \mathbb{N}$. In particular, Alice has a winning strategy for 2-Pile Nim if and only if $r_1 \neq r_2$. ■

In this proof, we relied on strong induction in this proof to account for any possible number of stones that the players could choose to take on each turn.

Main Takeaways:

- Sometimes, we need to select different base cases or structure the inductive step differently based on what we are trying to prove.
- When we start our induction with base case $n = k_0$, we can only conclude that $P(n)$ is true for $n \geq k_0$.
- In *Strong Induction* our inductive hypothesis assumes that the inductive statement is true for every value j between k_0 and k.
- Games have a structure that is amenable to inductive arguments.

Exercises

8.1 Use a case analysis to argue that we cannot represent $23 = 4x + 9y$ for any $x, y \in \mathbb{N}$.

8.2 Argue that every natural number $n \geq 50$ can be represented $n = 6x + 11y$ for some $x, y \in \mathbb{N}$. Also, show that 49 cannot be represented in this way.

8.3 We can also consider sums with more than two sizes. Argue that every natural number $n \geq 18$ can be represented $n = 5x + 8y + 11z$ for some $x, y, z \in \mathbb{N}$. Also, show that 17 cannot be represented in this way.

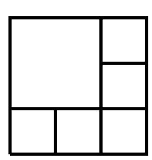
8.4 Give an example of $a, b \in \mathbb{N}$ for which there is no k_0 such that for all $n \geq k_0$ we can represent $n = ax + by$ for some $x, y \in \mathbb{N}$. (There are a lot of letters here. It may help to write out this sentence symbolically to better track the quantifiers and figure out what needs to be shown.

All of the preceding exercises relate to the following question:

> *Given some $a, b \in \mathbb{N}$, can we figure out the smallest k_0 such that for every $n \geq k_0$, we can represent $n = ax + by$ for some $x, y \in \mathbb{N}$?*

This is known as the *Frobenius Coin Problem*. In the case of two different "part sizes", there is a closed formula for the largest number that cannot be represented. Try to figure it out based on the examples we've provided. You can check your answer by looking up *Sylvester's Formula*.

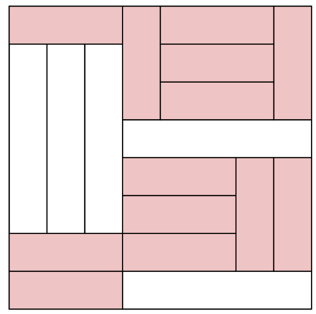
8.5 For which values of $n \in \mathbb{N}$ is it possible to divide a square into n smaller squares? For example, $n = 6$ is possible, as shown below.



Prove your answer is correct using strong induction.

8.6 Imagine that you have an infinite supply of 3×1 pink tiles and 5×1 white tiles.

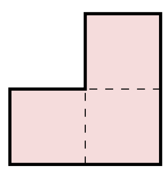
(a) For which $n \in \mathbb{N}$ is it possible to cover an $n \times n$ square with these tiles? For example, $n = 8$ is possible with the following tiling:



(b) Let T denote the set of all feasible square sizes you computed in part (a). Let's imagine the pink tiles are expensive, so we would like to find a tiling that uses as few 3×1 tiles as possible. For each $t \in T$, determine, with proof, this minimum necessary number of 3×1 tiles.

(c) Generalize your result from part (a) to rectangular areas. For which $(m, n) \in \mathbb{N} \times \mathbb{N}$ can you fully cover an $m \times n$ rectangular area with these tiles?

⋆ **8.7** For which $(m, n) \in \mathbb{N} \times \mathbb{N}$ is it possible to fully cover an $m \times n$ rectangular area with non-overlapping "L"-tiles (shown below).



8.8 Use strong induction to argue that 13^n can be written as a sum of two distinct perfect squares for all $n \in \mathbb{N}$.

8.9 Use strong induction to argue that 6^n can be written as a sum of three distinct perfect cubes for all $n \geq 2$

8.10 Suppose that a sequence $a_0, a_1, a_2, \ldots$ is defined recursively, with

$$a_0 = 1, \qquad a_n = \sum_{i=0}^{n-1} (n-1-i) \cdot a_i \quad \text{for all } n \geq 1.$$

Use strong induction to argue that $a_n = 2^{n-2}$ for all $n \geq 2$.

8.11 Consider the following variation on the game *Nim*:

> *Alice and Bob take turns selecting stones from a pile of size n, with Alice going first. On each of their turns, they may choose to take either 1 or 2 stones. A player loses if there are no stones left for them to take.*

(a) For which pile sizes n can Alice guarantee a win? Verify your claim by strong induction.

(b) Same question as part (a), but now players may take 1, 2, or 3 stones on their turn.

⋆ (c) Same question as part (a), but now there are 2 piles with size n. On a player's turn, they may select 1 or 2 stones from one pile (and none from the other).

8.12 In this exercise, we consider another game.

> *Alice and Bob have a chocolate bar with segments arranged in a $m \times n$ grid. They take turns (Alice going first) breaking the chocolate bar along one of the grid lines, separating it into two rectangular pieces. They select one of the pieces to eat, pass the other piece to the other player, and the game continues. The loser of the game, who must pay for the candy, is whoever ends up with the final 1×1 segment (that cannot be broken).*

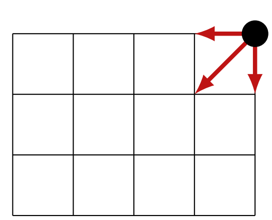
For which values of m, n does Alice have a winning strategy? (Hint: Have we seen this game before, just dressed up differently?)

8.13 Let's consider a variant of the game from Exercise 8.12:

> *Alice and Bob again take turns breaking the $m \times n$-segment chocolate bar. However, this time they only eat any 1×1 individual segments and pass all of the larger pieces to the other player. On each turn, they may break only one piece. The loser of the game is the first person who is not passed any chocolate.*

For which values of m, n does Alice have a winning strategy? Use strong induction to prove your answer is correct.

8.14 A stone is placed on the upper right square of an $m \times n$ grid. The instance with $m = 3$ and $n = 4$ is shown below.



Alice and Bob take turns moving the stone toward the lower left corner. On each turn, they can move the square one unit down, one unit left, or diagonally down left to the next corner (as depicted by the red arrows). A player wins if they move the stone to the lower left corner on their turn.

For which values of m, n does Alice have a winning strategy?

8.15 Consider the following game:

> *In front of Alice and Bob, there are two piles of candy containing m pieces and n pieces. On a player's turn, they select one of the piles and eats all of the candy in this pile. Then, they divide the other pile into two (non-empty) piles. A player loses if, on their turn, they are unable to complete this procedure (in other words, the pile that they choose not to eat contains one piece of candy, so it cannot be divided).*

For which values of m, n does Alice have a winning strategy? Use strong induction to prove your answer is correct.

⋆ **8.16** Consider the following game:

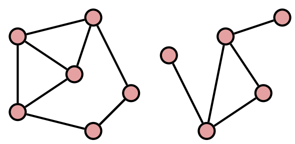
> *In front of Alice and Bob, there are k piles of candy with sizes $s_1 \geq s_2 \geq \ldots \geq s_k$. On a player's turn, they must eat the same number (≥ 1) of candies* from each of the piles. A player loses if they are left with no candy at the beginning of their turn.

Describe a property of $s_1, \ldots, s_k$ such that Alice has a winning strategy if and only if this property is satisfied. Verify that your property is correct using strong induction.

III Graphs and Relations

9. Graphs

The ability of computers to efficiently carry out large computations makes them the perfect tool for working with big data sets. To have any hope of making sense of this massive amount of data, we need to develop ways to keep the data organized, *data structures*, as well as design techniques, or *algorithms*, for efficiently carrying out computations on these data structures. One important data structure is a *graph*, which models connections between members of some set. We can visualize these connections as follows:



There are numerous places where graphs are used in practice that you are familiar with:

- Social networks such as Facebook maintain a graph of their users. Two accounts are linked together if they are friends. Facebook can use this connection data (along with a user's posts, location, etc.) to learn about a user's interests and more effectively target advertisements.
- Navigation sites such as *Google Maps* maintain a graph representation of the road network, where the roads connect together nearby addresses. When a user types a starting and ending address, Google uses the road graph to calculate an efficient travel route.
- The internet itself is a graph consisting of a large collection of web pages connected together by hyperlinks. Search engines traverse these links to understand the structure of this graph so they can reliably suggest appropriate pages in response to a user's queries.

In this lecture, we'll use the foundational mathematical tools such as logic, sets, and proof techniques we built up over the past couple of weeks to give a formal description of graphs and explore some of their basic properties.

9.1 Undirected Graphs

A graph consists of two pieces of information: an underlying collection of objects called the *vertices* or *nodes* (which we visualized as red circles) and connections between some pairs of vertices called *edges* (visualized as black lines between the circles). The edges convey a relationship between a pair of nodes. For example, in the Facebook network, there is a vertex for each user account, and there is an edge between two users who are friends. In a transportation network, there are vertices for each address or intersection, and the edges model road segments. We can formally describe these two graph components using set notation.

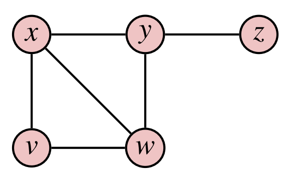
Definition 9.1 — Undirected Graph, Vertex/Node, Edge.

An *undirected graph* G is a pair of two finite sets $G = (V, E)$.

- V is a non-empty set of *vertices* or *nodes*.
- E is a set of *edges*. Each edge $e \in E$ is a subset of V containing two vertices.

This definition packages up all the information about a graph's structure into two sets. We can unpack the sets from this definition to draw the pictorial representation of the graph.

Example 9.1 Consider the undirected graph $G = (V, E)$ with $V = \{v, w, x, y, z\}$ and $E = \big\{\{v, w\}, \{w, x\}, \{v, x\}, \{x, y\}, \{y, z\}, \{y, w\}\big\}$. This satisfies the definition because each element of E is a set of two vertices. To draw this graph, we can first draw and label a circle representing each vertex. Then, we can iterate over the set of edges, and draw a line connecting the two vertices in that edge. We end up with the following picture. The exact positions of each of the vertices are not important; we only care about the structure of the connections between them.



The following terminology will be useful for talking about vertices and edges.

Definition 9.2 — Endpoint, Incidence, Neighbor.

An edge $e = \{u, v\} \in E$ has *endpoints* $u, v \in V$. We say that u and v are *incident* to e. We also say that u is a *neighbor* of v (and similarly v is a neighbor of u) in G since there is an edge between them.

We say that the graph is *undirected* because the edges treat their endpoints equally; the edge $\{v, w\}$ can be equivalently expressed as $\{w, v\}$ since the elements of a set are unordered. Later, we will introduce a different type of connection called an *arc* which differentiates the two endpoints and gives rise to *directed* graphs.

9.1.1 Neighborhoods and Degree

The neighbors of a vertex capture important information about the graph. Neighbors in the Facebook graph are a person's friends; it is reasonable to assume that neighbors share some underlying features (school, profession, interests). Neighbors in the Google Maps graph are adjacent addresses or intersections, which represent consecutive points we'd visit along a route. There are two more terms we use to discuss a vertex's neighbors.

Definition 9.3 — Neighborhood, Degree.
Given a vertex $v \in V$, its *neighborhood* in G, which we denote $N_G(v)$, is its set of neighbors:

$$N_G(v) = \{u \in V \mid \{u,v\} \in E\}.$$

The *degree* of v in G, which we write $\deg_G(v)$ is the size of v's neighborhood:

$$\deg_G(v) = |N_G(v)|.$$

Using these definitions, we can prove our first result about graphs.

Lemma 9.1 — The Handshake Lemma. In any undirected graph $G = (V,E)$,

$$\sum_{v\in V} \deg_G(v) = 2|E|.$$

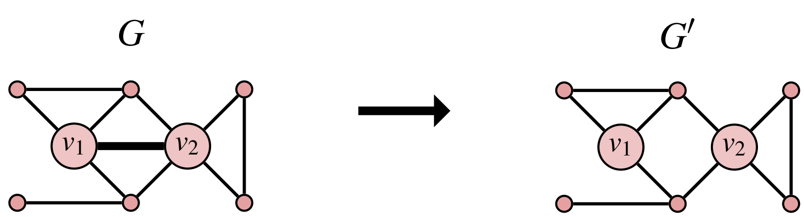
Imagine the graph's vertices represent people mingling at a party. We draw an edge between two people when they shake hands. At the end of the party, if we ask each person how many hands they shook and add these numbers, we obtain the degree sum. Intuitively, this will equal two times the number of handshakes because each shake is accounted for by its two members. In graph terminology, each edge contributes two units to this degree sum, one for each endpoint. We can formalize this idea with a short proof by induction.

Proof. We argue by induction on m, using the inductive statement $P(m) =$"Every undirected graph $G = (V,E)$ with $|E| = m$ has $\sum_{v\in V} \deg_G(v) = 2m$".

Base Case: Letting $m = 0$, $P(0)$ concerns graphs with 0 edges. Each vertex in such a graph has degree 0, as it is not connected to any other vertex. Thus, the degree sum is $0 = 2m$.

Inductive Hypothesis: Given any $k \in \mathbb{N}$, let us assume that $P(k)$ is true: any undirected graph with k edges has degree sum $2k$.

Inductive Step: Consider an arbitrary graph $G = (V,E)$ with $k+1 \geq 1$ edges, and let us notate one of these edges $\{v_1, v_2\} \in E$. Imagine that we delete $\{v_1, v_2\}$ from G. This would give us another graph $G' = (V,E')$ with the same vertex set, but with only k edges.



Since G' has k edges, we can apply the inductive hypothesis to conclude that its degrees sum to $2k$. Now, how do the degrees of G differ from G'? The degrees of v_1 and v_2 are each one more in G because of the additional edge $\{v_1, v_2\}$. The degree of each other vertex is the same since they have the same incident edges in G and G'. Thus,

$$\sum_{v \in V} \deg_G(v) = 2 + \sum_{v \in V} \deg_{G'}(v) \overset{\text{(IH)}}{=} 2 + 2k = 2(k+1).$$

Since G was chosen arbitrarily, we conclude that all undirected graphs with $k+1$ edges have degree sum $2(k+1)$, meaning $P(k+1)$ is true.

Conclusion: By the Principle of Induction, we have established the lemma. ■

We will see more examples of inductive proofs on graphs in the next lecture.

9.2 Directed Graphs

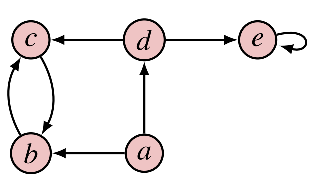
The definition that we have given for graphs is very natural for some of the settings we've described. However, it is not directly amenable to others. Consider the graph of web pages on the internet, where we wish to connect pages that link to each other. Suppose that a Wikipedia page includes a link to the arXiv page of a research article. Then, we wish to include an edge between these pages. However, the arXiv page will not have a link back to Wikipedia. In this case, the edge only travels one way. This is how the edges in a *directed graph* behave. To formally describe these directed edges, or *arcs*, we will need a way to order their endpoints. We can do this using a Cartesian product.

Definition 9.4 — Directed Graph, Arc.
A *directed graph* G is a pair of two finite sets $G = (V, A)$.

- V is a non-empty set of *vertices* or *nodes*.
- A is a set of *arcs*. Each arc $a \in A$ is an ordered pair $a = (u, v) \in V \times V$.

Notice that this definition allows for two new possibilities. First, there may two arcs (u, v) and (v, u) between the same pair of vertices oriented in opposite directions. Also, we can have an arc $(v, v) \in E$ where both vertices are the same, which we call a *self-loop* on a vertex V. We visualize an arc (u, v) in a directed graph using an arrow that points from u to v.

Example 9.2 Consider the directed graph $G = (V, A)$ with $V = \{a, b, c, d, e\}$ and $A = \big\{(a,b), (a,d), (b,c), (c,b), (d,c), (d,e), (e,e)\big\}$. Notice that each element of A is an ordered pair of vertices (i.e. is in $V \times V$). We visualize this graph:



Many of our definitions for undirected graphs have natural analogues for directed graphs.

Definition 9.5 — In/Out-Neighborhood, In/Out-Degree.

Given an arc $(u,v) \in A$, we say that u is an *in-neighbor* of v and v is an *out-neighbor* of u.

The *in-neighborhood* of vertex v is the set of its in-neighbors: $N_{\text{in},G}(v) = \{u \in V \mid (u,v) \in A\}$.

The *in-degree* of v is the size of its in-neighborhood: $\deg_{\text{in},G}(v) = |N_{\text{in},G}(v)|$.

The *out-neighborhood* of v is the set of its out-neighbors: $N_{\text{out},G}(v) = \{w \in V \mid (v,w) \in A\}$.

The *out-degree* of v is the size of its out-neighborhood: $\deg_{\text{out},G}(v) = |N_{\text{out},G}(v)|$.

We leave it as an exercise (Exercise 9.7) to write and prove a result similar to Lemma 9.1 for directed graphs.

9.3 Paths and Cycles

As we progress through the semester, we will define many properties of graphs and structures that can be found within a graph. We will end today's lecture by introducing a few of these that are particularly important. To motivate these ideas, let's return to our example of the Google Maps graph, where the vertices are all of the intersections, and two intersections are linked if we can drive between them without passing through another intersection.

How can we formalize a driving route using this graph terminology? In a driving route, we begin at our current location, following along road segments (arcs) and passing through many intersections (vertices) on our way to our destination. We can completely identify the route that we took by the sequence of intersections that we passed through. This sequence of vertices is referred to as a *path* in a graph.

Definition 9.6 — (Simple) Path.

In an Undirected Graph: A path P of length k in an undirected graph $G = (V,E)$ is a sequence of $k+1$ vertices $P = (v_0, v_1, \dots, v_k) \in V^{k+1}$ such that $\{v_{i-1}, v_i\} \in E$ for each $i \in [k]$, and all such edges are distinct.

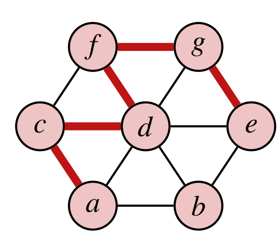
In a Directed Graph: A path P of length k in a directed graph $G = (V,A)$ is a sequence of $k+1$ vertices $P = (v_0, v_1, \dots, v_k) \in V^{k+1}$ such that $(v_{i-1}, v_i) \in A$ for each $i \in [k]$, and all such arcs are distinct.

In either case, the path is *simple* if $\{v_0, v_1, \dots, v_k\}$ are all distinct.

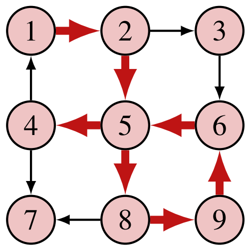
Some texts will instead define paths as the sequence of edges/arcs that are traveled along; however, both of these definitions convey the same information. These definitions are best illustrated by some examples.

Example 9.3

- In the undirected graph on the right, (a,c,d,f,g,e) is a simple path, which we visualize using the thick red edges.
- (a,d,b,g,f) is not a path. There is not an edge between consecutive vertices b and g.
- (a,d,b,d,e) is not a path. Although each adjacent pair of vertices are endpoints of an edge, the edge $\{b,d\}$ is repeated.



Example 9.4



- In the directed graph on the left, $(1,2,5,8,9,6,5,4)$ is a path, which we visualize using the thick red arrows. This path is *not simple* because it revisits the vertex 5.
- $(5,4,7,8,9)$ is not a path. Although there are arcs between each pair of consecutive vertices in this sequence, the arc between 7 and 8 is pointing in the wrong direction.

When a path ends at the same vertex where it started, we call it a cycle.

Definition 9.7 — (Simple) Cycle.

In either an undirected graph or a directed graph, a cycle of length $k \geq 1$ is a path of length k with the additional property that $v_0 = v_k$.

We say that a cycle is simple if $\{v_1, \dots, v_k\}$ are all distinct.

Example 9.5

- In the undirected graph from Example 9.3, the sequences (a,c,d,a), (d,f,g,e,d), and (a,c,d,f,g,e,b,a) are all simple cycles. (c,d,f,g,e,d,a,c) is a non-simple cycle since it returns to the vertex d twice.
- In the directed graph from Example 9.4, the sequences $(1,2,5,4,1)$ and $(5,8,9,6,5)$ are both simple cycles. $(1,2,5,8,9,6,5,4,1)$ is a non-simple cycle that revisits vertex 5.

Being able to efficiently locate paths and cycles in a graph is important for procedures (or *algorithms*) for network planning and routing, among myriad other applications. The following lemma gives us a sufficient criterion to guarantee that a graph has a cycle.

Lemma 9.2 Let $G = (V,E)$ be an undirected graph with $\deg_G(v) \geq 2$ for all $v \in V$. Then, G contains a cycle.

Proof. Suppose that $P = (v_0, \dots, v_k)$ is a simple path in G with the maximum possible length. Since $\deg_G(v_0) \geq 2$, there must be some $u \in V \setminus \{v_1\}$ such that $\{u, v_0\} \in E$. If $u \notin \{v_2, \dots, v_k\}$, then $(u, v_0, \dots, v_k)$ would be a simple path that is longer than P, contradicting the maximality of P. Thus, $u = v_j$ for some $j \in 2, \dots, k$, and $C = (u, v_0, v_1, \dots, v_{j-1}, u)$ is a cycle in G. ■

Main Takeaways:

- *Graphs* are objects consisting of a set of *vertices/nodes* and a set of *edges/arcs*.
- In an *undirected* graph, edges are described by (unordered) sets of two vertices. In a *directed* graph, edges are described by (ordered) pairs of vertices.
- The *degree* of a vertex (in an undirected graph) is its number of incident edges.
- Paths and cycles are sequences of neighboring vertices.

Exercises

9.1 For each of the following formal descriptions of graphs, draw a picture that visualizes their nodes and edges.

(a) $G_1 = (V_1, A_1)$, $\quad V_1 = \{a, b, c, d\}$, $\quad A_1 = \{(a,b), (b,a), (b,c), (d,c)\}$

(b) $G_2 = (V_2, A_2)$, $\quad V_2 = \{1, 2, 3, 4, 5\}$, $\quad A_2 = \{(1,3), (2,4), (4,1), (2,2), (4,2), (3,3)\}$

What is interesting about the vertex 5 in this example?

(c) $G_3 = (V_3, A_3)$, $\quad V_3 = \{a, b, c, x, y, z\}$, $\quad A_3 = \{(a,x), (a,y), (a,z), (b,x), (b,y), (b,z), (c,x), (c,y), (c,z)\}$

This is the graph from the infamous *utilities puzzle*, where you are asked to connect three utilities (a, b, c) to three residences (x, y, z) without any of the lines touching. This is not possible, because the graph is *non-planar*; no matter how it is drawn, one arc will need to cross over another arc. We will talk more about graph *planarity* in a few lectures.

(d) $G_4 = (V_4, E_4)$, $\quad V_4 = \{\spadesuit, \clubsuit, \diamondsuit, \heartsuit\}$, $\quad E_4 = \{\{\spadesuit, \clubsuit\}, \{\spadesuit, \heartsuit\}, \{\spadesuit, \diamondsuit\}, \{\clubsuit, \heartsuit\}, \{\clubsuit, \diamondsuit\}, \{\heartsuit, \diamondsuit\}\}$

This undirected graph is called a *4-clique* because it contains edges between each pair of its 4 vertices.

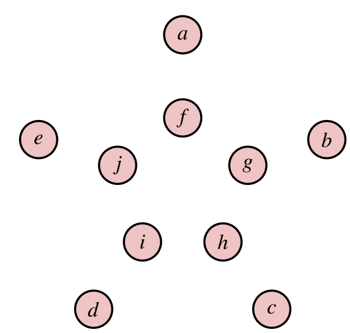
(e) $G_5 = (V_5, E_5)$, $\quad V_5 = \{1, 2, 3, 4, 5, 6, 7\}$, $\quad E_5 = \{\{1,2\}, \{1,3\}, \{1,4\}, \{5,6\}, \{5,7\}, \{6,7\}\}$

This graph is split into two "pieces" called *connected components*. We will discuss connected components when we learn about relations.

(f) $G_6 = (V_6, E_6)$, $\quad V_6 = \{a, b, c, d, e, f, g, h, i, j\}$

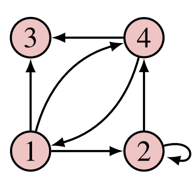
$E_6 = \{\{a,b\}, \{a,e\}, \{b,c\}, \{c,d\}, \{d,e\}, \{a,f\}, \{b,g\}, \{c,h\}, \{d,i\}, \{e,j\}, \{f,h\}, \{f,i\}, \{g,i\}, \{g,j\}, \{h,j\}\}$

This is a graph known as the *Petersen Graph*, which has become notorious for some of its strange properties; it serves as a small counterexample for many facts about graphs that we wish were true. For a nice visualization of the graph arrange the vertices as shown on the right.

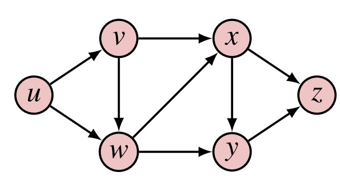


9.2 Write down a formal description of each of the following graphs.

(a)

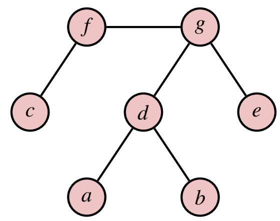


(b)



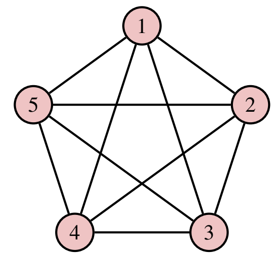
This is an example of a *directed acyclic graph (DAG)*, which we'll discuss soon.

(c)



This is an example of a *tree*, which we'll discuss next lecture.

(d)



This graph is called a *5-clique* because it contains edges between each pair of its 5 vertices.

9.3 Our definition of a graph only requires that the two inputs are sets. This means that any objects can act as the nodes, not just letters, numbers, or symbols. It also means that we can describe the edge set using set builder notation, rather than explicitly listing each of the edges. These ideas are closely related to *binary relations*, which we will study later in the course.

Draw the following graphs with more exotic descriptions.

(a) $G_1 = (V_1, E_1), \qquad V_1 = [9], \qquad E_1 = \{\{i, j\} \subseteq V_1 \mid |i - j| = 2\}$

(b) $G_2 = (V_2, A_2), \qquad V_2 = \mathcal{P}(\{a, b, c\}), \qquad A_2 = \{(S, T) \in V_2 \times V_2 \mid S \subseteq T, |T \setminus S| = 1\}$

This graph is an example of a *Hasse diagram* for $\subseteq$.

9.4 Consider the following multisets of natural numbers. For each, decide whether its elements can be the degrees of the vertices in a *simple* undirected graph. If they can be, draw a graph with these degrees. If they cannot be, give a brief explanation of why.

(a) $\{\!|1, 1, 2, 3, 3|\!\}$

(b) $\{\!|1, 1, 2, 2, 3|\!\}$

(c) $\{\!|1, 2, 2, 3, 4|\!\}$

(d) $\{\!|1, 1, 2, 3, 5|\!\}$

(e) $\{\!|1, 2, 3, 4, 4|\!\}$

If you are interested in reading more about this, the *Erdős-Gallai theorem* gives a characterization of which multisets of degrees are possible.

9.5 Argue that any undirected graph contains an even number of vertices with odd degree. (Hint: Use Lemma 9.1)

9.6 Argue that in any directed graph, the sum of in-degrees $\sum_{v \in V} \deg_{\text{in},G}(v)$ is equal to the sum of out-degrees $\sum_{v \in V} \deg_{\text{out},G}(v)$ which is equal to $|A|$.

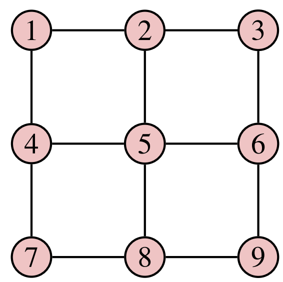
This result is the analogue of Lemma 9.1 for directed graphs.

9.7 For each of the following questions, it may help to work out the answer directly for some small values of n and then try to spot a pattern to figure out the general answer.

(a) What is the maximum number of arcs that an directed graph with n vertices can have?

(b) What is the maximum number of edges that an undirected with n vertices can have?

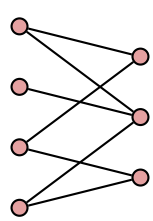
9.8 How many simple cycles are there in the following undirected graph? List them. (For the purpose of this exercise, we'll consider two cycles to be the same if they contain the same set of vertices, so $(1,2,5,4,1)$ and $(4,5,2,1,4)$ would be two "names" for the same cycle.)



9.9 Suppose that $G=(V,E)$ is an undirected graph in which $\deg_G(v) \le 2$ for all $v \in V$. Argue that G consists of a union of *vertex-disjoint* paths/cycles. In other words, we can make a list of paths/cycles in G such that:

- Each edge $e \in E$ has its endpoints appearing consecutively in *exactly one* of these paths/cycles.
- Each vertex $v \in V$ appears in *exactly one* of these paths/cycles.

9.10 We say that an undirected graph is *bipartite* if we can *partition* its vertices V into two sets L and R (that is, $V = L \cup R$ and $L \cap R = \varnothing$) such that each edge $\{u,v\} \in E$ has one endpoint in L and one endpoint in R. We often visualize bipartite graphs by lining up the L vertices in a column on the left and the R vertices in a column on the right, so that all edges cross through the gap in the middle. For example:



(a) Show that the undirected graph $G=(V,E)$ with the following set description is bipartite by partitioning its vertices into sets L and R that satisfy the above property.

$V = \{a,b,c,d,e,f,g,h\}$ $\qquad$ $E = \big\{\{a,c\},\{a,f\},\{b,e\},\{b,f\},\{c,d\},\{c,g\},\{d,e\},\{g,h\}\big\}$

(Hint: If you put a in L, how can you figure out where other vertices need to go?)

(b) Draw the graph from part (a) using the “two-column” layout described above.

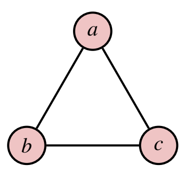
(c) A *triangle* is a cycle with length 3. Prove that if a graph is bipartite, it cannot contain a triangle.

(d) Show that the *converse* of part (c) is not true. In other words, argue that even if a graph does not contain a triangle it may not be bipartite.

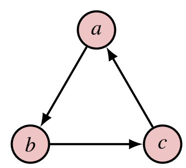
(e) Can you describe a property $P(G)$ of the cycles of a graph G such that $P(G)$ is true if and only if G is bipartite? We will revisit this question soon when we talk about induction on graphs.

⋆ **9.11** Given an undirected graph $G = (V, E)$, we can *orient* it by assigning a direction to each of its edges. Below, we show two different orientations of a small undirected graph.

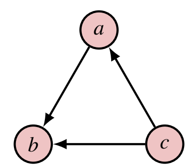
Undirected Graph:



Orientation 1:



Orientation 2:



Note that Orientation 1 has a cycle (a, b, c, a), while Orientation 2 does not. We say that Orientation 2 is a *directed acyclic graph (DAG)*.

(a) Consider the following algorithm to orient a graph:

Algorithm 1 ORIENT ACYCLIC

Input: Undirected graph $G = (V, E)$

```
Number the vertices of the graph 1,...,|V|
for each edge {u,v} ∈ E do
    Orient {u,v} so that its tail has a higher number than its head.
```

Argue that this algorithm always produces an orientation that is a directed acyclic graph. (Hint: Use a proof by contradiction.)

◇ (b) Given an undirected graph with a cycle, design an algorithm that will orient the graph so that it has a directed cycle. Argue that your algorithm is correct. (Hint: One possible solution is an algorithm that is based on a breadth-first search.)

10. Induction on Graphs

In this lecture, we will use induction to establish properties of two special families of graphs.

10.1 Trees

Trees are perhaps the most important family of graphs in computer science. They form the backbone of many efficient data structures such as search trees and heaps. They also play a prominent roll in a wide variety of algorithms for applications ranging from searching to text compression.

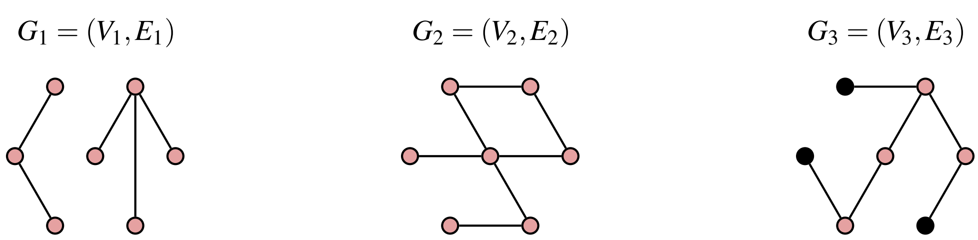
Trees are characterized by two properties. First, they are *connected*, meaning their edges link together all of the vertices into a single structure called a *connected component*.

Definition 10.1 — Connected Graph.
An undirected graph $G = (V, E)$ is *connected* if, for any two vertices $u, v \in V$, there is a path in G between u and v.

Additionally, a tree does not have any cycles.

Definition 10.2 — Tree.
An undirected graph $G = (V, E)$ is a *tree* if it is connected and has no cycles.

Example 10.1 Consider the three graphs shown below.



G_1 is not a tree because it is not connected; it has two separate connected components. G_2 is also not a tree. Although it is connected, it contains a cycle. G_3 is a tree because it is connected and contains no cycles.

Trees get their name from their structure. Different paths of edges extend outward and split off like the branches of a tree. Extending this analogy, we call the degree-1 vertices at the end of each of these branches a *leaf*.

Definition 10.3 — Leaf.
The vertices in a tree with degree 1 are called its *leaves*.

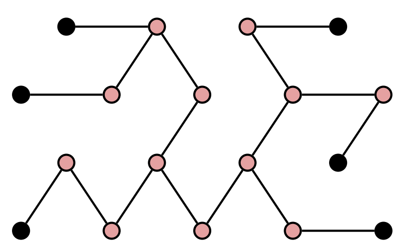
The leaves in G_3 from the preceding example are colored black. The following lemma, which will be helpful in our next inductive proof, says that a tree must have a leaf.

Lemma 10.1 Every tree $T = (V,E)$ with $|V| \geq 2$ has a leaf.

Proof. In the previous lecture, we proved that if every vertex in a graph G has degree ≥ 2, then G must contain a cycle. The contrapositive of this statement says that if a graph G does not contain a cycle, it must have at least one vertex with degree < 2. By definition, trees are acyclic, so must have such a vertex. In addition, since trees are connected, each vertex must have degree ≥ 1 (otherwise they would have paths to no other vertices). Thus, a tree must have at least one vertex with degree 1, a leaf. ■

10.1.1 Counting Edges in Trees

We would like to find a formula that relates the number of vertices in a tree to the number of edges. Our example G_3 contains 7 vertices and 6 edges. The larger example below contains 18 vertices and 17 edges.



Armed with these two examples, we can confidently propose the following theorem.

Theorem 10.1 Let $T = (V,E)$ be a tree. Then $|E| = |V| - 1$.

We would like to prove this theorem using induction, but how should we do this? To use our ideas from the past lectures, we should start by developing an inductive statement, which should be expressed in terms of some variable n over the natural numbers. In induction, we want small values of n to correspond to the simplest (base) cases, and larger values to correspond to more challenging examples. Here, the trees can grow more complicated when they have more vertices, so we'll let $n = |V|$. Since the theorem doesn't make any assumptions about the tree, we will need to show that it holds for all trees. For this reason, we should include a quantification over trees in the inductive statement:

$$P(n) = \text{“Every tree } T = (V,E) \text{ with } n \text{ vertices has } n-1 \text{ edges.”}$$

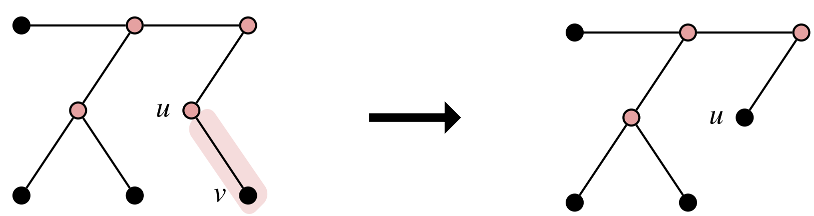
We are ready to write the proof.

Proof. We'll establish the claim by induction on n, using the inductive statement written above.

Base Step: We start from $n = 1$ (since we only consider graphs with a non-empty vertex set). The only tree with 1 vertex has $0 = 1 - 1$ edges.

Inductive Hypothesis: We assume that $P(k)$ is true for some $k \geq 1$. That is, any tree with k vertices has $k - 1$ edges.

Inductive Step: We must argue that every tree with $k + 1$ vertices has k edges. Let's consider an arbitrary such tree $T = (V,E)$. In order to invoke the inductive hypothesis, we will need to find a way to reduce to thinking about a graph with only k vertices. To do this, we can use Lemma 10.1, which tells us that there must be some $v \in V$ that is a leaf. Thus, v is incident to exactly one edge $\{u,v\} \in E$. Suppose that we remove vertex v and edge $\{u,v\}$ from the tree to obtain a graph $T' = (V',E')$.



T' is still a tree; each pair of nodes in V' is connected by the same path as in T, and removing an edge from T cannot introduce a cycle. Moreover, we removed one vertex to obtain T', so $|V'| = |V| - 1 = k$. But then, T' is exactly a tree that is referenced by the inductive hypothesis, so we have that $|E'| = k - 1$. However, E' was obtained from E by deleting one edge, so $|E| = |E'| + 1 = k$, completing the inductive step.

Conclusion: Since we have established the base and inductive steps, we conclude by the principle of induction that every tree $T = (V,E)$ has $|E| = |V| - 1$. ■

10.2 Planar Graphs

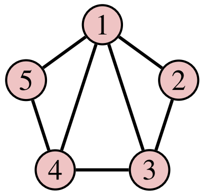
The second family of graphs that we will consider today are *planar* graphs.

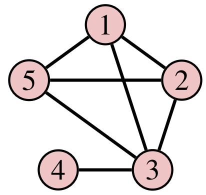
Definition 10.4

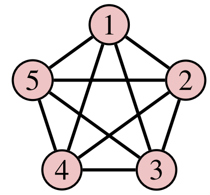
An undirected graph $G = (V,E)$ is *planar* if it can be drawn (in two dimensions) so that none of its edges intersect.

Example 10.2 Consider the three graphs shown below.

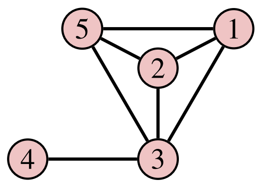
$G_4 = (V_4, E_4)$ $\qquad$ $G_5 = (V_5, E_5)$ $\qquad$ $G_6 = (V_6, E_6)$





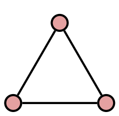


G_4 is a planar graph; its representation here does not have any crossing edges. G_5 is also planar. Although the visualization of the graph above has the edge $\{1,3\}$ cross the edge $\{2,5\}$, we can rearrange the vertices to eliminate this crossing, shown below (Note the existential quantifier in the definition: *there exists* a way to draw the graph ...).

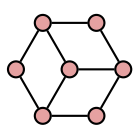


G_6 is not a planar graph. No matter how it is drawn, it will always have intersecting edges. This fact is not at all obvious, but can be shown using the result that we will prove next (see Exercise 10.10).

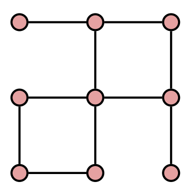
Notice how the edges in the planar graph G_4 split the white space of the page into 4 different regions: the space inside the triangle with vertices $\{1,2,3\}$, the space inside triangle $\{1,3,4\}$, the space inside triangle $\{1,4,5\}$ and all of the space outside of the graph. We call these regions the *faces* of the graph and use F to represent the set of faces. Similar to what we did for trees, we would like to find a formula that relates the number of vertices, edges, and faces of planar graphs. Both G_4 and G_5 have $|V| = 5$, $|E| = 7$, and $|F| = 4$. We can draw some more planar graphs with varying sizes to collect some more data.



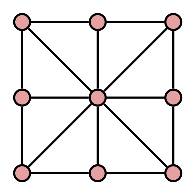
$|V| = 3, |E| = 3, |F| = 2$



$|V| = 7, |E| = 9, |F| = 4$



$|V| = 9, |E| = 10, |F| = 3$



$|V| = 9, |E| = 16, |F| = 9$

If we examine this data carefully, we can spot the following pattern. The number of vertices plus the number of faces is always two more than the number of edges. This result is known as Euler's Formula.

Theorem 10.2 — Euler's Formula. Let $G = (V, E)$ be a connected planar graph with face set F. Then, $|V| - |E| + |F| = 2$.

We will again use induction to prove this theorem. This time it will be more convenient to induct on the number of edges in the graph.

Proof. We'll establish the claim by induction on m, using the inductive statement,

$$P(m) = \text{"Any connected planar graph } G = (V,E) \text{ with } |E| = m \text{ edges has } |E| - |V| + 2 \text{ faces."}$$

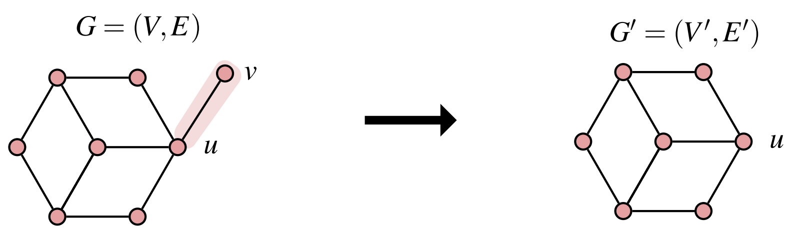
Base Step: Consider a graph with $m = 0$ edges. It must have at least one vertex (we always assume the vertex set is non-empty). Moreover, the graph cannot have more than one vertex, as this would require it to include an edge to be connected. The only possibility is the graph with just a single vertex. This graph has $|E| - |V| + 2 = 0 - 1 + 2 = 1$ face, as required.

Inductive Hypothesis: We assume that $P(k)$ is true for some $k \in \mathbb{N}$. That is, any connected planar graph $G' = (V', E')$ with k edges has $|E'| - |V'| + 2$ faces.

Inductive Step: We consider an arbitrary connected planar graph $G = (V,E)$ with $k+1$ edges. In order to apply the inductive hypothesis, we would like to find a way to remove one edge from this graph. We consider two separate cases.

<u>Case 1:</u> G contains a vertex with degree 1.

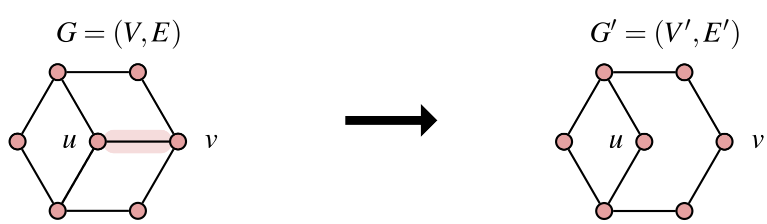
Call this degree-1 vertex v, and let $e = \{u,v\} \in E$ be its incident edge. We can delete edge e and vertex v from the graph to obtain a new graph $G' = (V', E')$.



G' is still planar (removing an edge cannot introduce an edge intersection) and connected (each remaining pair of vertices is still connected by the same path as before), and it has $|E'| = |E| - 1 = k$. Then, G' satisfies the criteria of the inductive hypothesis, so we may conclude that it has $|E'| - |V'| + 2 = (|E| - 1) - (|V| - 1) + 2 = |E| - |V| + 2$ faces. However, removing the edge $\{u,v\}$ and the vertex v did not change the number of faces, so G must also have $|E| - |V| + 2$ faces.

<u>Case 2:</u> Every vertex in G has degree at least 2.

From our earlier lemma, G must contain a cycle. Select an arbitrary edge $\{u,v\}$ in this cycle and remove it (but not its endpoints) to obtain a new graph $G' = (V', E')$.



G' is still planar (removing an edge cannot introduce an edge intersection) and connected (this takes a little more work to show; the crux of the argument is that whenever the edge $\{u,v\}$

appeared in a path connecting vertices in G, we can replace it by the other edges in its cycle). Since we removed one edge, $|E'| = |E| - 1 = k$. Thus, G' satisfies the criteria of the inductive hypothesis, so we may conclude that it has $|E'| - |V'| + 2 = (|E| - 1) - |V| + 2 = (|E| - |V| + 2) - 1$ faces. However, removing the edge $\{u, v\}$ from G caused the two faces that it separated to merge. Therefore, G' has one fewer face than G, meaning G has $|E| - |V| + 2$ faces.

In both cases, we showed that the desired formula holds for our graph G. Since these cases are exhaustive, and since we considered an arbitrary graph G with $k+1$ edges, we conclude that $P(k+1)$ is true.

Conclusion: Since we have established the base and inductive steps, we conclude by the principle of induction that every connected planar graph $G = (V, E)$ has $|E| - |V| + 2$ faces. ■

Main Takeaways:

- We can perform induction on other objects (e.g.) by having the variable n represent the "size" of the object (e.g. number of vertices/edges) and choosing an inductive statement that quantifies over all objects with that size.
- Trees are connected graphs with no cycles. They contain $|V| - 1$ edges.
- A planar graph can be drawn in the plane without any intersecting edges.
- *Euler's Formula* tells us that in any connected planar graph $G = (V, E)$ with f faces, $|V| - |E| + f = 2$.

Exercises

10.1 Use induction to argue that every tree with at least one edge must have at least 2 leaves. This is a strengthening of Theorem 10.1.

10.2 Use induction to argue that every undirected (but not necessarily connected) graph with n vertices and at least n edges contains a cycle.

10.3 Use induction to argue that every simple graph on n vertices with more than $\frac{n}{2}$ edges has a path of length 2.

10.4

(a) How many different trees are there with vertices $V = \{1, 2, 3\}$? Draw them.

(b) How many different trees are there with vertices $V = \{1, 2, 3, 4\}$? Draw them.

(c) There are 125 different trees with vertices $V = \{1, 2, 3, 4, 5\}$ (if you're feeling very ambitious, you could try to draw them all). Can you guess a formula for the number of trees on n vertices? We will see how to prove this formula later in the semester.

10.5

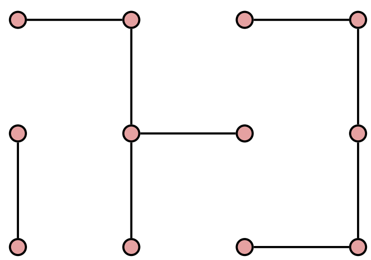
(a) Using the two properties in the definition of a tree, argue that in a tree $T = (V, E)$, there is *exactly* 1 path between any two of its vertices $u \neq v \in V$. (You do not need induction for this.)

(b) Use part (a) argue that if any edge is added to a tree, it introduces a cycle.

(c) Strengthen the result from part (b) by arguing that exactly one cycle is introduced. (Hint: use a proof by contradiction.)

(d) Is it true that if two edges are added to a tree, exactly two cycles are introduced? Justify your answer with either a proof or a counterexample.

10.6 A undirected (but not necessarily connected) graph with no cycles is called a *forest*. It gets this name because each of its, potentially many, connected components is a tree. One forest is shown below.



(a) By drawing a few more examples of forests, guess a formula for the number of edges in a forest with c connected components.

(b) Prove that your formula is correct by induction. (Hint: One way to conceive this proof is as induction on the number of connected components. In the inductive step, can you think of a way to decrease the number of connected components by 1?)

10.7 In this exercise, you'll generalize Euler's formula for graphs with multiple connected components, similar to what was done for *forests* in Exercise 10.6.

(a) By experimenting with some small examples, come up with a formula that relates $|V|$, $|E|$, $|F|$, and $|C|$ = the number of connected components in a planar graph $G = (V, E)$.

(b) Verify that your formula from part (a) is correct using induction. (Hint: Similar to Exercise 10.6, the easiest strategy will be to do induction on the number of connected components.)

10.8 Consider the following claim and its supplied "proof."

Claim: Let $G = (V, E)$ be a graph where each vertex $v \in V$ has $\deg(v) \geq 1$. Then G is connected.

Proof. We argue the claim by induction on the number of vertices in the graph, with inductive statement

$P(n) =$ "Every undirected graph $G = (V, E)$ with n vertices, each with degree ≥ 1, is connected."

Base Step: When $n = 1$, there is a single vertex in the graph, which is trivially connected.

Inductive Hypothesis: We assume that $P(k)$ is true for some $k \geq 1$, meaning every undirected graph on k vertices where each vertex has degree ≥ 1 is connected.

Inductive Step: Given a graph $G = (V,E)$ with k vertices, each with degree ≥ 1, we can grow it to a graph with $k+1$ vertices as follows. First, we select a vertex $v \in V$. Then, we create a new vertex w and draw an edge from v to w. This gives a new graph $G' = (V \cup \{w\}, E \cup \{\{v,w\}\})$ which has $k+1$ vertices. Each of these vertices has degree ≥ 1: this was true for all vertices in V even before we added edge $\{v,w\}$, and w has degree 1 because it is connected to v. By the inductive hypothesis, G is connected, so has a path between each pair of its vertices. To show that G' is connected, all that we need to do is find a path between w and each vertex $u \in V$. To do this, we simply take a path $(u, \ldots, v)$ in G and add w at the end. Thus, G' is a graph with $k+1$ vertices each with degree ≥ 1, and we have shown that G' is connected, meaning $P(k+1)$ is true.

Conclusion: By the principle of induction, we conclude that $P(n)$ is true for all $n \geq 1$, establishing the claim. ■

(a) Give an example graph that shows that the claim is false.

(b) Explain the mistake that is made in this proof by induction.

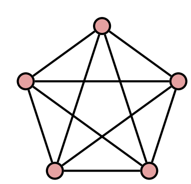
⋄ **10.9** The two main results from this lecture (Theorems 10.1 and 10.2) are very similar, in that they both given an equation that relates the counts of components in a certain families of graphs. In this exercise, you'll establish an even stronger connection.

(a) Argue that every tree is planar by describing a way to lay out its vertices and edges such that none of them intersect. (Hint: a breadth first search of the tree will be helpful here.)

(b) How many faces does any tree have? Use this to show that when we apply Theorem 10.2 to a tree, we get back Theorem 10.1.

⋆ **10.10**

In this exercise, you'll use Euler's formula to derive an inequality that holds for all planar graphs and then use this inequality to prove that the *complete graph* on 5 vertices (often denoted K_5), shown on the right, is not planar.



Suppose that $G = (V,E)$ is a connected planar graph with $|E| \geq 3$.

(a) Given a face in G, at least how many edges need to be on the boundary of each face? (Hint: draw some pictures.)

(b) Given an edge in G, of at most how many faces can it be a boundary? (Hint: draw some pictures.)

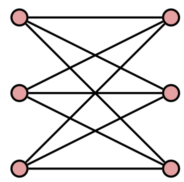
(c) Use parts (a) and (b) to derive an inequality between the $|F|$ and $|E|$ in G. (Hint: think about counting the number of (face, boundary edge) pairs in two different ways.)

(d) Substitute your inequality from part (c) into Euler's Formula to obtain an inequality involving $|V|$ and $|E|$. This inequality must be true of any planar graph.

(e) Use your inequality from part (d) to show that K_5 is not planar.

⋆ **10.11** In this exercise, you'll argue that another graph is not planar by using similar ideas from Exercise 10.10. The graph which we'll consider is the *complete bipartite* graph with part sizes $(3,3)$, often abbreviated $K_{3,3}$.

To draw this graph, we create two columns of 3 vertices each. Then, we connect every vertex in the left column with every vertex in the right column, but *do not* draw any connections within the left column or within the right column. We arrive at the following figure.



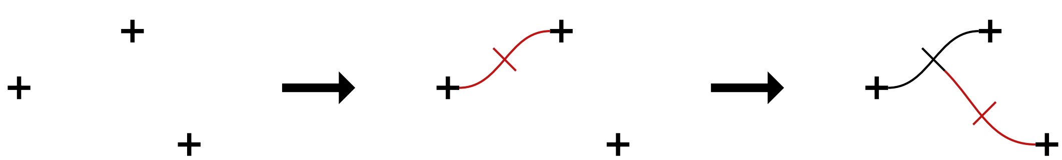
(a) Argue that $K_{3,3}$ has no cycles with exactly three edges. This means that every cycle in this graph has length at least 4.

(b) For sake of contradiction, suppose that $K_{3,3}$ is a planar graph with $|F|$ faces. Using a similar idea to Exercise 10.10 part (c) and your observation from part (a), derive an inequality that relates $|F|$ and $|E|$.

(c) Substitute your inequality from part (c) into Euler's Formula to obtain an inequality involving $|V|$ and $|E|$ and use this to derive a contradiction. From here, we can conclude that $K_{3,3}$ is not planar.

Interestingly, these two examples of non-planar graphs are, in a sense, the only examples. Theorems proven by Kuratowski and Wagner show that within any non-planar graph, we can find a hidden "copy" (this word needs a more careful definition) of one of these two graphs.

10.12 In this exercise, we'll look at a cool application of Euler's Formula to analyze a game called "Brussels Sprouts." At the start of the game, there are p plus signs scattered around a piece of paper. We think of the four ends of each plus signs as terminals that we would like to connect. On a player's turn. They attempt to identify two unused terminals that they can connect with an edge without intersecting any previously-drawn edges. If they can draw such an edge, they do, and they add a small bar crossing the middle of this edge that creates two new terminals. An illustration of two possible moves in the game is shown below.

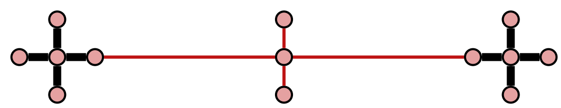


A player loses the game when they are unable to make a move (that is, they cannot find two unused terminals that can be connected without intersecting a previous edge). We would like to know which player will win the game for different values of p.

(a) Draw out all of the possible sequences of moves that can be made on a game instance with $p = 1$ plus sign. You should end up with three possible end configurations. How many moves were made in each of these configurations?

(b) Trace through a few games when there are $p = 2$ plus signs. How many moves are made in these games?

By playing around with these small examples, you should see that it seems like the number of moves that the game lasts does not change when different moves are made. In light of this, we can conjecture that the number of moves that a game will last is a function only of the number p of plus signs. To do this, we will view the game as the construction of a planar graph. We'll treat each plus sign as a connected component with 5 vertices. When we draw new edges in the graph, we add new edges and vertices as in the following figure.



(c) At the beginning of the game, how many vertices does the graph have? How many edges does it have? How many unused terminals are there?

(d) In each move, how many new vertices are added? How many new edges are added? How does the number of unused terminals change?

(e) Explain why the graph that we are drawing will remain planar throughout the game.

⋆ (f) Argue that throughout the game the number of unused terminals is always greater than or equal to the number of faces in the graph.

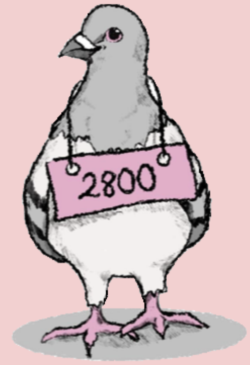
⋆ (g) At the end of the game (i.e., when there are no more available moves) argue that the graph must be connected.

(h) At the end of the game, what is the relationship between the number of unused terminals and the number of faces?

(i) Suppose that at the end of the game, there have been m moves. Express the number of vertices, edges, and faces in the resulting graph as functions of m and p.

(j) Plug your results from part (i) into Euler's Formula to determine the number of moves that a game with p initial plus signs will last.

(k) Based on your answer to part (j), describe which player will win the game.



11. Binary Relations

Two lectures ago, we saw how graphs can be used to model relationships between elements in a set. In particular, the vertices of a graph represent an underlying set of objects, and the arcs in a directed graph (or the edges in an undirected graph) represent some type of connection between a pair of vertices. In today's lecture, we'll introduce another way to formalize these types of connections: binary relations. In the language of mathematics, sets and objects serve as nouns, while relations as adjectives and adverbs. They allow us to qualify the relationships between elements in a set. In doing so, they provide a lens through which we can understand inherent structures within sets. Later in the course, we will learn about the verbs in this mathematical language: functions.

In this lecture, we will define relations and some of their basic properties and look at many examples. In the next two lectures, we will look at two special families of relations, equivalence relations and partial orders, that are pervasive throughout mathematics.

11.1 Defining Binary Relations

Relations are used to describe *relationships* between elements of an underlying set X, referred to as the *ground set*. That is, given two elements $x,y \in X$, a binary relation R tells us whether x is related to y in some particular way. As a *binary* relation, R will always describe relationships between pairs of elements of X. Thus, we'd expect the formal definition of a binary relation to involve the Cartesian product $X \times X = X^2$.

Definition 11.1 — Binary Relation.

A *binary relation* R on a set X is a subset of X^2.

Equivalently, we may view R as assigning a value in $\{\text{True}, \text{False}\}$ to each pair (x,y) based on whether or not this pair satisfies the desired relationship (i.e., is an element of R).

R There are two natural ways to generalize the definition of a relation.

1. Binary relations can define a relationship between elements of two different sets. In this case, a relation on sets X_1 and X_2 would be a subset of $X_1 \times X_2$.
2. Relations need not be binary. An n-ary relation on sets $X_1, \dots, X_n$ is a subset of $X_1 \times \cdots \times X_n$.

In this course, we'll only consider binary relations on a single set, as described in Definition 11.1.

Most of the time, we use an *infix notation* when we use binary relations, writing the relation symbol in between the two set elements when they are related. For example, we'd write "$x\,R\,y$" to represent that $(x,y) \in R$. Often, we will use symbols such as $\{\sim, \equiv, \approx, \succeq\}$ rather than letters to represent relations. Although you may not have been aware of it, you are already familiar with using this infix notation with some common arithmetic relations (often called "logical operators" in CS).

Example 11.1 The "$>$" operator is a relation on $\mathbb{N}$ (or many other sets of numbers: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$). It contains all pairs whose first argument is greater than its second argument, $\big\{(1,0),(2,0),(2,1),\dots\big\}$. We write $7 > 4$ (for example) to express the fact that the ordered pair $(7,4)$ belongs to this relation.

Similarly, the other logical operators $\{=,<,\geq,\leq,\neq\}$ are also relations on these sets of numbers. However, these operations are far from the only examples of relations. The criteria in Definition 11.1 are very general, allowing for any arbitrary subset of the pairs of elements from any set X. The next example illustrates some more relations defined on a small finite set.

Example 11.2 Consider the ground set $X = \{1,2,3,4,6\}$. The following are all relations on X.

- The relation T consists of all pairs whose entries are exactly three apart: $x\,T\,y \Leftrightarrow |x-y| = 3$. Expressed as a set, $T = \big\{(1,4),(4,1),(3,6),(6,3)\big\}$.
- The relation B consists of pairs of elements in X where the first entry is at least 2 more than the second entry: $x\,B\,y \Leftrightarrow x-y \geq 2$. Expressed as a set, $B = \big\{(3,1),(4,1),(6,1),(4,2),(6,2),(6,3),(6,4)\big\}$.
- The relation S consists of pairs whose entries sum to 6: $x\,S\,y \Leftrightarrow x+y = 6$. Expressed as a set, $S = \big\{(2,4),(4,2),(3,3)\big\}$.
- The relation Q consists of pairs whose product is a perfect square: $x\,Q\,y \Leftrightarrow \exists\, k \in \mathbb{Z}.\, xy = k^2$. This relation includes 7 pairs.

There are many more examples of relations that are defined on other sets X.

Example 11.3

- The subset operation, $\subseteq$, defines a relation on a set of sets (for example, $X = \mathcal{P}(\mathbb{N})$).
- Let $\mathcal{F}(\mathbb{N})$ be the set of all *finite* subsets of $\mathbb{N}$. Then, we may define an "equal cardinality relation" E with $S\,E\,T \Leftrightarrow |S| = |T|$ for all $S,T \in \mathcal{F}(\mathbb{N})$. In a few weeks, we will see how to extend this relation to include infinite sets.

- The arc set A in a directed graph $G=(V,A)$ is a relation on V, since it is a subset of V^2.
- We can view $\Rightarrow$ as a binary relation over the set of compound propositions formed from a fixed set of atomic propositions: given compound propositions c_1 and c_2, $c_1 \Rightarrow c_2$ if and only if every truth assignment that makes c_1 true also makes c_2 true.

As these examples demonstrate, the definition of binary relations is simple and flexible (or, as mathematicians say, "general") enough to capture many useful examples in a variety of settings. For this reason, relations are a foundational building block of math. However, the definition of a relation, on its own, is not very useful (after all, every subset of these Cartesian products gives a relation). Instead, we will introduce many special properties of relations that can be used to categorize relations into useful classes.

11.2 Four Properties of Relations

For the rest of this lecture, we'll discuss four properties that we can use to describe and categorize binary relations. We'll practice working with these properties by checking them for some of the relations that we introduced earlier. In the next two lectures, we'll introduce two special classes of relations that are defined in terms of these.

The reflexivity property says that *every* element in the ground set is related to itself.

Definition 11.2 — Reflexivity.
A binary relation R on set X is *reflexive* if

$$\forall\, x \in X. \qquad x\,R\,x.$$

The symmetry property says that the inclusion of a pair does not depend on the order of its arguments.

Definition 11.3 — Symmetry.
A binary relation R on set X is *symmetric* if

$$\forall\, x,y \in X. \qquad x\,R\,y \;\Rightarrow\; y\,R\,x.$$

The anti-symmetry property says that the only time that the relation holds "in both directions" is when the elements are equal.

Definition 11.4 — Anti-Symmetry.
A binary relation R on set X is *anti-symmetric* if

$$\forall\, x,y \in X. \qquad x\,R\,y \wedge y\,R\,x \;\Rightarrow\; x = y.$$

Finally, the transitivity property says that *chains* of related elements can be "collapsed down".

Definition 11.5 — Transitivity.
A binary relation R on set X is *transitive* if

$$\forall\, x,y,z \in X. \qquad x\,R\,y \wedge y\,R\,z \;\Rightarrow\; x\,R\,z.$$

Let's determine which properties hold for some relations we introduced earlier in the lecture.

Example 11.4 We consider the relation $<$ on $\mathbb{N}$.

- $<$ is not reflexive: $1 \not< 1$.
- $<$ is not symmetric: $3 < 6$ but $6 \not< 3$.
- $<$ is anti-symmetric: This property holds *vacuously* since there are no $n_1, n_2 \in \mathbb{N}$ for which both $n_1 < n_2$ and $n_2 < n_1$.
- $<$ is transitive: Given natural numbers $n_1, n_2, n_3 \in \mathbb{N}$, if $n_1 < n_2$ and $n_2 < n_3$, then $n_1 < n_3$.

Example 11.5 We consider the $\Rightarrow$ relation on the set of compound propositions formed from the atomic propositions $\{p, q\}$, which we denote by $C[p, q]$.

- $\Rightarrow$ is reflexive. Given any compound proposition $c \in C[p, q]$, the proposition $c \Rightarrow c$ is true, which we conclude from the truth table of $\Rightarrow$.
- $\Rightarrow$ is not symmetric: Consider the compound propositions $(p \wedge q)$ and $(p \vee q)$. Note that $(p \wedge q) \Rightarrow (p \vee q)$; however, $(p \vee q) \not\Rightarrow (p \wedge q)$.
- $\Rightarrow$ is not anti-symmetric: Consider the compound propositions $c_1 = \neg(p \wedge q)$ and $c_2 = \neg p \vee \neg q$. From De Morgan's Law, we know that these propositions are equivalent (meaning $c_1 \Rightarrow c_2$ and $c_2 \Rightarrow c_1$). However, they are not equal (since they are built using different logical operations).
- $\Rightarrow$ is transitive: Given any propositions $c_1, c_2, c_3 \in C[p, q]$, if $c_1 \Rightarrow c_2$ and $c_2 \Rightarrow c_3$, then $c_1 \Rightarrow c_3$. We can verify this by constructing a truth table and highlighting the rows where both $c_1 \Rightarrow c_2$ and $c_2 \Rightarrow c_3$ are true. In these highlighted rows, $c_1 \Rightarrow c_3$ is true.

c_1	c_2	c_3	$c_1 \Rightarrow c_2$	$c_2 \Rightarrow c_3$	$c_1 \Rightarrow c_3$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Example 11.6 We consider the "equal cardinality" relation E on $\mathcal{F}(\mathbb{N})$.

- E is reflexive. Any set $S \subseteq \mathbb{N}$ has $|S| = |S|$.
- E is symmetric. If $|S| = |T|$ then $|T| = |S|$.
- E is not anti-symmetric: $|\{1\}| = |\{2\}|$ (and so $|\{2\}| = |\{1\}|$ by symmetry) but $\{1\} \neq \{2\}$.
- E is transitive. Suppose that $S, T, U \subseteq \mathbb{N}$ with $|S| = |T|$ and $|T| = |U|$. By the transitivity of equality, $|S| = |U|$.

E is an *equivalence relation* on $\mathcal{F}(\mathbb{N})$, which we will discuss more in the next lecture.

Example 11.7 We consider the subset relation $\subseteq$ on $\mathcal{P}(\mathbb{N})$.

- $\subseteq$ is reflexive: Given any $S \subseteq \mathbb{N}$, $S \subseteq S$.
- $\subseteq$ is not symmetric: $\{1\} \subseteq \{1,2\}$ but $\{1,2\} \not\subseteq \{1\}$. A single counterexample suffices to disprove the universally quantified ($\forall$) statement in Definition 11.3.
- $\subseteq$ is anti-symmetric: The only way that $S \subseteq T$ and $T \subseteq S$ is if every element in S is in T and every element of T is in S, so $S = T$.
- $\subseteq$ is transitive: Suppose that $S \subseteq T$ and $T \subseteq U$. Then, given any $s \in S$, $s \in T$ since $S \subseteq T$ and then $s \in U$ since $T \subseteq U$. Thus, $S \subseteq U$.

$\subseteq$ is a *partial order* on $\mathcal{P}(\mathbb{N})$, which we will discuss more in two lectures.

Main Takeaways:

- A binary relation on set X is a subset of X^2. We usually think about a relation as defining a connection between elements of X.
- Relations can be classified according to different properties: reflexivity, symmetry, anti-symmetry, transitivity.

Exercises

11.1 Determine whether each of the relations from Example 11.2 (T, B, S, Q) is:

(a) Reflexive (b) Symmetric (c) Anti-symmetric (d) Transitive

11.2 A relation R on set X is *irreflexive* if $\forall x \in X.\, \neg(x\,R\,x)$.

(a) Express this definition in words.

(b) Give two examples of irreflexive relations over *different* ground sets.

(c) Determine whether the following statement is true of false.

If a relation R on set X is not reflexive, then it is irreflexive.

Justify your answer with a proof or a counterexample.

(d) Is the converse of the statement from part (c) true or false? Justify your answer with a proof or a counterexample.

11.3 A relation R on X is *asymmetric* if $\forall x, y \in X.\, x\,R\,y \Rightarrow \neg(y\,R\,x)$.

(a) Express this definition in words.

(b) Give two examples of asymmetric relations over *different* ground sets.

(c) Prove the following claim, which uses the irreflexivity property defined in Exercise 11.2.

A relation R on X is asymmetric if and only if it is irreflexive and anti-symmetric.

11.4 Given a relation R on set X, we define its inverse as:

$$R^{-1} = \{(y,x) \in X^2 \mid (x,y) \in R\}.$$

(a) Describe (in words) how to find the inverse of a relation.

(b) Let $X = \{1,2,3,4\}$ and define relation $R = \{(1,3),(2,4),(3,3),(4,1),(4,2)\}$ on X. What is R^{-1}?

(c) Argue that the inversion of a relation *preserves* reflexivity. That is, argue that if R is a reflexive relation on set X, then R^{-1} is also a reflexive relation on X.

(d) Similarly to part (b), argue that inversion preserves symmetry, anti-symmetry, and transitivity.

(e) Argue that a relation R on set X is symmetric if and only if $R = R^{-1}$.

11.5 Given two relations R and S on set X, we define their composition as:

$$S \circ R = \{(x,z) \in X^2 \mid \exists\, y \in X.\ (x,y) \in R \wedge (y,z) \in S\}$$

(a) Describe (in words) how to find the composition of two relations.

(b) Let $X = \{1,2,3,4\}$. Define relations $R = \{(1,3),(1,1),(2,4)\}$ and $S = \{(3,4),(1,1),(2,2)\}$ on X. Find the compositions $S \circ R$, $R \circ S$, and $R \circ R$.

(c) We say that a property is *preserved under composition* if whenever two relations R and S on set X both have that property, then their composition $S \circ R$ does as well. Determine whether each of the four properties of relations (reflexivity, symmetry, anti-symmetry, transitivity) is preserved under composition.

(d) Argue that a relation R on set X is transitive if and only if $R \circ R \subseteq R$.

11.6 While attending this lecture, Hector accidentally mis-copied some symbols in the definitions of symmetric and transitive relations.

(a) In his definition, Hector wrote that a binary relation R on set X is symmetric if

$$\forall\, x,y \in X. \qquad x\,R\,y \Leftrightarrow y\,R\,x.$$

(He wrote $\Leftrightarrow$ instead of $\Rightarrow$) Is Hector's definition equivalent to Definition 11.3? Explain why or why not.

(b) In his definition, Hector wrote that a binary relation R on set X is transitive if

$$\forall\, x,y,z \in X. \qquad x\,R\,y \wedge y\,R\,z \;\Leftrightarrow\; x\,R\,z.$$

(Again, he wrote $\Leftrightarrow$ instead of $\Rightarrow$) Is Hector's definition equivalent to Definition 11.5? Explain why or why not.

11.7 Luna is looking over the properties of relations and makes the following observation:

> *If a relation R is both transitive and symmetric, then it is automatically reflexive. Suppose that $x\,R\,y$. By symmetry, $y\,R\,x$. Then, by transitivity, $x\,R\,y$ and $y\,R\,x$ implies that $x\,R\,x$, so R is reflexive.*

Luna's observation is not correct. Give a counterexample to show this and explain the mistake in her reasoning.

11.8 Given a relation R on set X, define its complement $\overline{R} = X^2 \setminus R$. Determine whether each of the following implications is true or false. Justify your answer with a proof or a counterexample.

(a) R is reflexive $\Rightarrow \overline{R}$ is reflexive.

(b) R is symmetric $\Rightarrow \overline{R}$ is symmetric.

(c) R is anti-symmetric $\Rightarrow \overline{R}$ is anti-symmetric.

(d) R is transitive $\Rightarrow \overline{R}$ is transitive.

11.9 Let $X = \{1,2,3\}$. Define relation on X (expressed as a set of pairs) that satisfy the following sets of properties:

(a) R_1 is reflexive and symmetric, but not transitive.

(b) R_2 is reflexive and transitive, but not symmetric.

(c) R_3 is symmetric and transitive, but not reflexive.

(d) R_4 is symmetric and anti-symmetric.

(e) R_5 is symmetric but not anti-symmetric.

(f) R_6 is neither symmetric nor anti-symmetric.

(g) R_7 is anti-symmetric but not symmetric.

(h) R_8 is transitive but not anti-symmetric.

11.10 For each of the following relations, determine whether it is reflexive, symmetric, anti-symmetric, and/or transitive (that is, answer all four questions for each).

(a) The relation $\neq$ on $\mathbb{N}$.

(b) The relation $\leq$ on $\mathbb{N}$.

(c) The relation L on $\mathcal{F}(\mathbb{N})$ with $S\,L\,T \Leftrightarrow |S| \leq |T|$.

(d) The relation $\succ$ on $\mathcal{P}(\mathbb{N})$ with $A \succ B$ if and only if the smallest element of A is greater than the largest element of B.

(e) The relation $\sim$ on $\mathbb{N}^2$ with $(x_1,y_1) \sim (x_2,y_2) \Leftrightarrow x_1 = x_2$ or $y_1 = y_2$.

(f) The relation $\approx$ on $\mathbb{Z}$ with $x \approx y \Leftrightarrow |x-y| < 3$.

12. Equivalence Classes

In the last lecture, we introduced binary relations on a set. We also discussed four properties — reflexivity, symmetry, anti-symmetry, and transitivity — that we can use to classify relations. In today's lecture, we'll look at a special family of binary relations, equivalence relations, that satisfy three of these properties. The idea of *equivalence* is a generalization of equality ($=$) that signifies that two objects share some desired properties (while perhaps differing in other properties). For this reason, equivalence classes are frequently used in math and computer science to help classify and categorize objects.

12.1 Equivalence Relations

Definition 12.1 — Equivalence Relation.
A binary relation $\equiv$ on a set X is an *equivalence relation* if it is reflexive, symmetric, and transitive.

In this definition, "$\equiv$" is one possible choice of symbol to represent the relation. It is easy to see that the three properties cited in the definition hold for "$=$"; in fact, they form the basis of our intuitive understanding of equality. In addition to equality, we have already seen one example of an equivalence relation.

Example 12.1 The "equal cardinality relation" E on the set of all finite subsets of $\mathbb{N}$, $\mathcal{F}(\mathbb{N})$, is an equivalence relation. We proved in the previous lecture that this relation is reflexive, symmetric, and transitive.

Here is another example of an equivalence relation.

Example 12.2 Consider the relation $=_1$ defined on $\mathbb{Z}^2$ with $(x_1,x_2) =_1 (y_1,y_2)$ if and only if $x_1 = y_1$. That is, this relation simply compares the first arguments of the ordered pairs while ignoring the second arguments. To show that this is an equivalence relation, we show that it satisfies the three required properties:

- $=_1$ is reflexive: For all $(x_1,x_2) \in \mathbb{Z}^2$, we must have $(x_1,x_2) =_1 (x_1,x_2)$ since $x_1 = x_1$.
- $=_1$ is symmetric: For all $(x_1,x_2),(y_1,y_2) \in \mathbb{Z}^2$, if $(x_1,x_2) =_1 (y_1,y_2)$, then $x_1 = y_1$. By the symmetry of equality, $y_1 = x_1$, meaning $(y_1,y_2) =_1 (x_1,x_2)$.
- $=_1$ is transitive: For all $(x_1,x_2),(y_1,y_2),(z_1,z_2)$, if $(x_1,x_2) =_1 (y_1,y_2)$ (so $x_1 = y_1$) and $(y_1,y_2) =_1 (z_1,z_2)$ (so $y_1 = z_1$), by the transitivity of equality, $x_1 = z_1$, so $(x_1,x_2) =_1 (z_1,z_2)$.

In these examples, two elements can be equivalent even if they are not the same. Rather, equivalence only requires the equality of some attribute(s) of the objects. The E relation is concerned only with the cardinality of the subsets; it ignores which numbers they contain. The $=_1$ relation looks only at the first coordinate of the pairs while ignoring the second coordinate. In this way, there may be a group of different elements that an equivalence relation all deems to be equivalent. We introduce the notion of an *equivalence class* to discuss these groups of elements.

12.2 Equivalence Classes and Partitions

Definition 12.2 — Equivalence Class.

Given an equivalence relation $\equiv$ on set X, and an element $x \in X$, the *equivalence class* of x under $\equiv$, denoted $[x]_\equiv$ includes all elements of X that are equivalent to x:

$$[x]_\equiv \ = \ \{y \in X \mid y \equiv x\}.$$

Example 12.3

- $\big[\{1,2\}\big]_E$ is the set of all 2-element subsets of $\mathbb{N}$. It is a subset of $\mathcal{P}(\mathbb{N})$.
- $[\varnothing]_E = \{\varnothing\}$; the only subset of $\mathbb{N}$ with cardinality 0 is $\varnothing$.
- $\big[(0,3)\big]_{=_1} = \{(x,y) \in \mathbb{Z}^2 \mid x = 0\}$, all integer ordered pairs whose first coordinate is 0.

Notice that one equivalence class (i.e., a subset of X) may be represented in multiple ways. For example, the set of 2-element subset of $\mathbb{N}$ that we described as $\big[\{1,2\}\big]_E$ above could also be written $\big[\{3,4\}\big]_E$. However, we could not write this set as $\big[\{3,4,5\}\big]_E$, as this would give us the set of all 3-element subsets instead. This observation forms the basis of the following lemma.

Lemma 12.1 Suppose that $\equiv$ is an equivalence relation on X, and $x_1,x_2 \in X$. Then exactly one of the following is true:

1. $[x_1]_\equiv \ = \ [x_2]_\equiv$ (the equivalence classes coincide)
2. $[x_1]_\equiv \ \cap \ [x_2]_\equiv \ = \ \varnothing$ (the equivalence classes are disjoint)

Proof. We use a case analysis, where we split into two cases based on whether or not $x_1 \equiv x_2$.

Case 1: $x_1 \equiv x_2$

We argue that $[x_1]_\equiv \ = \ [x_2]_\equiv$. As this is an equality of sets, we must show containment in both directions. First, suppose that y is an arbitrary element in $[x_1]_\equiv$. Then, by definition, $y \equiv x_1$.

Now, by transitivity of $\equiv$, since $y \equiv x_1$ and $x_1 \equiv x_2$ (by our assumption), then $y \equiv x_2$, meaning $y \in [x_2]_\equiv$. Thus, $[x_1]_\equiv \subseteq [x_2]_\equiv$. The reverse containment can be established in a similar manner (which is left as an exercise).

Case 2: $x_1 \not\equiv x_2$

We argue that $[x_1]_\equiv \cap [x_2]_\equiv = \varnothing$. For sake of contradiction, suppose that there were some y belonging to this intersection. Then, $y \in [x_1]_\equiv$, so $y \equiv x_1$ (and by symmetry $x_1 \equiv y$). Similarly, $y \in [x_2]_\equiv$, so $y \equiv x_2$. Now, by transitivity $x_1 \equiv y$ and $y \equiv x_2$ implies that $x_1 \equiv x_2$, contradicting our assumption in this case. Hence, the intersection must be empty.

Since these cases cover all possibilities, we have established the lemma. ■

12.3 The Quotient of an Equivalence Relation

We can use Lemma 12.1 to gain a better understanding of the collection of equivalence classes of an equivalence relation. This set is referred to as the *quotient*.

Definition 12.3 — Quotient.

Given an equivalence relation $\equiv$ on set X, the quotient of X under $\equiv$, denoted "$X/\equiv$", is the set of all equivalence classes of X under $\equiv$:

$$X/\equiv \;\; = \Big\{ S \in \mathcal{P}(X) \mid S = [x]_\equiv \text{ for some } x \in X \Big\}.$$

There are many different sets flying around in these definitions, so it's important that we pause and take stock of what types of objects we are dealing with. Each of the equivalence classes within the quotient is a subset of X, so belongs to $\mathcal{P}(X)$. The quotient itself is a set whose elements are these subsets. As a subset of all possible subsets of X, the quotient belongs to $\mathcal{P}\big(\mathcal{P}(X)\big)$. In fact, the quotient is a very special collection of subsets called a *partition*.

Definition 12.4 — Partition.

Given any set S, a collection of subsets $A_1, A_2, \ldots \in \mathcal{P}(S)$ forms a *partition* of S if:

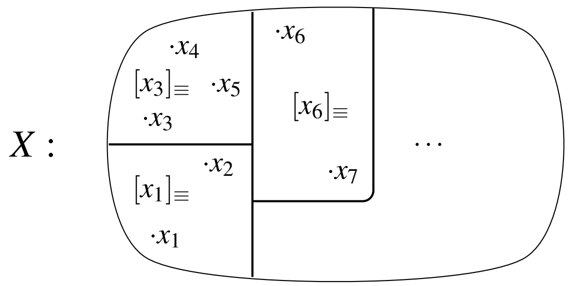
1. $A_i \cap A_j = \varnothing$ for all $i \neq j$ ($A_1, A_2, \ldots$ are *pairwise disjoint*)
2. $\bigcup_{i \geq 1} A_i = S$ ($A_1, A_2, \ldots$ *cover* S)

Equivalently, each $s \in S$ belongs to *exactly* one subset A_i.

We can use the fact that every element $x \in X$ belongs to its own equivalence class $[x]_\equiv$ and Lemma 12.1 to obtain the following result about the quotient set.

Corollary 12.1 If $\equiv$ is an equivalence relation on set X, then $X/\equiv$ is a partition of X.

As a metaphor, we can visualize the X as a cake. Then, $\equiv$ carves (partitions) this cake into pieces (equivalence classes), and the quotient $X/\equiv$ is the set of these pieces.



Often, the quotient set of an equivalence relation has a "natural" relationship (or *bijection*) to some other set. Understanding these bijections is a focus of the mathematical area of *(abstract) algebra*.

Example 12.4

- There is one equivalence class in $\mathcal{F}(\mathbb{N})/E$ for each $n \in \mathbb{N}$ containing all of the n-element subsets of $\mathbb{N}$.

$$\mathcal{F}(\mathbb{N})/E \;=\; \Big\{T \in \mathcal{P}(\mathcal{F}(\mathbb{N})) \mid T = \{S \in \mathcal{F}(\mathbb{N}) \mid |S| = n\} \text{ for some } n \in \mathbb{N}\Big\},$$

- There is one equivalence class in $\mathbb{Z}^2/=_1$ for each $x \in \mathbb{Z}$ containing all ordered pairs with first coordinate x.

$$\mathbb{Z}^2/=_1 \;=\; \Big\{S \in \mathcal{P}(\mathbb{Z}^2) \mid S = \{(x,y) \in \mathbb{Z}^2 \mid x = z\} \text{ for some } z \in \mathbb{Z}\Big\},$$

12.4 Connected Components

As a final example of an equivalence relation, we will revisit a concept of graph theory that we briefly touched on earlier, connected components. Recall the following definition for *connected* vertices.

Definition 12.5 — Connected.

In an undirected graph $G = (V,E)$, two vertices $u, v \in V$ are connected if G contains a path from u to v.

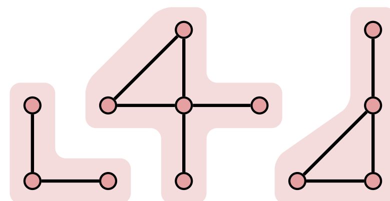
Notice that connectedness is a relation on V; given any two vertices, they are either connected or they are not. Moreover, we can show that connectedness is an equivalence relation by checking the three properties.

Reflexivity: Given any vertex $v \in V$, the path (v) with 0 edges takes us from vertex v to v. Trivially, each vertex is connected to itself.

Symmetry: Suppose that vertices u and v are connected in G. Then, there must be a path $(u, v_1, \dots, v_k, v)$ from u to v in G. As this is an undirected graph, we can consider the edges in the opposite order, so that $(v, v_k, \dots, v_1, u)$ is a path from v to u. By definition, v and u are connected in G.

Transitivity: Suppose that u is connected to v and v is connected to w in G. Then, there is a path along edges in G from u to v and from v to w. Concatenating these paths gives a path[1] from u to w, meaning u is connected to w.

Now, let's think about the equivalence classes of the connectivity relation. Given any vertex $v \in V$, its equivalence class includes all vertices that it can reach along a path of edges in G. This is called the *connected component* of v. We visualize the three connected components in an example graph in the following figure.



Notice that the connected components partition the vertices. The set of all connected components is the quotient set of the connectivity equivalence relation.

Main Takeaways:

- Equivalence relations are reflexive, symmetric, and transitive.
- An equivalence relation on set X *partitions* the elements of X into subsets called equivalence classes.
- The quotient of an equivalence relation $\equiv$ on a set X, notated $X/\equiv$ is the set of all equivalence classes $[x]_\equiv$.
- An undirected graph can be partitioned into *connected components*. There are no edges between different connected components in a graph.

Exercises

12.1 Let us define a relation $\approx_3$ on $\mathcal{P}(\mathbb{N})$ such that for all $S,T \in \mathcal{P}(\mathbb{N})$, $S \approx_3 T$ if and only of $S \cap [3] = T \cap [3]$.

(a) Argue that $\approx_3$ is an equivalence relation.

(b) What is $|\mathcal{P}(\mathbb{N})/\approx_3|$? List one representative from each of these equivalence classes.

We can generalize $\approx_3$ to an equivalence relation $\approx_n$ for each $n \in \mathbb{N}$, where for all $S,T \in \mathcal{P}(\mathbb{N})$, $S \approx_n T$ if and only of $S \cap [n] = T \cap [n]$.

(c) What is $|\mathcal{P}(\mathbb{N})/\approx_n|$? Your answer should depend on n.

(d) Argue that for all $S,T \in \mathcal{P}(\mathbb{N})$, $S = T$ if and only if $S \approx_n T$ for all $n \in \mathbb{N}$.

---

[1]There is a slight subtlety here that we must address to be fully formal. Recall that our definition of paths requires that each pair of consecutive vertices be joined by a *distinct* edge. It is possible that one edge belongs to both of the paths we concatenate. In this case, we can remove the edges between and including one of these repeating edges from the concatenation to obtain a path.

12.2 Let $C[p_1, p_2, p_3, p_4]$ denote the set of all compound propositions that can be expressed through a sequence of logical operations applied to p_1, p_2, p_3, and p_4.

(a) Argue that logical equivalence, $\equiv$, is an equivalence relation on $C[p_1, p_2, p_3, p_4]$.

(b) How many equivalence classes does $\equiv$ have?

$\star$ (c) List one representative of each equivalence class.

12.3 Given $n \in \mathbb{N}^+$, we define the *congruence modulo n* equivalence relation $\equiv_n$ on $\mathbb{Z}$ such that $x \equiv_n y$ if and only if $y - x = c \cdot n$ for some $c \in \mathbb{Z}$. (In many texts, you will see $x \equiv_n y$ written "$x \equiv y \pmod{n}$".)

(a) Argue that congruence modulo n is an equivalence relation on $\mathbb{Z}$ for all $n \in \mathbb{N}$.

For the remainder of the exercise, we focus on the special case $n = 5$.

(b) Find the equivalence class $[0]_{\equiv_5}$?

(c) How many equivalence classes belong to the quotient $\mathbb{Z}/\equiv_5$? Give a description of each of them.

We will often abbreviate these quotients as $\mathbb{Z}/n\mathbb{Z}$, which we call the set of *residue classes modulo n*. These residue classes form the basis for *modular arithmetic*, which is explored more in the next exercise.

12.4 In this exercise, we'll consider how to define addition and multiplication on the equivalence classes in $\mathbb{Z}/n\mathbb{Z}$ (that is, equivalence classes under congruence modulo n). This is a key tool in areas of math such as number theory and cryptography.

(a) Suppose that $a_1, a_2, b_1, b_2 \in \mathbb{Z}$ with $a_1 \equiv_n b_1$ and $a_2 \equiv_n b_2$. Argue that $(a_1 + a_2) \equiv_n (b_1 + b_2)$.

This tells us that given two equivalence classes $[a]_{\equiv_n}$ and $[b]_{\equiv_n}$ in $\mathbb{Z}/n\mathbb{Z}$, we can choose any representative $a' \in [a]_{\equiv_n}$ and any representative $b' \in [b]_{\equiv_n}$, and their sum $a' + b'$ will always belong to the same equivalence class, which is the equivalence class that includes $a' + b'$. For this reason, we define addition on $\mathbb{Z}/n\mathbb{Z}$ by the rule

$$[a]_{\equiv_n} \mathbf{+} [b]_{\equiv_n} := [a+b]_{\equiv_n}.$$

There's some subtlety here to appreciate. The "$+$" on the right side of this definition is the usual addition on the integers (i.e., the operation that is a function $\mathbb{Z}^2 \to \mathbb{Z}$). The bold "$\mathbf{+}$" on the left side of this definition describes a rule for adding *equivalence classes* (i.e. a function $(\mathbb{Z}/n\mathbb{Z})^2 \to \mathbb{Z}/n\mathbb{Z}$).

(b) Similar to above, suppose that $a_1, a_2, b_1, b_2 \in \mathbb{Z}$ with $a_1 \equiv_n b_1$ and $a_2 \equiv_n b_2$. Argue that $(a_1 \cdot a_2) \equiv_n (b_1 \cdot b_2)$.

For this reason, we can define multiplication on $\mathbb{Z}/n\mathbb{Z}$ by the rule

$$[a]_{\equiv_n} \cdot [b]_{\equiv_n} := [ab]_{\equiv_n}.$$

12.5 Many programming languages define the *modulo* operation (mod). Given $z \in \mathbb{Z}$ and $n \in \mathbb{N}$, $z \bmod n$ is the unique integer i such that $0 \leq i < n$ and $z \equiv_n i$.

(a) Calculate each of the following:

i. $75 \bmod 5$

ii. $75 \bmod 4$

iii. $80 \bmod 13$

iv. $-12 \bmod 7$

(b) Argue that $a \equiv_n b \Leftrightarrow a \bmod n = b \bmod n$.

12.6 In a directed graph $G = (V,A)$, we say that vertices $v, w \in V$ are *strongly connected* if G contains a path from v to w *and* a path from w to v.

Argue that strong connectivity is an equivalence relation on V.

We call the equivalence classes the *strongly connected components* of G.

12.7 Given two sets A and B, we define their *symmetric difference*

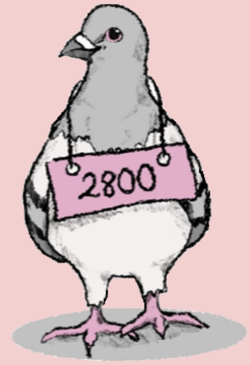
$$A \triangle B := (A \setminus B) \cup (B \setminus A).$$

(a) Compute the symmetric difference of $A = \{0,1,2,3,4,5,6\}$ and $B = \{1,3,5,7,9,11\}$.

(b) Define the relation $\sim$ on $\mathcal{P}(\mathbb{N})$ such that for any $A, B \subseteq \mathbb{N}$, $A \sim B$ if and only if $A \triangle B$ is a finite set. Argue that $\sim$ is an equivalence relation.

(c) Argue that $\mathcal{F}(\mathbb{N}) = \{S \in \mathcal{P}(\mathbb{N}) : |S| = n \text{ for some } n \in \mathbb{N}\}$, the set of all finite subsets of $\mathbb{N}$, forms an equivalence class of $\mathcal{P}(\mathbb{N})$ under $\sim$. (This entails arguing that all finite subsets belong to the same equivalence class, and all infinite subsets do not belong to this class.)

(d) Is $\sim$ an anti-symmetric relation? Provide either a proof or a counterexample.



13. Partial Orders

In the previous lecture, we studied equivalence relations. We saw that their three properties — reflexivity, symmetry, and transitivity — made equivalence a natural generalization of $=$. In today's lecture, we'll look at another special family of relations, *partial orders*, which generalize the behavior of $\leq$. We'll also introduce a graph called a *Hasse diagram* that allows us to visualize the structure of a partially ordered set.

13.1 Partial Orders

Partial orders generalize the $\leq$ relation on $\mathbb{N}$. They provide us with a way to talk about elements of a set being "smaller" or "larger" than other elements for some suitable definition of "small" and "large". Just as we saw for equivalence relations, the definition of a partial order cites three properties of relations.

Definition 13.1 — Partial Order, Poset.

A binary relation $\preceq$ on a set X is a *partial order* if it is reflexive, anti-symmetric, and transitive.

We refer to the pair $(X, \preceq)$ as a *partially ordered set*, or a *poset*.

Note that in this definition, "$\preceq$" is a symbol that we are using to represent the relation; however, it is just as valid to choose a different symbol, as our examples will demonstrate.

Example 13.1 We can formally define the $\leq$ relation on $\mathbb{Z}$ such that $x \leq y$ if and only if there is some $n \in \mathbb{N}$ such that $x + n = y$. We must verify that, under this definition, $\leq$ is reflexive, anti-symmetric, and transitive.

Reflexive: Given any $x \in \mathbb{Z}$, $x \leq x$ because $x + 0 = x$ and $0 \in \mathbb{N}$.

Anti-Symmetric: Consider any $x, y \in \mathbb{Z}$ such that $x \leq y$ and $y \leq x$. By definition, there must be $n, m \in \mathbb{N}$ such that $x + n = y$ and $y + m = x$. Substituting the second equation into the first, we obtain $y + m + n = y$, so $m + n = 0$. The only possibility is for $m = n = 0$, meaning $x = y$.

Transitivity: Consider any $x, y, z \in \mathbb{Z}$ such that $x \leq y$ and $y \leq z$. By definition, there must be $n, m \in \mathbb{N}$ such that $x + n = y$ and $y + m = z$. Substituting the first equation into the second, we obtain $x + (m+n) = z$. Since $m + n \in \mathbb{N}$, we conclude that $x \leq z$.

We conclude that $(\mathbb{Z}, \leq)$ is a poset.

Later we'll see that this is a very special type of poset, a *total order*. Before that, we'll give two more examples (and one non-example) of posets.

Example 13.2 Let's consider two different relations on $\mathcal{F}(\mathbb{N})$, the set of all finite subsets of $\mathbb{N}$.

The subset relation $\subseteq$ on $\mathcal{F}(\mathbb{N})$ is a partial order. We verified that $\subseteq$ is reflexive, anti-symmetric, and transitive two lectures ago.

Now, let's define a relation L on $\mathcal{F}(\mathbb{N})$ such that $S\,L\,T$ if and only if $|S| \leq |T|$. Although L is reflexive and transitive (verify this), it is not a partial order because it is not anti-symmetric. Note that $\{1\}\,L\,\{2\}$ because $|\{1\}| = 1 \leq 1 = |\{2\}|$. Similarly, $\{2\}\,L\,\{1\}$. However, $\{1\} \neq \{2\}$.

Our last example considers a different way (besides $\leq$) to order the positive integers.

Example 13.3 The divisibility relation "$|$" on $\mathbb{N}^+$ is defined such that $k \,|\, n$ (read "k divides n") if and only if there is some $c \in \mathbb{N}$ such that $n = c \cdot k$.

- $|$ is reflexive: Given any $n \in \mathbb{N}^+$, $n = 1 \cdot n$, so $n \,|\, n$.
- $|$ is anti-symmetric: Suppose that $k \,|\, n$ and $n \,|\, k$ for some $n, k \in \mathbb{N}^+$. Then, $n = c_1 \cdot k$ and $k = c_2 \cdot n$ for some $c_1, c_2 \in \mathbb{N}$. But then, $n = c_1 \cdot c_2 \cdot n$, so $c_1 \cdot c_2 = 1$. The only possibility is if $c_1 = c_2 = 1$, meaning $n = k$.
- $|$ is transitive: Suppose that $k \,|\, n$ and $n \,|\, m$ for some $k, n, m \in \mathbb{N}^+$. Then $n = c_1 \cdot k$ and $m = c_2 \cdot n$ for some $c_1, c_2 \in \mathbb{N}$. But then, $m = c_1 \cdot c_2 \cdot k$. By closure of $\mathbb{N}$ under multiplication, $c_1 \cdot c_2 \in \mathbb{N}$, so by definition $k \,|\, m$.

Thus, $|$ is a partial ordering on $\mathbb{N}^+$.

13.1.1 Hasse Diagrams

To visualize a poset, we can use a directed graph called a *Hasse diagram*. We will make use of the following definition to describe the arcs in a Hasse diagram.

Definition 13.2 — Covering.

Given a partial order $\preceq$ on a set X, we say that an element $x \in X$ *is covered by* an element $y \in X \setminus \{x\}$ (equivalently, y covers x) if:

- $x \preceq y$,
- There is no $z \in X \setminus \{x, y\}$ such that $x \preceq z$ and $z \preceq y$.

Said another way, x is covered by y if y is a "next largest" element in X after x. For example, in the poset $(\mathbb{Z}, \leq)$, x is covered by $x + 1$ for each $x \in \mathbb{Z}$.

Notice that the covering property is defined on two elements of the set X, and it either holds or does not hold for that pair of elements. Hence, covering is another relation on the set X. Now, we can describe how to construct a Hasse diagram.

Definition 13.3 — Hasse Diagram.

Given partial order $\preceq$ on a set X, the Hasse diagram for $(X,\preceq)$ is the directed graph $G=(V,A)$ with vertex set $V=X$ and with arc set

$$A=\{(x,y)\in X^2 \mid x \text{ is covered by } y\}.$$

When possible, we typically choose locations for the vertices in a Hasse diagram so that all of the arcs point (generally) upwards. When this is done, sometimes the arrowheads the arcs are omitted. We will omit the circles around each node so that the figures do not become too crowded. Hasse diagrams can be invaluable for understanding the structure of a poset.

Example 13.4 The Hasse diagram for $\big([4],\leq\big)$ is a single column.

$$\begin{tikzcd}
4 \\
3 \arrow[u] \\
2 \arrow[u] \\
1 \arrow[u]
\end{tikzcd}$$

This is an example of a *linearly*- or *totally-ordered* poset. The relation for a total order satisfies an additional totality property.

Definition 13.4 — Total Relation.

A binary relation R on a set X is *total* if for all $x,y\in X$, either $x\,R\,y$ or $y\,R\,x$ (or both).

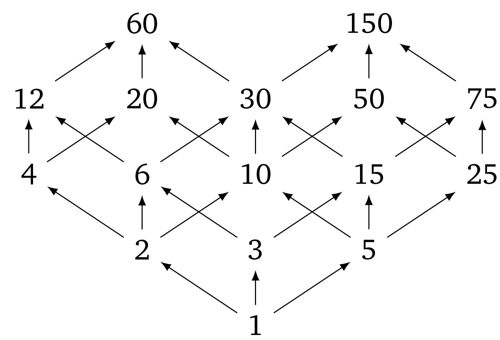
We can also draw Hasse diagrams for the other partial orders we discussed.

Example 13.5 Consider the $\subseteq$ relation on the set $\mathcal{P}(\{a,b,c\})$. This poset is often denoted B_3 (the *boolean* poset on three elements). We visualize its Hasse diagram:

$$\begin{tikzcd}
& \{a,b,c\} & \\
\{a,b\} \arrow[ur] & \{a,c\} \arrow[u] & \{b,c\} \arrow[ul] \\
\{a\} \arrow[u] \arrow[ur] & \{b\} \arrow[ul] \arrow[ur] & \{c\} \arrow[ul] \arrow[u] \\
& \varnothing \arrow[ul] \arrow[u] \arrow[ur] &
\end{tikzcd}$$

If you look carefully, you'll see that the twelve lines in this diagram form the edges of a cube.

Example 13.6 Consider the poset $(\mathbb{N}^+, |)$. Although we cannot visualize this entire (infinite) poset, we show the Hasse diagram of some of its elements below.



These two Hasse diagrams can highlight some of the structural similarities between posets with the subset and divisibility relations.

Main Takeaways:

- Partial orders are reflexive, anti-symmetric, and transitive relations that generalize the behavior of $\leq$.
- A *Hasse diagram* is a diagram that helps to visualize a partially ordered set, or *poset*.
- Equivalence relations are reflexive, symmetric, and transitive relations that generalize the behavior of $=$.

Exercises

13.1 Argue that every Hasse diagram is a DAG (directed acyclic graph).

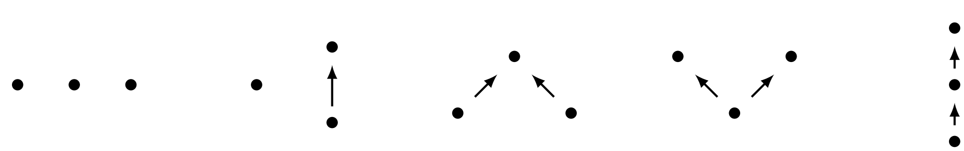
13.2 Draw the Hasse diagrams for the divisibility relation on the following sets.

(a) $S_1 = \{1,2,4,8,16\}$

(b) $S_2 = \{1,2,3,4,6,9,12,18,36\}$

(c) $S_3 = \{3,6,7,10,12,14,21,30,35,49,70\}$

13.3 Draw the Hasse diagrams of the sixteen distinct 4-element poset "shapes". You can use dots to represent the elements and line segments to represent the covering relations. For example, the five 3-element posets are shown below.



13.4 In this exercise, you'll consider possible ways to order pairs in $\mathbb{Z}^2$. For each of these possibilities, determine whether it is a partial order, providing either a proof or a counterexample.

(a) Intersection order: $(x_1, y_1) \preceq_I (x_2, y_2) \Leftrightarrow x_1 \leq x_2$ and $y_1 \leq y_2$

(b) Union order: $(x_1, y_1) \preceq_U (x_2, y_2) \Leftrightarrow x_1 \leq x_2$ or $y_1 \leq y_2$

(c) Lexicographic order: $(x_1, y_1) \preceq_L (x_2, y_2) \Leftrightarrow x_1 < x_2$ or $(x_1 = x_2$ and $y_1 \leq y_2)$

13.5 Two useful notions in a poset $(X, \preceq)$ are *chains* and *antichains*. A *chain* is a totally ordered subset $C \subseteq X$: for each $c_1, c_2 \in C$, either $c_1 \preceq c_2$ or $c_2 \preceq c_1$. An *antichain* is a subset $A \subseteq X$ of pairwise incomparable elements; for any distinct elements $a_1, a_2 \in A$, $a_1 \not\preceq a_2$ and $a_2 \not\preceq a_1$. A chain C (antichain A) is *maximal* if no additional element can be added to C (A) to make a larger chain (antichain).

(a) Argue that each 2-element subset of a poset X is either a chain or an antichain.

(b) List two maximal chains and two maximal antichains from the poset in Example 13.5.

(c) Let C be a chain and A an antichain in X. Argue that $|A \cap C| \leq 1$.

(d) Do any two maximal antichains in a poset have the same number of elements? Provide a proof or counterexample. Also answer this question for maximal chains.

13.6 In this exercise, you'll explore a unique feature of the equality relation "$=$" on any set.

(a) Given any set S, use set builder notation define the equality relation as a set of pairs.

(b) Argue that equality is the unique relation on the set S that is both a partial order and an equivalence relation.

13.7 Let $C[p_1, p_2, p_3, p_4]$ denote the set of all compound propositions containing only atomic propositions from $\{p_1, p_2, p_3, p_4\}$.

(a) Argue that $\Rightarrow$ is *not* a partial order over $C[p_1, p_2, p_3, p_4]$. Here, we say that $c_1 \Rightarrow c_2$ when this implication is true for all possible truth assignments of $\{p_1, p_2, p_3, p_4\}$.

In the previous lecture, you argued that logical equivalence ($\equiv$) is an equivalence relation on $C[p_1, p_2, p_3, p_4]$.

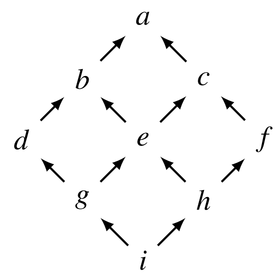
$\star$ (b) Argue that $\Rightarrow$ is a partial order over the quotient $C[p_1, p_2, p_3, p_4]/\Leftrightarrow$, where we say that $[c_1]_\equiv \Rightarrow [c_2]_\equiv$ if $c_1 \Rightarrow c_2$ (explain why the choice of representative from each equivalence class does not matter).

$\star$ **13.8** In a directed graph $G = (V, A)$, we say that vertices $v, w \in V$ are *strongly connected* if G contains a path from v to w *and* a path from w to v. In the previous lecture, you argued that strong connectivity is an equivalence relation on V. We call the equivalence classes the *strongly connected components* of G.

Let $\mathcal{K}$ be the set of strongly connected components. We say that a component $K \in \mathcal{K}$ *reaches* another component $K' \in \mathcal{K}$ if there are vertices $v \in K$ and $w \in K'$ such that G contains a path from v to w. Argue that the *reaches* property defines a partial order on $\mathcal{K}$.

13.9 Given a poset $(X, \preceq)$, and two elements $x, y \in X$, we say that z is an *upper bound* of x and y if $x \preceq z$ and $y \preceq z$. We say that z is the *least upper bound* or *join* of x and y (which we denote $z = x \vee y$) if z is an upper bound of x and y and for every upper bound u of x and y, $z \preceq u$.

(a) Find the join of each of the pair of elements in poset with the following Hasse diagram.

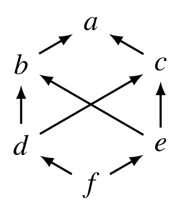


Similarly, we say that z is a *lower bound* of x and y if $z \preceq x$ and $z \preceq y$. We say that z is the *greatest lower bound* or *meet* of x and y (which we denote $z = x \wedge y$) if z is a lower bound of x and y and for every lower bound ℓ of x and y, $\ell \preceq z$.

(b) Find the meet of each pair of elements in the poset from part (a).

A poset $(X, \preceq)$ is a *lattice* if every pair of elements $x, y \in X$ has both a join and a meet.

(c) Explain why a poset with the following Hasse diagram is not a lattice.

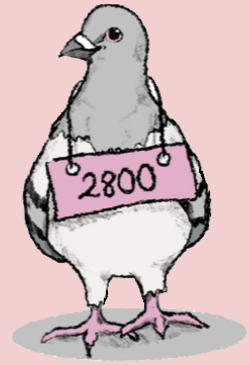


(d) Argue that $(\mathbb{Z}, \leq)$ is a lattice. Given any $x, y \in \mathbb{Z}$, what are $x \vee y$ and $x \wedge y$?

(e) Argue that $(\mathcal{P}(\mathbb{N}), \subseteq)$ is a lattice. Given any $S, T \in \mathcal{P}(\mathbb{N})$, what are $S \vee T$ and $S \wedge T$?

(f) Argue that $(|, \mathbb{N}^+)$ is a lattice. Given any $m, n \in \mathbb{N}^+$, what are $m \vee n$ and $m \wedge n$?

IV Functions and Cardinality



14. Functions

Along with sets and relations, functions are one of the most important building blocks of mathematics. Functions are the verbs in our mathematical language. They describe actions, or transformations, that we can perform on a set. You are likely familiar with functions in two contexts. In mathematics classes, you have most likely associated functions with curves graphed in the x,y-plane, and classes such as algebra, trigonometry, and calculus use the shapes of these curves to study the properties of functions. On the other hand, in computer science, we use functions to encapsulate our code. We define functions as routines that take in some input parameters, carry out a calculation and produce a return value. In this lecture, we'll give a more formal definition of a function that unifies both of these perspectives. In the coming lectures, we'll look at some fundamental properties of functions and how they can be used to reason about the sizes, or cardinalities, of (potentially infinite) sets.

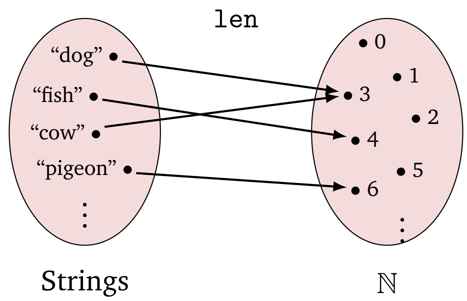
14.1 The Definition of a Function

A function describes a transformation from objects in one set (its inputs) to objects in another set (its outputs). To help us formalize this definition, we will consider two examples.

Example 14.1

- The function $f(x) = x^2$ takes in any real number input $x \in \mathbb{R}$ and returns another real number which is the square of x. For example, if we pass input 2 into f, we receive back output $f(2) = 2^2 = 4$.
- We can pass any string into the Python function `len`, and it will return the number of characters in that string. For example, if we pass input "pigeon" into `len`, it will return `len(‘‘pigeon’’) = 6`.

What do these examples have in common? Each of them takes an input value from a particular set ($\mathbb{R}$ for f, the set of all strings for `len`). We call this input set the *domain* of the function. Then, the function returns one particular output value from another set ($\mathbb{R}$ for f, $\mathbb{N}$ for `len`). We call this output set the *codomain* of the function. We can view the function as describing a way to map each element in the domain over to an element of the codomain. We'll use "blobs and arrows" diagrams to help visualize this mapping. For example, the mapping of the length function looks like:



The left blob depicts the domain set, with each point representing one of its elements. Similarly, the right blob depicts the codomain set. Each point in the domain has an outgoing arrow that points to the codomain element to which the function maps it.

From this picture, we see that we need to specify three things to formally define a function: the domain, the codomain, and the mapping. The first two are easy to formalize, they are just sets. Suppose that we use D to represent the domain and C to represent the codomain. We can express this using the notation $f : D \to C$ (often called the *signature* of the function), which is read "f maps D to C". To encode the mapping, we will need a way to represent each of the arrows. From our discussion of graphs, we can represent them as ordered pairs (d,c) with $d \in D$ and $c \in C$. In this way, the mapping f is a subset of $D \times C$.

However, not any subset will do. A *well-defined* function should be able to accept any input from the domain, and each input should be sent to exactly one output. These properties place restrictions on which subsets of $D \times C$ are allowed. We summarize this in the following collection of definitions.

Definition 14.1 — Domain, Codomain, Function, Well-Defined.

Given a *domain* set D and a *codomain* set C, a *function* $f : D \to C$ describes a mapping from D to C. To be a *well-defined* function $f \subseteq D \times C$ must have the following two properties.

Totality: Every element of the domain is mapped by f.

$$\forall\, d \in D, \quad \exists\, c \in C. \quad (d,c) \in f.$$

Determinism: The mapping of each domain element is unique.

$$\forall\, d \in D, \quad \forall\, c_1, c_2 \in C. \quad (d,c_1) \in f \wedge (d,c_2) \in f \implies c_1 = c_2.$$

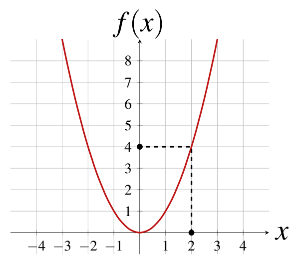
In this course, whenever we talk about a function, we will implicitly assume that it is well-defined. When you define a function, you should check that it satisfies these two properties. These properties tell us that for each $d \in D$ there is *exactly* one pair $(d,c) \in f$, which tells us that f sends the domain element d to the codomain element c. We typically use the notation "$f(d) = c$" as opposed to "$(d,c) \in f$", as this better illustrates the mapping of the function. We call this codomain element $c \in C$ the *image* of the domain element $d \in D$.

Definition 14.2 — Image.

Given a function $f : D \to C$ and an element $d \in D$, the *image* of d under f is the codomain element to which f maps d (i.e., $f(d) \in C$).

In math classes, we often associate a function with its *plots*, which visualizes it in the x,y coordinate plane.

Example 14.2 The function $f(x) = x^2$ is visualized as a parabola.



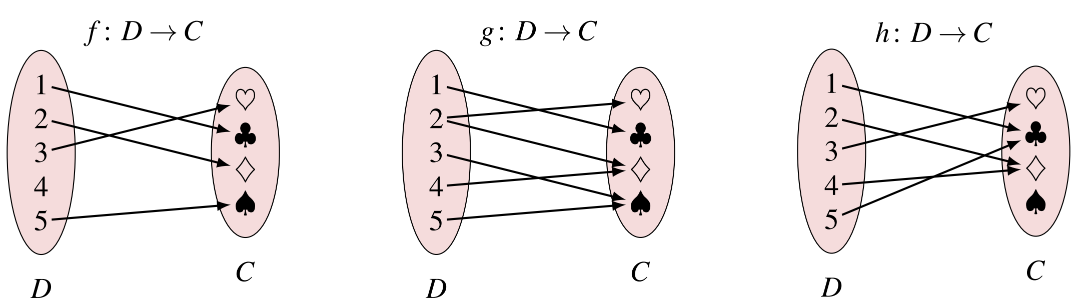
In this plot, the domain ($\mathbb{R}$) is visualized by the x-axis, and the codomain ($\mathbb{R}$) is visualized by the y-axis. The red points along the parabolic curve in this plot are exactly the members of the subset of $\mathbb{R} \times \mathbb{R}$ that define this function.

The plot gives a convenient way to visualize the mapping. Given any real number $x \in \mathbb{R}$, we locate x on the x-axis, trace vertically to the function curve, and then trace horizontally to the y-axis. The location we land at on the y-axis is $f(x)$, the image of x under f. The picture illustrates finding $f(2) = 4$ using this tracing process.

You may be familiar with the "vertical line test" for determining whether a plot is a function. This test is really checking that the function is well-defined; if a graph passes the vertical line test, then each x-value in the domain has exactly one associated y-value in the codomain (each vertical line intersects the function exactly once).

There is also an easy way to check whether a function represented as a "blobs and arrows" diagram is well-defined: it should have exactly one arrow leaving each element in the domain. Of course, this can only be practically checked when the domain and codomain are small enough that we can fully depict the mapping.

Example 14.3 Consider the following three arrow diagrams that represent mappings from domain $D = \{1,2,3,4,5\}$ to codomain $C = \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$.



f is not a (well-defined) function since it violates the totality property (it does not map 4 to any suit in C). Similarly, g is not a function since it violates the determinism property (it maps 2 to both $\heartsuit$ and $\diamondsuit$). h is a function since each number in D maps to exactly one suit in C.

Note that the properties of a function impose restrictions on the domain "coordinate" of $f \subseteq D \times C$; however, they do not place similar restrictions on the codomain. In our parabola example, the codomain element -1 was not mapped to by any domain element (perfect squares of real numbers must be non-negative), and the codomain element 1 was mapped to by two domain elements, -1 and 1. In the function h from the previous example, $\spadesuit$ is not mapped to, while both $\clubsuit$ and $\diamondsuit$ are mapped to twice. This asymmetry of the definition is (among other things) what makes functions interesting to study.

In the next lecture, we will discuss special functions that do impose these "backward" properties on their codomains.

14.2 The Range of a Function

Another important set associated with a function is its *range*.

Definition 14.3 — Range.

Given a function $f\colon D \to C$, its *range* consists of all codomain elements to which f maps a domain element. That is,

$$\text{range}(f) = \left\{c \in C \mid f(d) = c \text{ for some } d \in D\right\}.$$

Equivalently, the range consists of all $c \in C$ that are images of $d \in D$ under f.

Example 14.4

- The range of $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ is the non-negative real numbers $\mathbb{R}_{\geq 0}$.
- The range of h from Example 14.3 is $\{\heartsuit, \clubsuit, \diamondsuit\}$.
- The range of the function $f\colon \mathbb{Z} \to \mathbb{Z}$ with $f(z) = 2z$ is the even integers.
- The range of the function $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x$ is $\mathbb{R}$.
- The range of the function $f\colon \mathbb{N}^+ \to \mathbb{N}$ with $f(n) = n^0$ is $\{1\}$.

It is a common pitfall to confuse the range and the codomain. Remember that the codomain is something that we must specify when we define the function; it is part of the function signature. The range is a set that we can construct after the fact by reasoning about the images of the domain elements. Note that the range is a subset of the codomain.

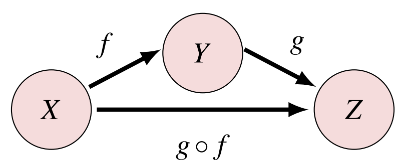
14.3 Function Composition

To conclude this lecture, we will talk about an operation[1] that we can perform on functions, composition. Intuitively, a function defines a rule for moving from one set to another. Therefore, the application of two functions in a row, one going from a set X to a set Y and another going from Y to a set Z, realizes a mapping from X to Z. As we will argue below, this mapping is, in fact, a function, called the *composition* (or *composite function*) of the two original functions.

Definition 14.4 — Composition.
Given two functions $f\colon X \to Y$ and $g\colon Y \to Z$, their *composition*, denoted $g \circ f$, is a function $g \circ f\colon X \to Z$ defined such that for a given $x \in X$,

$$(g \circ f)(x) = g(f(x)).$$

Let's unpack this definition. Since f is a function from $X \to Y$, $f(x) = y$ for some $y \in Y$. Then, since g is a function from $Y \to Z$, $g(f(x)) = g(y) \in Z$. We can visualize the composition of functions as a "short-cutting arrow" that bypasses the middle set.



One thing that this picture highlights is that the composition notation can seem "backwards": the function that is applied first is on the right side of the "$\circ$", and the function that is applied second is on the left side of the "$\circ$". This is a byproduct of the fact that we write the name of the function on the left.

To verify that $g \circ f$ is well-defined, we use the fact that both f and g are.

Lemma 14.1 The composition $g \circ f$ of $f\colon X \to Y$ and $g\colon Y \to Z$ given in Definition 14.4 is a well-defined function $X \to Z$.

Proof. We must verify that the two properties of a well-defined function hold for $g \circ f$.

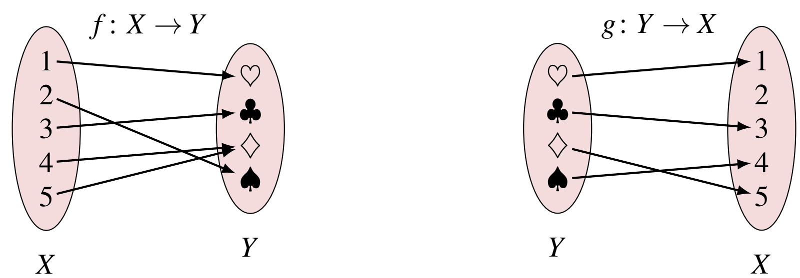
Totality: Given any $x \in X$, there is some $y \in Y$ for which $f(x) = y$ (by totality of f). In addition, there is some $z \in Z$ for which $g(y) = z$ (by totality of g). Then, $(g \circ f)(x) = g(f(x)) = g(y) = z$, so z is the image of x under $g \circ f$. Since x was arbitrary, we conclude that $g \circ f$ is total.

Determinism: Suppose that for a given $x \in X$, both $(x, z_1), (x, z_2) \in (g \circ f)$ for $z_1, z_2 \in Z$. By our definition of composition, this means that there must be some $y \in Y$ so that $(x, y) \in f$ (i.e., $y = f(x)$) both $(y, z_1), (y, z_2) \in g$. However, this implies that $z_1 = z_2$ (by the determinism of g). Thus, $g \circ f$ maps deterministically. ■

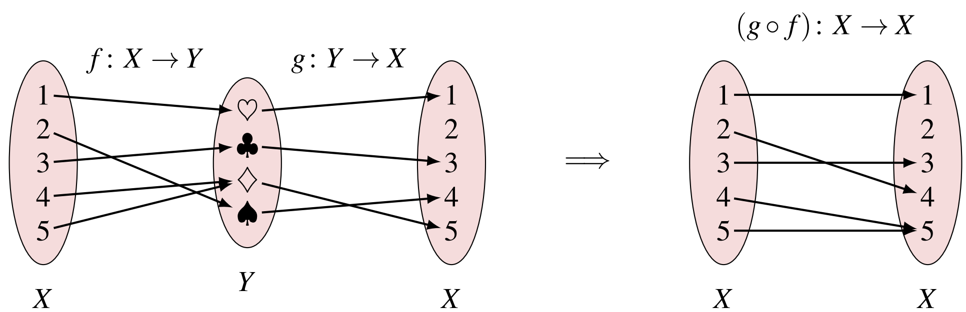
[1] Now that we have defined functions, we can give a more formal description of an operation. Namely, a binary operation on a set S is a function $S \times S \to S$. That is, a binary operation maps two elements of a set to another element of that set. For example, addition is a binary operation that maps two real numbers to another real number.

Critically, the composition $g \circ f$ is only defined when the codomain of f is the domain of g. For functions represented by arrow diagrams, we find the composition by "linking the arrows".

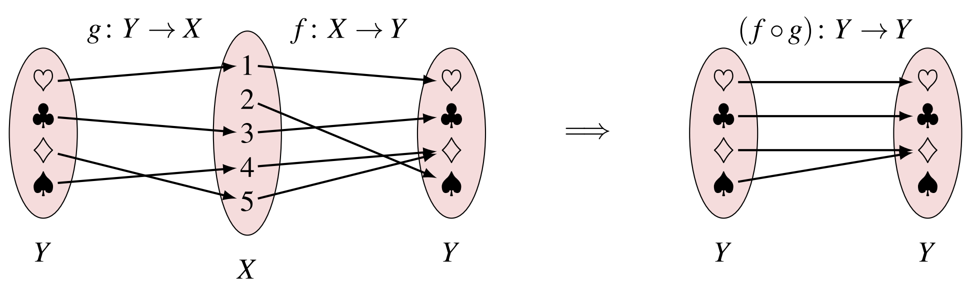
Example 14.5 Consider the functions



Since the codomain of f, Y, is the domain of g, the composition $g \circ f : X \to X$ is defined. We may visualize this composition:



Similarly, since the codomain of g, X, is the domain of f, the composition $f \circ g : Y \to Y$ is defined. We may visualize this composition:



For the algebraic functions that we are more used to, we can compute the composition by plugging the first (right) function in as the argument to the second (left) function.

Example 14.6 Suppose that the functions $f, g : \mathbb{R} \to \mathbb{R}$ are defined such that $f(x) = 3x - 2$ and $g(x) = (x+1)^2$. Then,

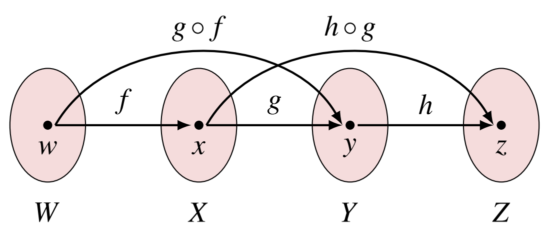
$$(g \circ f)(x) = g(f(x)) = g(3x-2) = \big((3x-2)+1\big)^2 = (3x-1)^2 = 9x^2 - 6x + 1,$$
$$(f \circ g)(x) = f(g(x)) = f\big((x+1)^2\big) = 3\big((x+1)^2\big) - 2 = 3(x^2+2x+1) - 2 = 3x^2 + 6x + 1.$$

In the next lecture, we will study the inversion of functions, where we will seek to "undo" the mapping of a function by composing it with another function. For now, we end by establishing one more important property of function composition.

Lemma 14.2 Function composition is *associative*: Given functions $f\colon W \to X$, $g\colon X \to Y$, and $h\colon Y \to Z$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Proof. To argue that two functions are equal, we must show that they have the same domain and codomain, and that they agree on the mapping of each domain element. By the definition of composition, both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ have domain W and codomain Z. For the last point, we'll let w be an arbitrary element of W. We must argue that $\big(h \circ (g \circ f)\big)(w) = \big((h \circ g) \circ f\big)(w)$. By the totality of f, g, h there is some element $x \in X$ such that $f(w) = x$, some element $y \in Y$ such that $g(x) = y$, and some element $z \in Z$ such that $h(y) = z$.



By the definition of function composition, we have $(g \circ f)(w) = g(f(w)) = g(x) = y$, and $(h \circ g)(x) = h(g(x)) = h(y) = z$, so that

$$\big(h \circ (g \circ f)\big)(w) = h\big((g \circ f)(w)\big) = h(y) = z = (h \circ g)(x) = (h \circ g)(f(w)) = \big((h \circ g) \circ f\big)(w).$$

■

Take some time to read this proof carefully and make sure you appreciate each step.

Main Takeaways:

- A function defines a mapping from an input set (domain) to an output set (codomain).
- To be well-defined, a function must identify a unique element of the codomain that corresponds to each element of the domain.
- The range of a function is the subset of its codomain that is mapped; in other words, it is the set of images of the domain.
- Function composition allows us to chain together the application of multiple mappings.

14.1 Consider sets $A = \{a, b\}$ and $B = \{1, 2, 3\}$. List,

(a) All functions $A \to B$, expressed as subsets of $A \times B$.

(b) All functions $B \to A$, expressed as subsets of $B \times A$.

14.2 Consider sets $S = \{\heartsuit, \diamondsuit, \clubsuit, \spadesuit\}$ and $N = \{1,2,3,4,5\}$. For each of the following sets, draw its corresponding arrow diagram, and determine whether it describes a function $S \to N$. For any that do not, explain why not.

(a) $\Big\{(\heartsuit,3),(\spadesuit,2),(\diamondsuit,4),(\clubsuit,1)\Big\}$

(b) $\Big\{(\heartsuit,2),(\clubsuit,2),(\heartsuit,4),(\spadesuit,1)\Big\}$

(c) $\Big\{(\heartsuit,3),(\clubsuit,3),(\spadesuit,3),(\diamondsuit,3)\Big\}$

(d) $\Big\{(\spadesuit,5),(\clubsuit,1),(\diamondsuit,2)\Big\}$

14.3 Consider the function f with domain $\mathbb{N}$ and with $f(n) = n^3 - 2n + 6$. List three possible codomains for f.

14.4 Let $S = \{1,2,3,4,5\}$, and define functions $f, g\colon S \to S$ with

$$f(x) = 6 - x, \qquad g(x) = \begin{cases} x+1 & x < 5 \\ 1 & x = 5. \end{cases}$$

Find the following compositions:

(a) $f \circ g$

(b) $g \circ f$

(c) $f^2 = f \circ f$

(d) g^2

(e) $f \circ g \circ f$

15. ’Jectivity and Inverses

In this lecture, we will discuss two important properties of some functions, injectivity and surjectivity. With these properties, we can reason about when the mapping of a function can be reversed, or *inverted*. These new concepts will provide us with useful tools as we progress in the semester. In the short term, they will connect to the pigeonhole principle and our definition of set cardinality that will conclude this module. Beyond this, invertible functions establish *bijective correspondences* between sets, a powerful tool that can illumine similarities between seemingly disparate mathematical objects.

15.1 Two Properties of Functions

Last lecture, we saw that well-defined functions have two properties that allow us to conceptualize them as mappings between sets. *Totality* assured us that every element of the domain (i.e., the input set) was prescribed a mapping. *Determinism* assured us that each domain element was sent to exactly one codomain (output) element.

We also saw that these same guarantees do not exist for elements of the codomain. It is possible that some element in the codomain is not the image of any domain element (i.e., the function’s range may be a strict subset of its codomain). It is also possible that one codomain element is mapped to by multiple elements of the domain. The properties of *injectivity* and *surjectivity* describe functions that do not encounter these obstacles.

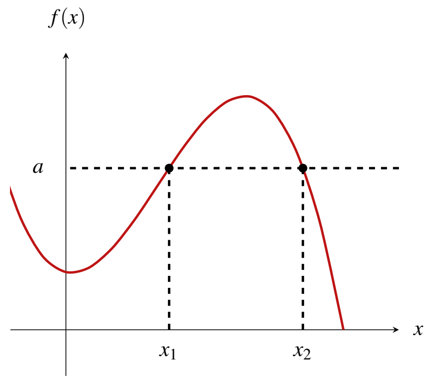
15.1.1 Injectivity

Definition 15.1 — Injectivity.

A function $f\colon D \to C$ is *injective*, or *one-to-one*, if each element of D maps to a distinct element of C. That is,

$$\forall\, d_1 \in D, \forall\, d_2 \in D.\ \big(d_1 \neq d_2 \quad \Rightarrow \quad f(d_1) \neq f(d_2)\big).$$

You may be familiar with the “horizontal line test” for checking whether a function is injective. If any horizontal line $y = a$ intersects the plot of a function more than once, then there must be two distinct x-values $x_1 \neq x_2$ for which $f(x_1) = f(x_2) = a$, certifying that the function is *not* injective.



Example 15.1

- The function $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ is not injective, since $f(2) = f(-2) = 4$.
- The function $f\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$ is injective, as we will prove below.
- The function $f\colon \mathbb{N} \to \mathbb{N}$ with $f(n) = 2n$ is injective. Given $n_1 \neq n_2 \in \mathbb{N}$, $f(n_1) = 2n_1 \neq 2n_2 = f(n_2)$.

A function represented with arrow diagrams is injective if each codomain element is pointed to by at most one arrow.

Example 15.2 Consider the following arrow diagrams that represent mappings from domain $D = \{1,2,3,4\}$ to codomain $C = \{a,b,c,d\}$.

Here, f is not injective since $f(2) = f(4) = c$. g is injective since each codomain element has one incoming arrow; no two domain elements map to the same codomain element.

This example illustrates that it suffices to exhibit a single counterexample (two distinct domain elements with the same image) to conclude that a function is *not* injective. To argue that a function is injective, it is often more convenient to use the following logically equivalent form of the definition:

$$\forall\, d_1 \in D,\, \forall\, d_2 \in D.\ \big(f(d_1) = f(d_2) \quad \Rightarrow \quad d_1 = d_2\big).$$

The implication in this expression is the *contrapositive* of the implication in the original definition. Thus, to prove that a function is injective, we can take arbitrary $d_1, d_2 \in D$, assume that $f(d_1) = f(d_2)$, and then use the definition of f to show that we must have $d_1 = d_2$.

Example 15.3 We argue that $f\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$ is injective. Suppose that $f(x_1) = f(x_2)$. By the definition of f, $x_1^2 = x_2^2$, so $0 = x_1^2 - x_2^2 = (x_1 + x_2)(x_1 - x_2)$. Then, one of these two factors must be 0. If $x_1 + x_2 = 0$, then we must have $x_1 = x_2 = 0$ as $x_1, x_2 \geq 0$. Otherwise, if $x_1 - x_2 = 0$, then $x_1 = x_2$. In either case, $x_1 = x_2$, so f is injective. ■

15.1.2 Surjectivity

Definition 15.2 — Surjectivity.
A function $f\colon D \to C$ is *surjective*, or *onto*, if each element of C is the image of some element in D

$$\forall\, c \in C, \quad \exists\, d \in D. \quad f(d) = c.$$

Equivalently, a function is surjective when its range is equal to its codomain.

Example 15.4

- The function $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ is not surjective, since $-1 \in \mathbb{R}$ is not the image of any real number under f.
- The function $f\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$ is surjective. For each $a \geq 0$, $\sqrt{a} \in \mathbb{R}_{\geq 0}$ has $f(\sqrt{a}) = (\sqrt{a})^2 = a$.
- The function $f\colon \mathbb{N} \to \mathbb{N}$ with $f(n) = 2n$ is not surjective, as all odd natural numbers fall outside of its range.
- From Example 15.2, f is not surjective as no number in D maps to b. g is surjective as each element in C has an incoming arrow.

To argue that a function is *not* surjective, it suffices to locate (with proof) one codomain element that is not in its range. To prove that a function is surjective, consider an *arbitrary* codomain element $c \in C$ and locate some $d \in D$ for which $f(d) = c$. The two stages in this argument are (1) arguing that the input you construct is in D and (2) arguing that it actually maps to c.

15.1.3 Bijectivity

Definition 15.3 — Bijectivity.
A *bijective* function $f\colon D \to C$ is both injective and surjective.

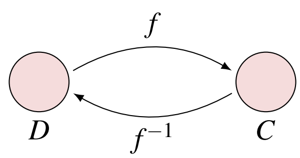
Another name for a bijective function is a *one-to-one correspondence*, as it provides a pairing between elements of the domain and codomain. The function g from Example 15.2 is a good example of this. The four arrows in this diagram pair each number of the domain with a distinct letter in the codomain, and vice versa. In the next section, we will draw a connection between bijections and invertible functions.

15.2 Inverse Functions

Before we can discuss function inversion, we must introduce a special family of functions.

Definition 15.4 — Identity Functions.
Given a set S, the identity function on S, denoted id_S, is a function $S \to S$ that maps each element to itself. That is, $\mathrm{id}_S(s) = s$ for each $s \in S$.

On their own, identity functions seem rather boring and pointless. However, they are useful in the context of function composition. The identity function gives us a language for describing when a composition of functions returns us to where we started. This is exactly the goal of function inversion. When we take the inverse of a function, we are reversing its mapping. That is, we are finding a rule that associates to each element of the codomain $c \in C$ an element of $d \in D$ which maps to it. We would like to understand when this rule is itself a function $f^{-1}: C \to D$, which we call the inverse of f.



Definition 15.5 — Inverse Function, Invertible.
Given a function $f: D \to C$, a function $g: C \to D$ is the *inverse* of f, denoted $g = f^{-1}$, if

$$g \circ f = \mathrm{id}_D \qquad \text{and} \qquad f \circ g = \mathrm{id}_C.$$

If such a function g exists, we say that f is *invertible*.

Recall that two functions are equal if they have the same domains and codomains, and if they agree on the mapping of all domain elements. Thus, to prove that two functions $f, g: D \to C$ are equal, show that for an *arbitrary* element $d \in D$, $f(d) = g(d)$.

Example 15.5 Consider the functions $f, g: \mathbb{R} \to \mathbb{R}$ with $f(x) = 3x + 2$ and $g(x) = \frac{x-2}{3}$. We argue that $g = f^{-1}$. First, for any $x \in \mathbb{R}$,

$$(g \circ f)(x) = g(f(x)) = g(3x+2) = \frac{(3x+2)-2}{3} = x,$$

so $g \circ f = \mathrm{id}_{\mathbb{R}}$. Similarly, for any $x \in \mathbb{R}$,

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{x-2}{3}\right) = 3\left(\frac{x-2}{3}\right) + 2 = x,$$

so $f \circ g = \mathrm{id}_{\mathbb{R}}$.

The following theorem describes the connection between bijective functions and inverses.

Theorem 15.1 A function $f\colon D \to C$ is invertible if and only if it is bijective.

Proof. We must establish both directions of this implication.

For the forward direction, suppose that f is invertible. We must argue that f is bijective, which amounts to showing that it is both injective and surjective.

Injective: Using the contrapositive, suppose that $f(d_1) = f(d_2)$ for some elements $d_1, d_2 \in D$. By the determinism property of f^{-1}, we have

$$f^{-1}(f(d_1)) = f^{-1}(f(d_2)) \quad \Rightarrow \quad (f^{-1} \circ f)(d_1) = (f^{-1} \circ f)(d_2) \quad \Rightarrow \quad d_1 = d_2.$$

Here, the first implication uses the definition of function composition, and the second implication follows from the definition of f^{-1}, which tells us that $f^{-1} \circ f = \mathrm{id}_D$.

Surjective: Given any $c \in C$, note that $f^{-1}(c) \in D$ (since D, the domain of f, is the codomain of f^{-1}). Since $f \circ f^{-1} = \mathrm{id}_C$, we have $f(f^{-1}(c)) = c$, so we have exhibited an element of D that f maps to c.

For the reverse direction, suppose that f is bijective. We must construct the inverse of f. Let us define the function $g\colon C \to D$ such that for each $c \in C$, $g(c)$ is the element $d \in D$ such that $f(d) = c$. We are guaranteed that there is such a d since f is surjective. Moreover, we are guaranteed that there is only one such d since f is injective. Hence, g is well defined. To show that $g = f^{-1}$ we compose it with f. Given any $d \in D$,

$$(g \circ f)(d) = g(f(d)) = d,$$

since (applying the definition of g) d is the unique $d \in D$ that f maps to $f(d)$. Hence, $g \circ f = \mathrm{id}_D$. Also, given any $c \in C$,

$$(f \circ g)(c) = f(g(c)) = c,$$

since g maps c to the element $d \in D$ that f maps to c. Hence, $f \circ g = \mathrm{id}_C$, so $g = f^{-1}$. ■

In light of Theorem 15.1, there are two common ways to prove that a function f is bijective.

1. Argue that f is both injective and surjective.
2. Identify another function $g\colon C \to D$ and argue that $g \circ f = \mathrm{id}_D$ and $f \circ g = \mathrm{id}_C$.

Main Takeaways:

- An *injective* function maps each element of its domain to a *distinct* element of the codomain.
- A *surjective* function maps to *every* element of its codomain; that is, its codomain is its range.
- A function is *invertible* when it can be composed with another function to give the identity function.
- There is a strong connection between the properties of injectivity and surjectivity and the invertibility of a function.

Exercises

15.1 Consider the functions $f(x) = \frac{3x+4}{x-2}$ and $g(x) = \frac{2x+4}{x-3}$. Is it true that $f = g^{-1}$? Be careful in your answer. (*Hint:* What other information do we need to determine whether a function is an inverse?)

15.2 Argue that if $g = f^{-1}$, then $f = g^{-1}$. Use this to conclude that $(f^{-1})^{-1} = f$. We say inversion is an *involution* since applying it twice brings us back where we started.

15.3 Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x,y) = (x+y, x-y).$$

Note here that we drop the extra parentheses of the ordered pair, using $f(x,y)$ to represent the correct but annoying notation $f\big((x,y)\big)$.

(a) Argue that f is injective.

(b) Argue that f is surjective.

(c) Since f is bijective, Theorem 15.1 tells us that f is invertible. What is $g = f^{-1}$? Verify that your g satisfies both properties of being an inverse.

15.4 Argue that if a function has an inverse, this inverse is unique. As a hint, suppose that both g_1 and g_2 are inverses of f and use properties of inverses and composition to conclude that $g_1 = g_2$.

⋆ **15.5** Let X and Y be non-empty sets. Argue that there exists an injective function $f\colon X \to Y$ if and only if there exists a surjective function $g\colon Y \to X$. (Hint: In both directions of the proof, you'll need to use one function to construct the other.)

15.6 In this exercise, we reason about functions $f\colon D \to D$ with the property that $f^2 = f \circ f = f$.

(a) Argue that such an f is invertible if and only if $f = \mathrm{id}_D$.

(b) Give an example of a non-invertible function with this property.

15.7 When we discussed set operations, we noticed that the Cartesian product is not an associative operation, since $A \times (B \times C) \neq (A \times B) \times C$. However, the difference between these sets is easily remedied.

Describe (with proof) a bijection between $(A \times B) \times C$ and $A \times (B \times C)$. The existence of such a bijection affords us the ability to write $A \times B \times C$ (which we understand as the set of all *ordered triples*) without much consternation. We can generalize this to construct *ordered tuples* with any number of elements.

15.8 Suppose that S is a finite set. Describe a bijection between $\mathcal{P}(S)$ and

$$\{0,1\}^{|S|} = \underbrace{\{0,1\} \times \ldots \times \{0,1\}}_{|S| \text{ copies}}.$$

Include in your answer a proof that your function is a bijection.

⋆ **15.9** Let S be a non-empty set and $s \in S$. Describe (with proof) two bijections $f\colon \mathcal{P}(S) \to \mathcal{P}(S)$ with the property that for any $A \subseteq S$, $s \in A \Leftrightarrow s \notin f(A)$.

⋆ **15.10** Given a function $f\colon D \to C$, we can define a second function $F\colon \mathcal{P}(D) \to \mathcal{P}(C)$ such that

$$F(S) = \{c \in C \,|\, \exists\, d \in S.\ f(d) = c\}$$

for all $S \subseteq D$.

(a) Describe in words the behavior of F.

(b) Argue that f is injective if and only if F is.

(c) Argue that f is surjective if and only if F is.

15.11 Given a function $f\colon D \to C$, we can define a second function $\reflectbox{$f$}\colon C \to \mathcal{P}(D)$ such that

$$\reflectbox{f}(c) = \{d \in D \,|\, f(d) = c\}$$

for all $c \in C$. The elements in the range of $\reflectbox{$f$}$ are called the *pre-images* of f.

(a) Describe in words the behavior of $\reflectbox{$f$}$.

(b) Argue that each $d \in D$ belongs to exactly one set in range($\reflectbox{$f$}$).

(c) If f is injective, what does this tell us about the elements in range($\reflectbox{$f$}$)?

(d) If f is surjective, what does this tell us about the elements in range($\reflectbox{$f$}$)?

(e) If f is invertible, argue that $\reflectbox{$f$}(c) = \{f^{-1}(c)\}$ for each $c \in C$.

15.12 Give an example of two functions $f, g\colon X \to X$ (you should also specify the set X) with the following properties:

- $f \circ g = g \circ f$,
- $f \neq g$,
- $f \neq g^{-1}$.

15.13 In this question, you'll argue that function composition preserves injectivity and surjectivity. As common setup, we'll consider two function $f\colon X \to Y$ and $g\colon Y \to Z$.

(a) Prove that if f and g are both injective, then $g \circ f$ is also injective.

(b) Prove that if f and g are both surjective, then $g \circ f$ is also surjective.

(c) Conclude that if f and g are both bijective, then $g \circ f$ is also bijective. In this case, show that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

15.14 Here, we'll continue our exploration of the relationship between injectivity, surjectivity, and composition from the previous question. Again, $f\colon X \to Y$ and $g\colon Y \to Z$.

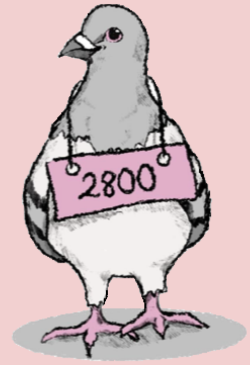
(a) Give an example where f is injective, g is not injective, and $g \circ f$ is not injective.

(b) Give an example where g is injective, f is not injective, and $g \circ f$ is not injective.

(c) Prove or disprove the following claim: If $g \circ f$ is injective, then f is injective.

(d) Prove or disprove the following claim: If $g \circ f$ is injective, then g is injective.

(e) Repeat parts (a)-(d) for surjective and bijective functions.



16. The Pigeonhole Principle

Two proof techniques are central to many arguments in discrete math and computer science: induction and the pigeonhole principle, the focus of today's lecture. Although its basic form can appear obvious, clever applications of the pigeonhole principle appear in many of the most beautiful and elegant proofs. In today's lecture, we'll introduce the pigeonhole principle and its generalized form and use it to establish some results in arithmetic and graph theory.

16.1 The (Basic) Pigeonhole Principle

Each week, the 2800 course staff orders food for our grading session. There are 13 grading sessions over the course of a semester, and there are 10 restaurants that we can order from. Is it possible for us to order from a different restaurant each time?

Hopefully, you can quickly answer this question "no." There are more sessions than restaurants, so we will run out of restaurant options before we hold all of the grading sessions and we will need to repeat a restaurant at some point in the semester. The reasoning that we used here did not depend on the specifics of this application. We did not use the fact that we were ordering for grading sessions or anything about the restaurants. Rather, this reasoning simply used the fact that we were assigning elements of a larger set (the sessions) to elements of a smaller set (the restaurants). In other words, we uncovered a fact about a function between two sets. This fact is called the *pigeonhole principle* and can be written more formally as follows.

Lemma 16.1 — The (Basic) Pigeonhole Principle.

Let $f\colon X \to Y$ be a function between two finite sets with $|X| > |Y|$. Then, there are $x_1, x_2 \in X$ such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$. That is, f is *not* injective.

In our example, X is the set of grading sessions, Y is the set of restaurants, and f maps each grading session to the restaurant where the food was ordered. The pigeonhole principle tells us that at two different grading sessions ($x_1 \neq x_2$), we must order from the same restaurant ($f(x_1) = f(x_2)$).

Proof. We will argue the contrapositive. Suppose that f is injective. This means that each $y \in Y$ is mapped to by at most one $x \in X$. However, f is total, so this accounts for every element in X. Thus, $|X| \leq |Y|$. ■

The pigeonhole principle gets its name from imagining that the domain elements are pigeons that are landing in a collection of pigeonholes (the codomain elements). When there are more pigeons than holes, then two pigeons must land in the same hole. Often, the trick of applying the pigeonhole principle is identifying what plays the role of the pigeons and what plays the role of the holes. We will start with more straightforward applications and work toward more involved examples at the end of the lecture.

Example 16.1 We will prove that if 6 distinct integers between 1 and 10 are selected, two of these integers will sum to 11.

Let our 6 chosen integers be the pigeons, and let the 5 sets $\{1,10\}$, $\{2,9\}$, $\{3,8\}$, $\{4,7\}$, $\{5,6\}$ be the holes. We will map each integer to the set where it is a member. Since there are more pigeons (6) than holes (5), the pigeonhole principle tells us that two of our chosen integers belong to the same set. However, the two elements in each of these sets sum to 11. ■

We just used the pigeonhole principle to prove an existential statement ("There exist two integers that sum to 11."). Normally, we prove a "there exists" statement by giving an example; however, this was not the case here. We do not know which two integers specifically were chosen sum to 11. In light of this, we say that the pigeonhole principle is a *non-constructive* proof technique; it assures us that something exists without actually finding it!

16.2 The Generalized Pigeonhole Principle

Notice that the (basic) pigeonhole principle tells us that when the domain is larger than the codomain, we can always find two different domain elements that map to the same codomain element. However, this doesn't mean that there are *exactly* two such elements. There may be many more; in our example, we could have ordered from *Taste of Thai* every week (much to the delight of many TAs).

Sometimes, we can establish a stronger guarantee. Consider the following example: There are 49 undergraduate TAs in 2800. We want to argue that at least 5 of them were born in the same month. We cannot derive this from Lemma 16.1; we will need another version.

Theorem 16.1 — Generalized Pigeonhole Principle. Let $f: X \to Y$ be a function between two finite sets with $|X| > k \cdot |Y|$ for some $k \in \mathbb{N}$. Then, there are at least $k+1$ distinct elements $x_1, x_2, \ldots x_{k+1} \in X$ such that $f(x_1) = f(x_2) = \cdots = f(x_{k+1})$.

The proof of this theorem is similar to Lemma 16.1 and left as an exercise (Exercise 16.1).

We can use this theorem to prove our earlier statement. We'll let the 49 TAs be the pigeons, the 12 months be the holes, and we'll map each TA to their birth month. Noting that $49 > 48 = 4 \cdot 12$, we may use Theorem 16.1 to conclude that at least $4+1=5$ TAs were both in the same month.

We say that Theorem 16.1 *generalizes* Lemma 16.1 because the lemma is a special case of the theorem. In particular, we exactly recover the statement of the lemma if we fix $k = 1$. By allowing k to take any arbitrary value, we obtain a stronger statement that can be applied in more settings.

16.3 More Examples

We will end the lecture by looking at three more examples of applying the pigeonhole principle. The first is another arithmetic example, and the latter two come from graph theory.

Example 16.2 We argue that if 4 distinct integers are chosen between 1 and 60 (inclusive), then two of them must differ by at most 19.

Here, we let the pigeons be the four chosen integers. The holes will be "thirds" of the range 1-60: namely, $\{1,\ldots,20\}$, $\{21,\ldots,40\}$, and $\{41,\ldots,60\}$. Our function f will map each selected integer to the set that contains it. Then, the pigeonhole principle tells us that two of our chosen integers come from the same set. Since each of these sets consists of 20 contiguous integers, the difference between these two integers can be at most 19. ■

Example 16.3 We argue that in any undirected graph with at least two vertices, two vertices must have the same degree.

As an initial attempt, we can let the pigeons be the vertices and the holes be the possible degrees. The degree of each vertex can be any integer between 0 (if the vertex is not incident to any edges) and $|V|-1$ (if the vertex is connected to every other vertex). However, this gives $|V|$ pigeons and $|V|$ holes, which is not good enough to apply Lemma 16.1. We will need to be more clever. We do this by distinguishing two cases.

Case 1: Every vertex has an incident edge.

In this case, no vertex has degree 0, meaning the possible degrees are $\{1,\ldots,|V|-1\}$, $|V|-1$ possibilities. Since there are more vertices than possible degrees, we may use the pigeonhole principle to conclude that two vertices have the same degree.

Case 2: There is a vertex $v \in V$ with no incident edges.

In this case, no vertex can have degree $|V|-1$, since this would imply that it was connected to v. Again, there are only $|V|-1$ possible degrees ($\{0,\ldots,|V|-2\}$), so we use the pigeonhole principle to conclude that two vertices have the same degree.

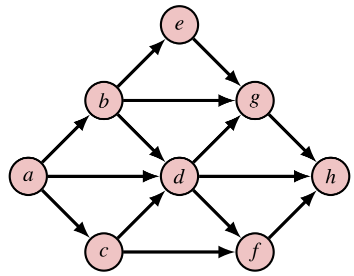
As our two cases exhaust all possibilities, we have proven the claim. ■

Our final example will concern directed acyclic graphs (DAGs).

Definition 16.1 — DAG.

A directed graph $G = (V,A)$ is a *DAG* if it does not contain any *directed cycles*. That is, there is no cycle $(v_0, v_1, \ldots, v_k)$ with $(v_{i-1}, v_i) \in A$ for each $i \in [k]$.

One DAG is drawn below.



Lemma 16.2 If $G = (V,A)$ is a directed acyclic graph, then there is a vertex $v \in V$ with no outgoing arcs.

In our example DAG, h is a vertex with no outgoing arcs.

Proof. For the sake of contradiction, suppose there is no such $v \in V$. That is, every vertex in V has an outgoing arc. Now, select any vertex $v_0 \in V$. By our assumption, there is some arc $(v_0, v_1) \in A$ leaving v_0. We can again apply the assumption to locate an arc $(v_1, v_2) \in A$ leaving v_1. We may repeat this line of reasoning a total of $|V|$ times, identifying a *walk* (a path which may have repeated edges) $(v_0, v_1, \ldots, v_{|V|})$ in G. Traveling along this walk makes $|V|+1$ visits to vertices; however, there are only $|V|$ vertices in the graph. By the pigeonhole principle, the walk must visit a vertex twice, so must contain a cycle. This contradicts the fact that G is a DAG. Hence, there must be a vertex $v \in V$ with no outgoing arcs. ■

Main Takeaways:

- The pigeonhole principle says that if X is a set of pigeons and Y is a set of pigeonholes, with $|X| > |Y|$, then any function mapping pigeons to pigeonholes will assign more than one pigeon to some pigeonhole.
- The generalized pigeonhole principle says that if X is a set of pigeons and Y is a set of pigeonholes, with $|X| > k \cdot |Y|$ then any function mapping pigeons to pigeonholes will assign more than k pigeons to some pigeonhole.
- To apply the pigeonhole principle, we must carefully describe the pigeons, holes, and the mapping between them.

Exercises

16.1 Following the outline of the proof of Lemma 16.1, give a proof of Theorem 16.1.

16.2 Here, we suggest an alternate approach to establish results from Example 16.1. Fill in the remaining details to complete the proof.

Call the chosen integers $x_1, \ldots, x_6$. Then, define $y_1, \ldots, y_6$ with $y_i = 11 - x_i$. Now, we apply the pigeonhole principle with pigeons $x_1, \ldots, x_6, y_1, \ldots, y_6$. Consider the possible values of each of these 12 integers.

16.3 Suppose that 51 distinct integers from $[100] = \{1, \dots, 100\}$ are selected. Argue that (at least) two of the chosen integers are consecutive. (Hint: There are two main approaches to this exercise. One is similar to Example 16.1, and the other is similar to Exercise 16.2. Try to find them both.)

16.4 Here, we ask you to generalize the result from the previous exercise.

(a) Argue that if $n+1$ distinct integers are chosen from $[2n]$, then two are consecutive.

(b) Argue that if $n+1$ integers are chosen from $[3n]$, then two have a difference of at most 2.

(c) Generalize the result from parts (a) and (b) to discuss selecting $n+1$ integers from $[kn]$.

(d) Where have we seen the special case $k = 1$ of part (c)?

(e) Argue that if $2n+1$ distinct integers are chosen from $[3n]$, then three are consecutive.

(f) How can you generalize part (e)?

16.5 In the most recently published report, the student population at Cornell is 25,582. What is the largest number n such that we can be guaranteed that at least n Cornell students share the same birthday? (Don't forget February 29.)

16.6 A restaurant pays its employees every other Friday. Argue that each year, there is at least one month where the employees receive three paychecks. Can you strengthen this result?

16.7 A team bicycle race includes 25 teams with 4 riders each. All the riders begin the race together, and the winning team is the first to have *all four* of its riders complete the race. After how many riders have finished the race can we be guaranteed to know the winning team?

⋆ **16.8** Consider a sequence of n integers $x_1, x_2, \dots, x_n$. Argue that there are integers $1 \leq i \leq j \leq n$ such that $x_i + x_{i+1} + \ldots + x_j$ (that is, some sum of consecutive numbers) is divisible by n.

(Hint: Consider the partial sums $S_k = \sum_{i=1}^{k} x_i$ of this sequence.)

⋆ **16.9** For his birthday, Alfred receives a box of 35 assorted cookies. Over the next 24 days, Alfred eats all the cookies, having at least one cookie each day. Argue that there is a group of consecutive days where Alfred eats exactly 12 cookies.

(Hint: The solution that we have in mind for this exercise combines ideas from Exercises 16.2 and 16.8.)

16.10 A donut shop bakes 8 each of 9 different flavors of donuts. They arrange these 72 donuts into 6 one-dozen (12 donut) boxes. Argue that there is a box that contains (at least) two each of two different flavors of donut.

16.11 Argue that in any set of 14 integers, there are two whose sum or difference is a multiple of 25.

16.12 Suppose that $f\colon X \to Y$ is a function between two *finite* sets with $|X| = |Y|$. Argue that f is injective if and only if f is surjective.

16.13 In this exercise, we consider an application of the result from Example 16.3.

(a) There are n guests at a party. Throughout the evening, pairs of guests introduce themselves to one another and exchange a handshake. Supposing that no pair of guests shake hands more than once, argue that two guests shook the same number of hands.

(b) Now, suppose that there are $2n$ guests who arrive as couples. We assume that the two members of each couple do not shake hands (they should already know each other). Again, argue that two guests shook the same number of hands.

⋆ (c) (A puzzle by Lars Bertil Owe) Consider the special case of part (b) with $n = 5$ couples. At the end of the evening, one woman surveys the other nine participants and observes that each of them shook a different number of hands. How many hands did she shake?

(d) Generalize the result from part (c) to n couples.

16.14 Must a directed graph have two vertices with the same in-degree? Justify your answer with a proof or a counterexample.

16.15 If $G = (V,A)$ is a DAG, argue that there is a vertex $v \in V$ with no incoming arcs.

16.16 Given a directed graph $G = (V,A)$ with $|V| = n$, a *topological sort* of G orders its vertices in a way such that for each arc $(u,v) \in A$, u is listed before v. For example, a topological sort of the graph pictured in the lecture lists the vertices in alphabetical order.

(a) Argue that if G has a topological sort, then G is a DAG. (Hint: Use the contrapositive.)

(b) Use induction and Exercise 16.15 to argue that if G is a DAG, then G has a topological sort.

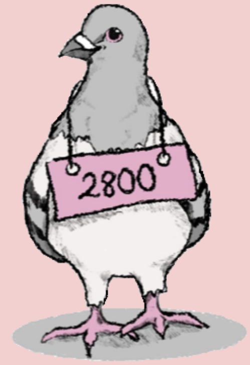
⋄ **16.17** Here, we consider how the pigeonhole principle can be applied to geometric problems. To do this, we'll consider the problem of placing points in a 1×1 square.

(a) Argue that if 5 points are placed in this square, then there are two points at most $\frac{1}{\sqrt{2}} \approx 0.71$ units apart.

(Hint: think about dividing the square into smaller pieces.)

(b) Argue that if 10 points are placed in this square then there are two points at most $\frac{\sqrt{2}}{3} \approx 0.47$ units apart.

⋆ (c) Use the generalized pigeonhole principle to argue that if 9 points are placed in this square then there are two points at most $\frac{1}{2\cos(\frac{\pi}{12})} = \frac{\sqrt{6}-\sqrt{2}}{2} \approx 0.52$ units apart.



17. Cardinality

When we defined sets, we introduced the notation $|X|$ to refer to the number of elements in a *finite* set X. This provides us with a way to compare the sizes of finite sets, we simply look at how many elements they contain. Can we do something similar for infinite sets? Are there infinite sets that are larger than other infinite sets? How do we even think about the size of an infinite set? We will answer these questions over the next two lectures. As descriptors of maps *between* sets, functions will serve as a crucial tool in our study of set *cardinality*.

We will describe the cardinality of infinite sets in two steps. First, we will define two relations, an *equivalence relation* and a *partial order*, that will tell us when one set is larger than or equal in cardinality to (respectively) another set. Then, we'll assign labels, called *cardinal numbers*, to the equivalence classes that describe the cardinality of these sets.

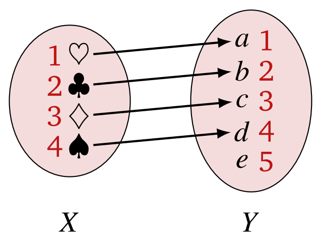
We can visualize these two steps as placing all sets into a hierarchy. Each level in the hierarchy contains every set with a particular cardinality and is labeled with a cardinal number. Level 0 contains the only set with 0 elements, the empty set. Level 1 contains all 1-element sets. The top portion of the hierarchy will include the infinite sets. Our cardinality relations will tell us whether a set sits at the same level or above another set in this hierarchy.

$$
\begin{array}{rc}
 & \vdots \\
2: & \boxed{\text{2-element sets}} \\
 & \rotatebox{90}{$\leq$} \\
1: & \boxed{\text{1-element sets}} \\
 & \rotatebox{90}{$\leq$} \\
0: & \boxed{\varnothing}
\end{array}
$$

17.1 The Cardinality Relations

To develop a cardinality relation, we must find a way to determine if "$|X| \leq |Y|$" (read "X is no larger than Y") for any (possibly infinite) sets X and Y. When X and Y are finite, there is an easy way to do to this. We can count the elements in each set and compare these counts. This doesn't generalize well to infinite sets, since it's unclear how to count infinitely many elements. We'll have more success if we look at this counting from a different perspective.

Consider the sets $X = \{\spadesuit, \clubsuit, \heartsuit, \diamondsuit\}$ and $Y = \{a,b,c,d,e\}$. We can count that X has 4 elements and Y has 5 elements, so Y is larger. When we count, we assign numbers $1,2,\ldots$ to each element in these sets. We compare the counts by pairing up the elements with the same numbers. We can pair every element of X with an element of Y and still have an element of Y left unpaired. This tells us that X must be smaller. This pairing is really describing a function $X \to Y$ that maps the X elements to the Y elements with the same number.



These arrows in the above picture visualize an *injective* function $X \to Y$. We can use this observation to propose the following lemma.

Lemma 17.1 Given finite sets X and Y, $|X| \leq |Y|$ if and only if there is an injection $f\colon X \to Y$.

Proof. For the forward direction of this implication, we will assume that $|X| \leq |Y|$, and we must show that there is an injection $f\colon X \to Y$. We can do this by constructing such an injection. To do this, we will formalize the "numbering and pairing" strategy described above.

When we number the elements in X, we are describing a bijection $e_X\colon X \to \big[|X|\big]$ (which we call an *enumeration* of set X). Similarly, counting the elements of Y in some order gives us a bijection $e_Y\colon Y \to \big[|Y|\big]$. We'll use these enumerations to define our injection $f\colon X \to Y$. Given any $x \in X$, we define $f(x) = e_Y^{-1}(e_X(x))$. Our assumption $|X| \leq |Y|$ ensures that $\big[|X|\big] \subseteq \big[|Y|\big]$. Thus, $e_X(x)$ lies in the codomain of e_Y, ensuring that f is well-defined. f is injective because given any $x_1,x_2 \in X$, if $f(x_1) = f(x_2)$, we have $e_Y^{-1}(e_X(x_1)) = e_Y^{-1}(e_X(x_2))$. e_Y^{-1} is invertible (its inverse is e_Y) so must be injective. Thus, $e_X(x_1) = e_X(x_2)$. Similarly, e_X is injective, so $x_1 = x_2$.

We have already seen the reverse implication, albeit in a different form. If we take its contrapositive, we get, "If X and Y are finite sets with $|X| > |Y|$, then there is no injection $f\colon X \to Y$." This is the pigeonhole principle that we proved last lecture! ■

Lemma 17.1 asserts a fact about finite sets; notice that we have only defined the notation $|X|$ for finite sets X. Nevertheless, we will use the result of this lemma as the basis of our definition of the cardinality relation[1].

[1] We must specify on which set this relation is defined. We'd like to use the "set of all sets", but this cannot exist. We will sweep aside this issue; there will always be a suitably large set containing all the sets we'll care about.

Definition 17.1 — Cardinality Relation.

Given two (possibly infinite) sets X and Y, $|X| \leq |Y|$ (read "X is no larger than Y") if and only if there is an injection $f: X \to Y$.

If $|X| \leq |Y|$ and $|Y| \leq |X|$, we write $|X| = |Y|$ (read "X and Y have the same cardinality").

17.1.1 Alternate Definitions

Our definition of the cardinality relation was based on injective functions; however, this was not the only possibility. We could have instead used surjective functions. Consulting our diagram from our motivating example, we see that since X is smaller than Y, we are unable to find a surjection $f: X \to Y$; however, it is not hard to construct a surjection from a larger set (Y) to a smaller set (X). By this line of reasoning, we can similarly define $|X| \leq |Y|$ if and only if there is a surjection $Y \to X$ (notice the switching of the order).

These definitions are, in fact, equivalent. This is a consequence of a property of functions that was the subject of Lecture Exercise 15.5:

> Let X and Y be non-empty sets. Then, there exists an injective function $f: X \to Y$ if and only if there exists a surjective function $g: Y \to X$.

Using these equivalent definitions, we see that there are four different ways to argue that $|X| = |Y|$; two choices for each direction of the inequality:

1. Find an injection $f: X \to Y$ and an injection $g: Y \to X$.
2. Find a surjection $f: X \to Y$ and a surjection $g: Y \to X$.
3. Find an injection $f_1: X \to Y$ and a surjection $f_2: X \to Y$.
4. Find an injection $g_1: Y \to X$ and a surjection $g_2: Y \to X$.

Of course, if we could find a bijection $f: X \to Y$, this could play the role of both f_1 and f_2 in option 3 (and similar for a bijection $g: Y \to X$ in option 4). However, is the converse of this also true? That is, between any two sets with equal cardinality, must there be a bijection? An affirmative answer is provided by the Schröder-Bernstein Theorem.

Theorem 17.1 — Schröder-Bernstein Theorem. Given sets X and Y, if there is an injection $f: X \to Y$ and a surjection $g: X \to Y$, then there is also a bijection $h: X \to Y$.

We will relegate the proof of this theorem to the lecture postscript.

Lemma 17.2 The relation $|X| = |Y|$ on a set of sets $\mathcal{S}$ given by $|X| = |Y| \Leftrightarrow$ there is a bijection $f: X \to Y$ is an equivalence relation.

Lemma 17.3 Let E denote the equivalence relation from Lemma 17.2. Then, the relation L on $\mathcal{S}/E$ given by $|[X]_E| \leq |[Y]_E| \Leftrightarrow$ there is an injective function $f: X \to Y$ is a partial order.

Together, these lemmas, whose proofs are left as exercises, formalize our "hierarchy of cardinalities" illustration from earlier. The equivalence classes are the levels of this hierarchy, and the L relation from Lemma 17.2, which in addition to being a partial order is actually a *total* order, stacks these equivalence classes from the "smallest cardinality" sets to "larger cardinality" sets. For the rest of today's lecture, and in the next lecture, we will use these cardinality relations to compare and classify the cardinalities of some infinite sets.

17.2 Countable Sets

As a first example of an infinite set, we will consider the natural numbers, $\mathbb{N}$. What is $|\mathbb{N}|$? It cannot be finite: Suppose, for sake of contradiction, that $|\mathbb{N}| = n$ for some $n \in \mathbb{N}$. This means that there is an injection $f : \mathbb{N} \to [n]$. Then, the *restriction* of f onto $[n+1]$ (that is, the function that agrees with f on all mappings but has domain $[n+1]$ instead of $\mathbb{N}$) must also be injective, violating the pigeonhole principle. Thus, we must introduce a new cardinal number to describe $|\mathbb{N}|$. As a convention, we let $|\mathbb{N}| = \aleph_0$ ("aleph null"), using the Hebrew letter "aleph". We declare sets with cardinality $\aleph_0$ to be *countably infinite*.

Definition 17.2 — Countably Infinite, Countable.

A set S is *countably infinite* if $|S| = |\mathbb{N}| = \aleph_0$.

A set S is *countable* if it is finite or countably infinite. In other words, S is countable if it has the same cardinality as a subset of $\mathbb{N}$

We can think of countable infinity as the "smallest" infinity. In other words, if $|S| < \aleph_0$ (meaning $|S| \le |\mathbb{N}|$ but $|\mathbb{N}| \not\le |S|$), then $|S| = n \in \mathbb{N}$, so S is a finite set.

For the rest of this and the next lecture, we will go about comparing the cardinalities of different infinite sets. The following fact will be useful.

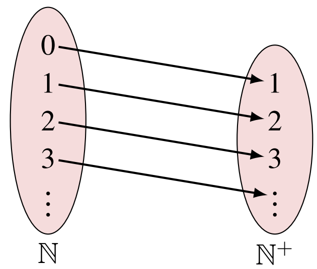
Lemma 17.4 Suppose that $X \subseteq Y$. Then, $|X| \le |Y|$.

Proof. Applying the definition, we must find an injection $f\colon X \to Y$. However, $X \subseteq Y$, so every element in X is also in Y. Therefore, the obvious choice is to have f map each element to itself. We call this the *inclusion* map; although it behaves just like the identity, it asserts the inclusion of set X in set Y. ■

When we consider infinite sets, much of our intuition about cardinality is no longer true.

Fact 1: We can add/remove finitely many elements from an infinite set without affecting its cardinality.

Example 17.1 We consider the sets $\mathbb{N} = \{0, 1, \ldots\}$ and $\mathbb{N}^+ = \{1, 2, \ldots\}$. Here, $\mathbb{N}$ includes the element 0, but $\mathbb{N}^+$ does not, so one might assume that $|\mathbb{N}|$ should be bigger. Nevertheless, $|\mathbb{N}| = |\mathbb{N}^+| = \aleph_0$. To prove this, we argue the two inequalities separately. $|\mathbb{N}^+| \le |\mathbb{N}|$ by Lemma 17.4. To argue that $|\mathbb{N}| \le |\mathbb{N}^+|$ we must exhibit an injection $f : \mathbb{N} \to \mathbb{N}^+$. One such injection is $f(n) = n+1$: note that $f(n_1) = f(n_2) \implies n_1 + 1 = n_2 + 1 \implies n_1 = n_2$.



As we construct this function from "top to bottom" we use the fact that the codomain is infinite to ensure that we will always be able to select another element of $\mathbb{N}$ to map to that hasn't yet been used. This ensures injectivity.

A more extreme version of the first fact is also true.

Fact 2: We can add a countable number of elements to an infinite set without affecting its cardinality.

Example 17.2 Let E be the set of even natural numbers, $E = \{0, 2, 4, \ldots\}$. We'll argue that $|E| = |\mathbb{N}|$. Again, $|E| \leq |\mathbb{N}|$ by Lemma 17.4. To argue that $|\mathbb{N}| \leq |E|$ consider the map $f : \mathbb{N} \to E$ given by $f(n) = 2n$. Note that $f(n_1) = f(n_2) \implies 2n_1 = 2n_2 \implies n_1 = n_2$, so f is an injection. By definition, $|\mathbb{N}| \leq |E|$, so $|E| = |\mathbb{N}|$.

Main Takeaways:

- To reason about the size of infinite sets, we will need to expand the notion of cardinality beyond the number of elements.
- We say that set X has cardinality no larger than set Y, $|X| \leq |Y|$, when there is an injection $f : X \to Y$.
- Properties of functions and the Schröder-Bernstein Theorem provide additional characterizations of cardinality.
- A *countably infinite* set has the same cardinality as the natural numbers.

17.3 Exercises

17.1 Prove Lemma 17.2.

17.2 Prove Lemma 17.3.

17.3 Argue that $|\mathbb{Z}| = |\mathbb{N}|$. In your proof, construct an explicit bijection $\mathbb{Z} \to \mathbb{N}$.

17.4 Suppose that X and Y are both countable sets. Argue that $X \cup Y$ is countable. Rather than appealing to Fact 2, give a proof where you construct any necessary injections/bijections.

⋆ **17.5** Formalize Facts 1 and 2 in the special case where "infinite" is replaced by "countably infinite". For each, you should have a formal statement (which defines the cast of mathematical objects, states your assumptions, and conclusion) along with a proof.

17.6 Give examples of countably infinite sets A and B (different in each part) for which:

(a) $A \setminus B$ and $B \setminus A$ are both finite.

(b) $A \setminus B$ is finite and $B \setminus A$ is infinite.

(c) $A \setminus B$ and $B \setminus A$ are both infinite.

17.7 Find (with proof) a bijection that shows that each of the following pairs of sets have the same cardinality.

(a) $S_1 = [100]$, $\quad S_2 = \{n \in \mathbb{N} \mid 200 < n <= 400, n \text{ even } \}$.

(b) $S_1 = [15]$, $\quad S_2 = [3] \times [5]$

(c) $S_1 = \mathbb{N}$, $\quad S_2 = \{\text{odd natural numbers}\}$

(d) $S_1 = \{\text{integer multiples of } 4\}$, $\quad S_2 = \{\text{integer multiples of } 3\}$

17.8 Consider sets A_1, A_2, and B.

(a) Does $|A_1| = |A_2|$ imply that $|A_1 \times B| = |A_2 \times B|$? Provide a proof or a counterexample.

(b) Does $|A_1 \times B| = |A_2 \times B|$ imply that $|A_1| = |A_2|$? Provide a proof or a counterexample.

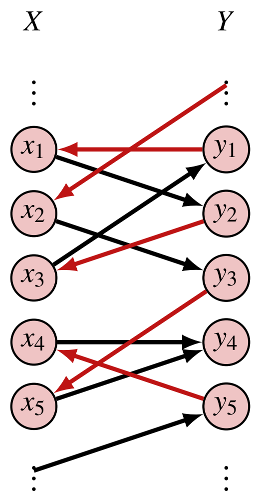
17.4 Postscript: Proving the Schröder-Bernstein Theorem

Let us recall the statement of the Schröder-Bernstein Theorem. We are given two sets X and Y and told that there is an injective function $f : X \to Y$ and a surjective function $g : X \to Y$.

We can use these functions to define a directed graph (potentially with infinitely many vertices) as follows:

- The vertices of the directed graph will be the set $X \sqcup Y$. Here, this square union symbol represents the "disjoint union", which means that the set will contain all elements of X and Y, including two *distinguishable* copies of each element $s \in X \cap Y$ (the "X" copy and the "Y" copy). To visualize the graph, we will think about drawing the vertices from X in a column on the left and the vertices from Y in a column on the right.
- For each $x \in X$, we will draw a "forward" black arc $(x, g(x))$.
- For each $x \in X$, we will draw a "backward" red arc $(f(x), x)$.

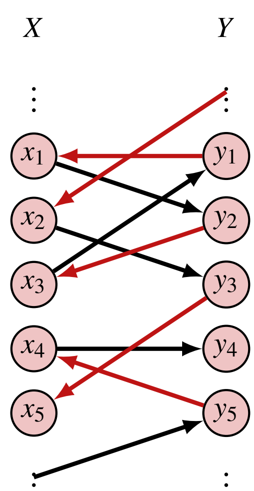
A "slice" of one possible graph is shown on the following page.



We can make the following observations about this graph:

- Since the red arcs (reversed) correspond to the mappings of an injective function, there is exactly one incoming red arc to each vertex in X and at most one outgoing red arc from each vertex in Y.
- Since the black arcs correspond to the mappings of a surjective function, there is exactly one outgoing black arc from each vertex in X and at least one incoming black arc to each vertex in Y.

If a vertex $y \in Y$ has multiple incoming black arcs (as y_4 does in the above picture, we will (arbitrarily) select one to retain in the graph and delete all the others. For example, in the above graph, we can select the edge (x_4, y_4) and remove the edge (x_5, y_4), leaving us



In this modified graph, every vertex has exactly one incoming arc and at most one outgoing arc, so the graph decomposes into a vertex-disjoint union of paths and cycles. There are four possible forms that these paths and cycles can take.

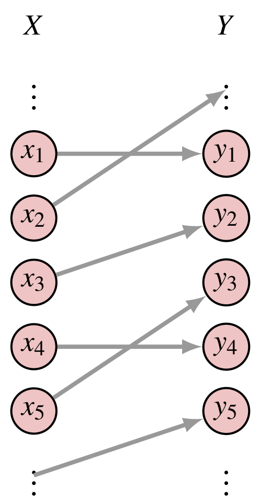
1. A (finite length) cycle, such as $(x_1, y_2, x_3, y_1, x_1)$ in the above picture.
2. A path with a fixed ending vertex in X that extends infinitely backward. The path ending

$(\ldots, x_2, y_3, x_5)$ in the above picture exemplifies such a path. Such a path arises from the "arc deletion" step described above.

3. A path with a fixed ending vertex in Y that extends infinitely backward. The path ending $(\ldots, y_5, x_4, y_4)$ in the above picture exemplifies such a path. Such a path arises for each codomain element $y \in Y$ outside of the range of f.

4. A path that extends infinitely in both directions.

Notice that the paths and cycles alternate between red and black arcs (since black arcs always start in X and end in Y and red arcs start in Y and end in X). We can use this observation to define a bijection $h \colon X \to Y$ as follows. For any $x \in X$ on a path/cycle in cases 1,2, or 4. We define $h(x)$ as its in-neighbor from its incident red arc. For any $x \in X$ on a path in case 3, we define $h(x)$ as its out-neighbor from its incident black arc. The "slice of " h in our example graph is shown below.



This associates each $x \in X$ with exactly one $y \in Y$, so gives a well-defined function $X \to Y$. This function h is a bijection since each $y \in Y$ belongs to one such path/cycle and will be the in-neighbor along a red arc (in cases 1,2,4) or out-neighbor along a black arc (in case 3) to one $x \in X$. ■

18. Countable and Uncountable Sets

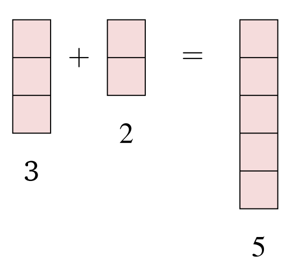
In the last lecture, we saw how we could use functions to compare the cardinality of sets. We also defined the collection of *countably infinite* sets (those with the same cardinality as $\mathbb{N}$) and their corresponding *cardinal number* $\aleph_0$. In today's lecture, we'll continue our study of cardinality. We'll see another, perhaps more surprising, example of a countable set with a clever proof. We'll also give an example of proving that a set is *uncountable*. Both of the strategies that we will use will involve a literal tilting of our perspective, as they utilize the diagonals of tabular representations of sets.

18.1 Cardinal Numbers

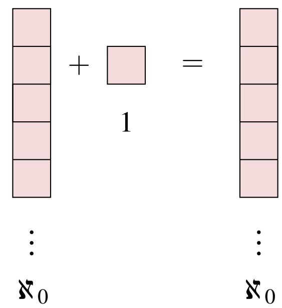
Recall that the cardinal numbers are the labels that we give to the possible cardinalities of sets. Each natural number $n \in \mathbb{N}$ is a cardinal number since it describes the cardinality of the set $[n]$. In addition, the infinite cardinal number $\aleph_0 = |\mathbb{N}|$. At the end of the lecture, we will introduce another infinite cardinal number. Just as with regular (natural, rational, real) numbers, we can define arithmetic operations on the cardinal numbers. These will allow us to reason about the size of the set that results from some set operations.

18.1.1 Addition

Consider the arithmetic equation "3 + 2 = 5." If we interpret these numbers as cardinal numbers (representing the size of sets) we can read this equation as saying that when we add together (i.e. take the union of) a set with 3 elements with a (disjoint) set with 2 elements, we obtain a set with 5 elements. We visualize,



We can think about extending this addition operation to include $\aleph_0$. We can visualize $\aleph_0$ as counting a column with infinitely many boxes (one for each natural number). Then, when we add finitely many boxes to this infinite column, we still have an infinite column.

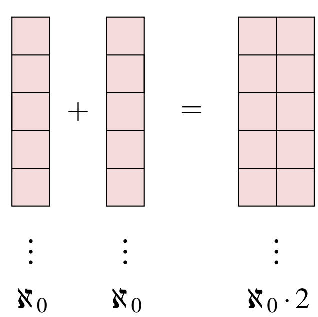


This same idea works for any (finite) number of boxes in the second column, so we can conclude that

$$\aleph_0 + n = \aleph_0 \qquad \text{for all } n \in \mathbb{N}.$$

Notice that these box diagrams are really just a visualization of our function reasoning from before. When we line up all of the elements into a single column, we are implicitly giving a way to enumerate them (from top down). We'll use this style of reasoning throughout today's lecture because it will save us from needing to write down complicated descriptions of functions but still provide a clear picture of how to formalize the argument.

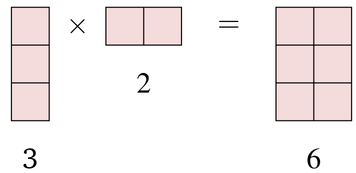
We still need to define the addition operation for $\aleph_0 + \aleph_0$. From the last lecture, we can guess that the answer should be $\aleph_0$, but how do we visualize this? We cannot simply stack two infinitely long columns on top of each other. Intuitively, since the columns are infinitely long, we cannot lift one high enough to place it on the other. Instead, we can think about gluing together these columns into a 2-column wide, infinitely long array.



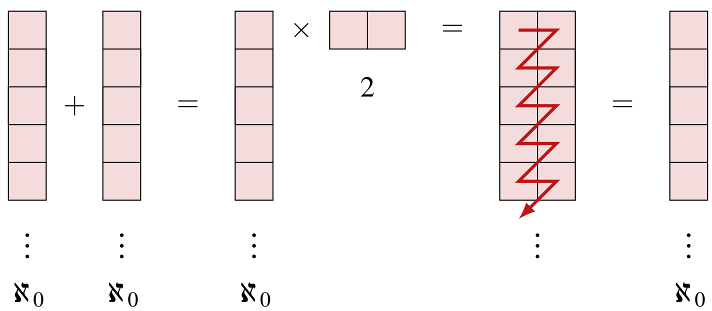
This lets us recast this problem in terms of multiplication of $\aleph_0$ by a natural number.

18.1.2 Multiplication

How can we think about the equation "$3 \cdot 2 = 6$" in terms of set sizes? One natural way is using the Cartesian product. When we take the Cartesian product of a set with 3 elements and a set with 2 elements, we obtain a set with 6 pairs. We visualize,



When the first column is infinite, we will obtain an infinitely long array with multiple columns. To show that this array has size $\aleph_0$, we need a way to iterate over its cells so that we visit each cell exactly once. We cannot do this by iterating over the first column and then iterating over the second column. The first column is infinitely long, so we will never reach the bottom and thus never jump over and visit the cells in the second column. If we instead visit all cells in each row (left to right) before moving to the next row, we can guarantee that we will eventually visit each cell (exactly once). Thus,

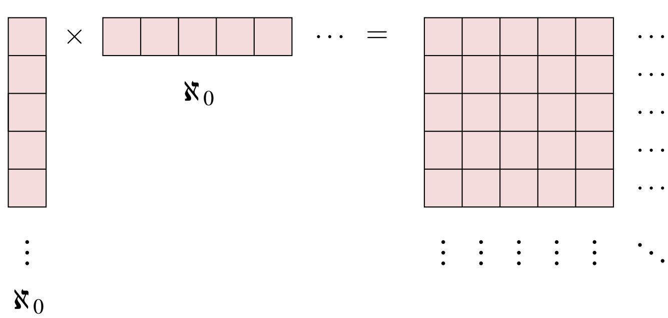


We can record this observation as the fact

$$\aleph_0 \cdot n = \aleph_0 \qquad \text{for all } n \in \mathbb{N}^+.$$

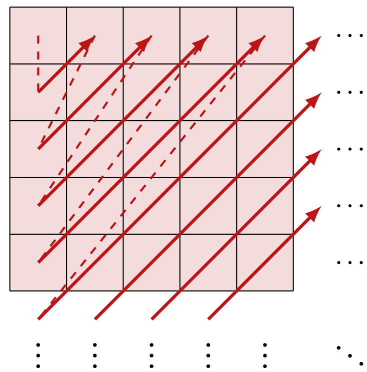
18.2 Cartesian Products of $\mathbb{N}$

Next, we would like to calculate $\aleph_0 \cdot \aleph_0$. Using our intuition that the multiplication of cardinal numbers corresponds to taking Cartesian products of sets, the answer should be $|\mathbb{N} \times \mathbb{N}|$. In our visualization, this will be the number of boxes in a grid that expands infinitely in both the horizontal and vertical directions.



How many cells are in this grid? Is this still countable? To answer this question in the affirmative, we would need to find some way to enumerate all of these cells. We saw before that we cannot do this by working one column at a time; we would never leave the first column. Now, we can also no longer work one row at a time (that is, we cannot first visit the first cell in every column, then the second cell in every column, etc.); since there are infinitely many columns, we would never leave the first row. Instead, we must find a method that will allow us to make progress on both the rows and columns simultaneously.

Suppose that we split up the array into diagonal bands going from the bottom left to the top right. Since the grid has fixed left and upper edges, each of these bands contains finitely many cells. We can enumerate a band by starting at its cell in the leftmost column and walking along the band upward and to the right until we reach the cell in the top row. Then, we can proceed downward to the next band.



This procedure will eventually visit each square in the array (exactly once), so describes an enumeration of $\mathbb{N}\times\mathbb{N}$. We record this fact in the following lemma.

Lemma 18.1 $\mathbb{N}\times\mathbb{N}$ is countably infinite. That is, $|\mathbb{N}\times\mathbb{N}| = (\aleph_0)^2 = \aleph_0$.

In fact, we can generalize this further (see Exercise 18.5). A similar diagonal trick can be used to enumerate larger Cartesian products such as $\mathbb{N}\times\mathbb{N}\times\mathbb{N}$, showing that these are countably infinite as well. We express this fact symbolically as

$$(\aleph_0)^n = \aleph_0 \qquad \text{for all } n \in \mathbb{N}^+.$$

At this point, every operation that we have performed on a countably infinite set with the hope of making it bigger has still left us with a countably infinite set. This could lead us to believe that countable infinity is the only infinity; in other words, $\aleph_0$ is the largest cardinal number. Interestingly, this is not the case; there are many larger infinities than $\aleph_0$. Next, we will introduce another technique, called *diagonalization*, that will allow us to show this.

18.3 Diagonalization

As our next object of study, we will take $\mathcal{P}(\mathbb{N})$, the set of all subsets of natural numbers. This is a very large set! It contains:

- The singleton set $\{n\}$ for each $n\in\mathbb{N}$. The injection $\mathbb{N}\to\mathcal{P}(\mathbb{N})$ given by $n\mapsto\{n\}$ establishes that $|\mathbb{N}|\le|\mathcal{P}(\mathbb{N})|$.
- The sets of all even numbers, odd numbers, multiples of 3, etc.,
- The set of all prime numbers,

- The set of all square numbers,
- The set of all odd numbers less than 500 and all perfect cubes greater than 1000,

among many other sets. We will show that this set is so large that it cannot be enumerated. It is our first example of an *uncountable* set.

Theorem 18.1 $\mathcal{P}(\mathbb{N})$ is uncountable. That is, $|\mathcal{P}(\mathbb{N})| \not\leq |\mathbb{N}|$.

By the equivalent cardinality definition presented in the last lecture, it suffices to argue that there is no surjection $f\colon \mathbb{N} \to \mathcal{P}(\mathbb{N})$. As both $\mathbb{N}$ and $\mathcal{P}(\mathbb{N})$ are infinite sets, there are infinitely many candidate functions that we would need to check (i.e. verify are not surjections). Just as usual, since we cannot check each function directly, we will instead argue that an *arbitrary* such function is not surjective.

Proof. Consider an arbitrary function $f\colon \mathbb{N} \to \mathcal{P}(\mathbb{N})$. For each $n \in \mathbb{N}$, $f(n)$ is a subset of $\mathbb{N}$. We can visualize each of these subsets as a row in a table whose columns are labeled with $\mathbb{N}$ and whose cells in column j indicate whether $j \in f(n)$. For example, the set $S = \{0,2,3,6\}$ corresponds to the table row:

	0	1	2	3	4	5	6	7	8	...
S	Y	N	Y	Y	N	N	Y	N	N	...

Now, consider the complete table (extending infinitely in both directions), where the i'th row corresponds to set $f(i)$. Note that our example illustrates one possible choice of the initial sets, but the argument does not rely on these specific choices.

i	$f(i)$	0	1	2	3	4	5	6	...
0	$\{0,2,3,6\}$	Y	N	Y	Y	N	N	Y	...
1	{ odd numbers }	N	Y	N	Y	N	Y	N	...
2	$\varnothing$	N	N	N	N	N	N	N	...
3	{ prime numbers }	N	N	Y	Y	N	Y	N	...
4	$\{1,4,5,21,37\}$	N	Y	N	N	Y	Y	N	...
5	{ square numbers }	Y	Y	N	N	Y	N	N	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Using this table, we will construct a new *diabolical* subset $D \subseteq \mathbb{N}$ with $D = \{i \in \mathbb{N} \mid i \notin f(i)\}$. In other words, to determine whether an element i should be placed in D, we check whether it was included in $f(i)$. If $i \in f(i)$, we exclude it from D; otherwise, if $i \notin f(i)$, we include it in D. Now, observe that D is not the same set as any of the $f(i)$. In our running example,

	0	1	2	3	4	5	...
D	N	N	Y	N	N	Y	...

D was constructed in a way that its table row will differ from the $f(i)$'s table row in the "diagonal cell" in column i (highlighted pink in the figure). Thus, f is not surjective since D lies outside of its range. Since this is true for an arbitrary $f\colon \mathbb{N} \to \mathcal{P}(\mathbb{N})$, $|\mathcal{P}(\mathbb{N})| \not\leq |\mathbb{N}|$, so $\mathcal{P}(\mathbb{N})$ is uncountable. ■

This style of proof, first introduced by Georg Cantor is called *diagonalization* since the key step utilized the diagonal of the table to construct the diabolical set. If you take an algorithms or theory of computation course in the future, you will likely see diagonalization again, where it is used to prove that certain problems cannot be solved by a computer.

Since $\mathcal{P}(\mathbb{N})$ is uncountable, we need to introduce a new symbol. The conventional choice is $\mathfrak{c}$, representing the cardinality of the *continuum* ($\mathbb{R}$). This choice follows from the fact that $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$. This proof requires mathematical ideas beyond our scope. In the exercises (see Exercise 18.7), we outline a proof that the real numbers are uncountable. There is also an exercise (Exercise 18.11) that explores the connection between exponential expressions with cardinal numbers and the number of functions between two sets.

Main Takeaways:

- Many of the operations that change the size of a finite set do not have the same effect on infinite sets. These include adding finitely many elements, adding countably many elements, and taking Cartesian products.
- To prove that a set is countable, it suffices to describe an enumeration procedure that visits each element exactly once.
- When an enumeration procedure requires multiple steps, we need to ensure that early steps do not go on forever; otherwise, we will never get to later steps.
- Diagonalization is a proof technique that can be used to argue that a set is uncountable.

18.4 Exercises

18.1 In our argument that $\aleph_0 \cdot \aleph_0 = \aleph_0$, we used the fact that the grid had a fixed upper and left edges to conclude that each diagonal band included only finitely many elements. This structure is not present in $\mathbb{Z} \times \mathbb{Z}$; there is no upper left corner of this grid since it extends infinitely in all directions. Nevertheless, argue that $\mathbb{Z} \times \mathbb{Z}$ is countable by describing an enumeration (do not simply cite that $|\mathbb{Z}| = \aleph_0$).

18.2 Argue that the set of rational numbers $\mathbb{Q}$ is countable. (Hint: Describe a function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Q}$. Then, use the previous exercise.)

18.3 In this exercise, we consider explicit bijections $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$.

(a) First, suppose that we number the rows and columns of our infinite array with 0 at the top and left. Find the formula for the bijection $f\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ given by our diagonal bands technique; that is, cell (i, j), should be the $f(i, j)$'th cell that is visited on our zigzagging path.

Recall that we do not need a bijection to establish equal cardinality. We can also do this using two injections.

(b) Describe a relatively simple injection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$. This tells us that $|\mathbb{N}| \leq |\mathbb{N} \times \mathbb{N}|$.

⋄ (c) Describe an injection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$. This tells us that $|\mathbb{N} \times \mathbb{N}| \leq |\mathbb{N}|$. (Hint: one approach is to use the fact that each natural number has a unique prime factorization.)

18.4 Here, we consider the following alternate enumeration of $\mathbb{N} \times \mathbb{N}$. We proceed in phases 0, 1, 2, ... In phase 0, we visit cell $(0,0)$. Then, in each other phase i, we carry out the following two steps.

1. Visit cells $(0,i),(1,i),\ldots,(i-1,i)$
2. Visit cells $(i,i),(i,i-1),\ldots,(i,0)$

(a) Draw a diagram that shows the order which the cells are visited.

(b) The description ensures that no cell is visited more than one time in the same phase. Therefore, to conclude that this is a valid enumeration, it suffices to show that each cell is visited in exactly one phase. To do this, define a function $p\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that $p(i,j)$ is the phase in which cell (i,j) is visited.

⋆ **18.5** Here, we lay some intuition for how to argue that finite Cartesian products of countable sets are countable. Think about the set $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$, which we can visualize as a 3-dimensional arrangement of cells extending infinitely in three directions. Describe a way to enumerate the cells in this arrangement. Try to generalize either the diagonal bands technique or the procedure described in Exercise 18.4.

18.6 Here, we reason about a countable unions of sets.

(a) Argue that a countable union (that is, a union of countably many sets) of countable (disjoint) sets is countable. (Hint: One way to do this is by describing a surjection from $\mathbb{N} \times \mathbb{N}$ onto this union.)

(b) Use the previous part to conclude that $\mathcal{F}(\mathbb{N})$, the set of all *finite* sets of $\mathbb{N}$ is countable.

(c) Where does the diagonalization argument break down when we try to apply it to $\mathcal{F}(\mathbb{N})$?

18.7 One way to formalize the real numbers is as the set of all possible sequences of decimal digits $\{0,1,2,3,4,5,6,7,8,9\}$ with finitely many digits to the left of a "central" decimal point, and infinitely many digits to the right of this decimal point that do not end with an infinite sequence of 9s. Here, the last condition ensures each sequence leads to a unique real number by avoiding the "$1.\overline{0} = 0.\overline{9}$" problem. Using this definition of the real numbers, give a diagonalization proof that the interval $[0,1) = \{r \in \mathbb{R} \mid 0 \leq r < 1\}$ is uncountable. In your proof, be sure to point out where you are addressing the "infinite sequence of 9s" technicality.

⋆ **18.8** Describe an explicit bijection between the open interval $(0,1) = \{r \in \mathbb{R} \mid 0 < r < 1\}$ and the closed interval $[0,1] = \{r \in \mathbb{R} \mid 0 \leq r \leq 1\}$. (Hint: separately handle a countably infinite subset of these intervals, using our techniques to add finitely many elements without affecting the cardinality.)

18.9 Mimic the diagonalization proof to argue that for any set S, $|\mathcal{P}(S)| \not\leq |S|$. By repeatedly taking the power set of $\mathbb{N}$, we can conclude that there are at least countably infinitely many infinities. In fact, there are way more!

18.10 Let B^∞ be the set of all infinite binary sequences. For example, the repeating sequences $(0,0,0,\ldots)$, $(1,0,1,0,1,\ldots)$ are both in B^∞. Define the relation $\approx_f$ on B^∞ such that two sequences $s_1 \approx_f s_2$ if and only if s_1 and s_2 disagree in only finitely many positions.

(a) Argue that $\approx_f$ is an equivalence relation.

⋆ (b) Argue that each equivalence class of B^∞ under $\approx_f$ is countably infinite.

⋆ (c) Argue that $B^\infty/\approx_f$ is uncountable. (Hint: Use a proof by contradiction and Exercise 18.6.)

⋄ (d) Use the result of this exercise to develop a solution to the following puzzle:

> There are a countable number of people standing in a line, numbered $0,1,\ldots$. Each of them is wearing either a red or a white hat. They are facing in such a way that person i can see the hats of every person $j > i$, but cannot see their own hat or the hats of anyone before them. Each person (in order $0,1,\ldots$) is asked to identify the color of their hat. If they each guess, we'd expect half of them (so infinitely many people) to be wrong. How can they devise a strategy that ensures that only finitely many of them are wrong?

⋄ (e) Can you improve your strategy from part (d) so that at most one person (the first person) makes a mistake? You can use the fact that each person hears each of the other people's guesses and whether they were right or wrong. (Hint: The solution we have in mind uses modular arithmetic.)

⋆ **18.11** In the lecture, we drew connections between arithmetic operations on cardinal numbers and counting sets after performing a set operation. We saw that addition of cardinal numbers corresponds to unions of disjoint sets and that multiplication of cardinal numbers corresponds to Cartesian products of sets. Another natural operation to consider is exponentiation, which we do in this exercise.

(a) Show that for finite sets X,Y with $|X| = n$ and $|Y| = m$, the number of functions $X \to Y = m^n$. It may help to write out a few examples.

We can generalize this to infinite cardinal numbers as well, letting b^a count the number of functions $A \to B$ where $|A| = a$ and $|B| = b$. We use the notation B^A as a shorthand for the set of all functions $A \to B$. In the case that $A = [n]$, we will often drop the brackets and write B^n (and similarly write n^A if $B = [n]$).

(b) Note that our previous notation $\mathbb{N}^2 = \mathbb{N} \times \mathbb{N}$ is sensible by describing a bijection between $\mathbb{N} \times \mathbb{N}$ (pairs of natural numbers) and $\mathbb{N}^2$ (functions $[2] \to \mathbb{N}$).

(c) Describe a bijection between $\mathcal{P}(\mathbb{N})$ and $2^{\mathbb{N}}$ (all functions $\mathbb{N} \to [2]$). Explain how this shows that $2^{\aleph_0} = \mathfrak{c}$.

(d) Argue that $4^{\aleph_0} = \mathfrak{c}$ by establishing a bijection between the functions $\mathbb{N} \to [4]$ and the functions $\mathbb{N} \to [2]$. (Hint: Given a function $f\colon \mathbb{N} \to [4]$, express each of its outputs as a binary number. Collect this information in a table, and use this to describe a function $g\colon \mathbb{N} \to [2]$.)

(e) Generalize your argument from part (d) to show that $n^{\aleph_0} = \mathfrak{c}$ for all $n \in \mathbb{N} \setminus \{0,1\}$. (Hint: Separately identify an injection $2^{\mathbb{N}} \to n^{\mathbb{N}}$ and an injection $n^{\mathbb{N}} \to 2^{\mathbb{N}}$.

Finally, we'll consider $(\aleph_0)^{\aleph_0}$.

(f) Describe an injection $2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$. This tells us that $\mathfrak{c} = 2^{\aleph_0} = |2^{\mathbb{N}}| \leq |\mathbb{N}^{\mathbb{N}}| = (\aleph_0)^{\aleph_0}$.

(g) Describe an injection $\mathbb{N}^{\mathbb{N}} \to 2^{\mathbb{N}}$. This tells us that $(\aleph_0)^{\aleph_0} \leq \mathfrak{c}$. (Hint: This argument should use ideas from part (e) along with ideas from the diagonal bands argument.)

Combining with the previous part, we conclude that $(\aleph_0)^{\aleph_0} = \mathfrak{c}$.

(h) Many of the algebraic properties that we are used to (distributivity, laws of exponents, etc.) also hold for cardinal numbers. Using this, write our a simpler, algebraic proof that $(\aleph_0)^{\aleph_0} = \mathfrak{c}$.

(i) As an extra challenge, try to argue that $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$. This will require ideas from Exercise 18.7, in addition to the earlier parts of this exercise.

18.12 Here, you'll argue that $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$. Here, our diagonal bands argument won't work because the rows and columns of the "table" cannot be enumerated.

(a) Find one bijection $X \times Y \to Z$, where $|X| = |Y| = |Z| = \mathfrak{c}$

(b) Why is this single bijection from part (a) sufficient to establish this result?

(c) Give a second algebraic proof that $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$.

V

Combinatorics

19. Counting Rules

The study of *combinatorics* centers on counting sets of discrete mathematical objects with various properties. Combinatorics serves as a bridge between many different areas of math; historically, it has been an under-appreciated area that has come into prominence as it fills in the cracks that exist between the older disciplines like algebra, geometry, and analysis. More relevant to computer science, techniques from combinatorics show up regularly in runtime analysis, complexity theory, and randomized algorithms, which require us to count the number of possible states to bound how much time, space, or randomness our computation will need. For our purposes, we will use combinatorics to revisit many of the objects (sets, functions, relations, etc.) that we studied earlier in the course.

Of course, when the set that we are concerned with is finite, one way to determine its cardinality is to list its items. However, doing this is often tedious and unnecessary: one does not need to write out every possible license plate number to count the possibilities. We will see that often the beauty of the examples comes not from the answer, but from the counting process itself. In this first lecture, we'll introduce some of the fundamental rules for counting. While they likely will feel very intuitive, combining these rules together gives us the power to enumerate large and complicated sets of objects.

19.1 The Multiplication Rule

Most of the time in combinatorics, rather than thinking of the objects themselves, we will think about a process that we can use to generate the objects. Said differently, we will think about performing a series of steps, each of which will fix some attribute of the object that we are constructing. Then, to determine the size of the set, we reason about how many "choice paths" we could have taken.

Example 19.1 A New York license plate includes three letters followed by four numbers.

H	X	M	7	1	2	8
A-Z	A-Z	A-Z	0-9	0-9	0-9	0-9

We can break down the process of constructing a license plate into 7 decisions; each decision fixes the next symbol on the plate. There are 26 choices for the first letter, 26 choices for the second letter, and 26 choices for the third letter. Similarly, there are 10 choices for each of the 4 numbers. To get the total number of possible license plates, we *multiply* the number of choices for each decision.

$$\begin{array}{ccccccccccccc} \boxed{\text{A-Z}} & \rightarrow & \boxed{\text{A-Z}} & \rightarrow & \boxed{\text{A-Z}} & \rightarrow & \boxed{\text{0-9}} & \rightarrow & \boxed{\text{0-9}} & \rightarrow & \boxed{\text{0-9}} & \rightarrow & \boxed{\text{0-9}} \\ 26 & \cdot & 26 & \cdot & 26 & \cdot & 10 & \cdot & 10 & \cdot & 10 & \cdot & 10 \end{array}$$

Performing this multiplication, we find that there are 175,760,000 possible license plates.

This example shows that when we make decisions *sequentially*, we multiply the number of choices we had at each decision. This principle is known as the multiplication rule.

Multiplication Rule:

Suppose that we can enumerate a set S using an n-step decision process where:

- There are c_1 choices for the first decision.
- *Regardless of our first choice*, there are c_2 choices for the second decision.
- *Regardless of all previous choices*, there are c_i choices for the i'th decision for all $i \leq n$.

If each possible sequence of choices results in a different element of S, then,

$$|S| = c_1 \cdot c_2 \cdot \ldots \cdot c_n = \prod_{i=1}^{n} c_i.$$

Example 19.2

- If $|A| = n$ and $|B| = m$, then $|A \times B| = n \cdot m$. We can generate an element of $A \times B$ by choosing its first entry $a \in A$ ($c_1 = n$ choices) and then choosing its second entry $b \in B$ ($c_2 = m$ choices).
- If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. We can generate a subset $S \in \mathcal{P}(A)$ by separately deciding whether or not ($c_i = 2$ choices) to include each element of A in S.

Although this was true of our first examples, it is not necessary that the choices in the second (and all subsequent) steps be the same regardless of the choice in the first step. To use the multiplication rule, we need only that there is the same *number* of choices. This phenomenon is illustrated by our next example, which concerns a new object called a *permutation*.

Definition 19.1

Given a finite set S, a *permutation* of S lists all of its elements in a particular order.

Alternatively, if $|S| = n$, one can view a permutation of S as a bijection $\sigma : S \to [n]$ where $\sigma(s)$ is the position of s in the ordering.

Example 19.3 How many permutations are there of the set $S = \{a,b,c,d\}$? Two possible permutations are $[a,b,c,d]$ and $[b,d,c,a]$, but there are many others. We can count them using the multiplication rule, where in each decision, we choose the element that is listed next.

- We can place any of the four letters in the first position of the permutation ($c_1 = 4$).
- Next, we consider the second position. There are four letters, but we cannot repeat the letter written in the first position (a permutation lists each element of the set exactly once). Hence, there are $c_2 = 3$ choices for the second letter.
- Similarly, there are $c_3 = 2$ choices for the third letter, the two letters that did not appear in the first two positions.
- After we have decided the first three letters, there is only $c_4 = 1$ letter left to place in the last position, the one that we have not used.

In all, we find that there are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ permutations of set S.

Since permutations are pervasive in combinatorics, we have a special notation to count them.

Definition 19.2 — Factorial.
We use the notation $n!$ (read "*n factorial*") for the number of permutations of an n-element set.

By extending the reasoning from Example 19.3, we can obtain the closed formula

$$n! = \prod_{i=1}^{n} i.$$

We can also define the factorial function by the recurrence:

$$0! = 1, \qquad n! = n \cdot (n-1)! \quad \text{for all } n \geq 1$$

We can reason about this formula as follows. In the base case, there is one way to arrange zero elements (the empty list). In the recursive case, we can arrange n elements by selecting the first element ($c_1 = n$ choices) and then choosing an ordering of the remaining $n-1$ elements (inductively, $c_2 = (n-1)!$ choices).

19.2 Addition/Subtraction Rules

A key requirement for using the multiplication rule was that earlier choices did not affect the *number* of available later choices. However, this is often not the case. Sometimes, a decision that we make early in the enumeration process can affect the number of choices that are available later on. In these cases, we can use the addition rule to split our counting problem into several smaller problems, whose results are aggregated in the end.

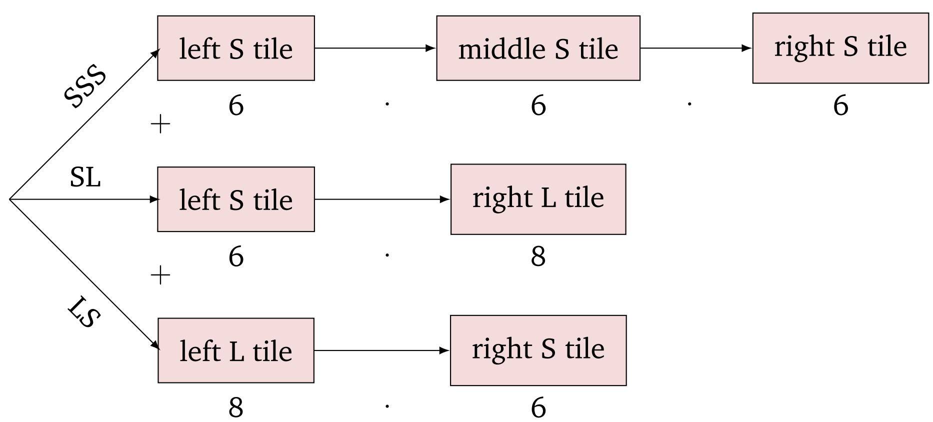
Addition Rule:
Suppose that we can partition a set S into subsets $S_1, \ldots S_n$. Then,

$$|S| = \sum_{i=1}^{n} |S_i|.$$

Here, the keyword in this rule is *partition*: we must be guaranteed that each element of S belongs to *exactly* one subset S_i so that we do not miss or double-count any element. In later lectures, we will use a more fine-grained approach to count elements from partially overlapping sets known as the *Inclusion-Exclusion Rule*.

Example 19.4 An artist wishes to decorate the top 1×3 inch section of their desk nameplate with mosaic tiles. They have 6 colors of small (1×1 inch) tiles and 8 different colors of large (1×2 inch) tiles. How many possible ways are there to decorate their nameplate?

We can partition the set of possible arrangements based on which size of tiles is used. There are three possibilities: either the artist uses 3 small tiles, a small tile followed by a large tile, or a large tile followed by a small tile. Each of these cases is disjoint, and they partition all of the possible arrangements. For each case, we can compute the number of arrangements using the multiplication rule. We have,



Applying the addition rule, there are $216 + 48 + 48 = 312$ possibilities.

This example already illustrates that we often will need to use multiple rules in conjunction to count a particular set. The next example provides another example of this, where we must carefully account for the different cases.

Example 19.5 In how many permutations of $[5] = \{1, 2, 3, 4, 5\}$ is the number 2 not next to 3?

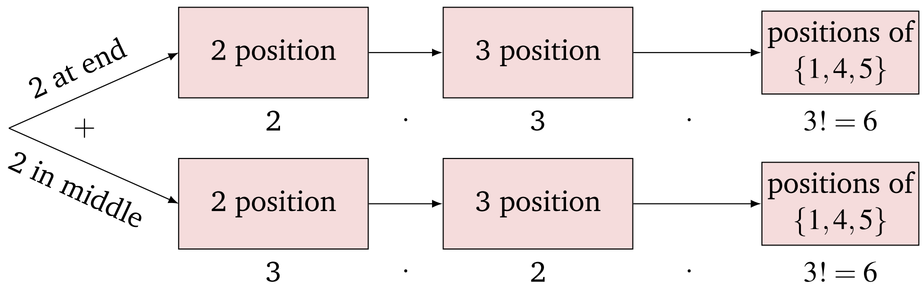
To construct these permutations, we'll choose the position of each number, starting with 2, then 3, and then the rest. There are five possible positions of 2, which we break into two cases:

Case 1: 2 is at an end of the permutation.

There are two possible locations for the 2 (left and right ends), then three positions for the 3 (anywhere but where the 2 is and right next to it), then six possible permutations of the remaining three digits in the remaining three positions.

Case 2: 2 is in the middle of the permutation.

There are three possible locations for the 2, then two positions for the 3 (not where the 2 is or to either side of it), then six possible permutations of the remaining three digits in the remaining three positions.



In total, there are 72 arrangements.

In fact, there was an easier way to count the arrangements in this example using the so-called subtraction rule.

> **Subtraction Rule**:
>
> Let S be a set with $A \subseteq S$. Then,
> $$|A| = |S| - |S \setminus A|.$$

From this rule, we see that another way to count the arrangements from the previous example is to first count the total number of arrangements of the five digits ($5! = 120$) and then subtract out the arrangements where 2 is next to 3. To count these arrangements, we will make use of yet another counting rule, the bijection rule.

19.3 The Bijection Rule

The bijection rule here is really just a restatement of an observation that we made in an earlier lecture when we were discussing cardinality.

> **Bijection Rule**:
>
> Suppose that there are finite sets S and T for which there is a bijection $f : S \to T$. Then, $|S| = |T|$.

In earlier lectures, we were mostly interested in the converse of this fact: that between two sets with equal cardinality, we can always find a bijection. In combinatorics, we use this rule differently. Often, we will have a set S that cannot be easily counted and a set T that can be. If we can find a bijection $S \to T$, we can use the easier methods on T as a proxy to count S. As our first example of this, we'll return to our running example.

Example 19.6 We wish to count the permutations of $[5]$ where 2 is next to 3. We partition these permutations into two sets based on whether the 2 is to the left or right of the 3.

First, we'll count the permutations with 2 to the immediate left of 3. Rather than enumerating these permutations, we make the following observation: since the 2 will be to the immediate left of the 3, we can push the 2 and the 3 together into a single symbol "23".

$$(1\,,4\,,2\,,3\,,5) \quad \mapsto \quad (1\,,4\,,23\,,5)$$

This maps the permutations of $[5]$ with 2 to the immediate left of 3 to the permutations of $\{1,23,4,5\}$. To see that this map is a bijection, we note that it has an obvious inverse; namely, the map that sticks a comma in between the "2" and the "3" in "23". Therefore, these sets have the same cardinality: $4! = 24$.

To count the permutations with 2 immediately to the right of 3, we again appeal to the bijection rule. Note that the map that switches the positions of the 2 and 3 in the permutation describes a bijection to the set from the first case. Therefore, this second set also includes 24 permutations, giving 48 permutations overall.

Applying the subtraction principle again shows that there are $120 - 48 = 72$ permutations of $[5]$ where 2 is not next to 3. This example illustrates that there are often many different procedures for counting the same set, some of which may be easier than others. In the next lecture, we'll leverage this idea when we discuss *combinatorial proofs*.

19.4 The Division Rule

So far, our counting rules have allowed us to determine the size of a set by finding a multi-step process to construct all of its elements and carefully reasoning about the number of possible choices that could be made at each step of the process. A crucial part of this technique was guaranteeing that each element of the set could be reached by *exactly* one sequence of choices. The division rule allows us to generalize this technique.

Division Rule:

Suppose that there is a function $f : S \to T$ that is a k-to-1 correspondence (that is, f is a surjection and each $t \in T$ is the image of exactly k elements of S). Then, $|T| = \frac{|S|}{k}$.

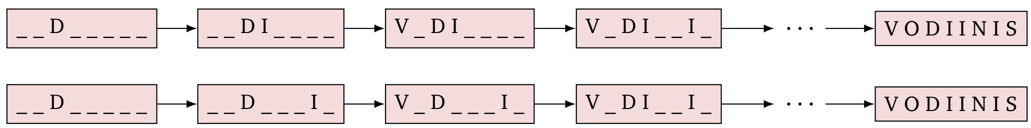
Said differently, a k-to-1 correspondence maps k distinct elements of S to each element of T, meaning S must be k times larger than T. Typically, we use this rule rather informally, where S is the set of choice sequences in our generating process (so is easier to count using addition and multiplication rules). Thus, to apply the division rule, we must argue that exactly k of these choice sequences generates each element of our desired set T.

As a first example, we'll consider counting permutations of a multiset. Recall that, unlike a set, a multiset can contain duplicate elements. Thus, the permutations of a multiset are all distinct arrangements of its (possible duplicate) elements, where we treat duplicated elements as indistinguishable.

Example 19.7 We count the number of different ways we can rearrange the letters in the word "DIVISION". Note that this word has 8 letters, so we can think about generating arrangements by placing the letters one at a time and appealing to the multiplication rule.

- There are 8 positions where we can place the "D".
- After the "D" is placed, there are 7 positions where we can place the first "I".
- Then, there are 6 remaining positions for the "V"...

There are $8! = 40,320$ possible ways that this process can unfold. However, note that not all of them result in a different arrangement, as illustrated by the following example:



It does not matter which of the multiple "I"s, end up in which position in the final arrangement. Therefore, each resulting arrangement can be arrived at by $3! = 6$ different choice processes, corresponding to the 6 orders that the "I"s can be filled in. By the division rule, there are $\frac{40,320}{6} = 6,720$ distinct arrangements of the letters in "DIVISION".

We record the following fact. Its proof generalizes the reasoning from the previous example.

Lemma 19.1 Suppose that M is a multiset that contains distinct elements $\{a_1, \dots, a_k\}$, and with the multiplicity of each a_i (that is, the number of times a_i appears in M) equal to n_i. Then, the number of permutations of M is

$$\frac{(n_1 + n_2 + \dots + n_k)!}{n_1!\, n_2!\, \dots\, n_k!}.$$

We will use the division rule next lecture to count subsets, or *combinations*, of a set.

Main Takeaways:

- When we are counting a set of objects, it is often helpful to think about a process for generating the objects, rather than the objects themselves.
- The multiplication rule allows us to count steps that happen sequentially, while the addition rule allows us to count steps that happen in disjoint cases.
- A permutation assigns an ordering to a (finite) set of objects. The factorial function counts the number of permutations.
- Bijections preserve the size of finite sets, so can often provide a way to convert a counting problem to a simpler one.
- The division rule can be used to account for multiple "choice paths" leading to the same object.

Exercises

19.1

(a) How many 3-digit numbers are there?

(Do not include 3-digit sequences such as "013" that begin with "0" in this count.)

(b) How many 3-digit numbers are *not* multiples of 5?

(c) How many 3-digit numbers do not contain any digit more than once?

(d) How many 3-digit numbers do not contain two *consecutive* copies of the same digit?

19.2 Suppose that we create 5-digit (or 4-digit if the leftmost digit is a 0) numbers using each of the digits $\{0,1,2,4,5\}$ exactly once. How many such numbers are

(a) Odd?

(b) Less than $30,000$?

(c) Multiples of 5?

(d) Multiples of 25?

(e) Multiples of 4?

19.3 In how many permutations of $[7]$ does

(a) 6 appear as the last element?

(b) 1 appear to the left of 2?

(c) Both (a) and (b)?

(d) Neither (a) nor (b)?

19.4 A chessboard is an 8×8 grid of squares. A rook occupies one square on the chess board and can *attack* any piece in its row or column.

(a) How many ways can 8 rooks be placed so that no rook can attack another?

(b) How many ways can 7 rooks be placed so that no rook can attack another?

(Hint: Start with your answer to part (a). How can you turn one of these arrangements into an arrangement of 7 rooks?)

⋆ (c) How many ways can 6 rooks be placed so that no rook can attack another?

For an extra challenge, try counting the arrangements of other numbers of rooks.

19.5 Suppose that S is a set with n elements.

(a) How many binary relations are there on set S?

(b) How many of these binary relations are reflexive?

(c) How many of these binary relations are symmetric?

(d) How many of these binary relations are anti-symmetric?

19.6 10 men and 10 women have signed up for a community ballroom dance class.

(a) How many different ways can the instructor assign the participants into 10 (man, woman) pairs?

(b) Suppose instead that 11 women sign up for the class. Assuming that the instructor dances with the unpaired woman, how many different ways are there to pair the participants now?

19.7 Eight people are to be seated at a round dining table.

(a) How many distinct seating arrangements are possible? Here, we define two seating arrangements to be non-distinct if each person has the same person sitting on their left and the same person sitting on their right (regardless of which chair they are sitting in).

(b) Same question, but now we define two seating arrangements to be non-distinct if each person is sitting next to the same two people (not necessarily on the same sides).

19.8 How many different arrangements are there of the letters in each of the following words?

(a) "COMPUTER"

(b) "SCIENCE"

(c) "DISCRETE"

(d) "STRUCTURES"

(e) "IRREFLEXIVE"

19.9 Recall that a number $d \in \mathbb{N}^+$ is a *divisor* of a number $n \in \mathbb{N}$ if there is some $c \in \mathbb{N}$ such that $dc = n$. How many divisors are there of:

(a) $35 = 5 \cdot 7$?

(b) $24 = 2^3 \cdot 3$?

(c) $600 = 2^3 \cdot 3 \cdot 5^2$?

19.10 A *partial permutation* of a set S is an ordered list of some *but not necessarily all* of its elements. For example, both $[b,d]$ and $[a,d,c]$ are partial permutations of $\{a,b,c,d\}$.

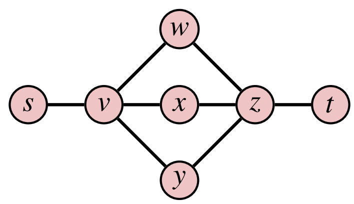
(a) How many partial permutations of $[5]$ include exactly 3 elements?

(b) Generalize your answer from part (a). How many partial permutations of $[n]$ include exactly k elements, where $k,n \in \mathbb{N}$ with $k \leq n$? You should end up with a very simple answer in terms of factorial functions.

(c) Write an expression for the total number of partial permutations of $[n]$.

(d) Write out all of the partial permutations of $[3]$. Does the number that you found agree with your formula from part (c)?

19.11 How many $s-t$ paths are there in the following undirected graph that travel along each edge at most once (but may revisit a vertex any number of times)?



20. Binomial Coefficients

In the previous lecture, we talked about five rules that are helpful for counting sets without the need to enumerate all of their elements: the multiplication rule, addition rule, subtraction rule, bijection rule, and division rule. In this lecture, we'll use these rules (particularly the division rule) to derive a formula for counting subsets, or *combinations*, of a set with a particular size. The answers to this counting question are referred to as the *binomial coefficients*. Binomial coefficients are one of the "big four" sets of coefficients in combinatorics (we'll see others in later lectures), so we'll explore some of their properties and associated formulas, most notably Pascal's formula and the binomial theorem.

20.1 Counting Combinations

In combinatorics, we often refer to a subset of a set S as a *combination*, because it tells us one particular way of combining together elements of S. Crucially, combinations are unordered; we care only about the contents of the set and not the order that they were chosen. This contrasts with *permutations*, where both the contents and the order matter. One of the most basic tasks we can consider is counting the number of combinations of a certain size.

Example 20.1 We count the number of subsets (combinations) of $[6]$ with size 4. Since this is a small example, we can list all 15 of these subsets:

$$\begin{array}{ccccc} \{1,2,3,4\} & \{1,2,3,5\} & \{1,2,3,6\} & \{1,2,4,5\} & \{1,2,4,6\} \\ \{1,2,5,6\} & \{1,3,4,5\} & \{1,3,4,6\} & \{1,3,5,6\} & \{1,4,5,6\} \\ \{2,3,4,5\} & \{2,3,4,6\} & \{2,3,5,6\} & \{2,4,5,6\} & \{3,4,5,6\} \end{array}$$

How can we do this more systematically? Let's consider the map φ that takes a permutation of $[6]$ and maps it to a subset of size four by selecting the four leftmost elements.

$$[1,4,5,2,6,3] \quad \overset{\varphi}{\mapsto} \quad \{1,2,4,5\}$$

Note that φ will map to the same set regardless of how we arrange the first four elements in the permutation. Similarly, it will map to the same set regardless of how we arrange the last two elements of the permutation. Thus, there are exactly $4! \cdot 2!$ permutations that φ maps to each subset. By the division rule, the number of subsets of size 4 is $\frac{6!}{4! \cdot 2!} = \frac{720}{24 \cdot 2} = 15$.

By generalizing this example, we obtain the following lemma.

Lemma 20.1 The number of subsets of size k (k-combinations) of a set of size n is $\dfrac{n!}{k!(n-k)!}$.

Since combinations are so pervasive throughout combinatorics, we introduce a special notation called a *binomial coefficient* to shorthand these expressions.

Definition 20.1 — Binomial Coefficient.
Given $k \le n \in \mathbb{N}$, the *binomial coefficient* $\binom{n}{k}$ is the number of k-combinations of a set of n elements. That is,

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}.$$

Example 20.2 The CS department will offer 15 undergraduate electives in the spring. Assuming that each student may enroll in at most three of these classes and that there are no prerequisites or time conflicts, how many subsets of courses can appear on a schedule?

We can use the addition rule, separating into the schedules that contain exactly 0,1,2, or 3 of these classes. There is $\binom{15}{0} = \frac{15!}{0! \cdot 15!} = 1$ way to choose 0 classes. There are $\binom{15}{1} = \frac{15!}{1! \cdot 14!} = 15$ ways to choose 1 class. There are $\binom{15}{2} = \frac{15!}{2! \cdot 13!} = 105$ ways to choose 2 classes. Finally, there are $\binom{15}{3} = \frac{15!}{3! \cdot 12!} = 455$ ways to choose 3 classes. In total, there are $1 + 15 + 105 + 455 = 576$ possible schedules.

20.2 Properties of Binomial Coefficients

Next, we'll derive two useful properties of binomial coefficients. Both of these properties can be verified algebraically by plugging into Definition 20.1 and simplifying, but we will take a different approach. We will look for *combinatorial proofs*.

Combinatorial Proofs:

To verify an identity $E_1 = E_2$ (for some expressions E_1 and E_2) using a *combinatorial proof*, find a set that can be counted in two different ways. One way should give rise to E_1 and the other should give rise to E_2. Then, since we have counted the same set in two different ways, the resulting counts must be the same, proving $E_1 = E_2$.

The first property we'll prove concerns a symmetry property of the binomial coefficients.

Lemma 20.2 For all $n, k \in \mathbb{N}$ with $0 \le k \le n$, $$\binom{n}{k} = \binom{n}{n-k}.$$

Proof. By Definition 20.1, we know that the binomial coefficient on the left side of this identity counts the number of subsets of $[n]$ with size k. Thus, we must find a way to count these subsets that gives rise to the right side of the identity. Note that we can also obtain a k element set by choosing the $n-k$ elements that it does *not* include (and then placing the remaining k elements into the set). There are $\binom{n}{n-k}$ ways to select these elements. ■

The second property concerns a sum of binomial coefficients.

Lemma 20.3 For all $n \in \mathbb{N}$, $$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Proof. We'll argue that both sides count the elements of $\mathcal{P}([n])$. This is true of the right-hand side by the multiplication rule. We can construct any subset of $[n]$ by choosing to include or exclude each of the n elements. For the left-hand side, note that we can also select a subset of $[n]$ by first choosing its size k (an integer between 0 and n) and then selecting a subset of that size. By Definition 20.1, there are $\binom{n}{k}$ subsets of size k for each k. Since the possible sizes partition the set of subsets, we may apply the addition rule to obtain the expression on the left-hand side. ■

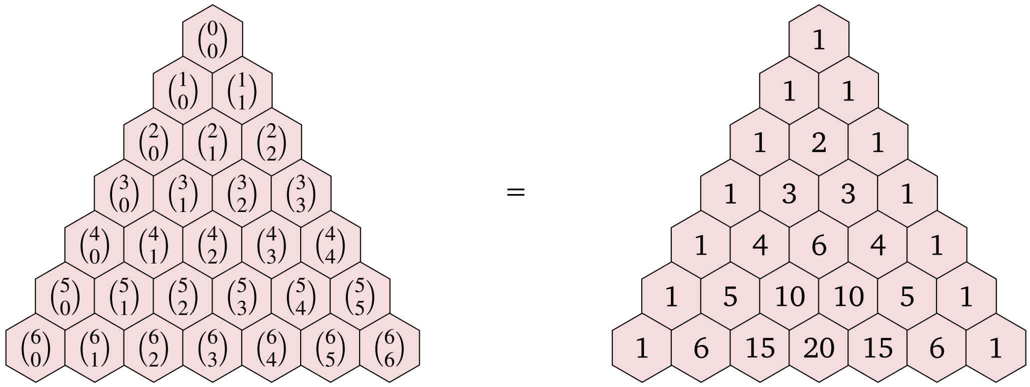
The third property that we will establish is known as *Pascal's Formula*.

Lemma 20.4 — Pascal's Formula.

For all $n, k \in \mathbb{N}$ with $1 \le k < n$, $$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. We again provide a combinatorial proof, which by Definition 20.1 reduces to arguing that the righthand side counts the number of subsets of $[n]$ of size k. To select such a subset, we can first decide whether it will include the element n. If it does, we must select $k-1$ additional elements from the $n-1$ that remain. There are $\binom{n-1}{k-1}$ ways to do this. Otherwise, if the subset does not include n, we must select k of the remaining $n-1$ elements, which we can do in $\binom{n-1}{k}$ ways. The result follows from the addition rule. ■

Pascal's Formula gives us a recursive way to calculate the binomial coefficients from those with a smaller number on top. We can visualize this recurrence by representing the binomial coefficients as a triangle, with rows corresponding to values of n and cells in the row corresponding to different values of k. This diagram is called Pascal's Triangle.



In the diagram, the fact that $\binom{n}{0} = \binom{n}{n} = 1$ for each n allows us to fill 1s along the outer edges, and the recurrence tells us that each interior cell is the sum of the two cells above it. There are many other patterns hidden in Pascal's triangle. We mention a few of them in the exercises.

20.3 The Binomial Theorem

One key use of binomial coefficients is in the *Binomial Theorem*, which gives a formula for expanding an expression of the form $(x+y)^n$. We rewrite this as a product of n terms:

$$(x+y)^n = \underbrace{(x+y)\,(x+y)\,\ldots\,(x+y)}_{n \text{ times}}.$$

Now, we consider expanding this product. Each of the 2^n terms in this product will be obtained by choosing either the x or the y from each of the n factors, so will have the form $x^k y^{n-k}$ for some $0 \le k \le n$. Now, we reason about how many copies of this $x^k y^{n-k}$ term there will be. This is exactly the number of ways that we could select k of the n factors to contribute the x's (with the other $n-k$ contributing the y's), so $\binom{n}{k}$. Since this is true regardless of the value of k, we obtain the following identity.

Theorem 20.1 — Binomial Theorem.

For all $n \in \mathbb{N}$ and all $x, y \in \mathbb{R}$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k\, y^{n-k}.$$

The preceding paragraph gives some combinatorial justification for this theorem. We give a formal proof by induction on n.

Proof. As our base case, when $n=0$, we have $(x+y)^0 = 1 = \binom{0}{0} x^0 y^0$. As our inductive hypothesis, we assume that the identity holds for n (as in the theorem statement) and consider the case $n+1$. We have,

$$
\begin{aligned}
(x+y)^{n+1} &= (x+y)\cdot(x+y)^n \\
&= (x+y)\cdot \sum_{k=0}^{n} \binom{n}{k} x^k\, y^{n-k} && \text{(Inductive Hypothesis)} \\
&= \sum_{k=0}^{n} \binom{n}{k} x^{k+1}\, y^{n-k} + \sum_{k=0}^{n} \binom{n}{k} x^k\, y^{n+1-k} && \text{(Distributing)} \\
&= \sum_{k=1}^{n+1} \binom{n}{k-1} x^k\, y^{n+1-k} + \sum_{k=0}^{n} \binom{n}{k} x^k\, y^{n+1-k} && (k \mapsto k-1 \text{ in first sum}) \\
&= \binom{n}{n} x^{n+1}\, y^0 + \binom{n}{0} x^0\, y^{n+1} + \sum_{k=1}^{n} \left[\binom{n}{k-1} + \binom{n}{k} \right] x^k\, y^{n-k+1} \\
&= x^{n+1} + y^{n+1} + \sum_{k=1}^{n} \binom{n+1}{k} x^k\, y^{n-k+1} && \text{(Lemma 20.4)} \\
&= \sum_{k=0}^{n+1} \binom{n+1}{k} x^k\, y^{n+1-k}.
\end{aligned}
$$

Thus, the formula holds for $n+1$, so by induction holds for all $n \in \mathbb{N}$. ■

We can expand $(x+y)^n$ using the coefficients in the n'th row of Pascal's triangle. For example,

$$(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.$$

We can also use the binomial theorem to obtain many interesting formulas. First, setting $x = y = 1$ gives another proof of Lemma 20.3:

$$2^n = (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^k\, 1^{n-k} = \sum_{k=0}^{n} \binom{n}{k}.$$

Now, suppose that $x = -1$ and $y = 1$. The binomial theorem tells us that for all $n > 0$,

$$0 = (-1+1)^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \quad \Longrightarrow \quad \sum_{\substack{k=0 \\ k \text{ even}}}^{n} \binom{n}{k} = \sum_{\substack{k=0 \\ k \text{ odd}}}^{n} \binom{n}{k}.$$

Every finite (non-empty) set has the same number of subsets with even cardinality as with odd cardinality. We walk through a bijective proof of this in the exercises.

Main Takeaways:

- The binomial coefficient $\binom{n}{k}$ expresses the number of k-element subsets a set of size n.
- Pascal's triangle illuminates many useful identities involving the binomial coefficients.
- A combinatorial proof establishes an identity by counting the same set in two ways.
- The binomial theorem gives a compact formula for expanding powers of a binomial expression.

Exercises

20.1 Vincent is trying to count the number of subsets of $[n]$ with size at most 3 and suggests the following:

> *We'll use a two-stage process to generate these subsets. First, we'll select a subset S of size exactly 3. There are $\binom{n}{3}$ ways to do this. Then, for each subset S, all subsets $T \subseteq S$ have size at most 3. There are $2^3 = 8$ such subsets T for each S. By the multiplication rule, there are $8 \cdot \binom{n}{3}$ subsets of $[n]$ with size at most 3.*

(a) Explain the error in Vincent's approach. Will Vincent's formula undercount or overcount these subsets?

(b) Find the actual formula for the number of subsets of $[n]$ with size at most 3.

20.2 Use induction on n to argue that Definition 20.1 satisfies Pascal's recurrence formula for all $n,k \in \mathbb{N}$ with $k \leq n$. In your proof, you may assume that $\binom{n}{n} = 1$ for all $n \in \mathbb{N}$. (Hint: You'll need a way to handle both quantifiers in this claim. Quantification over n is handled by the induction. This means that the inductive statement will need to be quantified over k.)

20.3 By observing the entries in Pascal's Triangle, we can notice that it has a left-right mirror symmetry.

(a) Write a mathematical identity in terms of binomial coefficients that describes this symmetry. Make sure that you carefully quantify any variables that you use in your identity.

(b) Use Definition 20.1 to algebraically verify your identity.

(c) Give a combinatorial proof of your identity.

20.4 Give a combinatorial proof that $k \cdot \binom{n}{k} = n \cdot \binom{n-1}{k-1}$.

(Hint: Think about different ways to select a team with one member designated as its captain.)

20.5 Give a combinatorial proof that $\binom{n}{k} - \binom{n-2}{k} = \binom{n-1}{k-1} + \binom{n-2}{k-1}$.

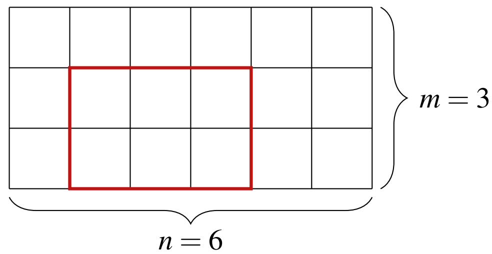
(Hint: Think about a set of n elements with two particular elements a and b. Count some particular subsets of this set in two different ways.)

20.6 Suppose that $A \subseteq C$ with $|A| = a$ and $|C| = c$.

(a) How many sets B are there such that $A \subseteq B \subseteq C$?

(b) Given any b with $a \leq b \leq c$, how many B are there such that $A \subseteq B \subseteq C$ and $|B| = b$?

20.7 Consider an $m \times n$ grid. Find a formula (in terms of m and n) for the number of rectangles that can be drawn along the lines of this grid. One such rectangle is highlighted in red below.


m = 3
n = 6

20.8 A city's streets are arranged in a grid pattern with intersections identified by integer-ordered pairs (x,y). A street-sweeping vehicle starts at intersection $(0,0)$ and cleans 1 block (that is, one road connecting two adjacent intersections) every 10 minutes. The sweeper only moves to the right (increasing x-coordinate) and up (increasing y-coordinate). Suppose that after $10(x+y)$ minutes, the sweeper is at intersection (x,y). Find an expression (in terms of x and y) for how many possible paths the sweeper could have taken to arrive here. Give combinatorial reasoning for your answer. (Hint: it may help to draw out some small examples and try to develop a recurrence.)

⋆ **20.9** Suppose that S is a finite set that contains n elements.

(a) If $A \subseteq S$ with $|A| = k$, how many sets B are there with $A \subseteq B \subseteq S$? Explain your answer.

The following process generates pairs of subsets $(A,B) \in \mathcal{P}(S) \times \mathcal{P}(S)$ with $A \subseteq B$:

1. Select the size k of the subset A.
2. Select k elements of S to add to A.
3. Select a set B with $A \subseteq B \subseteq S$.

(b) Write an expression that counts the number of pairs of subsets (A,B) with $A \subseteq B \subseteq S$ that is based on the above process. Explain how you obtained your answer.

(c) Use the Binomial Theorem to simplify the expression from part (b) as much as possible.

(d) The expression from part (c) suggests that there is another process for generating these pairs of subsets. Give a brief description of this process.

(e) Finally, let's generalize the ideas from this exercise. Given any finite set S with $|S| = n$, and any $j \in \mathbb{N}$, how many ways are there to select subsets

$$S_1 \subseteq S_2 \subseteq \ldots \subseteq S_j \subseteq S\,?$$

20.10 Here, we explore some patterns in Pascal's triangle.

The first diagonal band in from the outer "1"s band (that is, $\binom{n}{1}$ for each n) includes the positive natural numbers.

(a) Give an inductive proof of this fact using Lemma 20.4 and the fact that $\binom{n}{0} = 1$ for each $n \in \mathbb{N}$ (do not appeal to Definition 20.1).

The second diagonal band (that is, $\binom{n}{2}$ for each n) includes the *triangular numbers*: 1,3,6,10....

(b) Explain why this is a reasonable name for this sequence.

(c) Use Definition 20.1 to find a closed formula for $\binom{n}{2}$. This should be a polynomial in n.

(d) Verify that your formula from part (d) is correct by induction on n. Your proof should make use of Lemma 20.4 and part (a), but not Definition 20.1).

20.11 Use the Binomial Theorem to expand the following polynomials.

(a) $(x+y)^7$

(b) $(x+2)^3$

(c) $(2x+3y)^4$

(d) $(x+\frac{1}{x})^5$

⋆ (e) $(x+(y+z))^3$ Use the Binomial Theorem multiple times until no parentheses remain.

20.12 Given $n \in \mathbb{N}$, let $\mathcal{E} \subseteq \mathcal{P}([n])$ include all subsets with even cardinality and $\mathcal{O} \subseteq \mathcal{P}([n])$ include all subsets with odd cardinality. Here, we'll walk through a bijective proof that $|\mathcal{E}| = |\mathcal{O}|$ using a technique called *pairing*.

(a) Describe a (relatively simple) way to pair up the elements of $\mathcal{P}([n])$ that ensures that one element of each pair is in $\mathcal{E}$ and the other is in $\mathcal{O}$.

(Hint: Think about which sets contain the element n.)

(b) Argue that this pairing gives a bijection between $\mathcal{E}$ and $\mathcal{O}$. As part of your answer, give an explicit description of this bijection.

(c) We can use the bijection rule to conclude that $|\mathcal{E}| = |\mathcal{O}|$. Thus, how many even-cardinality subsets are there of $\mathcal{P}([n])$?

⋆ **20.13** Given any $n \in \mathbb{N}$ and any $k_1, \ldots, k_m \in \mathbb{N}$ with $k_1 + \ldots + k_m = n$, the *multinomial coefficient* $\binom{n}{k_1,\ldots,k_m}$ counts the ways to split n (distinguishable) elements into m groups of size $k_1, \ldots, k_m$.

(a) The multinomial coefficients are a *generalization* of the binomial coefficients. How can we express $\binom{n}{k}$ as a multinomial coefficient? Explain your answer by describing what "groups" are being created.

(b) Use the multiplication rule to find a formula for the multinomial coefficient $\binom{n}{k_1,\ldots,k_m}$ in terms of binomial coefficients.

(c) Simplify your answer to part (a) to a formula involving only factorials.

(d) Explain the connection between multinomial coefficients and multiset permutations.

We can generalize the binomial theorem to the *multinomial theorem*, which says that for each $k_1 + \ldots + k_m = n$, the coefficient on $x_1^{k_1} \ldots x_m^{k_m}$ in $(x_1 + \ldots + x_m)^n$ is $\binom{n}{k_1,\ldots,k_m}$.

⋆ **20.14** Give a combinatorial proof of the following identity:

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n},$$

which is a special case of the *Vandermonde convolution* identity.

(Hint: Imagine choosing a subset of $2n$ elements by first splitting objects into two piles with n objects each and then choosing some objects from each pile. Think about what the variable k can represent.)

⋄ **20.15** We can use calculus together with the Binomial Theorem to find more identities involving the binomial coefficients. Throughout this exercise, we'll consider the function

$$f(x) = (x+1)^n.$$

(a) Use the Binomial Theorem to write the expanded form of $f(x)$.

Plugging in $x = 1$ into our original definition of $f(x)$ and the expanded for from part (a) should recover the identity from Lemma 20.3.

(b) Evaluate $f'(1)$ in both definitions of $f(x)$ to find a simplification of

$$\sum_{k=1}^{n} k\binom{n}{k}.$$

(c) Evaluate $f'(-1)$ in both definitions of $f(x)$ to find a simplification of

$$\sum_{k=1}^{n} (-1)^k \cdot k\binom{n}{k}.$$

⋆ (d) Evaluate $f''(1)$ in both definitions of $f(x)$, and use this to find a simplification of

$$\sum_{k=1}^{n} k^2\binom{n}{k}.$$

There is extra work that needs to be done here using your results from earlier parts.

⋆ (e) Evaluate $F(1)$ (that is, the anti-derivative of $f(x)$ evaluated at 1) in both definitions of $f(x)$, and use this to find a simplification of

$$\sum_{k=0}^{n} \frac{1}{k+1}\binom{n}{k}.$$

21. The Inclusion-Exclusion Rule

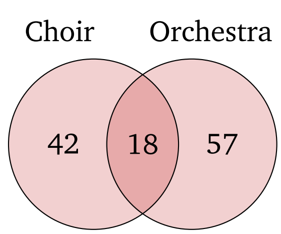
The Addition Rule allows us to calculate the size of a set S by separating its elements into a collection of subsets $S_1, \ldots, S_k$, calculating the size of each subset, and adding these sizes. This works when $S_1, \ldots, S_k$ form a *partition* of S. However, any element $s \in S$ that belongs to multiple subsets will be counted multiple times, and the Addition Rule will return the wrong result. To remedy this, we will introduce the *Inclusion-Exclusion Rule*, a generalization of the Addition Rule that accounts for partial overlaps between various subsets.

21.1 Deriving the Inclusion-Exclusion Rule

To motivate the Inclusion-Exclusion Rule, consider the following scenario.

Example 21.1 A music teacher directs two ensembles, a choir and an orchestra. They would like to know how many total students they have so that they know how many dress outfits need to be ordered. They look at their class lists and see that 60 students have signed up for choir and 75 students for orchestra. Is this enough information for them to place the order?

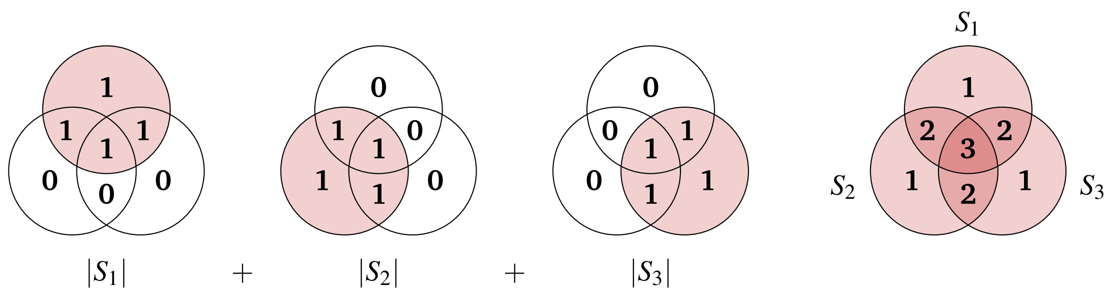
At first thought, it may appear that 135 outfits are needed. However, this assumes that the students in choir are disjoint from the students in orchestra. What about a student who is in both groups? They do not need two of the same dress outfit; one outfit that they will wear to the performances of both groups will suffice. Thus, from the total of 135 students, the teacher must subtract one outfit for each student who is in both the choir and orchestra. If there are 18 such students, the number of outfits needed is $60 + 75 - 18 = 117$.



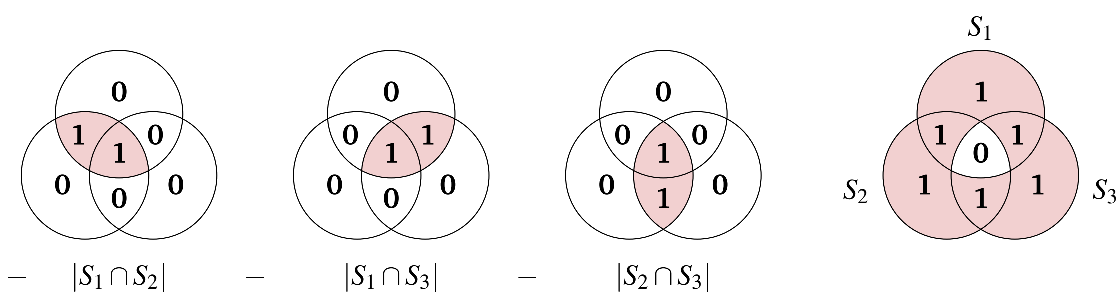
In the set-up for the Inclusion-Exclusion rule, we are given a collection of sets $S_1, \ldots, S_n$ and wish to calculate the size of their union $|\bigcup_{i \in [n]} S_i|$. In our example, S_1 contained the choir students, and S_2 contained the orchestra students, and we were computing the total number of students. The $n = 2$ case of Inclusion-Exclusion says that

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|.$$

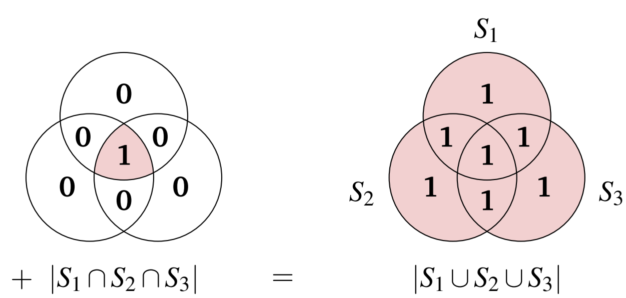
Next, let's think about the case of $n = 3$ sets. We wish to compute $|S_1 \cup S_2 \cup S_3|$. To start, we can simply add the sizes of these three sets. We'll use a three-circle Venn diagram to visualize our calculation, where each segment is labeled with a bold number representing how many times it is counted. We have,



Just as before, we have multiply-counted the elements that belong to multiple sets. To correct for this, we will subtract the size of each pairwise set intersection.



Now, we see that our subtraction was a little heavy-handed; we are no longer counting the elements that belong to all three sets. As a final correction step, we add back these elements.



Thus, our final Inclusion-Exclusion formula for $n = 3$ sets is

$$|S_1 \cup S_2 \cup S_3| = \Big(|S_1| + |S_2| + |S_3|\Big) - \Big(|S_1 \cap S_2| + |S_1 \cap S_3| + |S_2 \cap S_3|\Big) + |S_1 \cap S_2 \cap S_3|.$$

For any larger cases, the number of terms in this expression grows very quickly, Therefore, we introduce a more compact sum notation

$$\left|\bigcup_{i\in[3]} S_i\right| = \sum_{j=1}^{3}(-1)^{j-1}\sum_{\substack{I\subseteq[3]\\|I|=j}}\left|\bigcap_{i\in I} S_i\right|.$$

Let's unpack this notation. In the middle expression, the sum over j tells us to consider intersections of all sizes; $j=1$ corresponds to the sets S_i themselves and $j=3$ corresponds to the three-way intersection $S_1\cap S_2\cap S_3$. The $(-1)^{j-1}$ expresses that we need to switch between corrections that add set sizes and corrections that subtract set sizes. Then, the inner sum has us consider all possible j-way intersections of the $n=3$ sets. We can generalize this formula to hold for an arbitrary $n\in\mathbb{N}^+$.

Theorem 21.1 — Inclusion-Exclusion Rule. Given any n sets $S_1,\ldots,S_n$

$$\left|\bigcup_{i\in[n]} S_i\right| = \sum_{j=1}^{n}(-1)^{j-1}\sum_{\substack{I\subseteq[n]\\|I|=j}}\left|\bigcap_{i\in I} S_i\right|.$$

Note that plugging in $n=2$ to this formula recovers our formula for the two-set case.

Proof. To prove this claim, it suffices to argue that each element in the union is accounted for a net (after all the additions and subtractions) one time by the formula on the right side. To do this, consider an arbitrary element s in this union, and let $m\geq 1$ be the number of subsets in which s appears. Note that,

- s appears in $m=\binom{m}{1}$ sets.
- s appears in $\binom{m}{2}$ intersections of pairs of sets.
- More generally, s appears in $\binom{m}{j}$ intersections of j sets.

Thus, the number of times that s is counted on the right side is

$$\sum_{j=1}^{m}(-1)^{j-1}\binom{m}{j} \quad = -\sum_{j=1}^{m}\binom{m}{j}(-1)^{j}\cdot 1^{m-j} \quad = 1-\sum_{j=0}^{m}\binom{m}{j}(-1)^{j}\cdot 1^{m-j} \quad = 1.$$

Here, the last equality follows from the Binomial Theorem (with $x=-1$, $y=1$). ■

21.2 An Application of Inclusion-Exclusion

For the rest of the lecture, we will look at an application of the Inclusion-Exclusion rule to count permutations with a certain property, known as *derangements*.

Definition 21.1 — Derangement.

A permutation of $[n]$ is a derangement if there is no element $i\in[n]$ that appears in the i'th position.

For example, $(4,1,2,3)$ is a derangement of $[4]$, but $(2,4,3,1)$ is not since 3 appears in the third position.

To calculate the number of derangements, we will use a variant of the formula from Theorem 21.1. We'll think about $S_1,\dots,S_n$ as encoding "bad events"; each contains elements that have some undesirable property. Then, we wish to count the "good" elements, those that avoid all of the undesirable properties. If we let S be the set of all elements, then the set of good elements has size

$$\left|\bigcap_{i\in[n]}(S\setminus S_i)\right| = |S| - \left|\bigcup_{i\in[n]} S_i\right| = |S| + \sum_{j=1}^{n}(-1)^j \sum_{\substack{I\subseteq[n]\\|I|=j}}\left|\bigcap_{i\in I} S_i\right|.$$

Lemma 21.1 For each $n\in\mathbb{N}^+$, the number of derangements of $[n]$ is $n!\cdot\sum_{j=0}^{n}\frac{(-1)^j}{j!}$.

Proof. We'll define a set of n bad events $S_1,\dots,S_n$, where S_i includes those permutations with i in their i'th position. We'll use our variant of the Inclusion-Exclusion rule to count the permutations that avoid all of these bad events. To do this, we must understand the sizes of various intersections.

- For each singleton set $I=\{i\}$, the permutations in S_i have i in their i'th position, but can freely arrange the other $n-1$ elements in the remaining $n-1$ positions. Thus $|S_i|=(n-1)!$.
- For each two-element set $I=\{i,j\}$, the permutations in $S_i\cap S_j$ must have i in their i'th position and j in their j'th position, but can freely arrange the other $n-2$ elements in the remaining $n-2$ positions. Thus $|S_i\cap S_j|=(n-2)!$.
- By the same logic, any k-way intersection of bad event sets has cardinality $(n-k)!$.

Noting that the total number of permutations of $[n]$ is $n!$, we find that the number of derangements is

$$n!+\sum_{j=1}^{n}(-1)^j\sum_{\substack{I\subseteq[n]\\|I|=j}}\left|\bigcap_{i\in I}S_i\right| = n!+\sum_{j=1}^{n}\binom{n}{j}(-1)^j\cdot(n-j)! = n!\sum_{j=0}^{n}\frac{(-1)^j}{j!}$$

Here, the last equality pulls the $n!$ into the sum as the $j=0$ term, uses the definition of the binomial coefficient $\binom{n}{j}=\frac{n!}{j!(n-j)!}$ to further simplify each term in the sum, and then pulls out the common factor of $n!$. ■

The fraction of permutations that are derangements is $\sum_{j=0}^{n}\frac{(-1)^j}{j!}$. As $n\to\infty$, this fraction approaches the infinite series $\sum_{j=0}^{\infty}\frac{(-1)^j}{j!}=\frac{1}{e}\approx 0.368$.

Main Takeaways:

- The Inclusion-Exclusion rule provides a way to determine the size of a union of partially overlapping sets.
- We can rearrange the Inclusion-Exclusion formula to count the number of elements that avoid a set of undesired properties, which is useful in many applications.
- *Derangements* are permutations of $[n]$ where no element i appears in the i'th position.

Exercises

21.1 A pizzeria sells pizza and chicken wings. Last night, there were 190 orders placed. Of these orders, 45 included both pizza and wings, and 32 included only wings (no pizza).

(a) How many orders included only pizza (no wings)?

(b) How many orders included pizza (with or without wings)?

21.2 An art studio teaches three classes: watercolors, photography, and sculpture. There are 24 students in the watercolor class, 20 students in the photography class, and 16 students in the sculpture class.

(a) Why is this not enough information to determine the total number of students at the studio?

(b) What is the maximum possible number of students at the studio? The minimum possible number?

(c) The same instructor teaches the photography and watercolor classes and reports that 9 students are taking both. With this additional information, how can we adjust lower and upper bounds on the total number of students from part (b)?

(d) Suppose additionally that we learn that 3 students are enrolled in all three classes. Now, how can we adjust lower and upper bounds from part (c)?

21.3 Given a set S, suppose that we can construct k subsets $S_1, \ldots, S_k$ such that:

- Each $s \in S$ is an element of at least one subset S_i.
- The intersection of any j subsets (for any $j \in [k]$) contains 3^{k-j} elements.

(a) Give an example of a set S and subset S_1 with these properties in the case $k = 1$. How many elements are in S?

(b) Give an example of a set S and subsets S_1, S_2 with these properties in the case $k = 2$. How many elements are in S?

(c) Find a general formula (in terms of k) for the size of S. Simplify your answer as much as possible.

21.4 How many integers between 1 and 600 (inclusive) are not divisible by 3, 4, or 5?

◇ **21.5** In this exercise, we'll generalize the ideas from Exercise 21.4 to obtain a formula for an important formula in number theory, the *Euler totient function* $\varphi(n)$.

Recall that the *gcd* of two natural numbers $m,n \in \mathbb{N}$ is the largest $d \in \mathbb{N}$ that divides both m and n. Then, we define

$$\varphi(n) = \Big|\{m \in [n] \mid \gcd(m,n) = 1\}\Big|.$$

That is, $\varphi(n)$ is the number of positive integers $m \le n$ that share no prime factors with n.

(a) Calculate $\varphi(n)$ for all $n \in [20]$.

Suppose that $p_1 < p_2 < \ldots < p_k$ are the distinct prime factors of n. For example, $n = 360 = 2^3 \cdot 3^2 \cdot 5$ would have $p_1 = 2$, $p_2 = 3$, and $p_3 = 5$. We would like to find a formula for $\varphi(n)$ in terms of $n, p_1, \ldots, p_k$.

(b) To find this formula (i.e., to count the set of m), we will use the "bad event" formulation of Inclusion-Exclusion. What will be our set of bad events?

Hint: There should be k of them.

(c) Find a formula for the size of S_i for each $i \in [k]$.

(d) Find a formula for the size of $S_i \cap S_i'$ for each $i \neq i' \in [k]$.

(e) Generalizing the results of parts (c) and (d), find a formula for $\left|\bigcap_{i \in I} S_i\right|$ where $I \subseteq [k]$.

Hint: The formula that we have in mind includes a product over $i \in I$.

⋆ (f) Plug your formula from part (e) into the Inclusion-Exclusion formula. Then, factor the resulting expression to obtain a nice formula for $\varphi(n)$.

(g) Does your formula agree with your calculations in part (a)?

21.6 In the game of FizzBuzz, players take turns counting up $1, 2, \ldots$, except:

- Whenever the number they are about to say is divisible by 3 (e.g. 12) *or* contains a 3 (e.g. 13), they instead say "Fizz".
- Whenever the number they are about to say is divisible by 5 (e.g. 25) *or* contains a 5 (e.g. 52), they instead say "Buzz".
- When *both* of the above conditions are met (e.g. 15, 54), they say "FizzBuzz".

Over the first 100 turns,

(a) How many times is "Fizz" said (either on its own, or as a part of the word "FizzBuzz")?

(b) How many times is "Buzz" said (either on its own, or as a part of the word "FizzBuzz")?

(c) How many times is "FizzBuzz" said?

(d) How many times is a number said?

⋆ (e) Also answer these questions for the first 1000 turns.

21.7 In a dance class, there are 12 (man,woman) pairs. How many ways can the instructor rearrange the pairs such that no one is dancing with their original partner?

21.8 Call a permutation of $[n]$ *follower-free* if it is never the case that i is written to the immediate left of $i+1$ for any $1 \leq i < n$. For example, $(1,3,5,4,2)$ is a follower-free permutation of $[5]$, but $(1,3,4,2,5)$ is not. Use Inclusion-Exclusion to derive a formula for the number of follower-free permutations of $[n]$.

21.9 Let $S = \{w \in \{0,1\}^* \mid |w| = 8\}$. How many strings in S contain the following substrings?

(a) 10

(b) 1100

(c) 1001

(d) 101

(e) 0000

21.10 A *fixed point* in a permutation of $[n]$ is a number i that appears in the i'th position. For example, $1,3$ and 4 are all fixed points of the permutation $\sigma = [1,5,3,4,2]$. Derangements are permutations with no fixed points, and we found a formula to count them in the lecture.

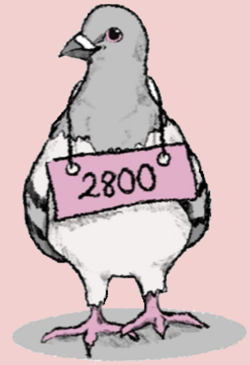
(a) Find a formula for the number of permutations of $[n]$ with *exactly one* fixed point.

Hint: Select one particular number (say 1) to be the fixed point and count the permutations that fix 1 but no other numbers with Inclusion-Exclusion. Then, multiply this count by n to consider each possible choice of the fixed point.

(b) Similarly, how many permutations of $[n]$ have *exactly two* fixed points?

(c) Generalize your result from parts (a) and (b). How many permutations have exactly k fixed points for any $k \in \{0,1,\ldots,n\}$?

(d) Call your formula from part (c), $F(n,k)$. Verify that $F(n,n) = 1$ and $F(n,n-1) = 0$ for all $n \in \mathbb{N}$ by plugging into your formula. Explain why both of these make sense.



22. Balls and Bins Problems

Over the next two lectures, we will think about counting functions with different properties. To do this, we'll use a popular framing in combinatorics of functions as arrangements of balls in bins. We'll view the domain elements as a set of balls and the codomain elements as a set of bins. A function maps each domain element to a unique codomain element, which corresponds to placing each ball into one bin.

Balls and bins problems are perhaps the most important for computer scientists. As we have mentioned numerous times, we can view computations as evaluating functions on data inputs. Therefore, being able to reason about the possible mappings will be crucial in analyzing randomized algorithms and understanding what we can expect from their performance. The prototypical example of this is the *hashing* problem, where items are stored in buckets with the hope that not too many items end up in any one bucket. You'll discuss hashing in greater detail in a data structures course. However, this *balls and bins* approach extends beyond just hashing. By being clever about what we view as the balls and what we view as the bins, we will see that many other problems also fit into this paradigm.

22.1 The Balls and Bins Setup

Throughout this and the next lecture, we'll use n to denote the number of balls (i.e., the size of the domain set) and m to denote the number of bins (i.e., the size of the codomain set). We will also consider two different types of balls and two different types of bins.

Distinguishable vs. Indistinguishable Balls:

In a setting with *distinguishable* balls, we imagine that each of the balls has a different color (we will also label them with different letters in our figures). Therefore, we can identify each individual ball by looking at their final arrangement. This contrasts with *indistinguishable* balls, which we imagine are all the same color. In this case, the only thing that distinguishes the arrangements is the *number* of balls that each bin contains. As an example, the two arrangements shown on the following page are different when the balls are distinguishable. However, they would be the same if the balls were made indistinguishable.

Distinguishable vs. Indistinguishable Bins:

In a setting with *distinguishable* bins, we imagine that each of the bins has been labeled with a different number. Therefore, the specific bin that each ball lands in makes a difference. This contrasts with *indistinguishable* bins, which we imagine are unlabeled and identical. In this case, the only thing that distinguishes the arrangements is how the balls are divided, and not the bin where each of the divisions is placed. As an example, the two arrangements shown below are different when the bins are distinguishable. However, they would be the same if the bins were made indistinguishable.

We will separately consider different combinations of these types of balls and bins, focusing on the cases of distinguishable balls and bins, indistinguishable balls with distinguishable bins, and (next lecture) distinguishable balls with indistinguishable bins. The case of indistinguishable balls and bins is more difficult and will not be considered in this course. In each of these settings, we wish to identify a formula for the number of functions (in general), the number of injective functions, and the number of surjective functions. By the end of the next lecture, we will be able to fill the following chart with these formulas.

n Balls	m Bins	Arrangements	Injective Arrangements	Surjective Arrangements
Distinguishable	Distinguishable			
Indistinguishable	Distinguishable			
Distinguishable	Indistinguishable			

22.2 Distinguishable Balls and Bins

As a warm-up, we'll consider the case of distinguishable balls and distinguishable bins. In this setting, each of the n balls has a different color $\{$red, gray, $\ldots\}$, and each of the m bins is labeled with a number from $[m]$. If you are asked to place each ball in a bin and then take a picture at the end, how many different pictures are possible?

In this case, we can use the Multiplication Rule, reasoning about one ball at a time. First, we choose which of the m bins receives the red ball. Next, we choose which of the m bins receives the blue ball. This repeats for all n balls. Thus, the total number of ways is m^n.

Now, suppose that we place the additional requirement that no bin can contain more than one ball. Viewing our arrangement as a function $\{\text{balls}\} \to \{\text{bins}\}$, this requirement imposes injectivity. As a first observation, we can note that this is impossible if $n > m$ (by the *Pigeonhole Principle*). Otherwise, we can again use the Multiplication Rule. First, we choose in which of the m bins to place the red ball. Next, we choose in which of the $m-1$ still-empty bins to place the blue ball. This repeats for all n balls. Giving us a total of

$$m \cdot (m-1) \cdot \ldots \cdot (m-n+1) = \frac{m!}{(m-n)!}$$

arrangements. As a convention, many people often *generalize* the definition of the binomial coefficients so that $\binom{m}{n} = 0$ when $n > m$. In this case, we can express the number of arrangements as $\binom{m}{n} \cdot n!$, to capture both of the above cases. This formula lends itself to another combinatorial interpretation of this problem. Namely, to select an injective map, we can first choose which n of the m bins will receive a ball. Then, we line the bins up and select a permutation over the balls to determine which bin gets which ball.

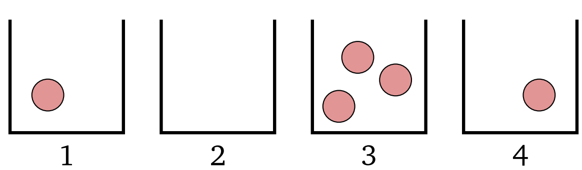
We could instead require that at least one ball be placed in each bin, which is equivalent to requiring that the $\{\text{balls}\} \to \{\text{bins}\}$ function is surjective. We will return to the question of counting these functions in the next lecture.

22.3 Indistinguishable Balls, Distinguishable Bins

Next, we'll consider the scenario where the balls are indistinguishable (they all have the same color), but the bins are distinguishable (labeled with numbers from $[m]$). How can we count these arrangements?

Finding a counting process is not as immediate in this case. We can't use the same trick of the Multiplication Rule to deal with the balls one at a time. Since the balls look the same, placing the first ball in bin 2 and the second ball in bin 5 will result in the same "picture" as if we placed the first ball in bin 5 and the second ball in bin 2. Moreover, we cannot use the Division Rule to account for these multiple paths to the same outcome, since this isn't consistent across the outcomes. For example, there's only one decision path that places all of the balls in bin 1. However, there are $n!$ different decision paths that place one ball in each bin when $n = m$. We will need a different approach. Amazingly, there is a nice bijective argument (often called the *stars and stripes* argument) that will allow us to count these arrangements. We'll illustrate this bijection with an example.

Example 22.1 Consider the case of $n = 5$ balls in $m = 4$ bins. One possible arrangement is



We can encode this arrangement with a string of '$*$'s and '$|$'s, where we interpret the stars as representing the balls and the stripes as delineating the bins. In this case, we have the string "$* \mid \mid * * * \mid *$".

Let's think more about this map φ from the arrangements to these strings. To understand about the range of this map, we note that

- Each string includes exactly n stars, 1 for each ball.
- Each string includes exactly $m-1$ stripes, 1 just after each bin except for the last.

In fact, φ describes a bijection between the arrangements and the strings that satisfy these two conditions. One could argue that this map is injective and surjective, but the easier way to see this is to note that this encoding process is reversible: given any string in this form, we can construct the corresponding arrangement by starting at bin 1, dropping a ball whenever we see a star, and moving to the next bin whenever we see a bar.

By the Bijection Rule, the number of arrangements is exactly the number of these strings. To generate these strings we can select $m-1$ of the $n+m-1$ positions to place stripes and fill the remaining positions with stars. Thus, there are $\binom{n+m-1}{m-1} = \binom{n+m-1}{n}$ such strings.

Again, we can ask about injective and surjective maps in this setting. Here, the arrangements that result from an injective map will have 1 ball each in n of the m bins. Thus, there are $\binom{m}{n}$ such arrangements; where again we are using the generalized binomial coefficient to ensure there are 0 arrangements when $n > m$.

For surjective maps, we again use the Bijection Rule. Note that an arrangement resulting from a surjective map has at least one ball in each bin. By removing one ball from each bin, we are left with an arbitrary arrangement of $n-m$ balls across the m bins. This removal of a ball from each bin is reversible, so it describes a bijection between the surjective arrangements and the arbitrary smaller arrangements. Using our result from above, there are $\binom{(n-m)+m-1}{m-1} = \binom{n-1}{m-1} = \binom{n-1}{n-m}$ such arrangements.

To complete this discussion, we present two applications that require us to carefully consider which objects serve as balls and as bins.

Example 22.2 A donut shop makes 7 different donut flavors. We'll count how many different boxes of a dozen (12) donuts are possible. Here, we're assuming that the position of the donuts in the box is irrelevant; we only care about how many of each flavor the box contains. We can view the $n = 12$ spots in the box as the "balls" and the $m = 7$ flavors as the bins since a donut arrangement tells how many of the spots are occupied by each flavor (read this carefully, as it's easy to confuse which is which!). Thus, there are $\binom{12+7-1}{7-1} = \binom{18}{6} = 18,564$ different boxes possible.

Suppose that there is an additional restriction that one donut of each flavor must be included. This stipulates that the map is surjective. Using our formula, there are $\binom{12-1}{7-1} = \binom{11}{6} = 462$ such boxes.

Example 22.3 We'll count the number of non-negative integer solutions to the equation $x+y+z=7$. Here, we think of the "balls" as the 7 units on the right side and the bins as the variables x, y and z. Applying our formula, there are $\binom{7+3-1}{3-1} = \binom{9}{2} = 36$ such solutions.

So far, we can fill in the following entries of our chart. We will finish filling out this chart in the next lecture.

n Balls	m Bins	Arrangements	Injective Arrangements	Surjective Arrangements
Distinguishable	Distinguishable	m^n	$n! \cdot \binom{m}{n}$	
Indistinguishable	Distinguishable	$\binom{n+m-1}{m-1}$	$\binom{m}{n}$	$\binom{n-1}{m-1}$
Distinguishable	Indistinguishable			

Main Takeaways:

- The balls and bins framework is a useful framework for reasoning about the number of functions with particular properties.
- The *stars and stripes* technique is used to reason about the setting with indistinguishable balls and distinguishable bins.
- When counting objects using the balls and bins framework, the first step is to (carefully) identify which items are the balls and which items are the bins.

Exercises

22.1 Consider a setting with $n = 4$ balls and $m = 4$ distinguishable bins.

(a) If the balls are distinguishable, how many arrangements are possible? Injective arrangements? Surjective arrangements (although we don't yet have a formula for this, it shouldn't be too hard to calculate this with the techniques we've learned)?

(b) If the balls are indistinguishable, how many arrangements are possible? Injective arrangements? Surjective arrangements?

(c) Repeat parts (a) and (b) with $n = 3$ balls and $m = 4$ bins.

(d) Repeat parts (a) and (b) with $n = 4$ balls and $m = 3$ bins.

22.2 In this question, we'll consider counting solutions to the equation $w + x + y + z = 22$ with $w, x, y, z \in \mathbb{N}$.

(a) Explain how we can interpret this as a balls and bins problem.

(b) Calculate the number of solutions to this equation.

(c) Calculate the number of solutions with the condition that $w, x, y, z > 0$.

(d) Calculate the number of solutions with the condition that $w, x, y, z > 1$.

(e) Calculate the number of solutions with the conditions $w > 0$, $x > 2$, $y > 1$, $z > 3$.

(f) Calculate the number of solutions with the condition $w \leq 3$.

22.3 Continuing in the spirit of the previous question, we'll consider counting the natural number solutions to equations of the form $x_1 + \ldots + x_m = n$ with upper bounds on each variable. To handle these restrictions, we'll use Inclusion-Exclusion.

As an example of this technique, we'll consider counting the natural number solutions to $x+y+z = 25$ with $x,y,z \leq 10$.

(a) How many solutions does this equation have without the restriction that $x,y,z \leq 10$?

(b) We'll use the bad events version of the Inclusion-Exclusion Rule. What are good choices for three bad events?

(c) Compute the size of each possible intersection of your bad events from part (c).

(d) Plug into the Inclusion-Exclusion formula to count the number of suitable solutions.

22.4 In this exercise, you'll apply the technique that you developed in the previous question to a new scenario.

At the end of the farmers' market, a stand has 5 peaches, 6 plums, 8 apples, and 4 pears. How many ways can they pack a basket of 12 fruits?

22.5 Let F_n denote the set of functions $[n] \to [n]$, and define the relation $\approx_R$ (R for right composition) on F_n such that

$$f_1 \approx_R f_2 \quad \Leftrightarrow \quad \text{there is a bijection } g\colon [n] \to [n] \text{ such that } f_2 = f_1 \circ g.$$

(a) Argue that $\approx_R$ is an equivalence relation on F_n.

(b) Argue that all injective (equivalently, surjective or bijective since the domain and codomain are the same) functions in F_n belong to the same equivalence class of $\approx_R$.

(c) Explain how we reframe calculating $|F_n/\approx_R|$ as a balls and bins problem.

(d) Find a closed formula for $|F_n/\approx_R|$ in terms of n. Verify that your formula is correct by describing all equivalence classes for $n = 1,2,3,4$.

22.6 A taqueria sells a combo of 4 make-your-own tacos. On each taco, the customer selects one of five proteins (chicken, steak, carnitas, fish, tofu), one of three salsas (mild, medium, hot), and any number of four other toppings (lettuce, cheese, sour cream, guacamole). We wish to calculate how many different combos are possible (assuming that the order that which the tacos are selected does not matter).

(a) Show that there are 240 possible ways to select the ingredients in a single taco.

Hector uses the Division Rule to calculate the total number of taco combos. He reasons:

> *There are 240 ways to select toppings for each taco. Thus, 240^4 counts the number of ways to select toppings for four tacos. Since we do not care about the order of the tacos, we should divide this by 4!, the number of permutations of the tacos. We find that there are $\frac{240^4}{4!} = 138{,}240{,}000$ possible combos.*

(b) Identify the mistake in Hector's reasoning. Has Hector over-counted or under-counted the number of combos?

(c) Compute, with explanation, the correct number of possible combos.

22.7 In the lecture, we considered only two possibilities for the balls: that all balls were distinguishable (unique) and that all balls were indistinguishable (identical). More generally, we could have *partial distinguishability*. For example, some of the balls can be red and some can be blue, and all balls with the same color are indistinguishable. While this is very tricky in general, we'll consider a special case in this exercise.

(a) How many arrangements are there of 4 red balls and 5 blue balls into 3 distinguishable bins?

(b) How many of these arrangements place at least 1 red ball in each bin?

(c) How many of these arrangements place at least 1 red and 1 blue ball in each bin?

(d) How many of these ways include at least 1 ball (red or blue) per bin?

Hint: Use Inclusion-Exclusion

23. The Stirling Numbers

In the previous lecture, we introduced the balls and bins framework for counting functions with certain properties. We derived most of the formulas in the case of distinguishable bins. Today, we will focus mainly on the case of distinguishable balls and indistinguishable bins. To handle this case, we will introduce a new important family of numbers known as the *Stirling Numbers (of the second kind)*.

23.1 Distinguishable Balls, Indistinguishable Bins

Recall that we use n to denote the number of balls. Since they are distinguishable, we imagine that each of the balls has a different color. We use m to denote the number of bins. As the bins are indistinguishable, the only thing that differentiates two arrangements is the sets of contents of the bins. We are not concerned with which set of balls ends up in which bin. For example, we would deem the following two arrangements to be identical since each has one empty bin, one bin with just a pink ball, and one bin with a gray ball and a red ball.

In this setting, the number of injective arrangements is straightforward. When there are more balls than bins, the answer is again 0 by the Pigeonhole Principle. When there are at least as many bins as balls, the answer is 1. We can't tell the bins apart, so the only injective arrangement is one that places each ball in its own (indistinguishable) bin!

Next, we'll reason about the surjective arrangements, as this reasoning will be needed for the general arrangements. We'll introduce the notation $\left\{ {n \atop m} \right\}$ to count these arrangements. These numbers $\left\{ {n \atop m} \right\}$ are called *Stirling Numbers (of the second kind)*.

Definition 23.1 — Stirling Numbers.
Given $m,n \in \mathbb{N}$, the notation $\left\{ n \atop m \right\}$, called a *Stirling Number (of the second kind)*, represents the number of ways to arrange n distinguishable balls into m indistinguishable bins such that no bin remains empty.

23.1.1 Stirling Number Recurrence Relation

To start, we will develop a recurrence relation for the Stirling Numbers. Later in the lecture, we will derive a closed formula. In our recurrence relation, we would like to express $\left\{ n \atop m \right\}$ in terms of other Stirling numbers with *smaller* entries. As the smallest possible entries are 0s, our base cases will need to define all Stirling Numbers that include at least one 0 entry. We do this with three base cases:

- $\left\{ 0 \atop 0 \right\} = 1$: there is one way to place no balls into no bins,
- $\left\{ 0 \atop i \right\} = 0$ for all $i > 0$: the bins are empty, so the arrangement is not surjective,
- $\left\{ i \atop 0 \right\} = 0$ for all $i > 0$: there is no bin in which to place a ball.

For the recurrence, we consider finding an expression for $\left\{ n \atop m \right\}$ in the case that $n,m \geq 1$. Since the balls are distinguishable, we can focus our attention on one particular ball (say the red ball). There are two possibilities for this ball:

Case 1: The red ball is in a bin with other balls.

In this case, if we remove the red ball, we still have a surjective arrangement. Thus, we can enumerate these arrangements using a two-step process. First, we select a surjective arrangement of $n-1$ balls (excluding the red ball) into the m bins. By definition, there are $\left\{ n-1 \atop m \right\}$ ways to do this. Note that after this step, there is at least one (distinguishable) ball in each bin, so the bins are now distinguishable by their contents. Next, we select one of these (now distinguishable) m bins to place the red ball. By the Multiplication Rule, we see that there are $m \cdot \left\{ n-1 \atop m \right\}$ arrangements in this case.

Case 2: The red ball is the only ball in its bin.

In this case, the other $n-1$ balls must be arranged among the remaining $m-1$ bins such that each of these bins contains at least one ball. Referring back to the definition, we see that the number of such arrangements is given by the Stirling Number $\left\{ n-1 \atop m-1 \right\}$.

Putting the two cases together, we obtain the recurrence:

$$\left\{ n \atop m \right\} = m \cdot \left\{ n-1 \atop m \right\} + \left\{ n-1 \atop m-1 \right\}.$$

This formula is very reminiscent of Pascal's Formula for the binomial coefficients, just with an extra factor of m on the first term. Before we find a closed formula for the Stirling Numbers, we will finish filling out the remaining entries in our balls and bins chart.

23.1.2 General Arrangements

In a general arrangement of distinguishable balls into indistinguishable bins, we must allow for the case that one or more of the bins remain empty. We can count these arrangements by using a two-step process.

1. Select the number k of bins that we will place balls in. We allow k to be any value between 0 and m.
2. Select a surjective arrangement of the n balls into k bins. By definition, this can be done in $\left\{ {n \atop k} \right\}$ ways.

As the number of possibilities in step 2 is different for different values of k, we must use the Addition Rule to aggregate these counts. We find that the number of general arrangements is

$$\sum_{k=0}^{m} \left\{ {n \atop k} \right\}.$$

23.1.3 Surjective Arrangements with Distinguishable Bins

We can also count surjective arrangements of distinguishable balls into distinguishable bins using a two-step process.

1. Select a surjective arrangement of n distinguishable balls into m indistinguishable bins. By definition, this can be done in $\left\{ {n \atop m} \right\}$ ways.
2. Label each of the bins with numbers from $[m]$. Note that each of the bins is distinguishable since it contains at least one distinguishable ball. Thus, there are $m!$ distinct ways to carry out this labeling (regardless of how the balls are arranged).

By the Multiplication Rule, we conclude that there are $m! \cdot \left\{ {n \atop m} \right\}$ such arrangements. At this point, we have found all of the entries in our balls and bins table.

n Balls	m Bins	Arrangements	Injective Arrangements	Surjective Arrangements
Distinguishable	Distinguishable	m^n	$n! \cdot \binom{m}{n}$	$m! \cdot \left\{ {n \atop m} \right\}$
Indistinguishable	Distinguishable	$\binom{n+m-1}{m-1}$	$\binom{m}{n}$	$\binom{n-1}{m-1}$
Distinguishable	Indistinguishable	$\sum_{k=0}^{m} \left\{ {n \atop k} \right\}$	$\begin{cases} 1 & n \le m \\ 0 & n > m \end{cases}$	$\left\{ {n \atop m} \right\}$

23.2 A Closed Formula for the Stirling Numbers

In order to derive a closed formula for the Stirling Numbers, we will count the number of surjective arrangements of distinguishable balls into distinguishable bins (i.e., the upper right cell of the table) in a different way using Inclusion-Exclusion. In particular, we will make use of the "bad event" formulation.

We'll let S be the set of all (not necessarily surjective) arrangements of n distinguishable balls into m distinguishable bins. Since we desire a surjective arrangement, we will define bad events

$S_1, \ldots, S_m$ so that S_i includes the arrangements that leave bin i empty. By the Inclusion-Exclusion Rule, we have

$$m! \cdot \left\{ n \atop m \right\} = \left| \bigcap_{i \in [m]} (S \setminus S_i) \right| = |S| + \sum_{j=1}^{m} (-1)^j \sum_{\substack{I \subseteq [m] \\ |I| = j}} \left| \bigcap_{i \in I} S_i \right|$$

Now, we note that

- The total number of arrangements $|S|$ is m^n.
- For each singleton set $I = \{i\}$, the arrangements in S_i leave bin i empty but can arbitrarily arrange the n balls among the remaining $m-1$ bins. Hence, $|S_i| = (m-1)^n$.
- For each two-element set $I = \{i, j\}$, the arrangements in $S_i \cap S_j$ leave bins i and j empty but can arbitrarily arrange the n balls among the remaining $m-2$ bins. Hence, $|S_i \cap S_j| = (m-2)^n$.
- By the same logic, any j-way intersection of bad event sets has cardinality $(m-j)^n$.

Plugging these observations into our formula from above, we find that

$$\begin{aligned} m! \cdot \left\{ n \atop m \right\} &= m^n + \sum_{j=1}^{m} (-1)^j \cdot \binom{m}{j} \cdot (m-j)^n \\ &= \sum_{j=0}^{m} (-1)^j \cdot \binom{m}{j} \cdot (m-j)^n. \end{aligned}$$

Dividing through by $m!$, we obtain our final closed formula:

$$\left\{ n \atop m \right\} = \frac{1}{m!} \sum_{j=0}^{m} (-1)^j \cdot \binom{m}{j} \cdot (m-j)^n.$$

Main Takeaways:

- The *Stirling Numbers* $\left\{ n \atop m \right\}$ count the number of surjective arrangements of n distinguishable balls into m indistinguishable bins.
- By reasoning about whether one particular ball is in its own bin, we can obtain a recurrence relation for the Stirling Numbers.
- We can use Inclusion-Exclusion to obtain a closed formula for the Stirling Numbers.

Exercises

23.1 We could have added a fourth column to our chart to count the number of bijective arrangements; however, we omitted it because the answers are rather uninteresting. Find, with explanation, the count of bijective arrangements in each of the three cases we considered.

23.2 Fill out the complete balls and bins chart for the following values of n and m. (Many of the entries were computed in Exercise 27.1.)

(a) $n = 4$, $m = 4$

(b) $n = 3$, $m = 4$

(c) $n = 4$, $m = 3$

23.3 Let F_n denote the set of functions $[n] \to [n]$, and define the relation $\approx_L$ (L for left composition) on F_n such that

$$f_1 \approx_L f_2 \quad \Leftrightarrow \quad \text{there is a bijection } g\colon [n] \to [n] \text{ such that } f_2 = g \circ f_1.$$

(a) Argue that $\approx_L$ is an equivalence relation on F_n.

(b) Argue that all injective (equivalently, surjective or bijective since the domain and codomain are the same) functions in F_n belong to the same equivalence class of $\approx_L$.

(c) Explain how we reframe calculating $|F_n/\approx_L|$ as a balls and bins problem.

(d) Find a closed formula for $|F_n/\approx_L|$ in terms of n. Verify that your formula is correct by describing all equivalence classes for $n = 1, 2, 3, 4$.

23.4 Find, with explanation, simpler formulas for the following Stirling numbers. (Hint: think of their combinatorial interpretation.)

(a) $\left\{ {n \atop 1} \right\}$

(b) $\left\{ {n \atop 2} \right\}$

(c) $\left\{ {n \atop n-1} \right\}$

(d) $\left\{ {n \atop n-2} \right\}$

23.5 In this question, we'll walk through how to use the recurrence relation for the Stirling Numbers to give an inductive proof of the closed formula

$$\left\{ {n \atop m} \right\} = \frac{1}{m!} \sum_{j=0}^{m} (-1)^j \cdot \binom{m}{j} \cdot (m-j)^n.$$

(a) First, argue that the three base cases satisfy the closed formula. Rather than arguing the second two cases by induction, you can argue that they hold for an *arbitrary* $i > 0$.

⋆ (b) Rather than performing a double induction on n and m, we will use ordinary induction on n only. Our predicate $P(n)$ will be that the Stirling number formula is correct for $\left\{ {n \atop m} \right\}$ for all $m \in \mathbb{N}$. We have already established the base case in part (a). For the inductive step, assume $P(n-1)$ is true. We must argue $P(n)$ is true. To do this, it suffices to verify the Stirling number formula is correct for $\left\{ {n \atop m} \right\}$ for an *arbitrary* $m \geq 1$ ($m = 0$ was handled in part (a)). Complete this argument (which is primarily algebra).

⋆ **23.6** Over the past two lectures, in the case of distinguishable balls, we've considered only two possibilities for the bins: that all bins were distinguishable (unique) and that all bins were indistinguishable (identical). More generally, we could have partial distinguishability. For example, some of the bins can be red and some can be blue, and all bins with the same color are indistinguishable. While this is very tricky in general, we'll consider a special case in this exercise.

(a) How many *surjective* arrangements are there of 8 distinguishable balls into 2 red bins and 3 blue bins?

(b) How many arrangements are there of 8 distinguishable balls into 2 red bins and 3 blue bins which place at least one ball in each *red* bin?

(c) How many arrangements are there of 8 distinguishable balls into 2 red bins and 3 blue bins which place at least one ball in each *blue* bin?

(d) How many ways are there to arrange 8 distinguishable balls into 2 red bins and 3 blue bins without any additional restrictions?

23.7 The *Bell numbers* B_n count the number of partitions of a set of n (distinguishable) items. Equivalently, B_n is the number of general arrangements of n distinguishable items into n indistinguishable bins.

(a) Use the balls and bins chart to write a closed formula for B_n.

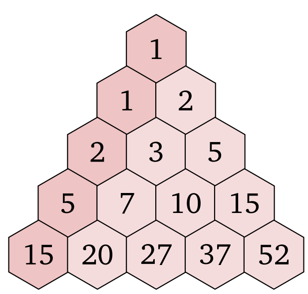
(b) Argue that the Bell Numbers satisfy the recurrence relation

$$B_0 = 1 \qquad\qquad B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k.$$

Hint: Let k be the number of balls that remain if we take away the bin containing ball n.

(c) Calculate B_n for $n \in [5]$.

We can calculate the Bell Numbers using the *Bell Triangle*. The first few rows are shown below.



The Bell numbers appear on the left edge of the triangle. To build the triangle, we use the following rules:

- Write in the two "1" entries at the top of the triangle.
- To find the next (light) cell in a row, add the numbers to its left and above it on the left.
- Once you reach the end of a row, copy that cell to the first (dark) cell in the next row.

⋆ (d) Give an inductive argument for why this procedure correctly computes the Bell numbers.

Hint: Argue that the m'th cell in the n'th row of the triangle (starting both counts from 0) contains the value

$$\sum_{j=0}^{m} \binom{m}{m-j} B_{n-j}.$$

Then use the recurrence relation from part (b) and the properties of binomial coefficients.

23.8 How many equivalence relations are there on a set S with n elements?

23.9 Give combinatorial proof that

$$\left\{ {n \atop m} \right\} = \sum_{k=m-1}^{n-1} \binom{n-1}{k} \left\{ {k \atop m-1} \right\}.$$

Hint: Our definition of the Stirling Numbers tells us what the left side counts. How can the right side describe another process to count the same set? To start, figure out what the variable k can represent.

VI Probability and Statistics

Probability in Computer Science

In the final section of the course, we will turn our attention to the study of probability. The field of probability centers around understanding the likelihood of different results that come about through random processes. While we will start off by examining some common toy examples in probability (flipping coins, drawing cards, and rolling dice), we will push much farther, exploring some of the strange, often counter-intuitive, but beautiful phenomena in this field. You may ask why learning probability is important for a computer scientist. Where are these ideas used? Here is a short, by no means comprehensive, list of some key applications of probability:

1. In the area of algorithm design, incorporating randomness in algorithms can increase their efficiency without sacrificing on accuracy. Quicksort, one of the *Top 10 Algorithms from the 20th Century*,[1] is a sorting algorithm that uses random decisions and has exceptional (expected) performance. The analysis of randomized algorithms like Quicksort requires an understanding of random variables and expectation. While we have similar efficient algorithms for sorting, there are many other problems whose best-known solutions require the use of randomness.

2. At their core, many machine learning approaches utilize ideas from probability and statistics. Theoretical guarantees for these methods, such as understanding the convergence of neural networks (i.e., how much training is required to obtain good predictive power) requires an analysis based on probabilistic tools like Chernoff bounds. We will discuss many probability bounds later in this section.

3. The burgeoning of big data applications gives technology companies access to vast amounts of information on which to draw conclusions. Having a working knowledge of probability and statistics allows one to think critically about the way that data are presented.

4. Although this is largely beyond the scope of our course, a lot of researchers in discrete mathematics rely on (continuous) probability as a tool to obtain deterministic results about the objects that they study. This approach, known as the *probabilistic method*, is

[1] Dongarra, J., & Sullivan, F. (2000). Guest Editors Introduction to the top 10 algorithms. Computing in Science & Engineering, 2(01), 22-23.

quite powerful, and can be used to argue about the existence of mathematical objects (such as graphs or encoding schemes) with particular properties, even when such objects are difficult to write down explicitly.

In its entirety, probability is a vast field that has deep connections with many areas of mathematics, including combinatorics, geometry, and analysis. We will barely scratch the surface in our whirlwind introduction, focusing mostly on understanding some fundamental terminology and classic examples. We'll begin by defining *probability spaces*, the setting where we study probability. With this architecture in place, we can build up the concepts of outcomes, events, and random variables, which help us to understand and quantify random experiments. We'll learn about some common distributions and their various properties such as expectation, variance, and correlation. Finally, we'll discuss some techniques to reason about the likelihood of observing certain phenomena and how we can use this to design statistical tests.

24. Probability Spaces

When you think about probability, your mind likely goes to some elementary examples of random chance: picking a random card from a deck, flipping a coin, or rolling dice. In each of these settings, there is some underlying randomness that precludes us from knowing ahead of time what will happen (which card will be selected, which side the coin will land on, or which number will be rolled). Thus, each of these scenarios realizes a *random experiment*, a situation whose result is dictated by randomness. In order to study random experiments, we will need to introduce a setting where we can formalize and quantify this randomness. We do this through the definition of a *probability space*, which we introduce in today's lecture.

24.1 Probability Spaces

As our motivating example for a probability space, we will think about drawing a card from a shuffled deck[1]. Here, the randomization comes from the fact that we do not know the order of the cards at the time that we must make our selection. Think about the time between when we pull our chosen card out of the deck and when we turn it over to look at it. At this point, we have no way of knowing which card we have chosen (the *outcome* of the experiment). There are 52 possibilities, one for each of the cards in the deck. This collection of possibilities comprises the *sample space* of our random experiment.

Definition 24.1 — Outcome, Sample Space.

The *outcomes* of an experiment correspond to the possible results after the randomness has been realized. The set of all possible outcomes of the random experiment is its *sample space*, often denoted Ω (Omega).

[1]In this course, when we refer to a deck of cards, we are referencing the 52-card *French-suited* deck. There are four suits of cards: spades ($\spadesuit$), clubs ($\clubsuit$), hearts ($\heartsuit$), and diamonds ($\diamondsuit$). The spades and clubs are "black" suits and the hearts and diamonds are "red" suits. In each suit, there are 13 cards of different *ranks*. There are nine *number cards* $2, 3, 4, 5, 6, 7, 8, 9, 10$; three *face cards* jack (J), queen (Q), and king (K); and an ace (A). Each card is named by the combination of its suit and its rank. For example, one card is the "seven of clubs" ($7\clubsuit$).

Using this new notation, the sample space of our card-drawing experiment is

$$\Omega = \{2\heartsuit, 3\heartsuit, \dots, K\heartsuit, A\heartsuit, 2\clubsuit, \dots, A\clubsuit, 2\diamondsuit, \dots, A\diamondsuit, 2\spadesuit, \dots, A\spadesuit\}.$$

While we may be interested in knowing whether a particular card is drawn, we can also think of settings where we may only need to know about certain features of the card we have selected. For example,

- Did we select a red card?
- Is the card we selected from the clubs suit?
- Did we select a face card?

Each of these questions partitions the sample space into two subsets: the set of cards for which the answer to the question is "yes", and the set for which the answer is "no." We are interested in knowing whether the outcome of the experiment falls within the "yes" subset. We refer to these subsets as *events*.

Definition 24.2 — Event, Event Space.

An *event* A is a subset of the sample space, $A \subseteq \Omega$. The *event space*, usually denoted $\mathcal{F}$, contains all possible events of our experiment. Note that $\mathcal{F} \subseteq \mathcal{P}(\Omega)$, and must have the following properties:

1. $\Omega \in \mathcal{F}$
2. $\mathcal{F}$ is *closed under complements*. That is, if $A \in \mathcal{F}$, then $(\Omega \setminus A) \in \mathcal{F}$
3. $\mathcal{F}$ is *closed under countable unions*. That is, if $A_0, A_1, \dots \in \mathcal{F}$, then $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}$

Note that properties 1 and 2 together ensure that $\varnothing = \Omega \setminus \Omega$ is an element of $\mathcal{F}$. By letting $A_i = \varnothing$ in property 3 for $i = 0$ and all $i \geq n$ for some $n \in \mathbb{N}$, we can see that the union of any finite number of sets in $\mathcal{F}$ belongs to $\mathcal{F}$.

The study of probability is concerned with reasoning about the likelihood of different events. When Ω is a countable set (in which case, we say that we are working on a *discrete probability space*), we will often consider every subset of Ω to be an event; that is, we'll take $\mathcal{F} = \mathcal{P}(\Omega)$: note that this satisfies the three required properties. Later in today's lecture, we'll look at an example where an alternate choice of the sample and event spaces is preferable.

In our card example, letting $\mathcal{F} = \mathcal{P}(\Omega)$ results in 2^{52} events corresponding to the 2^{52} different subsets of cards.

Example 24.1 To formalize our events from earlier,

- If A_1 represents the event of selecting a red card, then
$$A_1 = \{2\heartsuit, 3\heartsuit, \dots, K\heartsuit, A\heartsuit, 2\diamondsuit, \dots, A\diamondsuit\}.$$
- If A_2 represents the event of selecting a card from a clubs suit, then
$$A_2 = \{2\clubsuit, 3\clubsuit, 4\clubsuit, 5\clubsuit, 6\clubsuit, 7\clubsuit, 8\clubsuit, 9\clubsuit, 10\clubsuit, J\clubsuit, Q\clubsuit, K\clubsuit, A\clubsuit\}$$

- If A_3 represents the event of selecting a face card, then

$$A_3 = \{J\diamondsuit, Q\diamondsuit, K\diamondsuit, J\clubsuit, Q\clubsuit, K\clubsuit, J\heartsuit, Q\heartsuit, K\heartsuit, J\spadesuit, Q\spadesuit, K\spadesuit\}$$

Now that we have the vocabulary to discuss an experiment's outcomes and events, we can think about how to quantify probability; that is, how to assign a measure of likelihood to each event. We will do this using a function called a *probability function*.

Definition 24.3 — Probability Function.

A probability function $\Pr\colon \mathcal{F} \to \mathbb{R}_{\geq 0}$ describes the likelihood that the outcome of a random experiment belongs to a given event. A probability function must satisfy the following two properties:

1. **Total Probability**: $\Pr(\Omega) = 1$
2. **Countable Additivity**: Given a countable collection of *pairwise disjoint* events $A_0, A_1, \ldots$:

$$\Pr\Big(\bigcup_{i\in\mathbb{N}} A_i\Big) = \sum_{i\in\mathbb{N}} \Pr(A_i).$$

By letting $A_i = \varnothing$ in property 2 for $i = 0$ and all $i \geq n$ for some $n \in \mathbb{N}$ (note that $\varnothing$ is disjoint from every set by definition), we can obtain a statement about the probability of a union of finitely many disjoint events:

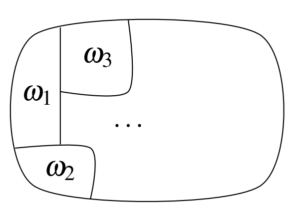
$$\Pr\Big(\bigcup_{i\in[n]} A_i\Big) = \sum_{i\in[n]} \Pr(A_i).$$

In the exercises, we ask you to verify that the probability of any event is between 0 and 1. This number represents how likely the event is: events with probability 0 will (almost) surely never happen, while events with probability 1 will (almost) surely happen.

When Ω is countable and $\mathcal{F} = \mathcal{P}(\Omega)$, it suffices to define Pr for each singleton event $\{\omega\}$ for $\omega \in \Omega$. We can then use the countable additivity property to compute the probability of any event $A \subseteq \Omega$. For notational convenience, we will often abbreviate the probability of a singleton event by omitting the brackets: $\Pr(\{\omega\}) = \Pr(\omega)$.

One way to visualize the probability function is to think of Ω as occupying one unit of area. Then, Pr tells us how much area each outcome $\omega \in \Omega$ takes up in Ω.

Ω :



Returning to our card example, if we assume that the cards were shuffled well, then we are equally likely to draw any of the cards. In this case, $\Pr(\omega) = \frac{1}{52}$ for each card ω in the deck. This is referred to as a *uniform* distribution over the cards and is a good default assumption. However, this is not the only possibility. Suppose that the deck is brand new, so in the default order in its box (say, $A\spadesuit, 2\spadesuit, \ldots, K\spadesuit, A\clubsuit, \ldots, K\clubsuit, A\heartsuit, \ldots, K\heartsuit, A\diamondsuit, \ldots, K\diamondsuit$). If we choose the top card, we are guaranteed (with probability 1) to have the ace of spades. Thus the probability function provides crucial information for calculating the likelihood of various events.

Definition 24.4 — Probability Space.
A probability space is a triple $(\Omega, \mathcal{F}, \Pr)$ consisting of a sample space Ω, an event space $\mathcal{F} \subseteq \mathcal{P}(\Omega)$, and a probability function $\Pr\colon \mathcal{F} \to \mathbb{R}_{\geq 0}$.

In any setting where we wish to reason about probability, the first thing we should do is identify these three components of the probability space in which we are working.

24.2 Calculating Probabilities

As we noted above, when Ω is countable and $\mathcal{F} = \mathcal{P}(\Omega)$, Pr is completely described by probabilities of the singleton outcome sets $\Pr(\omega)$ for $\omega \in \Omega$; Countable additivity gives us that

$$\Pr(A) = \sum_{\omega \in A} \Pr(\omega)$$

for each event $A \subseteq \Omega$. A particularly nice case is a uniform probability function, where $\Pr(\omega) = \frac{1}{|\Omega|}$ for each $\omega \in \Omega$. This allows us to simplify the above formula to

$$\Pr(A) = \frac{|A|}{|\Omega|}.$$

Therefore, the problem of calculating a probability reduces to two counting problems: we must count the sample space, and then we must count the event (that is, the elements of the sample space with a certain additional property). For this counting, we can utilize all of our techniques from combinatorics such as the Multiplication Rule, Addition Rule, etc.

Example 24.2 Suppose that we draw a card uniformly at random from the deck. Then,

- The probability of selecting a red card is $\Pr(A_1) = \frac{|A_1|}{|\Omega|} = \frac{26}{52} = \frac{1}{2}$.
- The probability of selecting a club is $\Pr(A_2) = \frac{|A_2|}{|\Omega|} = \frac{13}{52} = \frac{1}{4}$.
- The probability of selecting a face card is $\Pr(A_3) = \frac{|A_3|}{|\Omega|} = \frac{12}{52} = \frac{3}{13}$.

Let's look at a couple more examples of discrete probability spaces.

Example 24.3 A teacher selects three of five students – Andrew (a), Brianna (b), Caleb (c), Diego (d), and Eleanor (e) for an activity. Each subset of students is equally likely to be selected. We can model this random experiment using a probability space $(\Omega, \mathcal{F}, \Pr)$, where Ω consists of all $\binom{5}{3} = 10$ subsets of three students, $\mathcal{F} = \mathcal{P}(\Omega)$, and Pr has the uniform distribution so that $\Pr(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{10}$ for all $A \subseteq \Omega$. Then,

- To calculate the probability that Brianna is selected, note that the subsets containing Brianna are in bijective correspondence with the ways to select (the other) two students from $\{a, c, d, e\}$. Thus, the probability that Brianna is selected is $\frac{\binom{4}{2}}{10} = \frac{6}{10}$.

- To calculate the probability that Caleb and Eleanor are both selected, note that the subsets containing Caleb and Eleanor are in bijective correspondence with the ways to select one more student from $\{a,b,d\}$. Thus, the probability that Caleb and Eleanor are both selected is $\frac{\binom{3}{1}}{10} = \frac{3}{10}$.
- To calculate the probability that Diego is *not* selected, note that the subsets without Diego are exactly the subsets of $\{a,b,c,e\}$ with size 3. Thus, the probability that Diego is not selected is $\frac{\binom{4}{3}}{10} = \frac{4}{10}$.

Example 24.4 A fair coin is flipped until it lands on heads for the first time. We can model Ω as the set of strings $\left\{w \in \{H,T\}^* \mid w = T^iH \text{ for some } i \in \mathbb{N}\right\}$. As our event space, we may select $\mathcal{F} = \mathcal{P}(\Omega)$. Our probability measure is no longer uniform, though. Since the coin is fair, it is equally likely to land on heads or tails on each flip. Therefore, $\Pr(H) = \Pr(T^0H) = \frac{1}{2}$. One-fourth of the time, the coin will land tails on the first flip and then heads on the second flip, so $\Pr(TH) = \Pr(T^1H) = \frac{1}{4}$. Similarly, we find that $\Pr(T^iH) = \frac{1}{2^{i+1}}$ for all $i \in \mathbb{N}$. Using countable additivity, we can confirm that

$$\Pr(\Omega) = \Pr\Big(\bigcup_{i\in\mathbb{N}}\{T^iH\}\Big) = \sum_{i\in\mathbb{N}} \frac{1}{2^{i+1}} = \frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}} = \frac{1}{2}\cdot 2 = 1.$$

Here, we have used the infinite geometric series formula to simplify this infinite sum. Now, what is the probability that the coin is flipped an odd number of times? We may calculate this

$$\Pr\Big(\bigcup_{i\in\mathbb{N}}\{T^{2i}H\}\Big) = \sum_{i\in\mathbb{N}} \frac{1}{2^{2i+1}} = \frac{1}{2}\sum_{i\in\mathbb{N}}\frac{1}{4^i} = \frac{1}{2}\cdot\frac{1}{1-\frac{1}{4}} = \frac{1}{2}\cdot\frac{4}{3} = \frac{2}{3}.$$

Example 24.5 We'll consider the first roll in a game of Yahtzee. Five indistinguishable (fair) dice are rolled. One way that we can model this experiment is to consider the sample space of all multisets of $\{1,2,3,4,5,6\}$ containing 5 elements.

$$\Omega = \Big\{ \{\!\!\{1,1,1,1,1\}\!\!\}, \{\!\!\{1,2,3,4,5\}\!\!\}, \{\!\!\{2,2,3,3,3\}\!\!\}, \{\!\!\{1,2,2,4,6\}\!\!\}, \dots \Big\}.$$

Working with this sample space is difficult though. First of all, the size of $|\Omega|$ is not obvious (try calculating it). Beyond this, although the outcome of each die is uniform, these outcomes do not occur with equal probability. For example, there is only 1 way to roll $\{\!\!\{1,1,1,1,1\}\!\!\}$, but there are 5 ways to roll $\{\!\!\{1,1,1,1,2\}\!\!\}$ (any of the five dice can be the 2).

To make the setting easier, we'll "cheat" by expanding the sample space to all *sequences* of five dice rolls, $\Omega' = [6]^5$. In a way, we are making the dice distinguishable by having each coordinate refer to the number rolled by a particular die. Now, the induced distribution on this enlarged sample space is uniform, and we can do our calculations here. One caveat is that we must make sure the events we consider are expressible in our original sample space. That is, we cannot take $\mathcal{F} = \mathcal{P}(\Omega')$. Instead, $\mathcal{F}$ must have the property that for any $A \in \mathcal{F}$, if the sequence $(x_1,x_2,x_3,x_4,x_5) \in A$, then $(x_{\sigma(1)},x_{\sigma(2)},x_{\sigma(3)},x_{\sigma(4)},x_{\sigma(5)})$ is also in A for any permutation σ of $[5]$. (Note that this choice of $\mathcal{F}$ satisfies the required properties.) By doing this, we convert to a probability space with the uniform distribution, so that $\Pr(A) = \frac{|A|}{|\Omega'|} = \frac{|A|}{6^5} = \frac{|A|}{7776}$. Thus, we can calculate:

- The probability that at least one 1 is rolled: This is the *complementary event* (set complement in Ω) to the event that no 1 is rolled, which is easier to reason about. No 1 is rolled exactly when all 5 dice show one of the other five numbers. By the Multiplication Rule, there are 5^5 rolls that don't include a 1. Hence, our desired probability is $1 - \frac{5^5}{6^5} = \frac{4651}{7776} \approx 0.60$.
- The probability that five different numbers are rolled: We again use the Multiplication Rule. There are 6 possibilities for the first die, then 5 for the second die, etc. In total, there are $6! = 720$ rolls in this event, so it has probability $\frac{720}{7776} \approx 0.09$.
- The probability that exactly two 3s are rolled: Here, there are $\binom{5}{2} = 10$ ways to choose these two dice and 5^3 different possibilities for the other 3 dice. Thus, the probability of this event is $\frac{1250}{7776} \approx 0.16$.

Main Takeaways:

- We use a *probability space* $(\Omega, \mathcal{F}, \Pr)$ to model a random experiment. The *sample space* Ω is the set of possible outcomes, the *event space* $\mathcal{F}$ consists of subsets of outcomes that we can measure, and the *probability function* Pr assigns a likelihood to each event.
- The *(discrete) uniform probability distribution* assigns probability $\frac{1}{|\Omega|}$ to each singleton set $\{\omega\}$ with $\omega \in \Omega$, so that $\Pr(A) = \frac{|A|}{|\Omega|}$ for each $A \in \mathcal{F}$.
- When we can find a discrete probability space with a uniform probability distribution, we can convert questions about probability into questions about counting. Otherwise, we need other reasoning to calculate probabilities.

Exercises

24.1 Use the two properties of probability functions given in Definition 24.3 to establish these other properties:

(a) **Null Probability:** $\Pr(\varnothing) = 0$.

(b) **Monotonicity:** If $A \subseteq B$, then $\Pr(A) \leq \Pr(B)$.

(c) **Boundedness:** If $A \in \mathcal{F}$, then $0 \leq \Pr(A) \leq 1$.

(d) **Complement Probability:** If $A \in \mathcal{F}$, then $\Pr(\Omega \setminus A) = 1 - \Pr(A)$.

Since the complementary event comes up a lot in probability, we have a special notation for it: $\overline{A} = \Omega \setminus A$.

24.2 (a) Use the properties in Definition 24.3 to establish the following identity:

$$\Pr(A) + \Pr(B) = \Pr(A \cup B) + \Pr(A \cap B).$$

Since $\Pr(A \cap B) \geq 0$, we can rearrange this to the following inequality, called the *Union Bound*:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) \leq \Pr(A) + \Pr(B).$$

(b) Use induction to establish a generalized form of the union bound for any finite union of events.

⋆ **24.3** Use the two properties of an event space $\mathcal{F}$ given in Definition 24.2 to argue that $\mathcal{F}$ is closed under finite intersections. That is, if $A_1, \ldots A_n \in \mathcal{F}$, then $\bigcap_{i \in [n]} A_i \in F$.

(Hint: Use induction on n. It is more generally true that $\mathcal{F}$ is closed under *countable* intersections, but this is harder to show.)

24.4 Argue that we cannot define a uniform probability function over the natural numbers.

(Hint: Argue this by contradiction, considering what the value of $\Pr(\{1\})$ would be.)

24.5 Although the previous exercise shows that we cannot have a uniform probability function over the natural numbers, we can define a probability space with $\Omega = \mathbb{N}$.

In this question, the infinite geometric series formula will be useful:

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r} \qquad \text{for } -1 < r < 1.$$

Let us consider probability space $(\Omega, \mathcal{F}, \Pr)$ with $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$ and probability function given by $\Pr(n) = 2^{-1-n}$ for all $n \in \mathbb{N}$ and extended by countable additivity. (This should remind you of Example 24.4.) Compute the probability of each of the following events:

(a) $A_1 =$ the chosen number is less than 5.

(b) $A_2 =$ the chosen number is greater than 40.

(c) $A_3 =$ the chosen number is between 17 and 35 (inclusive).

(d) $A_4 =$ the chosen number is odd.

(e) $A_5 =$ the chosen number is not a multiple of three.

24.6 Argue that the restricted event space $\mathcal{F}$ described in Example 24.5 satisfies the properties in Definition 24.2.

24.7 A bag contains 5 marbles of different colors: red, blue, yellow, green, and black. For each of the following random experiments, give a formal description of its sample space, Ω, and compute its cardinality $|\Omega|$.

(a) You reach into the bag and grab two random marbles.

(b) You reach into the bag and grab one marble. Then, you reach into the bag a second time and grab another marble.

(c) You reach into the bag, grab a marble, look at its color, and place it back in the bag. You repeat this a second time.

24.8 Consider again the Yahtzee roll from Example 24.5. Compute the probability that the first roll includes each of the following patterns.

(a) Yahtzee (all five dice show the same number, e.g., $\{\!\{5,5,5,5,5\}\!\}$)

(b) Four of a kind (at least four dice show the same number, e.g., $\{\!\{1,1,1,1,6\}\!\}$)

(c) Three of a kind (at least three dice show the same number, e.g., $\{\!\{2,3,3,3,5\}\!\}$)

(d) Large straight (all five dice form a contiguous sequence, e.g., $\{\!\{1,2,3,4,5\}\!\}$)

(e) Small straight (four of the dice form a contiguous sequence, e.g., $\{\!\{2,3,3,4,5\}\!\}$)

(f) Full house (three of the dice show the same number, as do the other two dice, e.g., $\{\!\{2,2,2,6,6\}\!\}$)

24.9 Suppose that two (distinct) cards are drawn from a deck such that each pair is chosen with equal probability.

(a) Give a formal description of the probability space $(\Omega, \mathcal{F}, \Pr)$ associated with this random experiment.

Now, calculate the probabilities of each of the following events.

(b) A_1 = both chosen cards are 7s

(c) A_2 = both chosen cards are from the hearts suit

(d) A_3 = both chosen cards are from the same rank

(e) A_4 = both chosen cards are from the same suit

(f) $A_3 \cap A_4$

(g) $A_3 \cup A_4$

(h) A_5 = exactly one of the chosen cards is a face card

(i) A_6 = at least one of the chosen cards is a face card

(j) A_7 = exactly one of the cards is from the spades suit

(k) $A_6 \cap A_7$

(l) $A_6 \cup A_7$

25. Conditioning and Independence

In the previous lecture, we investigated the structure of a probability space and saw how to calculate the probabilities of events by reducing this to a counting problem. While we previously considered one event at a time, today we will focus on the relationships between multiple events as we study conditional probability and independence.

25.1 Conditioning

To motivate conditional probability, we consider two random experiments.

Example 25.1 Two fair dice, one gray and one red, are rolled. The probability that the red die lands on 4 is $\frac{1}{6}$. Now, suppose that before we look at the red die, we notice that the gray die landed on 3. Does this change the probability that the red die is a 4?

Example 25.2 A bag contains six marbles. Three are red and three are gray. You reach into the bag and grab one marble in each hand. The probability that your right hand holds a red marble is $\frac{1}{2}$. Now, suppose that with our right hand still closed, we open our left hand and see a gray marble. Does this change the probability that the right marble is red?

In the first example, it should hopefully be intuitive that the probability does not change; our observation was about a completely separate die. However, the probability does change in the second example; both hands grab a marble from a shared bag, so the colors they pull should be related. To formalize these ideas, we'll consider the sample space of each experiment.

For the dice, our sample space can be represented by the Cartesian product $\Omega = [6] \times [6]$, where the first coordinate represents the value of the red die and the second coordinate represents the value of the gray die. Each of the 36 outcomes is equally likely. We can visualize this with the following chart.

	⚀	⚁	⚂	⚃	⚄	⚅
⚀	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚁	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚂	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚃	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚄	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚅	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

Here, the six cells in the fourth column represent the event that the red die is a 4; the sum of the probabilities of these six cells is $\frac{1}{6}$. Now, what happens when we see that the gray die shows a 3? This allows us to eliminate some of the cells in the sample space as potential outcomes. We now know that the outcome of the experiment must be in the third row of the table.

	⚀	⚁	⚂	⚃	⚄	⚅
⚂	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

In this restricted chart, $\frac{1}{36}$ unit of probability "mass" belongs to the event where the red die is a 4. This is $\frac{1}{6}$ of the total probability mass in the restricted event (the sum of the six remaining cells is $\frac{1}{6}$). Hence, we may conclude that *conditioned on* knowing that the gray die shows a 3, the probability that the red die is 4 is $\frac{1}{6}$.

In general, *conditioning* describes this process of restricting our view of the sample space to reflect new information that we have gained. We introduce a new notation to express these conditional probabilities.

Definition 25.1

Given two events $A, B \in \mathcal{F}$, the conditional probability of A given B, denoted $\Pr(A \mid B)$, is the probability that our outcome belongs to A if we know that it belongs to B. If $\Pr(B) \neq 0$, we can calculate this conditional probability as

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

This definition formalizes our intuition from the dice example. Here, the denominator is the total probability mass of the observed event B (our restricted view), and the numerator is the amount of mass in this restricted view that also belongs to event A. After the conditioning, we are assured that our outcome falls in event B, so we should no longer consider any outcome outside of B.

Now, let's apply these ideas to the marble experiment. Again, we'll start by considering the state space. From the last lecture, we recall that our probability calculations will be easier if we can choose a set of outcomes that are all equally likely. To do this here, we need to be able to distinguish the three marbles of each color. We'll do this by assigning them labels $\{A, B, C\}$. Note that this is merely a tool we'll use to help with our analysis. These labels help us keep track of marbles throughout calculations, but will not factor into our final answer. With this labeling, we can now visualize the state space with the following chart.

L \ R	A	B	C	A	B	C
A		$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$
B	$\frac{1}{30}$		$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$
C	$\frac{1}{30}$	$\frac{1}{30}$		$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$
A	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$		$\frac{1}{30}$	$\frac{1}{30}$
B	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$		$\frac{1}{30}$
C	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	

Here, the rows represent the marble in our left hand, and the columns represent the marble in our right hand. The six diagonal entries are blacked out since the same marble cannot be in both of our hands (here is where we make use of the distinguishing labels). Now, we condition on the marble in our left hand being gray. This restricts our attention to the bottom three rows of the chart.

L \ R	A	B	C	A	B	C
A	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$		$\frac{1}{30}$	$\frac{1}{30}$
B	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$		$\frac{1}{30}$
C	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	$\frac{1}{30}$	

Here, $\frac{9}{30}$ units of probability mass out of the $\frac{15}{30}$ in the restricted view correspond to the event that the right marble is red. Therefore, the conditional probability is

$$\Pr(R \text{ red } \mid L \text{ gray}) = \frac{\frac{9}{30}}{\frac{15}{30}} = \frac{3}{5}.$$

A more straightforward way to arrive at this number is to think about the marbles being pulled from the bag separately. First, we pull out a gray marble with our left hand. In the bag, there remain 3 red and 2 gray marbles, to the probability that we select a red marble with our right hand is $\frac{3}{5}$. Notice that this conditional probability differs from the probability of $\frac{1}{2}$ before conditioning. This relationship between the probability of an event before and after conditioning on another event relates to the property of independence, which we'll study later in the lecture.

Before this, we'll look at one more application of conditional probability.

Example 25.3 You are speaking to some new friends, Alice and Bob, who tell you that they have two children, one in seventh grade and another in ninth grade. Assume that each child was equally likely to be born male or female (independent of the other child's sex). What is the probability that both children are male?

The sample space of this problem is $\{(M,M),(M,F),(F,M),(F,F)\}$, where the first coordinate represents the sex of the child in seventh grade. These four outcomes are equally likely, so the probability both children are male is $\frac{1}{4}$.

Now, Alice tells you that she has a son Charlie. What is the probability that both children are male?

Since we know that Alice has a son, we can eliminate the outcome corresponding to two female children. Then restricted sample space contains the three equally-likely outcomes $\{(M,M),(M,F),(F,M)\}$, so this probability is $\frac{1}{3}$.

Then, she tells you that Charlie is in seventh grade. What is the new probability?

Since Alice's younger child is male, we can eliminate outcome (F,M) from the sample space. We are left with two equally likely outcomes, $\{(M,M),(M,F)\}$, so this probability is $\frac{1}{2}$.

25.2 Independence

In the previous section, we gave two examples of randomized experiments. In the dice rolling experiment, the probability of rolling a 4 on the gray die was unaffected by whether the red die showed a 4. Said differently, the gray die is *independent* of the red die. On the other hand, in the marble experiment, the probability that we have chosen a red marble in our right hand increases when we condition on seeing a gray marble in our left hand; the color of one marble *depends* on the color of the other. We can formalize this notion of independence as follows:

Definition 25.2 — Independence.

Suppose that $A,B \in \mathcal{F}$ are events with probabilities $\Pr(A),\Pr(B) \neq 0$. Then, A is *independent* of B if

$$\Pr(A \mid B) = \Pr(A).$$

Equivalently (we ask you to verify this in the exercises),

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B).$$

Note that this second definition is symmetric with respect to A and B. Thus, when event A is independent of B, it must also be true that B is independent of A. For this reason, we simply say that events A and B are independent. Also, note that this second definition also allows us to consider events with probability 0; any event A with probability 0 is automatically independent of any other event B. For this reason, we more frequently utilize the second definition to check independence (although the intuition is clearer in the first definition).

In our dice example, $A = \{$ a red 4 is rolled $\}$ is independent of $B = \{$ a gray 3 is rolled $\}$ since

$$\Pr(A \cap B) = \Pr(\text{ red 4 and gray 3 }) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = \Pr(A) \cdot \Pr(B).$$

In our marbles example, $A = \{$ right marble red $\}$ is not independent of $B = \{$ left marble gray $\}$ since

$$\Pr(A \cap B) = \Pr(\text{ right marble red, left marble gray }) = \frac{9}{30} \neq \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \Pr(A) \cdot \Pr(B).$$

One way to think about independent events is that the randomness "controlling" each of them is distinct. In our dice example, we could separate the dice into two different random experiments (one where the red die is rolled and the other where the gray die is rolled) that are occurring simultaneously. Since these experiments are separated, the outcome of one experiment does not influence the other. However, this separation cannot happen in the marbles experiment. The bag of marbles is shared by both hands, so we cannot split the selection of each marble into its own random experiment.

While this notion of shared or separate randomness helps provide some intuition about when events are independent, it is not always so clear cut, as the following example illustrates. You should always fall back on the calculation to check independence.

Example 25.4 Two cards are drawn sequentially from a perfectly shuffled deck (each ordering is equally likely). Our state space is all $52 \cdot 51 = 2{,}652$ sequences of two distinct cards. Let A be the event that the first card is a face card. Let B be the event that the second card is red. We calculate $\Pr(A) = \frac{12}{52} = \frac{3}{13}$, and $\Pr(B) = \frac{26}{52} = \frac{1}{2}$. Now, we consider event $A \cap B$. We use the addition principle to divide this event into two cases:

Case 1: The first card is a red face card. There are 6 possibilities for this card. Regardless of our choice, there remain 25 red cards in the deck for our second choice. Thus, there are $6 \cdot 25 = 150$ outcomes in this case.

Case 2: The first card is a black face card. There are 6 possibilities for this card. Regardless of our choice, there remain 26 red cards in the deck for our second choice. Thus, there are $6 \cdot 26 = 156$ outcomes in this case.

In total, we find that,

$$\Pr(A \cap B) = \frac{306}{2{,}652} = \frac{3}{26} = \frac{3}{13} \cdot \frac{1}{2} = \Pr(A) \cdot \Pr(B).$$

Thus, these events are independent.

25.2.1 Independence with More than Two Events

We'd like to extend our notion of independence to more than two events. Intuitively, we'd like to say that a collection of events is independent if the probability of one of these events A does not change regardless of how we condition on the other events. To make this precise, it is more convenient to use the product form of the independence condition.

Definition 25.3 — Mutual Independence.
A finite collection of events $A_1, \ldots, A_n$ is *mutually independent* if for all subsets $S \subseteq [n]$,

$$\Pr\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \Pr(A_i).$$

This is a very strong (and involved) condition. For example, to prove that three events A_1, A_2, A_3 are mutually independent, one would need to check:

- $\Pr(A_1 \cap A_2) = \Pr(A_1) \cdot \Pr(A_2)$,
- $\Pr(A_1 \cap A_3) = \Pr(A_1) \cdot \Pr(A_3)$,
- $\Pr(A_2 \cap A_3) = \Pr(A_2) \cdot \Pr(A_3)$,
- $\Pr(A_1 \cap A_2 \cap A_3) = \Pr(A_1) \cdot \Pr(A_2) \cdot \Pr(A_3)$.

Nevertheless, mutually independent events (and their analog, mutually independent random variables that we will discuss soon) are both pervasive and of utmost importance in probability.

Example 25.5 Suppose that a fair coin is flipped n times, and let A_i be the event that the ith flip landed on heads. Then $\{A_1, \ldots, A_n\}$ are mutually independent. The sample space for this experiment is all 2^n sequences of $\{H, T\}$ with length n. All of these outcomes are equally likely. Now consider a subset $\{i_1, \ldots, i_k\} \subseteq \{1, \ldots, n\}$. The event $\bigcap_{j=1}^{k} A_{i_j}$ consists of all sequences in Ω with H in positions $i_1, \ldots, i_k$. The other $n-k$ positions can have either symbol, so there are 2^{n-k} outcomes in this event. Thus,

$$\Pr\Big(\bigcap_{j=1}^{k} A_{i_j}\Big) = \frac{2^{n-k}}{2^n} = \frac{1}{2^k} = \Big(\frac{1}{2}\Big)^k = \prod_{j=1}^{k} \Pr(A_{i_j}).$$

Our choice of subset $\{i_1, \ldots, i_k\}$ was arbitrary. Hence, this equality must hold for all subsets of events, meaning these events are mutually independent.

A somewhat weaker, but still useful, condition on multiple events is pairwise independence.

Definition 25.4 — Pairwise Independence.
A finite collection of events $A_1, \ldots, A_n$ is *pairwise independent* if for all $i \neq j \in [n]$,

$$\Pr(A_i \cap A_j) = \Pr(A_i) \cdot \Pr(A_j).$$

In words, this condition says that each pair of events in the set is independent (using the two-event definition). In the exercises, we ask you to verify that pairwise independence is implied by mutual independence. The following example shows that pairwise independent events need not be mutually independent.

Example 25.6 Again, we consider the experiment where we roll a pair of fair dice, one red and one gray. Let A_1 be the event that the red die shows an even number. Let A_2 be the event that the gray die shows an even number. Finally, let A_3 be the event that the sum of the dice is even. We visualize the sample space with the following chart.

	⚀	⚁	⚂	⚃	⚄	⚅
⚀	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚁	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚂	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚃	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚄	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
⚅	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

The outcomes corresponding to event A_3 are highlighted in pink. We have $\Pr(A_1) = \Pr(A_2) = \Pr(A_3) = \frac{18}{36} = \frac{1}{2}$. In addition, we have $\Pr(A_1 \cap A_2) = \Pr(A_1 \cap A_3) = \Pr(A_2 \cap A_3) = \frac{9}{36} = \frac{1}{4}$. Thus,

$$\Pr(A_1 \cap A_2) = \Pr(A_1) \cdot \Pr(A_2), \qquad \Pr(A_1 \cap A_3) = \Pr(A_1) \cdot \Pr(A_3), \qquad \Pr(A_2 \cap A_3) = \Pr(A_2) \cdot \Pr(A_3),$$

so the events are pairwise independent. However, note that $\Pr(A_1 \cap A_2 \cap A_3) = \frac{9}{36} = \frac{1}{4}$ (the outcomes in the lower right corner of the chart), so

$$\Pr(A_1 \cap A_2 \cap A_3) = \frac{1}{4} \neq \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \Pr(A_1) \cdot \Pr(A_2) \cdot \Pr(A_3),$$

so these events are not mutually independent.

Main Takeaways:

- When we condition on an event, we restrict the sample space to only the outcomes within that event.
- The notation $\Pr(A \mid B)$ denotes the probability of the event A conditioned on the event B. We calculate $\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$.
- When two events A and B are independent, conditioning on one of the events does not affect the probability of the other. We can confirm independence by checking that $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$.
- There are multiple notions of independence, including *mutual* and *pairwise* independence, to describe sets of more than two events.

Exercises

25.1 Let us return to the setting of a bag with three red and three gray marbles. Suppose that we remove two of the marbles (without looking at them, and place them into a smaller bag. Then, from this bag of two marbles, we remove one marble and see that it is gray. What is the probability that the other marble in this bag is red?

25.2 In the game blackjack, a player chooses whether to add more cards to their hand as they try to obtain a better score than the dealer. A crucial aspect of the game is that the player must make their decisions with knowledge of only one of the dealer's two starting carts. To reduce the risk exposure of the casino, the game is often played with multiple decks shuffled together. Here, we'll assume that 2 decks are being used. Suppose that you are dealt a hand $\{4\diamondsuit, \text{J}\clubsuit\}$.

(a) The dealer's face up card is A$\heartsuit$. What is the probability that they hold two aces?

(b) Suppose instead that the dealer keeps both cards face down but tells you (truthfully) that one of them is A$\heartsuit$. Does this alter the probability from part (a)?

(c) In both the settings from parts (a) and (b), how must we update this probability if we ask for another card and it is 2$\diamondsuit$?

(d) In both the settings from parts (a) and (b), how must we update this probability if we ask for another card and it is A$\heartsuit$?

25.3 A jewelry box has three drawers. One drawer contains two gold coins, one drawer contains two silver coins, and one drawer contains one gold and one silver coin. Each arrangement of the drawers is equally likely. You reach into the top drawer and pull out a silver coin. What is the probability that the other coin in this drawer is silver?

25.4 Argue that the two definitions of independence of a pair of events are equivalent when $\Pr(B) \neq 0$. That is, prove that $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B) \Leftrightarrow \Pr(A) = \Pr(A \mid B)$ under this assumption.

25.5 Given an event $A \in \mathcal{F}$, its *complementary event* $\overline{A}$ is $\Omega \setminus A$ (we assume that $\mathcal{F}$ is closed under complements). Argue that if $A, B \in \mathcal{F}$ are independent, then $\overline{A}$ and B are also independent.

25.6 Let $A, B, C \in \mathcal{F}$. Suppose that A and B are independent. Also, suppose that A and C are independent. In this question, we'll consider the relationship between events A and $B \cup C$.

(a) Give an example that shows that A and $B \cup C$ need not be independent.

(b) Under the additional assumption that A and $B \cap C$ are independent, argue that A and $B \cup C$ must be independent.

25.7 For a given probability space $(\Omega, \mathcal{F}, \Pr)$, define the binary relation $\sim$ on $\mathcal{F}$ such that $A \sim B$ if and only if A and B are independent. Determine whether $\sim$ has each of the following properties. Justify each answer with either a proof or a counterexample.

(a) Reflexive

(b) Symmetric

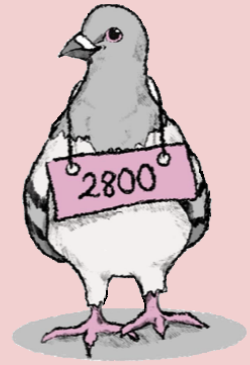
(c) Anti-Symmetric

(d) Transitive

25.8 Give a short argument that mutual independence implies pairwise independence.

$\star$ **25.9** Consider a random experiment where a fair coin is flipped 3 times. Define four events A_1, A_2, A_3, A_4 such that any three of these events are mutually independent but all four events are not mutually independent.

Hint: Try to mimic the structure of Example 25.6.



26. Bayes' Rule

26.1 Calculations with Conditional Probability

Conditioning is a very powerful tool for modeling uncertainty in real-world situations. It provides a framework for us to update our beliefs about an outcome as we gain information through observations. In fact, the Bayesian viewpoint of probability centers around this notion. A central idea of this approach is Bayes' Rule, which gives a formula to "reverse" the conditioning of two events. This has countless real-world applications, from medical diagnoses to spam detection. One tool that will be useful for Bayes' Rule calculations is the Law of Total Probability.

Lemma 26.1 — Law of Total Probability. Suppose that $A_1, A_2, \ldots, A_n \in \mathcal{F}$ form a partition of Ω with each $\Pr(A_i) > 0$. Then for any event $B \in \mathcal{F}$,

$$\Pr(B) = \sum_{i=1}^{n} \Pr(A_i) \cdot \Pr(B \mid A_i).$$

Proof. By applying the definition of conditional probability to each term of the RHS sum, we find that

$$\sum_{i=1}^{n} \Pr(A_i) \cdot \Pr(B \mid A_i) = \sum_{i=1}^{n} \Pr(A_i) \cdot \frac{\Pr(B \cap A_i)}{\Pr(A_i)} = \sum_{i=1}^{n} \Pr(B \cap A_i).$$

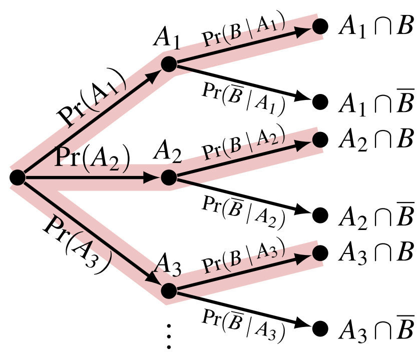
Since the events A_i partition Ω, they are pairwise disjoint. Thus, the intersections $(B \cap A_i)$ are also pairwise disjoint, as each of these sets is a subset of a different A_i. Thus, we may utilize the countable additivity property of probability (in reverse) to rewrite

$$\sum_{i=1}^{n} \Pr(B \cap A_i) = \Pr\left(\bigcup_{i=1}^{n} (B \cap A_i)\right) = \Pr\left(B \cap \bigcup_{i=1}^{n} A_i\right) = \Pr(B \cap \Omega) = \Pr(B).$$

Here, the second equality follows from the equivalence of these sets (we leave this as an exercise for you to verify). The third equality uses the fact that the events A_i partition Ω, so their union is Ω. Finally, the fourth equality follows since $B \subseteq \Omega$. ■

While this proof mathematically verifies the law, it provides little intuition. To gain this intuition, note that we can view the sum on the right side as a weighted average; the fact that the events A_i partition Ω implies that the probabilities $\Pr(A_i)$ sum to 1. Now, we can think about $\Pr(B)$ as representing the "concentration" of B in the entire sample space Ω. Using our "restriction" view of conditional probability, each $\Pr(B \mid A_i)$ represents the concentration of B in A_i. Thus, the Law of Total Probability tells us that if we mix solutions with concentrations $\Pr(B \mid A_i)$ in amounts proportional to $\Pr(A_i)$, we obtain a final solution with concentration $\Pr(B)$.

Another way to understand the Law of Total Probability is by thinking about the randomized experiment being performed in two stages. In the first stage, the randomness decides which of the A_i events the outcome belongs to. Then, in the second stage, the randomness fixes the outcome. Conditioned on choosing A_i in the first step (so restricting attention to event A_i) the probability that the final outcome belongs to event B is by definition $\Pr(B \mid A_i)$. We can visualize this two-stage experiment by the following flow chart.



The red paths represent the ways to end at an outcome in B. To compute the probability of event B, we take the product of the edge probabilities along each path, and then we sum the probabilities over all of the paths. This exactly recovers the Law of Total Probability.

One special case of the Law of Total Probability that we will use today partitions Ω into two pieces, an event $A \in \mathcal{F}$ and its complement $\overline{A} = \Omega \setminus A \in \mathcal{F}$:

$$\Pr(B) = \Pr(A) \cdot \Pr(B \mid A) + \Pr(\overline{A}) \cdot \Pr(B \mid \overline{A})$$

26.2 Bayes' Rule

Bayes' Rule is a formula that relates the conditional probabilities $\Pr(A \mid B)$ and $\Pr(B \mid A)$. Expanding both of these conditional probabilities, we have

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \qquad \Pr(B \mid A) = \frac{\Pr(A \cap B)}{\Pr(A)}.$$

Notice that both of these equations include $\Pr(A \cap B)$. Solving both equations for $\Pr(A \cap B)$ and equating them gives Bayes' Rule.

Lemma 26.2 — Bayes' Rule. Given any two events $A, B \in \mathcal{F}$ with $\Pr(A), \Pr(B) \neq 0$,

$$\Pr(B) \cdot \Pr(A \mid B) = \Pr(A) \cdot \Pr(B \mid A).$$

In its more familiar form, we have

$$\Pr(A \mid B) = \frac{\Pr(A) \cdot \Pr(B \mid A)}{\Pr(B)}.$$

Bayes' Rule, while simple to state, is a crucial tool for understanding conditional probability, which can often feel confusing or counter-intuitive. The following example illustrates such a scenario.

Example 26.1 A test for a rare genetic disease reports an accuracy of 99 percent. This means that if someone with the disease takes the test, there is a 99 percent chance of a positive result, and if someone without the disease takes the test, there is a 99 percent chance of a negative result. Based on historical medical records, we know that the disease affects 1 in every $40,000$ people. Suppose Marcie takes the test and gets a positive result. What is the probability that Marcie has the disease?

Before reading on, try to guess this probability. Since the test has such a high accuracy, it seems reasonable to assume that Marcie's test result is correct, meaning she almost certainly has the disease. However, this turns out not to be the case.

Let's use Bayes' Rule, where A is the event that an individual has the disease and B is the event that an individual tests positive for the disease. From the given information, we have that $\Pr(A) = \frac{1}{40{,}000}$ and $\Pr(B \mid A) = \frac{99}{100}$. However, we are not given $\Pr(B)$. We can calculate this using the Law of Total Probability. Since A and its complement, $\overline{A}$, partition Ω, we have

$$\Pr(B) = \Pr(A) \cdot \Pr(B \mid A) + \Pr(\overline{A}) \cdot \Pr(B \mid \overline{A}) = \frac{1}{40,000} \cdot \frac{99}{100} + \frac{39,999}{40,000} \cdot \frac{1}{100} = \frac{40,098}{4,000,000}.$$

Here, the $\Pr(B \mid \overline{A}) = \frac{1}{100}$ because there is a one percent chance that someone without the disease will test positive. Now, we have all of the information we need to apply Bayes' Rule.

$$\Pr(A \mid B) = \frac{\frac{1}{40{,}000} \cdot \frac{99}{100}}{\frac{40{,}098}{4{,}000{,}000}} = \frac{99}{40,098} \approx 0.0025.$$

There is less than a one percent chance that Marcie has the disease.

This example seems to defy our intuition. If the test is 99 percent accurate, why will it almost always return the wrong result to Marcie? The reason is that we are not assessing the accuracy of the test, we are assessing the *conditional* probability that the test is accurate when it returns a positive result. Now, the exceedingly rare probability of the disease makes the likelihood of a false positive result much greater than a true positive result. If you are still not convinced, imagine giving the test to every person in a city with a population of 4 million. We'd estimate that 100 people in the city have the disease, and 99 will receive a positive test result. Among the other $3,999,900$ people, $39,999$ will receive a false positive test result, so most of the positive test results are attributable to individuals without the disease. Let this example serve as a warning; while conditioning is a powerful technique, it must be used and interpreted carefully!

26.3 Bayesian Inference

Often in science, we are faced with some initial uncertainty about the state of a system. In the simplest example of this, we may have made some *hypothesis*, which can either be true or false. We can imagine that there is some underlying probability space $(\Omega, \mathcal{F}, \Pr)$ that models our uncertainty (that is, the various outcomes are all of the possible scenarios that we can envision). We can define A to be the set of outcomes where our hypothesis is true. Before we collect any data, we may have an initial guess as to what $\Pr(A)$ is. We call this the *prior probability* of A. As we gain more insight into the true state of the system, either by running experiments or making observations, some of the possible scenarios that we envisioned may no longer be relevant, and this allows us to update our belief about $\Pr(A)$. For example, an observation may correspond to an event B in our probability space. Once we know that we are within B, we can use Bayes' Rule to calculate

$$\Pr(A \mid B) = \frac{\Pr(A) \cdot \Pr(B \mid A)}{\Pr(B)} = \Pr(A) \cdot \frac{\Pr(B \mid A)}{\Pr(B)}.$$

This conditional probability $\Pr(A \mid B)$ is called the *posterior probability* of A conditioned on the evidence B. We obtain this from the *prior probability* by multiplying by the factor $\frac{\Pr(B|A)}{\Pr(B)}$, which models how likely this evidence would have arisen in event A versus in general. As we obtain more evidence, we can iterate this process. Our posterior probability from one iteration becomes the new prior when we condition on the new evidence. In doing this re-conditioning, we refine our beliefs about $\Pr(A)$ and gain confidence about whether our hypothesis is true.

Example 26.2 Suppose that we have two coins, labeled a and b. We know that one of these coins is fair (meaning the probability that it lands on heads is $\frac{1}{2}$), while the other is biased towards heads (the probability that it lands on heads is $\frac{3}{5}$). Initially, we don't know which coin is which. To model this, we can let the event[1] A be that coin a is the biased coin, so our prior probability is $\Pr(A) = \frac{1}{2}$.

Now, suppose we flip coin a and it lands on heads. We will call this event B_1. We can use Bayes' Rule to update our belief about $\Pr(A)$, calculating the posterior probability $\Pr(A \mid B_1)$:

$$\Pr(A \mid B_1) = \frac{\Pr(A) \cdot \Pr(B_1 \mid A)}{\Pr(B_1)} = \frac{\Pr(A) \cdot \Pr(B_1 \mid A)}{\Pr(A) \cdot \Pr(B_1 \mid A) + \Pr(\overline{A}) \cdot \Pr(B_1 \mid \overline{A})},$$

where the second equality applies the Law of Total Probability. Notice that $\Pr(A) = \frac{1}{2}$ and $\Pr(\overline{A}) = 1 - \frac{1}{2} = \frac{1}{2}$ from our prior. $\Pr(B_1 \mid A) = \frac{3}{5}$ since conditioning on event A tells us that coin a is the biased coin, so lands heads with probability $\frac{3}{5}$. $\Pr(B_1 \mid \overline{A}) = \frac{1}{2}$ since conditioning on event $\overline{A}$ tells us that coin a is the fair coin, so lands heads with probability $\frac{1}{2}$. Plugging these values into our formula from above gives us the posterior probability

$$\Pr(A \mid B_1) = \frac{\frac{1}{2} \cdot \frac{3}{5}}{\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{1}{2}} = \frac{6}{11}$$

It makes sense that the posterior probability is larger than the prior probability; having coin a land on heads is better explained when it is biased, so our confidence in the hypothesis A slightly increases.

[1] Here, we are vague about the probability space because we lack the tools to define it formally. You can imagine that the outcomes of this probability space are all pairs of infinite sequences of $\{H, T\}$ that could be generated by the two coins, along with the specification of which coin is which.

Suppose we now flip coin b and it lands on tails, which we denote by event B_2. How should we re-update our confidence in event A? We should not calculate $\Pr(A \mid B_2)$; this would ignore the insight we just gained from B_1. Rather, we should re-condition $\Pr(A \mid B_1)$ with this new evidence, computing a new posterior $\Pr(A \mid B_1 \cap B_2)$. To do this, we use Bayes' Rule, but only within the subset of probability space that includes B_1 (see Exercise 26.4):

$$\Pr(A \mid B_1 \cap B_2) = \frac{\Pr(A \mid B_1) \cdot \Pr(B_2 \mid A \cap B_1)}{\Pr(B_2 \mid B_1)}$$

This formula has the structure of Bayes' rule for $\Pr(A \mid B_2)$, except that the condition B_1 has been introduced on the right side of each probability. Notice that our previous posterior probability $\Pr(A \mid B_1) = \frac{6}{11}$ has taken the place of the prior $\Pr(A)$. Next, $\Pr(B_2 \mid A \cap B_1) = \Pr(B_2 \mid A)$ since the outcome of the first coin flip does not affect the outcome of the second flip. Moreover, since conditioning on event A tells us that coin b is fair, we have $\Pr(B_2 \mid A) = \frac{1}{2}$. By the Law of Total Probability (in the restricted probability space conditioned on event B_1), we may calculate the probability in the denominator

$$\begin{aligned}\Pr(B_2 \mid B_1) &= \Pr(A \mid B_1) \cdot \Pr(B_2 \mid A \cap B_1) + \Pr(\overline{A} \mid B_1) \cdot \Pr(B_2 \mid \overline{A} \cap B_1) \\ &= \tfrac{6}{11} \cdot \Pr(B_2 \mid A) + \left(1 - \tfrac{6}{11}\right) \cdot \Pr(B_2 \mid \overline{A}) \quad = \tfrac{6}{11} \cdot \tfrac{1}{2} + \tfrac{5}{11} \cdot \tfrac{2}{5} \quad = \tfrac{5}{11}.\end{aligned}$$

Thus,

$$\Pr(A \mid B_1 \cap B_2) = \frac{\frac{6}{11} \cdot \frac{1}{2}}{\frac{5}{11}} = \frac{3}{5}.$$

It is again sensible that our confidence that coin a is biased increases after observing the event B_2; having coin b land on tails is more likely if coin b is the fair coin, so this provides more evidence that points toward coin a being biased.

By continuing to observe more coin flips and re-condition, our posterior probability will continue to fluctuate based on new observations. Rather quickly, the confidence will tend sharply toward either 0 or 1 as the long-run average behavior of the coin concentrates toward its expected value. We will study these sorts of concentrations when we introduce random variables in the coming lecture.

Main Takeaways:

- The *Law of Total Probability* allows us to calculate the probability of an event B based on its conditional probability under different possible circumstances A_i.
- *Bayes' Rule* lets us to calculate conditional probabilities by "flipping the conditioning".
- In *Bayesian Inference*, we update our original belief about an event, known as its *prior* probability, by using Bayes' Rule to incorporate new observations to calculate the conditional *posterior* probability.
- Conditional probability is counter-intuitive. We must be careful when we are calculating or interpreting results involving conditioning.

Exercises

26.1 A hard candy factory has three production lines of different sizes which all feed over to a single packaging line. Line A is the largest line and produces $\frac{1}{2}$ of the candy, line B produces $\frac{1}{3}$ of the candy, and line C produces $\frac{1}{6}$ of the candy. We further describe which candy flavors are produced on each line.

- The production on line A is equally split across 4 flavors: lemon, cherry, orange, and pineapple.
- The production on line B is equally split across 3 flavors: cherry, grape, and orange.
- The production on line C is equally split across 3 flavors: lemon, cherry, and grape.

Suppose that a random piece of candy is chosen from the packaging line (such that all choices are equally likely). What is the probability that the flavor of the candy is:

(a) Lemon
(b) Pineapple
(c) Orange
(d) Grape
(e) Cherry

Now, suppose that a random piece of **cherry** candy is chosen from the packaging line. What is the probability that it was produced on:

(f) Line A
(g) Line B
(h) Line C

26.2 Let's return to the disease test in Example 26.1. Suppose that Marcie takes a second test and again receives a positive result. Now what is the probability that she has the disease? Repeat this calculation if Marcie receives a third positive result.

26.3 As an example of applying Bayes' Rule, we'll consider a calculation performed by a simple spam filter. Suppose that an email application is trying to decide whether to mark an incoming email as spam. It sees the phrase "exclusive offer" in the subject line. Based on historical data, the application knows that

- Around 1 out of every 8 incoming emails is tagged as spam.
- Around 1 in every $7{,}000$ email subject lines contains the phrase "exclusive offer."
- Of the messages tagged as spam, around 1 in every 950 has subject line containing "exclusive offer."

Suppose that we define the events

- $A =$ an incoming email is spam
- $B =$ an email's subject line contains "exclusive offer."

(a) How can we express the probability that the incoming message (with "exclusive offer" in its subject line) is spam in terms of events A and B?

(b) Use Bayes' Rule to estimate the probability from part (a).

26.4 Use the definition of conditional probability to verify following identities.

(a) $\Pr(A_1 \cap A_2 \mid B) = \Pr(A_1 \mid B) \cdot \Pr(A_2 \mid A_1 \cap B) = \Pr(A_2 \mid B) \cdot \Pr(A_1 \mid A_2 \cap B)$

(b) $\Pr(A \mid B_1 \cap B_2) = \frac{\Pr(A \mid B_1) \cdot \Pr(B_2 \mid A \cap B_1)}{\Pr(B_2 \mid B_1)}$

⋆ **26.5** Given a probability space $(\Omega, \mathcal{F}, \Pr)$ with events $A, B, C \in \mathcal{F}$ and $\Pr(B), \Pr(C) \neq 0$, suppose that wish to update our belief in the probability of event A based on our observations of events B and C (that is, we wish to calculate $\Pr(A \mid B \cap C)$. There are three different ways that we can do this.

1. First, we update the prior $\Pr(A)$ based on the observation of B to obtain posterior $\Pr(A \mid B)$. Then, we update $\Pr(A \mid B)$ based on the observation of C to obtain the new posterior $\Pr(A \mid B \cap C)$.
2. First, we update the prior $\Pr(A)$ based on the observation of C to obtain posterior $\Pr(A \mid C)$. Then, we update $\Pr(A \mid C)$ based on the observation of B to obtain the new posterior $\Pr(A \mid B \cap C)$.
3. We perform a single update of the prior $\Pr(A)$ based on the observation of $B \cap C$ to obtain posterior $\Pr(A \mid B \cap C)$.

Write out how the final posterior $\Pr(A \mid B \cap C)$ is being calculated in each of these approaches. Then, use the definition of conditional probability to show that all three of these expressions are equal.

26.6 Let's revisit the setting from Example 26.2 and continue to update our posterior on $\Pr(A)$ as we observe more coin flips.

(a) Suppose that coin b is flipped again and lands on heads (event B_3). Do you expect that $\Pr(A \mid B_1 \cap B_2 \cap B_3)$ will be greater than, less than, or equal to $\Pr(A \mid B_1 \cap B_2)$? Why?

Carry out this conditioning step to check if your intuition was correct.

(b) Suppose that coin a is flipped again and lands on heads (event B_4). Do you expect that $\Pr(A \mid B_1 \cap B_2 \cap B_3 \cap B_4)$ will be greater than, less than, or equal to $\Pr(A \mid B_1 \cap B_2 \cap B_3)$? Why?

Carry out this conditioning step to check if your intuition was correct.

(c) Finally, suppose that coin a is flipped again and lands on tails (event B_5). Do you expect that $\Pr(A \mid B_1 \cap B_2 \cap B_3 \cap B_4 \cap B_5)$ will be greater than, less than, or equal to $\Pr(A \mid B_1 \cap B_2 \cap B_3 \cap B_4)$? Why?

Carry out this conditioning step to check if your intuition was correct.

⋆ (d) Based on the result of Exercise 26.5 and the fact that the coin flips are independent, so the order in which they are observed does not matter, we can perform a single conditioning at the end of all coin flips based on the *number* of times each of the coins landed on heads and tails.

Suppose that coin a lands on heads h_a times and on tails t_a times. Similarly, coin b lands on heads h_b times and on tails t_b times. Call all of these observations the event B. Find a formula for $\Pr(A \mid B)$ in terms of h_a, h_b, t_a, and t_b. Verify that your formula agrees with the calculations in the lecture and in the earlier parts of this exercise.

⋆ (e) Simplify your formula from part (d) in the special case that the coins were flipped the same number of times (i.e., that $h_a + t_a = h_b + t_b$).

26.7 The CS department has given Matt and Alexandra each a bag of 20 pieces of candy to pass out to students on Halloween. One bag (the "good" bag) contains 15 Twix bars and 5 DumDums. The other ("bad") bag contains 12 DumDums and 8 Twix bars. Matt and Alexandra were equally likely to receive either bag.

Suppose that 4 students visit Matt's office and select a uniform random piece of candy (without replacement) from his bag. 3 students get a Twix bar and 1 student gets a DumDum.

(a) Conditioned on this observation, what is the posterior probability that Matt was given the good bag?

After this, suppose that 6 students visit Alexandra's office. Of these students, 3 students get a Twix bar and 3 get DumDums. You'll compute the posterior probability that Matt received the good bag in two ways.

(b) First, perform the calculation starting from our original uniform prior (that is, our assumption that Matt and Alexandra each got the good bag with probability $\frac{1}{2}$) and conditioning on the observation of all 10 students.

(c) Next, perform the calculation by starting from your posterior distribution from part (a) and performing another Bayesian update based on the 6 students who visited Alexandra's office. You should get the same answer as in part (b).

26.8 Alice and Bonnie are two up-and-coming tennis stars, and an exhibition match is organized to try and find out who is the higher-skilled athlete. A tennis *match* consists of up to three *sets*. The first athlete to win two sets is declared the winner of the match. We will assume that the competitor with higher skill will win any set with probability $\frac{2}{3}$ (mutually independent of their performance in any other sets). There are no ties.

(a) What is the probability that the higher-skilled athlete will win the exhibition match (that is, win two sets before their opponent does)?

As Alice and Bonnie have never competed against one another before, we have no way to know for certain who is the better athlete. However, based on their performance against other athletes, we estimate that Alice has higher skill with *prior probability* p.

Let us define events:

- $A =$ Alice has higher skill than Bonnie
- $M =$ Alice wins the exhibition match (that is, Alice wins two sets before Bonnie does)

For each of the remaining subparts, write (symbolically) the probability that we wish to compute in terms of events A and M (and M_2 for parts (d) and (e)). Calculate these probabilities using the formulas from the lecture. Your answers should be functions of p.

(b) What is the probability that Alice wins the exhibition match?

(c) If Alice wins the exhibition match, what is the *posterior probability* that she is the higher-skilled athlete?

(d) Suppose that Alice wins the exhibition match. Bonnie insists that this was luck and that she would win if a second match were played. Let us define event

$$M_2 = \text{Alice wins the second match}$$

Based on our new (*posterior*) probability that Alice is the higher-skilled player, what is the probability that Alice will win the second match?

(e) If Alice also wins the second match, how should we update our belief about $\Pr(A)$?

26.9 In machine learning, there are many different metrics used to evaluate the quality of a model. For simplicity, we'll consider a binary classifier (for example, a neural network that decides whether a picture contains a cat or not). One such evaluation metric is the **accuracy** of the network, which measures the probability that the network's classification is correct. Since it is untenable to reason about the network's decision on *every possible* image, we use *experimental probability*, computed from the test data of the classifier. That is,

$$\text{accuracy} = \frac{\#\text{ correct predictions}}{\#\text{ predictions}}$$

In this exercise, we'll consider the following two events (and their complements):

- C: A photo contains a cat
- Y: The classifier predicts that a photo contains a cat.

(a) Express the accuracy metric as a probability expression involving these events.

(b) Explain a scenario where a poor classifier can achieve high accuracy.

In light of this issue, we often consider other metrics.

- The **precision** of a classifier is the probability that its "yes" predictions are correct.
- The **recall** (or **sensitivity**) of a classifier is the probability that it detects (predicts "yes") a "yes" instance.
- The **specificity** of a classifier is the probability that correctly classifies a "no" instance.

(c) Express each of these other metrics as a probability in terms of events C and Y.

(d) Suppose that the test set for a classifier contains $1,000$ images, 300 of which include cats. The classifier predicts that 350 images include cats and has a precision of 80%. Compute the classifier's recall, accuracy, and specificity.

⋆ **26.10** It is the day of a criminal trial for a bank robbery, and the prosecution is presenting its case. First, the teller testifies that they observed the robber was missing the ring finger on their right hand. Sure enough, the defendant is also missing this finger. The prosecution reports that such a missing finger occurs with probability less than $\frac{1}{1000}$. Next, the teller testifies that the robber hit his head on a door frame on the way out of the bank, so was at least 6 foot 7, which occurs in the population with probability around $\frac{1}{1000}$. The prosecutor reasons on these two facts alone, there is less than a 1 in 1,000,000 chance that the defendant is innocent, so must be convicted beyond a reasonable doubt. Explain the flaw in this logic.

27. Discrete Random Variables

Many random experiments that we'd like to reason about are built up from smaller random decisions. To model all possible combinations of these decisions, the state space must be large and/or complicated. However, in many of these settings, we are only concerned with gaining insight about a particular "feature" of the outcome. For example,

- A coin is flipped 100 times and lands on heads 70 times. Can we quantify how likely it is that the coin is fair? Here, the space of all outcomes Ω consists of all 2^{100} flip sequences, but in each sequence, we only care about the number of heads.
- A class of 20 students is randomly arranged for a class photo. How likely is it that Jared will stand next to Maya? Here, Ω consists of the $20!$ different arrangements of the class. However, out of all of the information encoded in each permutation, we are concerned only with whether Jared is standing next to Maya (a single boolean "signal").

Random variables allow us to both simplify and quantify the outcomes of an experiment.

27.1 Defining Random Variables

In order to perform calculations on the outcomes of a random experiment, they must be numerical. When the outcomes are not numbers (for example, when they are more complicated mathematical objects like sets or sequences), we can use a function to map them to numbers. Such a function is called a random variable.

Definition 27.1 — (Discrete) Random Variable, Support.

A *random variable* X on a probability space $(\Omega, \mathcal{F}, \Pr)$ is a real-valued function $X : \Omega \to \mathbb{R}$.

The range of X, often denoted by R_X, is referred to as its *support*.

A *discrete* random variable has a countable support.

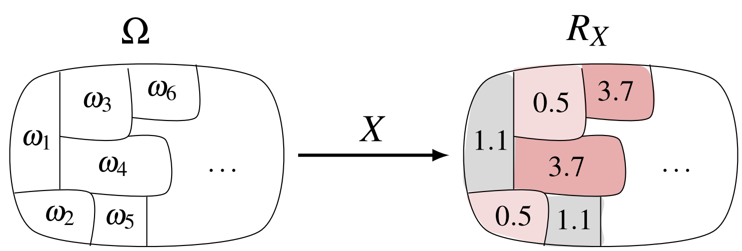
It is conventional to use uppercase letters to denote random variables and lowercase letters to denote the real values in their supports.

There are two main benefits that we get from random variables:

1. Random variables simplify the space of outcomes that we need to consider by mapping them down to one "feature". Two outcomes that share the same value of this feature are identical in the eyes of the random variable. Said differently, random variables provide a 1-dimensional representation of the sample space.
2. As we noted above, random variables convert the outcomes of an experiment to numbers, which allows us to perform meaningful calculations on the sample space.

27.1.1 Describing Discrete Random Variables

When we discussed probability functions on countable sample spaces, we thought of them as assigning some portion of the one unit of "probability mass" to each of the outcomes. We visualized this as a segmentation of Ω into differently sized chunks for each outcome. A random variable X describes a mapping of each outcome chunk $\omega_i \in \Omega$ to a real number.



From this picture, we see that we can think about the random variable as defining another probability distribution, this time on R_X. Each value in the support (i.e., outcome) is assumed on one or more of the chunks, which we interpret as fractions of the "probability mass". We call this mapping from the support to the interval $[0,1]$ the *probability mass function*.

Definition 27.2 — Probability Mass Function (pmf).

Given a random variable X on a probability space $(\Omega, \mathcal{F}, \Pr)$, the *probability mass function* $f_X : R_X \to [0,1]$ gives the probability that X takes on each value $r \in R_X$:

$$f_X(r) = \Pr\Big(\{\omega \in \Omega \mid X(\omega) = r\}\Big) \equiv \Pr(X = r).$$

To unpack this notation a little more, recall that Pr is a function whose domain consists of subsets of Ω. Therefore, "$X = r$" in the last expression is a shorthand that we use for the outcomes $\omega \in \Omega$ that X (a function $\Omega \to \mathbb{R}$) maps to r (or, the *pre-image* of r under X).

Another function that describes a random variable is its *Cumulative Distribution Function*.

Definition 27.3 — Cumulative Distribution Function (CDF).

Given a random variable X on a probability space $(\Omega, \mathcal{F}, \Pr)$, the *Cumulative Distribution Function* $F_X : \mathbb{R} \to [0,1]$ gives the probability that X assumes value at most $s \in \mathbb{R}$:

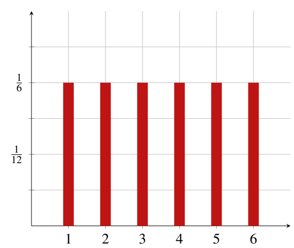
$$F_X(s) = \Pr\Big(\{\omega \in \Omega \mid X(\omega) \le s\}\Big) \equiv \Pr(X \le s).$$

Example 27.1 Two fair dice, one red and one gray, are rolled. This random experiment has $\Omega = [6]^2$, $\mathcal{F} = \mathcal{P}(\Omega)$, and uniform probability function $\Pr(A) = \frac{|A|}{36}$ for each $A \in \mathcal{F}$.

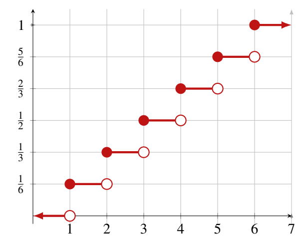
Let $G \colon \Omega \to \mathbb{R}$ be the random variable that represents the number shown on the gray die. If this is the second coordinate of each pair in the support, then $G(i,j) = j$ for each $(i,j) \in \Omega$. We visualize the values of G on the sample space in the leftmost figure. The support of G is $R_G = [6]$. The pmf f_G (which maps each value in the support to $\frac{1}{6}$) is shown in the histogram in the middle. The CDF F_G is visualized on the right.

	⚀	⚁	⚂	⚃	⚄	⚅
⚀	1	1	1	1	1	1
⚁	2	2	2	2	2	2
⚂	3	3	3	3	3	3
⚃	4	4	4	4	4	4
⚄	5	5	5	5	5	5
⚅	6	6	6	6	6	6

f_G



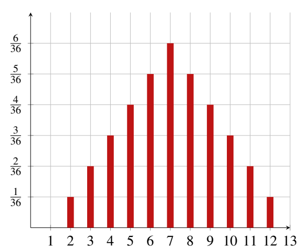
F_G



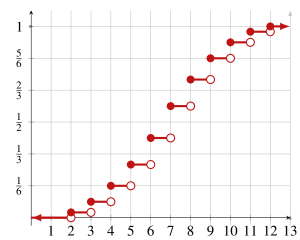
Let S be the random variable that represents the sum of the dice. Then, $S(i,j) = i + j$ for all $(i,j) \in \Omega$. We visualize the values of S on the sample space in the leftmost figure. The support of S is $R_S = \{2, 3, \ldots, 12\}$. The pmf $f_S(r) = \frac{6-|7-r|}{36}$ is visualized with a histogram in the middle figure. The CDF F_S is visualized on the right.

	⚀	⚁	⚂	⚃	⚄	⚅
⚀	2	3	4	5	6	7
⚁	3	4	5	6	7	8
⚂	4	5	6	7	8	9
⚃	5	6	7	8	9	10
⚄	6	7	8	9	10	11
⚅	7	8	9	10	11	12

f_S



F_S



There is a close relationship between the pmf and CDF of a discrete random variable. From the definitions, we can see that

$$F_X(s) = \Pr(X \leq s) = \Pr\Big(\bigcup_{\substack{r \in R_X \\ r \leq s}} \{X = r\} \Big) = \sum_{\substack{r \in R_X \\ r \leq s}} \Pr(X = r) = \sum_{\substack{r \in R_X \\ r \leq s}} f_X(r).$$

Here, the third equality follows from the countable additivity property of Pr. This relationship is illustrated in the plots of these functions; the location of each jump discontinuity in the CDF graph corresponds to a non-zero value of the pmf, and the size of the jump equals the value of the pmf at that point.

27.2 Discrete Distribution Families

The *distribution* of a discrete random variable is the information conveyed by its pmf; that is, the amount of probability mass that it assigns to each value in its (countable) support. There are many distributions that are ubiquitous in probability. We introduce a few of them here.

27.2.1 Discrete Uniform Distributions

Given integers $a < b \in \mathbb{Z}$, the discrete uniform distribution $\mathrm{Unif}\{a,b\}$ assigns equal probability to each integer in the interval $[a,b]$.

Definition 27.4 — Discrete Uniform Distribution.
If a random variable $U \sim \mathrm{Unif}\{a,b\}$ with $a < b \in \mathbb{Z}$, then $R_U = \{r \in \mathbb{Z} \mid a \le r \le b\}$, and $f_U(r) = \frac{1}{b-a+1}$ for all $r \in R_U$.

In this definition, we read the tilde "$\sim$" as "has the distribution", so $X \sim \mathrm{Unif}\{a,b\}$ means "X has the distribution $\mathrm{Unif}\{a,b\}$".

Example 27.2 Suppose $U \sim \mathrm{Unif}\{3,10\}$. Then $\Pr(U = r) = \frac{1}{8}$ for all $r \in \{3,4,5,6,7,8,9,10\}$.

27.2.2 Bernoulli Distributions

A Bernoulli random variable is a discrete random variable that has support $\{0,1\}$. We use the letter p to denote the probability that a Bernoulli random variable assumes value 1.

Definition 27.5 — Bernoulli Distribution.
If a random variable $X \sim \mathrm{Bern}(p)$ for some $p \in \mathbb{R}$ with $0 \le p \le 1$, then $R_X = \{0,1\}$ with:

$$f_X(1) = p, \qquad f_X(0) = 1 - p.$$

Example 27.3

- A fair coin is a $\mathrm{Bern}(\frac{1}{2})$ random variable, where the outcome 1 represents the coin landing heads and 0 represents the coin landing tails.
- A biased coin with probability p of landing heads is a $\mathrm{Bern}(p)$ random variable.

An important family of Bernoulli random variables are indicator random variables.

Definition 27.6 — Indicator Random Variable.
Given a probability space $(\Omega, \mathcal{F}, \Pr)$ and an event $A \in \mathcal{F}$, the *indicator random variable* for A, denoted $\mathbb{I}(A)$ is given by

$$\mathbb{I}(A)(\omega) = \begin{cases} 1 & \omega \in A, \\ 0 & \omega \notin A. \end{cases}$$

That is, $\mathbb{I}(A)$ assumes value 1 on (i.e., indicates) all outcomes ω in the event A.

Note that $\Pr(\mathbb{I}(A) = 1) = \Pr(\{\omega \in \Omega \mid \omega \in A\}) = \Pr(A)$. Hence, $\mathbb{I}(A) \sim \mathrm{Bern}\big(\Pr(A)\big)$.

27.2.3 Binomial Distributions

Binomial distributions have two parameters n and p. They are used to model the number of successes that are seen in n independent trials that are each successful with probability p.

Definition 27.7 — Binomial Distribution.

If a random variable $B \sim \text{Binom}(n,p)$ for some $n \in \mathbb{N}$ and some $p \in \mathbb{R}$ with $0 \le p \le 1$, then $R_B = \{0,1,\ldots,n\}$ with:

$$f_B(r) = \binom{n}{r} p^r (1-p)^{n-r} \qquad \text{for all } r \in R_B.$$

Notice that the Binomial Theorem ensures that f_B is a valid pmf (its values sum to 1), since

$$\sum_{r=0}^{n} \binom{n}{r} p^r (1-p)^{n-r} = (p+1-p)^n = 1^n = 1.$$

Suppose that a biased coin that lands heads with probability p is flipped n times, and let B be the number of times that it lands heads. Let us compute $f_B(r)$, the probability that the coin lands heads exactly r times. There are $\binom{n}{r}$ ways to select which of the flips land on heads, and each of these is equally likely. Given a fixed sequence (say the first r flips are heads and the next $n-r$ are tails), by the mutual independence of the coin flips, this occurs with probability

$$\left(\prod_{i=1}^{r} \Pr(\,i\text{'th flip heads})\right)\left(\prod_{j=r+1}^{n} \Pr(\,j\text{'th flip tails})\right) = p^r (1-p)^{n-r}.$$

Thus, $f_B(r) = \binom{n}{r} p^r (1-p)^{n-r}$. Since this is true for any $r \in \{0,1,\ldots,n\}$, $B \sim \text{Binom}(n,p)$. This is why the binomial distribution is so important.

27.2.4 Geometric Distributions

Geometric distributions have one parameter p and are used to model how long it takes for an event with probability p to take place. Unlike the previous distribution families we've introduced, geometric distributions have infinite support.

Definition 27.8 — Geometric Distribution.

If a random variable $G \sim \text{Geom}(p)$ for some $p \in \mathbb{R}$ with $0 < p \le 1$, then $R_G = \mathbb{N}^+$ with:

$$f_G(r) = p \cdot (1-p)^{r-1} \qquad \text{for all } r \in R_G.$$

Notice that the infinite geometric series formula ensures that f_G is a valid pmf:

$$\sum_{r=1}^{\infty} p \cdot (1-p)^{r-1} = p \cdot \sum_{r=1}^{\infty} (1-p)^{r-1} = p \cdot \sum_{r=0}^{\infty} (1-p)^r = p \cdot \tfrac{1}{1-(1-p)} = 1.$$

Suppose that a biased coin that lands heads with probability p is flipped until it lands on heads for the first time. Let G be the total number of times that it is flipped. Let us compute $f_G(r)$, the probability that the coin is flipped exactly r times. For this to happen, the first $r-1$ flips must have landed on tails, and the r'th flip must have landed on heads. By the mutual independence of the coin flips, this occurs with probability

$$\Pr(r\text{'th flip heads}) \cdot \left(\prod_{i=1}^{r-1} \Pr(i\text{'th flip tails}) \right) = p \cdot (1-p)^{r-1}.$$

Since this is true for any $r \in \mathbb{N}^+$, $G \sim \text{Geom}(p)$.

Main Takeaways:

- A *random variable* is a function from a sample space to the real numbers. It quantifies one particular attribute of the sample space.
- The *support* of a random variable is its range (so a subset of $\mathbb{R}$). *Discrete* random variables have countable support.
- The *distribution* of a random variable X describes how much probability it associates with various subsets of its support. The distribution of a discrete random variable can be described by its the pmf f_X and CDF F_X.
- There are several important families of distributions that arise frequently, such as the Discrete Uniform, Bernoulli, Binomial, and Geometric distributions.

27.3 Exercises

27.1 In one round of the National Hockey League (NHL) playoffs, two teams play each other in a best-of-7 series. That is, they play games against each other until one team has won 4 games. There are overtime rules that ensure that no game ends in a tie.

(a) Let us suppose that the two teams are evenly matched so that each team has probability $\frac{1}{2}$ of winning each game (mutually independent of all other games). Let N be the random variable representing the number of games that are played in the series. Find the support, pmf, and CDF of N.

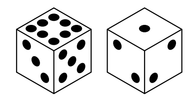
(b) Now, let's generalize the result from part (a). Instead of assuming that the teams are evenly matched, we will suppose that one team wins each game with probability p (mutually independent of all other games). If N again represents the number of games that are played in the series, find the support, pmf, and CDF of N. The latter two should have values expressed in terms of p.

27.2 Let us again consider the random experiment where two fair dice, one red and one gray, are rolled. We will represent the outcomes in $\Omega = [6]^2$ by pairs (i, j) where the first coordinate i represents the value of the red die and the second coordinate j represents the value of the gray die.

Determine the support, pmf, and CDF of each of the following random variables.

(a) $M = \max(i, j)$ (b) $D = i - j$ (c) $E = |i - j|$ (d) $I = \mathbb{I}(i > j)$

27.3



The Sicherman dice are a pair of 6-sided dice that are labeled differently from traditional dice. They were discovered in 1978 by the Buffalonian George Sicherman.

The first Sicherman die has faces labeled $\{1,3,4,5,6,8\}$.

The second Sicherman die has faces labeled $\{\!\!\{1,2,2,3,3,4\}\!\!\}$.

Suppose that S' is the random variable representing the sum of the values rolled on the Sicherman dice. Argue that S' has the same distribution as S, the sum of the rolls of two standard 6-sided dice.

The Sicherman dice are the only other pair of 6-sided dice with this property.

27.4 Suppose that three indistinguishable, fair 6-sided dice are thrown.

(a) Describe the (natural) probability space associated with this random experiment (that is, the outcomes should be multi-sets).

(b) Suppose that S_3 is the random variable representing the sum of the dice. Find the support, pmf, and CDF of S_3.

(c) Suppose that M_3 is the random variable representing the minimum value shown on any of the dice. Find the support, pmf, and CDF of M_3.

⋆ (d) Find a way to simulate rolling two distinguishable (red and gray) 6-sided dice using a single roll of these three indistinguishable dice. In other words, define two random variables $R,G\colon \Omega \to \mathbb{R}$ such that $\Pr(R = i \cap G = j) = \frac{1}{36}$ for all $(i,j) \in [6]^2$.

◇ **27.5** In this exercise, you will explore some properties that are true for all CDF functions. Throughout, suppose that $X : \Omega \to \mathbb{R}$ is a random variable on a probability space $(\Omega, \mathcal{F}, \Pr)$.

(a) What is $\lim_{s\to-\infty} F_X(s)$? Explain how you know this.

(b) What is $\lim_{s\to\infty} F_X(s)$? Explain how you know this.

(c) Argue that F_X is monotone non-decreasing. That is, for any $s_1 < s_2 \in \mathbb{R}$, $F_X(s_1) \le F_X(s_2)$.

(d) Argue that F_X is right continuous. That is, monotone non-decreasing. That is,

$$\lim_{s\to a^+} F_X(s) = F_X(a) \qquad \text{for all } a \in \mathbb{R}.$$

27.6 Given a probability space $(\Omega, \mathcal{F}, \Pr)$, suppose that $\mathbb{I}(A)$ and $\mathbb{I}(B)$ are the indicator random variables for two events $A, B \in \mathcal{F}$.

(a) Argue that $\mathbb{I}(A\cap B) = \mathbb{I}(A)\cdot\mathbb{I}(B)$.

(Recall that random variables are functions $\Omega \to \mathbb{R}$, so this amounts to showing that the expressions on both sides take the same value for each $\omega \in \Omega$.)

(b) Find a formula to express $\mathbb{I}(A\cup B)$ in terms of $\mathbb{I}(A)$, $\mathbb{I}(B)$, and $\mathbb{I}(A\cap B)$.

27.7 Given two discrete random variables X, Y on a common probability space $(\Omega, \mathcal{F}, \Pr)$, their sum $Z = X + Y$ is another random variable with $Z(\omega) = X(\omega) + Y(\omega)$ for each $\omega \in \Omega$.

(a) Use set builder notation to describe the support of Z.

(b) Given a real number $r \in R_Z$, how can we compute $f_Z(r)$?

27.8 Suppose that a fair coin is flipped repeatedly until it lands on heads twice. Let H_2 be the random variable that counts the number of flips up to and including the second heads.

(a) What is the support of H_2?

(b) What is the pmf of H_2? Verify that the pmf values sum to 1.

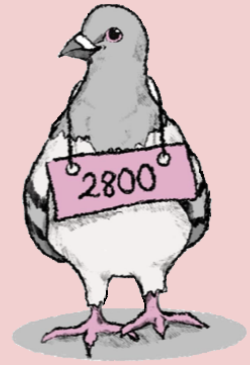
⋆ **27.9** In this exercise, we'll consider how to develop a recurrence to reason about the distribution of a random variable. To do this, let us consider a geometric random variable $G \sim \text{Geom}(p)$.

(a) Explain why the values of the CDF F_G at points in $\mathbb{N}$ satisfy the recurrence relation:

$$F_G(0) = 0, \qquad\qquad F_G(n+1) = p + (1-p)F_G(n).$$

(b) Use the recurrence relation from part (a) to argue that $F_G(n) = 1 - (1-p)^n$ for all $n \in \mathbb{N}$.

(c) Explain how part (b) allows us to conclude that $f_G(n) = p \cdot (1-p)^{n-1}$ for all $n \in \mathbb{N}^+$.



28. Expectation and Variance

In the last lecture, we introduced random variables, a tool that allows us to quantify the outcomes of a random experiment. That is, a random variable assigns values to each outcome in a sample space. While the full distribution of a random variable, conveyed through its pmf and CDF, can be useful, it is often sufficient to look at some of its *summary statistics*, values that help to describe important properties of the random variable. In today's lecture, we will introduce two of these summary statistics, *expectation* and *variance*.

28.1 Expectation

In many probabilistic settings, the underlying sample space and probability distribution can be very complicated. For example, card games played at the casino utilize a shuffled deck (or decks) of cards as their source of randomness, leading to factorially many possible outcomes. It would be impossible to capture all of the rich information in this distribution directly, so instead, we'd like a tool that can associate single meaningful numbers to this distribution. In a gambling game, a number we'd likely be interested in is how much we should expect to win (or more realistically lose) by taking a particular action. This quantity is referred to as an *expectation*. In other contexts, we refer to this quantity as an *expected value*, a *mean*, or even an *average*.

28.1.1 Definition

The expectation of a discrete random variable is the average of the values in its support, weighted by their probabilities.

Definition 28.1 — Expectation (Discrete Random Variable).

Suppose that X is a discrete random variable on a probability space $(\Omega, \mathcal{F}, \Pr)$. Then, its expectation $\mathbb{E}[X]$ is given by

$$\mathbb{E}[X] = \sum_{r \in R_X} r \cdot \Pr(X = r) = \sum_{r \in R_X} r \cdot f_X(r).$$

Example 28.1 Suppose that random variable D is the value of a fair dice roll. Since each of the six outcomes is equally likely (so occurs with probability $\frac{1}{6}$), we have

$$\mathbb{E}[D] = \sum_{r=1}^{6} r \cdot \Pr(D = r) = \sum_{r=1}^{6} \frac{r}{6} = \frac{21}{6} = \frac{7}{2}.$$

Example 28.2 — Bernoulli Random Variables. Recall that a Bernoulli random variable $X \sim \text{Bern}(p)$ assumes value 1 with probability p and 0 with probability $1-p$. Plugging into our discrete expectation formula, we find that

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1-p) = p.$$

A special case is indicator random variables. For any event $A \in \mathcal{F}$, $\mathbb{E}[\mathbb{I}(A)] = \Pr(A)$.

Example 28.3 — Geometric Random Variables. Suppose that $G \sim \text{Geom}(p)$. To calculate $\mathbb{E}[G]$ the expectation of a geometric random variable, we'll make use of the infinite geometric series formula, along with a useful "triangle summing trick". We have,

$$\mathbb{E}[G] = \sum_{r=1}^{\infty} r \cdot f_G(r) = \sum_{r=1}^{\infty} r \cdot p \cdot (1-p)^{r-1} = p \sum_{r=1}^{\infty} r \cdot (1-p)^{r-1}.$$

Let's focus on this last sum. It contains one copy of $(1-p)^0$, then two copies of $(1-p)^1$, three copies of $(1-p)^2$, and so on. We can arrange these in a triangle as shown below:

$$\begin{array}{lllll}
(1-p)^0 & & & & \\
+(1-p)^1 & +(1-p)^1 & & & \\
+(1-p)^2 & +(1-p)^2 & +(1-p)^2 & & \\
+(1-p)^3 & +(1-p)^3 & +(1-p)^3 & +(1-p)^3 & \\
\vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

The sum of the terms in each column of this triangle is a geometric series. If we pull out the common powers of $(1-p)$ from each of these columns, we end up with

$$\begin{aligned}
&\sum_{s=0}^{\infty}(1-p)^s \;+\; (1-p)\sum_{s=0}^{\infty}(1-p)^s \;+\; (1-p)^2\sum_{s=0}^{\infty}(1-p)^s \;+\; (1-p)^3\sum_{s=0}^{\infty}(1-p)^s \\
&= \left(\sum_{s=0}^{\infty}(1-p)^s\right)\left(1 + (1-p) + (1-p)^2 + (1-p)^3 + \dots\right) \\
&= \left(\sum_{s=0}^{\infty}(1-p)^s\right)^2 = \frac{1}{p^2}.
\end{aligned}$$

Thus, $\mathbb{E}[G] = p \cdot \frac{1}{p^2} = \frac{1}{p}$.

In the lecture exercises (Exercise 28.2, we walk through one calculation of the expectation of a binomial random variable. We will see a simpler way to reason about the expectation of binomial random variables in a few lectures.

28.1.2 The Law of the Unconscious Statistician

Sometimes, rather than reasoning about the expectation of a random variable itself, we might want to reason about the expectation of some function $f\colon \mathbb{R} \to \mathbb{R}$ applied to a random variable. We will see this later in the lecture, when we talk about variance. $g(X)$ is another random variable, which is the composition of $X\colon \Omega \to \mathbb{R}$ with $g\colon \mathbb{R} \to \mathbb{R}$. In other words, it maps each $\omega \in \Omega$ to $g(X(\omega))$. To calculate the expectation of a random variable in this form, we can appeal to the so-called "law of the unconscious statistician" (LOTUS). It gets this name because the statement feels so natural and its use so convenient that it is very often applied without much thought.

Lemma 28.1 — Law of the Unconscious Statistician (Discrete Random Variables).

Suppose that X is a discrete random variable on a probability space $(\Omega, \mathcal{F}, \Pr)$ and $g\colon \mathbb{R} \to \mathbb{R}$ any function. Then,

$$\mathbb{E}[g(X)] = \sum_{x \in R_X} g(x) \cdot f_X(x).$$

Proof. Define random variable $Y = g(X)$. Then,

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] = \sum_{y \in R_Y} y \cdot f_Y(y) = \sum_{y \in R_Y} \sum_{\substack{x \in R_X \\ g(x)=y}} g(x) \cdot f_X(x) = \sum_{x \in R_X} g(x) \cdot f_X(x)$$

Here, the second equality is an application of Definition 33.1. The third equality recognizes that all of the probability mass that Y assigns to y is probability mass that X assigned to $x \in g^{-1}(y)$. Finally, the last equality uses the fact that the double sum is exactly iterating over all $x \in R_X$. ■

Example 28.4 Consider a discrete random variable X with support $R_X = \{1, 3, 5, 10\}$ and with pmf described in the following table.

x	1	3	5	10
$f_X(x)$	0.2	0.35	0.35	0.1

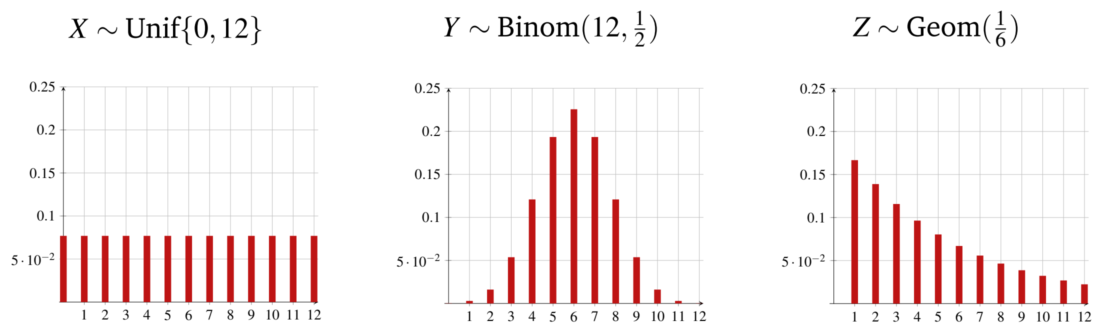
Using the definition of expectation and the LOTUS, we calculate

$$\mathbb{E}[X] = \sum_{r \in R_X} r \cdot f_X(r) = 1 \cdot 0.2 + 3 \cdot 0.35 + 5 \cdot 0.35 + 10 \cdot 0.1 = 4,$$
$$\mathbb{E}\left[X^2\right] = \sum_{r \in R_X} r^2 \cdot f_X(r) = 1 \cdot 0.2 + 9 \cdot 0.35 + 25 \cdot 0.35 + 100 \cdot 0.1 = 22.1.$$

Therefore, $\mathrm{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 = 22.1 - 16 = 6.1$. Taking the square root of this variance gives the standard deviation $\sigma(X) = \sqrt{6.1} \approx 2.47$. We'd expect the value of X to be approximately 2.47 away from its mean 4.

28.2 Variance

While the expectation provides some useful information about a random variable, it is by no means a complete description. For example, consider the following three random variables, whose pmfs are visualized below:



All three of these random variables have expectation 6; however, the shapes of their distributions are quite different. X evenly spreads its probability across $\{0, 1, \ldots, 12\}$. Y concentrates its probability around the mean 6. Z concentrates its probability at 1 but has infinite support with an exponentially small probability mass extending to arbitrarily large natural numbers. Beyond the expectation, it seems that another important summary statistic should provide richer information about the shape of the distribution of a random variable, with a specific emphasis on how close we'd expect a sample from a random variable to be to its mean. This is exactly the information conveyed by the *variance*.

28.2.1 Definition

The *variance* of a random variable is the expectation of its squared deviation from its mean.

Definition 28.2 — Variance, Standard Deviation.
Given a random variable X, its *variance*, denoted $\text{Var}(X)$ is given by

$$\text{Var}(X) = \mathbb{E}\left[\left(X - \mathbb{E}[X]\right)^2\right].$$

The *standard deviation* of X, denoted $\sigma(X)$ is the square root of its variance, $\sigma(X) = \sqrt{\text{Var}(X)}$.

Let's unpack this definition. The expression $X - \mathbb{E}[X]$ is a random variable representing how far samples from X are from the distribution mean $\mathbb{E}[X]$. We are mainly concerned with the magnitude of this distance, and not whether the value is larger or smaller than the expectation. To disregard this directionality, we square these distances.

Since the expectation defining the variance is taken over squared distances, it makes sense to take the square root of the variance to obtain a value in the same units as the original variable, the *standard deviation*. The standard deviation is a measure of how far we should expect the assumed value of X to be from its expectation.

28.2.2 Calculating the Variance

While we can directly use Definition 28.2 to compute the variance, we often use the following formula.

Lemma 28.2 $\operatorname{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2$.

Proof. We can derive this form of the variance by using the LOTUS with function $g(x) = (x - \mathbb{E}[X])^2$. We have,

$$
\begin{aligned}
\operatorname{Var}(X) &= \mathbb{E}\left[\left(X - \mathbb{E}[X]\right)^2\right] && \text{(Definition 28.2)} \\
&= \sum_{r \in R_X} \left(r - \mathbb{E}[X]\right)^2 \cdot f_X(r) && \text{(LOTUS)} \\
&= \sum_{r \in R_X} \left(r^2 - 2r \cdot \mathbb{E}[X] + \mathbb{E}[X]^2\right) \cdot f_X(r) \\
&= \sum_{r \in R_X} r^2 \cdot f_X(r) \; - \; 2\mathbb{E}[X] \sum_{r \in R_X} r \cdot f_X(r) \; + \; \mathbb{E}[X]^2 \sum_{r \in R_X} f_X(r) \\
&= \mathbb{E}\left[X^2\right] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2 && \text{(LOTUS)} \\
&= \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 && \blacksquare
\end{aligned}
$$

Example 28.5 — Bernoulli Random Variables. Suppose that $X \sim \text{Bern}(p)$. Note that $X^2 = X$, as $x^2 = x$ for each $x \in \{0, 1\}$, the support of a Bernoulli variable. Hence,

$$
\operatorname{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 = \mathbb{E}[X] - \left(\mathbb{E}[X]\right)^2 = p - p^2 = p(1-p).
$$

In the exercise (and in the next lecture), we walk through the calculations for the other discrete distribution families. In particular, we have

Geometric: $G \sim \text{Geom}(p)$, $\operatorname{Var}(G) = \frac{1-p}{p^2}$

Binomial: $B \sim \text{Binom}(n, p)$, $\operatorname{Var}(B) = np(1-p)$

Main Takeaways:

- The *expectation* of a random variable is a *summary statistic* that often reflects its "typical", or "long-term average", value.
- To compute the expectation of a discrete random variable, we sum up its values weighted by the probability mass function.
- The *Law of the Unconscious Statistician* (LOTUS) allows us to take the expectation of a function of a random variable.
- The *variance* is a summary statistic that measures how far a realization of a random variable is likely to be from its expectation. The *standard deviation* is the square root of the variance.

Exercises

28.1 Suppose that the heights (in inches) of 20 students in a 3rd-grade class are given in the following table.

Height (in)	46	48	51	53	57
Number of Students	3	6	7	3	1

Let H be a random variable that represents the height of a (uniformly) randomly chosen student.

(a) Find the probability mass function f_H of H. Use this to compute $\mathbb{E}[H]$.

Now, let's compute $\mathrm{Var}(H)$ in two ways.

(b) First, let's define an auxiliary random variable $A = (H - \mathbb{E}[H])^2$. Find the pmf f_A. Then, calculate $\mathbb{E}[A] = \mathrm{Var}(H)$.

(c) Next, use the LOTUS to calculate $\mathbb{E}\left[H^2\right]$ and then use this to calculate $\mathrm{Var}(H)$ using the formula from the lecture. Your answer should agree with part (b).

(d) What is the standard deviation of the heights of these students?

28.2 In this exercise, you'll compute the expectation of a binomial directly from its pmf.

(a) We will need to use the following in our calculation:

$$s \cdot \binom{n}{s} = n \cdot \binom{n-1}{s-1}.$$

Give a combinatorial proof of this identity.

(b) Suppose that $B \sim \mathrm{Binom}(n, p)$. Compute $\mathbb{E}[B]$ by plugging directly into Definition 28.1 and simplifying. You'll need to make use of the identity in part (a).

28.3 Suppose that $U \sim \mathrm{Unif}\{1, b\}$ for some $b \in \mathbb{N}$, $b \geq 2$.

(a) Find a formula for $\mathbb{E}[U]$ as a function of b.

(b) Find a formula for $\mathrm{Var}(U)$ as a function of b.

$\star$ (c) Generalize to the case where $U \sim \mathrm{Unif}\{a, b\}$ with $a < b \in \mathbb{Z}$. (Your answers should now be functions of both a and b.)

28.4 Suppose that two fair 6-sided dice are rolled and S is the random variable representing their sum. Recall from the previous lecture that S has support $R_S = \{2, \ldots, 12\}$ and pmf $f_S(r) = \frac{6-|s-7|}{36}$. Calculate $\mathbb{E}[S]$, $\mathrm{Var}(S)$, and $\sigma(S)$.

28.5 Suppose that X is a discrete random variable on a probability space $(\Omega, \mathcal{F}, \Pr)$, and define another random variable $Y = X + 5$ (that is, $Y(\omega) = X(\omega) + 5$ for all $\omega \in \Omega$).

(a) Find an expression for $\mathbb{E}[Y]$ in terms of $\mathbb{E}[X]$.

(b) Find an expression for $\text{Var}(Y)$ in terms of $\text{Var}(X)$.

28.6 In this exercise, you'll consider the variance of a sum of random variables. We will consider this further in the next lecture.

(a) Give an example of two random variables X and Y on a common probability space $(\Omega, \mathcal{F}, \Pr)$ with $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

(b) Now, give an example of two random variables X and Y on a common probability space $(\Omega, \mathcal{F}, \Pr)$ with $\text{Var}(X+Y) \neq \text{Var}(X) + \text{Var}(Y)$

28.7 Suppose that $G \sim \text{Geom}(p)$. In this exercise, you'll argue that $\text{Var}(G) = \frac{1-p}{p^2}$. In this argument, we'll use a more complicated variant of the "triangle summing trick" from the lecture.

(a) Use induction to prove that for each $r \in \mathbb{N}$, $r^2 = \sum_{s=1}^{r}(2s-1)$.

(b) Use the LOTUS to compute $\mathbb{E}\left[G^2\right]$.

As a first step, apply the result from part (a) to rewrite the expectation as a double sum. This is analogous to the "triangle summing trick", which uses the more banal identity $r = \sum_{s=1}^{r} 1$. Once this tricky step is done, the rest of the calculation requires applying the infinite geometric series property, some distribution, and two applications of the (original) "triangle summing trick".

(c) Use your result from part (b) to compute $\text{Var}(G)$.

⋆ **28.8** Suppose that $B \sim \text{Binom}(n, p)$. Expand on the ideas from Exercise **??** to argue that $\mathbb{E}\left[B^2\right] = np + n^2p^2 - np^2$. Use this to argue that $\text{Var}(B) = np(1-p)$. We will see an easier way to calculate this variance soon!

⋆ **28.9** The *Poisson Distribution* with rate parameter $\lambda \in \mathbb{R}^+$, denoted $\text{Pois}(\lambda)$, is another discrete distribution family. A random variable $P \sim \text{Pois}(\lambda)$ has support $R_P = \mathbb{N}$ and pmf

$$f_P(k) = \frac{\lambda^k e^{-\lambda}}{k!} \qquad \text{for each } k \in \mathbb{N}.$$

(a) Argue that this is a valid pmf.

$\left(\text{Hint: Use the power series expansion } e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}\right)$

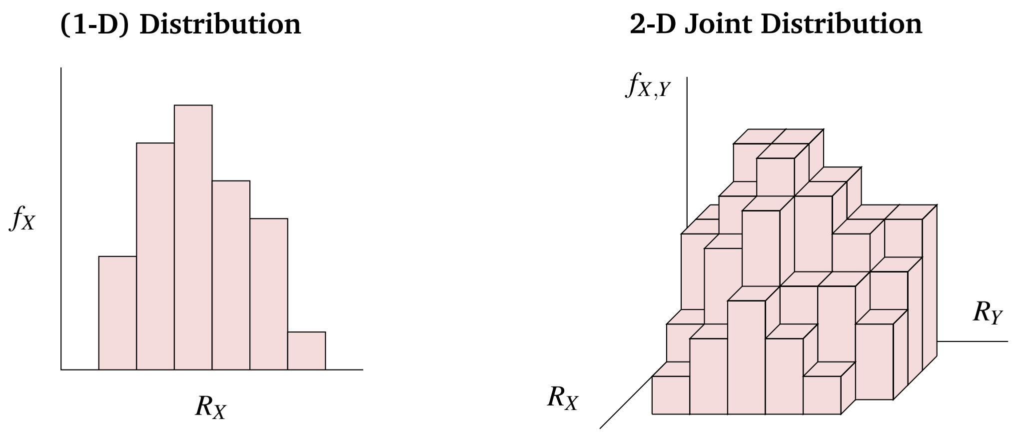
(b) What is $\mathbb{E}[P]$? (Your answer should be a function of λ.)

(c) Calculate $\mathbb{E}\left[P^2\right]$ and use this to find $\text{Var}(P)$. (Your answers should be functions of λ.)

29. Joint Distributions

When we learned about probability spaces, we started by considering events in isolation and moved on to discuss interactions between multiple events. In an analogous way, we'll consider the interactions between multiple random variables on the same probability space. These interactions are a focus of statistics, which seeks to identify and quantify relationships between observed phenomena. However, these ideas also work in another direction, allowing us to break down random variables into smaller parts in a way that simplifies many calculations. To begin, we'll introduce the concept of *joint distributions*, which provides a setting to reason about interactions between random variables.

29.1 Joint Distributions



Definition 29.1 — Joint Distribution (Discrete Random Variables).

Let X and Y be two discrete random variables on a shared probability space $(\Omega, \mathcal{F}, \Pr)$. The *joint distribution* of X and Y is a description of the probability that X and Y simultaneously assume certain values.

Their *joint pmf* $f_{X,Y} : R_X \times R_Y \to [0,1]$ has

$$f_{X,Y}(r_1,r_2) = \Pr\Big(\big\{\omega \in \Omega \mid X(\omega) = r_1, Y(\omega) = r_2\big\}\Big) \equiv \Pr(X = r_1 \cap Y = r_2)$$

for each $r_1 \in R_X$ and $r_2 \in R_Y$.

Their *joint CDF* $F_{X,Y} : \mathbb{R}^2 \to [0,1]$ has

$$F_{X,Y}(s_1,s_2) = \Pr\Big(\big\{\omega \in \Omega \mid X(\omega) \le s_1, Y(\omega) \le s_2\big\}\Big) \equiv \Pr(X \le s_1 \cap Y \le s_2).$$

for each $s_1, s_2 \in \mathbb{R}$. That is, $F_{X,Y}(s_1,s_2)$ is the probability that X assumes value at most s_1 and Y assumes value at most s_2 simultaneously.

A common term that often appears when discussing joint distributions is a *marginal distribution*. This is just the regular 1-dimensional distribution that we would get by considering one random variable and ignoring the other.

Definition 29.2 — Marginal Distribution.
Given a joint distribution $f_{X,Y}$ on a probability space $(\Omega, \mathcal{F}, \Pr)$, f_X and f_Y are its *marginal* distributions with respect to X and Y (respectively).

Example 29.1 We'll consider two probability spaces that describe choosing a random card from among four possibilities. In both probability spaces, each card is equally likely to be drawn, and the two sets of cards are

$$\Omega_1 = \big\{A♠, Q♣, A♡, Q♢\big\}, \qquad \Omega_2 = \big\{A♠, A♣, Q♡, Q♢\big\}.$$

On both probability spaces, we'll define two indicator random variables:

$$X = \mathbb{I}\big(\text{chosen card is an ace}\big),$$
$$Y = \mathbb{I}\big(\text{chosen card is black}\big).$$

Both probability spaces admit the same marginal distributions, as each has $f_X, f_Y \sim \text{Bern}\big(\frac{1}{2}\big)$. However, their joint distributions are different. We visualize,

Ω_1 :

X \ Y	red (0)	black (1)
Q (0)	Q♢	Q♣
A (1)	A♡	A♠

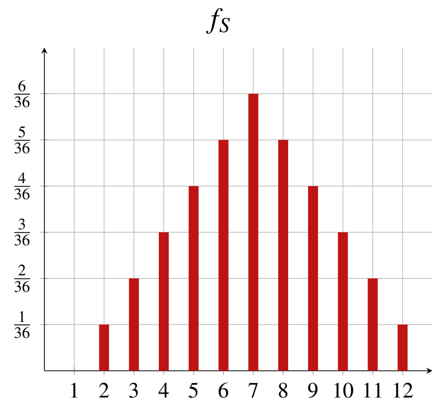
Ω_2 :

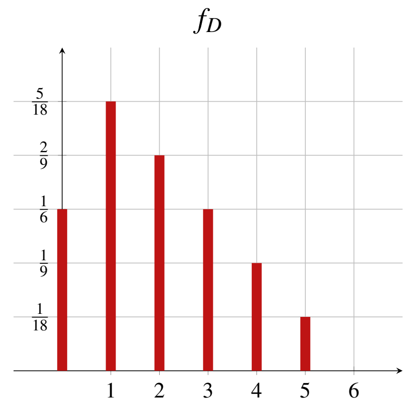
X \ Y	red (0)	black (1)
Q (0)	Q♢, Q♡	
A (1)		A♠, A♣

Thus, for Ω_1, $f_{X,Y}(0,0) = f_{X,Y}(1,0) = f_{X,Y}(0,1) = f_{X,Y}(1,1) = \frac{1}{4}$. However, for Ω_2, $f_{X,Y}(0,0) = f_{X,Y}(1,1) = \frac{1}{2}$ and $f_{X,Y}(0,1) = f_{X,Y}(1,0) = 0$.

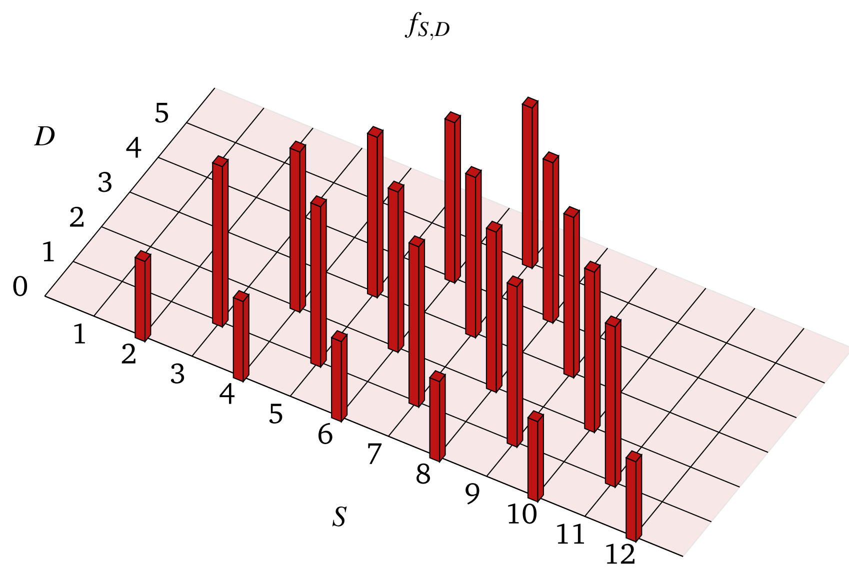
Example 29.2 Consider a random experiment where two (distinguishable) dice are rolled. We can model this random experiment with a probability space $(\Omega, \mathcal{F}, \Pr)$ where $\Omega = [6]^2$, $\mathcal{F} = \mathcal{P}(\Omega)$ and Pr is the uniform probability function that assigns $\Pr(A) = \frac{|A|}{36}$ for each $A \subseteq \Omega$.

On this probability space, let us define two random variables. S is the sum of the two dice rolls and D is the (absolute) difference of the two dice rolls. First, let us understand the marginal distributions of S and D. S has support $R_S = \{2, \ldots, 12\}$ and D has support $R_D = \{0, \ldots, 5\}$. Their pmfs are visualized below.





Now, let's consider the joint distribution of S and D. Note that each pair of distinct dice rolls appears with probability $\frac{1}{18}$ and corresponds to a pair of values (S, D). For example, the dice rolls $(4,5)$ and $(5,4)$ correspond to $S = 9$ and $D = 1$, so $f_{S,D}(9,1) = \frac{1}{18}$. Each pair of identical dice rolls appears with probability $\frac{1}{36}$. For example, the roll $(3,3)$ results in $f_{S,D}(6,0) = \frac{1}{36}$. Plotting the joint pmf, we obtain:



We can observe a close connection between the plot of the joint pmf and the marginal pmfs. Namely. if we imagine stacking all of the bars of the joint for one value of S, we obtain the height of the bar in the marginal distribution of S. Similarly, if we stack the bars for each value of D, we obtain the marginal distribution of D.

Stacking the bars in the joint pmf gives us the marginal distributions. By restricting our attention to certain "slices" of the joint pmf, we can obtain another type of distribution known as a *conditional distribution*.

Definition 29.3 — Conditional Distribution.

Given a probability space $(\Omega, \mathcal{F}, \Pr)$, a random variable X, and an event $A \in \mathcal{F}$, the *conditional distribution* of X on event A has support $R_{X|A} = \{r \in \mathbb{R} \mid X(\omega) = r \text{ for some } \omega \in A\}$ and pmf $f_{X|A} \colon R_{X|A} \to \mathbb{R}$ with $f_{X|A}(r) = \Pr(X = r|A)$.

Example 29.3 In the setting from Example 29.1, the conditional distribution of X given $Y = 1$ (i.e., the card is from a black suit) in Ω_1 is $\text{Bern}(\frac{1}{2})$. This is because of the two (equally-likely) black cards, exactly one is an ace. In Ω_2, the conditional distribution of X given $Y = 1$ is $\text{Bern}(1)$; in other words, under this conditioning, $X = 1$ with probability 1. Once we know that the chosen card is black, we are guaranteed that it is an ace.

Example 29.4 In the setting from Example 29.2, the conditional distribution of S given $D = 2$ is uniform over $\{4, 6, 8, 10\}$, as the four (equally-likely) pairs of rolls that have difference 2 are $\big\{\{1,3\}, \{2,4\}, \{3,5\}, \{4,6\}\big\}$.

The conditional distribution of D given $S = 6$ has support $\{0, 2, 4\}$ with $\Pr(D = 0 \mid S = 6) = \frac{1}{5}$ and $\Pr(D = 2 \mid S = 6) = \Pr(D = 4 \mid S = 6) = \frac{2}{5}$. Notice that there are five (equally likely) rolls that have sum 6: $\big\{(1,5), (2,4), (3,3), (4,2), (5,1)\big\}$. Of these rolls, one has difference 0, two have difference 2, and two have difference 4.

29.2 Independence of Random Variables

Example 29.1 help illustrate a new *independence* phenomenon, this time for random variables. In the probability space with Ω_1, there seems to be a decoupling between the likelihood of a drawn card being black and being an ace. More formally, the probability of selecting an ace remains the same $\frac{1}{2}$ whether we consider all of the cards, or just the black cards. Contrast this with Ω_2, where the likelihood that a drawn card is black is tightly coupled with whether it is an ace; once we see that it is an ace, we are guaranteed that it is black. We formalize the notion of independence with the following definition.

Definition 29.4 — (Mutual) Independence of Discrete Random Variables.

Two discrete random variables X and Y on a shared probability space $(\Omega, \mathcal{F}, \Pr)$ are *independent* if for all $r_1 \in R_X$ and all $r_2 \in R_Y$,

$$f_{X,Y}(r_1, r_2) = f_X(r_1) \cdot f_Y(r_2).$$

Equivalently, the events $\{X = r_1\}$ and $\{Y = r_2\}$ are independent.

Random variables $X_1, X_2, \ldots, X_n$ are *mutually independent* if for all $r_i \in R_{X_i}$, the events $\{X_1 = r_1\}, \{X_2 = r_2\}, \ldots \{X_n = r_n\}$ are mutually independent.

In the exercises (see Exercise 29.5), we ask you to establish an equivalent definition of the independence of random variables in terms of conditional distributions.

Example 29.5 Suppose that two fair dice, one red and one blue, are rolled. Then, the random variables R representing the value of the red die and B representing the value of the blue die are independent. This is because for any $r_1, r_2 \in \{1,2,3,4,5,6\}$,

$$f_{R,B}(r_1,r_2) = \Pr(R = r_1 \cap B = r_2) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = \Pr(R = r_1) \cdot \Pr(B = r_2) = f_R(r_1) \cdot f_B(r_2).$$

In this example, the independence is related to the fact that we can "divide the randomness" of the experiment into smaller, separate random events. This experiment is equivalent to running two separate experiments where one die is rolled. We can also combine a collection of simple independent random variables to come up with a more complicated distribution. As an example, the following lemma gives an alternate description of the binomial distribution.

Lemma 29.1 Suppose that $B_1, \dots, B_n$ are mutually independent $\text{Bern}(p)$ random variables. Then, their sum $B = B_1 + \ldots + B_n$ is a random variable with $B \sim \text{Binom}(n,p)$.

Proof. First, let's reason about the support of B. Since B is the sum of $\{0,1\}$-random variables, its support must be a subset of $\mathbb{N}$. The smallest value that B can assume is $B = 0$, which happens when each $B_i = 0$. The largest value that B can assume is $B = n$, which happens when each $B_i = 1$. Furthermore, B can assume any natural number $0 \le k \le n$ (for example, if $B_1, \dots, B_k = 1$ and $B_{k+1}, \dots, B_n = 0$), so $R_B = \{0, \dots, n\}$.

Now, let's reason about the pmf of B. To introduce some helpful notation, given any subset $S \subseteq [n]$, let us define $\mathbb{I}(S) = (b_1, \dots, b_n) \in \{0,1\}^n$ with

$$b_i = \begin{cases} 0 & i \notin S, \\ 1 & i \in S. \end{cases}$$

This gives us a compact way to construct a vector of the values of all of the B_i random variables. Now, given any $0 \le k \le n$, note that

$$f_B(k) = \sum_{\substack{S \subseteq [n] \\ |S| = k}} f_{B_1,\dots,B_n}\big(\mathbb{I}(S)\big).$$

This says that $B = k$ when exactly k of the B_i variables have value 1, and we must consider all possible subsets S of k of these variables. Now, by the mutual independence of the B_i, we have

$$f_{B_1,\dots,B_n}\big(\mathbb{I}(S)\big) = \prod_{i \in S} f_{B_i}(1) \prod_{j \in [n] \setminus S} f_{B_j}(0) = p^{|S|}(1-p)^{|[n] \setminus S|} = p^k (1-p)^{n-k}.$$

As this is the same value for all subsets S, we conclude that

$$f_B(k) = \binom{n}{k} p^k (1-p)^{n-k},$$

meaning $B \sim \text{Binom}(n,p)$. ■

Main Takeaways:

- A *joint distribution* describes the probability that two random variables defined on the same probability space *simultaneously* assume a particular value.
- Random variables are independent when we can decompose their joint pmf as a product of their marginal pmfs.
- The sum of mutually independent Bernoulli random variables is a binomial random variable.

Exercises

29.1 Given two discrete random variables X and Y on a common probability space $(\Omega, \mathcal{F}, \Pr)$ and two values $s_1, s_2 \in \mathbb{R}$, express the value $F_{X,Y}(s_1, s_2)$ of the joint CDF in terms of values of the joint pmf $f_{X,Y}$.

29.2 Suppose that X and Y are discrete random variables on a common probability space $(\Omega, \mathcal{F}, \Pr)$. Let $f_{X,Y}$ be their joint pmf and let f_X and f_Y be their marginal pmfs.

(a) Argue that for each $r_1 \in R_X$,

$$f_X(r_1) = \sum_{r_2 \in R_Y} f_{X,Y}(r_1, r_2).$$

(b) Establish a similar identity for f_Y.

29.3 In this exercise, we'll explore a cool consequence of Lemma 29.1.

(a) Use Lemma 29.1 to argue that if $B_1 \sim \text{Binom}(n_1, p)$ and $B_2 \sim \text{Binom}(n_2, p)$ are independent, and if $B = B_1 + B_2$, then $B \sim \text{Binom}(n_1 + n_2, p)$.

(b) Give an alternate proof of the claim from part (b) where you directly reason about the pmf of B. You may wish to use the identity:

$$\sum_{j=0}^{k} \binom{n}{j} \binom{m}{k-j} = \binom{n+m}{k},$$

which can be proven combinatorially.

(c) Explain why the independence condition is necessary in part (b). That is, give an example where $B_1 \sim \text{Binom}(n_1, p)$, $B_2 \sim \text{Binom}(n_2, p)$, $B = B_1 + B_2$, and $B \not\sim \text{Binom}(n_1 + n_2, p)$

29.4 Suppose that we roll two (fair, 6-sided) dice and let U be the larger number that is rolled and L be the smaller number that is rolled (if both numbers are the same, then $U = L$ is this common number).

(a) Compute the marginal distributions of U and L.

(b) Compute the joint distribution of U and L.

(c) For each $\ell \in [6]$, compute the conditional distribution of U given $L = \ell$.

(d) For each $u \in [6]$, compute the conditional distribution of L given $U = u$.

(e) Compute the conditional distribution of U given $U = L + 1$.

(f) Are U and L independent random variables? Explain how you reached this conclusion.

⋆ **29.5** Suppose that X and Y are random variables on a common probability space $(\Omega, \mathcal{F}, \Pr)$.

(a) Prove that if X and Y are independent, then for all $y \in R_Y$, $f_{X|Y=y} = f_X$ (i.e., the conditional distribution of X on event $Y = y$ is equal to the marginal distribution of X).

(b) Conversely, prove if $f_{X|Y=y} = f_X$ for all $y \in R_Y$, then X and Y are independent.

This equivalent criteria for independence of random variables is an analogue of our "conditioning doesn't effect probabilities" characterization of independent events.

29.6 Suppose that $G_1, \ldots, G_r \sim \text{Geom}(p)$ are mutually independent. Then, their sum $G = G_1 + \ldots + G_r$ is a random variable with a *negative binomial* distribution, $G \sim \text{NB}(r, p)$.

G has support $R_G = \{n \in \mathbb{N} \mid n \geq r\}$ and pmf f_G given by

$$f_G(k+r) = \binom{k+r-1}{r-1} p^r (1-p)^k.$$

for each $k \in \mathbb{N}$.

(a) Verify that f_G is a valid pmf.

(b) Give a combinatorial explanation for this pmf.

(c) Compute $\mathbb{E}[G]$ and $\text{Var}(G)$.

(d) Based on your answer to part (c), how many times should I expect to need to flip a fair coin before it comes up heads 4 times?

(e) What is the probability that the coin will land heads at least 4 times if it is flipped 8 times?

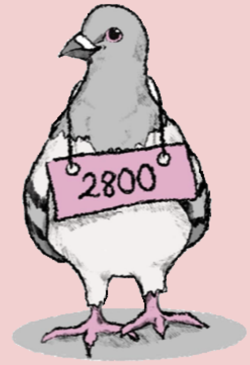
(Try calculating this in two ways: once using a Binomial distribution and once using a Negative Binomial distribution. Make sure that your answers agree.)

⋆ **29.7** Recall that a random variable P has the *Poisson Distribution*, $P \sim \text{Pois}(\lambda)$, if it has support $R_P = \mathbb{N}$ and pmf:

$$f_P(k) = \frac{\lambda^k e^{-\lambda}}{k!} \qquad \text{for each } k \in \mathbb{N}.$$

Suppose that $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ are independent, and let $Z = X + Y$. Argue that $Z \sim \text{Pois}(\lambda_1 + \lambda_2)$.

(Hint: Use the Binomial Theorem.)



30. Covariance and Correlation

Two lectures ago, we introduced two important *statistics*, expectation and variance, of a random variable. In the last lecture, we introduced joint distributions as a way to simultaneously reason about multiple random variables. In today's lecture, we will combine these ideas together to reason about statistics involving multiple random variables.

30.1 Linearity of Expectation

One of the most important facts in all of probability and statistics is the *linearity of expectation*. This gives us a nice formula to compute the expectation of *linear combinations* of random variables (that is, sums of random variables with constant coefficients).

Theorem 30.1 — Linearity of Expectation. Suppose that X and Y are random variables on a common probability space $(\Omega, \mathcal{F}, \Pr)$, and let $a, b \in \mathbb{R}$. Then,

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

Before we consider how to prove this theorem, we make two key observations. First, the theorem does not limit us to combinations of two random variables; we can use induction to generalize this theorem to any finite linear combination (that is sum of constants times random variables) of random variables (see Exercise 30.4). Second, note that the theorem does not make any assumptions about the independence of random variables X and Y:

Linearity of expectation holds for variables that are *not* independent!

Proof. Define random variable $Z = aX + bY$, so that $\mathbb{E}[aX + bY] = \mathbb{E}[Z]$. Applying the definition of expectation, we have

$$\mathbb{E}[Z] = \sum_{z \in R_Z} z \cdot f_Z(z).$$

Since $Z(\omega) = a \cdot X(\omega) + b \cdot Y(\omega)$ for all $\omega \in \Omega$, R_Z consists of real numbers that can be realized as linear combinations of values in R_X and R_Y:

$$R_Z = \Big\{ z \in \mathbb{R} \mid z = ax + by \text{ for some } x \in R_X \text{ and } y \in R_y \Big\}.$$

Thus, we may rewrite

$$\begin{aligned}
\sum_{z \in R_Z} z \cdot f_Z(z) &= \sum_{x \in R_X} \sum_{y \in R_Y} (ax+by) \cdot f_{X,Y}(x,y) \\
&= a \sum_{x \in R_X} \sum_{y \in R_Y} x \cdot f_{X,Y}(x,y) + b \sum_{x \in R_X} \sum_{y \in R_Y} y \cdot f_{X,Y}(x,y) \\
&= a \sum_{x \in R_X} x \sum_{y \in R_Y} f_{X,Y}(x,y) + b \sum_{y \in R_Y} y \sum_{x \in R_X} f_{X,Y}(x,y).
\end{aligned}$$

Here, the latter two lines use the distributive property to rearrange these summations. From here, let's focus on the inner summations. By countable additivity, we have

$$\sum_{y \in R_Y} f_{X,Y}(x,y) = \sum_{y \in R_Y} \Pr(X = x \cap Y = y) = \Pr\Big(\bigcup_{y \in R_Y} (\{X = x\} \cap \{Y = y\}) \Big)$$

Now, using the fact that the events $\{Y = y\}$ for $y \in R_Y$ partition Ω, we may continue the simplification,

$$= \Pr\Big(\{X = x\} \cap \bigcup_{y \in R_Y} \{Y = y\} \Big) = \Pr\Big(\{X = x\} \cap \Omega \Big) = \Pr(X = x) = f_X(x).$$

Similarly, we may simplify the second inner summation to $f_Y(y)$. Thus, we have,

$$\mathbb{E}[aX + bY] = a \sum_{x \in R_X} x \cdot f_X(x) + b \sum_{y \in R_Y} y \cdot f_Y(y) = a\mathbb{E}[x] + b\mathbb{E}[Y].$$

■

The linearity of expectation is extremely powerful and gets applied almost anywhere where you see both a sum and an expectation. As a first application, we can use the linearity of expectation to more easily calculate the expectation of a binomial random variable.

Example 30.1 — Binomial Random Variables. Suppose that $B \sim \text{Binom}(n, p)$. Recall that we can write $B = B_1 + \ldots + B_n$, where each $B_i \sim \text{Bern}(p)$. By linearity of expectation, we have

$$\mathbb{E}[B] = \mathbb{E}[B_1 + \ldots + B_n] = \mathbb{E}[B_1] + \ldots + \mathbb{E}[B_n] = np.$$

Next, we give a more "exotic" use of the linearity of expectation. In this example, reasoning about the expectation directly would be cumbersome (requiring us to divide the set of all 52! permutations of a deck of cards into many complicated cases). However, the linearity of expectation allows us to break this calculation down into more manageable pieces.

Example 30.2 Suppose that a deck of cards is shuffled (so all orders are equally likely) and the entire deck is dealt out into piles as follows. The first card starts the first pile. Then, whenever the next card has the same color as the previous card, it goes on its pile. Otherwise, when it has the opposite color of the previous card, it begins a new pile. Let X be the random variable that counts the number of piles. We'll compute $\mathbb{E}[X]$.

We'll define random variables $\{X_i\}_{i=1}^{51}$ such that $X_i = \mathbb{I}(\text{card } i \text{ has opposite color as card } i+1)$. Then, $X = 1 + X_1 + \ldots + X_{51}$ (1 for the first pile, and then each X_i indicates the start of a new pile). To compute each $\mathbb{E}[X_i]$, suppose we look at the color of the i'th card. Since each of the remaining 51 cards is equally likely to be the next card, and 26 of them have the opposite color as the i'th card, $\mathbb{E}[X_i] = \frac{26}{51}$. By linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}[1 + X_1 + \ldots + X_{51}] = 1 + \mathbb{E}[X_1] + \ldots + \mathbb{E}[X_{51}] = 1 + 51 \cdot \tfrac{26}{51} = 27.$$

30.2 Independence, Expectation, and Variance

Next, we'll show that the expectations and variances of independent random variables have additional nice properties. In particular, we will give an analog of the linearity of expectation for the variance of *independent* random variables. This result will rely on the following lemma.

Lemma 30.1 Suppose that X and Y are independent discrete random variables. Then,

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

The proof of this lemma has a similar structure to the proof of the linearity of expectation.

Proof. Define random variable $Z = XY$. Then,

$$\begin{aligned}
\mathbb{E}[XY] = \mathbb{E}[Z] &= \sum_{z \in R_Z} z \cdot f_Z(z) && \text{(Definition 33.1)} \\
&= \sum_{x \in R_X} \sum_{y \in R_Y} (xy) \cdot f_{X,Y}(x,y) && (\star) \\
&= \sum_{x \in R_X} \sum_{y \in R_Y} (xy) \cdot f_X(x) \cdot f_Y(y) && \text{(Independence)} \\
&= \Big(\sum_{x \in R_X} x \cdot f_X(x) \Big) \Big(\sum_{y \in R_Y} y \cdot f_Y(y) \Big) && \text{(Distributivity)} \\
&= \mathbb{E}[X] \, \mathbb{E}[Y]. && \text{(Definition 33.1)}
\end{aligned}$$

■

Note that this equality will always hold for independent random variables; however, it may also hold for a pair of random variables that are not independent, as we will discuss more later.

A consequence of this expectation formula is an analog of the linearity of expectation for *independent* random variables. Note here that independence is crucial; this equality does not hold in general!

Theorem 30.2 Suppose that X and Y are *independent* random variables, and $a, b \in \mathbb{R}$. Then,

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + b^2 \cdot \text{Var}(Y).$$

Proof.

$$\begin{aligned}
\operatorname{Var}(aX+bY) &= \mathbb{E}\left[(aX+bY)^2\right] - \big(\mathbb{E}[aX+bY]\big)^2 \\
&= a^2\cdot\mathbb{E}\left[X^2\right] + 2ab\cdot\mathbb{E}[XY] + b^2\cdot\mathbb{E}\left[Y^2\right] - \big(a\cdot\mathbb{E}[X]+b\cdot\mathbb{E}[Y]\big)^2 \\
&= a^2\cdot\mathbb{E}\left[X^2\right] + 2ab\cdot\mathbb{E}[XY] + b^2\cdot\mathbb{E}\left[Y^2\right] \\
&\quad - a^2\big(\mathbb{E}[X]\big)^2 - 2ab\cdot\mathbb{E}[X]\,\mathbb{E}[Y] - b^2\big(\mathbb{E}[Y]\big)^2 \\
&= a^2\cdot\operatorname{Var}(X) + b^2\cdot\operatorname{Var}(Y) + 2ab\Big(\mathbb{E}[XY]-\mathbb{E}[X]\,\mathbb{E}[Y]\Big) \\
&= a^2\cdot\operatorname{Var}(X) + b^2\cdot\operatorname{Var}(Y)
\end{aligned}$$

Here, the last line follows from the lemma (since X an Y are independent). ■

Example 30.3 — Binomial Random Variables. Suppose that $B\sim\text{Binom}(n,p)$. Since B has the same distribution as the sum of n mutually independent $\text{Bern}(p)$ random variables $B = B_1+\ldots+B_n$, we have,

$$\operatorname{Var}(B) = \operatorname{Var}(B_1+\ldots+B_n) = \operatorname{Var}(B_1)+\ldots+\operatorname{Var}(B_n) = np(1-p).$$

30.3 Covariance and Correlation

A related notion to variance is *covariance*. While the variance quantifies the distribution of one random variable, the covariance quantifies the joint distribution of two random variables.

Definition 30.1 — Covariance.
Given two random variables X and Y, their covariance, denoted $\operatorname{Cov}(X,Y)$, is given by

$$\operatorname{Cov}(X,Y) = \mathbb{E}\left[\big(X-\mathbb{E}[X]\big)\big(Y-\mathbb{E}[Y]\big)\right].$$

Note that the covariance is a generalization of the variance, as $\operatorname{Cov}(X,X)=\operatorname{Var}(X)$. Also, note that we can again use the linearity of expectation to obtain an alternate formula for covariance.

$$\begin{aligned}
\operatorname{Cov}(X,Y) &= \mathbb{E}\left[\big(X-\mathbb{E}[X]\big)\big(Y-\mathbb{E}[Y]\big)\right] \\
&= \mathbb{E}\left[XY - \mathbb{E}[Y]X - \mathbb{E}[X]Y + \mathbb{E}[X]\,\mathbb{E}[Y]\right] \\
&= \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] + \mathbb{E}[X]\,\mathbb{E}[Y] \\
&= \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].
\end{aligned}$$

Observe that Lemma 30.1 implies that independent random variables have covariance 0. A natural question to ask is what the covariance tells us about the random variables more generally. Let's look at our original definition. Suppose that $\operatorname{Cov}(X,Y)$ is positive. Then, when X is relatively large (i.e. greater than $\mathbb{E}[X]$), we'd expect Y to be relatively large (i.e. greater than $\mathbb{E}[Y]$) as well so that the product is positive. On the other hand, when $\operatorname{Cov}(X,Y)$ is negative, we expect larger values of X to coincide with smaller values of Y. This explains the sign of the covariance, but what about its magnitude? We cannot really interpret the magnitude of the covariance directly, as it is largely affected by how spread out the values of the random variables are (i.e., their variances). To account for this, we re-normalize the covariance using the standard deviations of each variable to give a new measure, known as the correlation.

Definition 30.2 — Correlation.

Given two random variables X and Y, their correlation, denoted $\text{corr}(X,Y)$, is given by

$$\text{corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma(X)\sigma(Y)}.$$

We can record the following facts about how to interpret the correlation.

- When $\text{corr}(X,Y) > 0$, X and Y are *positively correlated*. Larger values of X tend to coincide with larger values of Y. The maximum correlation value $\text{corr}(X,Y) = 1$ corresponds to perfectly positively correlated variables ($X = Y$).
- When $\text{corr}(X,Y) < 0$, X and Y are *negatively correlated*. Larger values of X tend to coincide with smaller values of Y. The minimum correlation value $\text{corr}(X,Y) = -1$ corresponds to perfectly negatively correlated variables ($X = -Y$).
- When $\text{corr}(X,Y) = 0$, X and Y are *uncorrelated*. There is no quantifiable relationship between the values of X and Y.

This notion of *uncorrelation* is a generalization of the independence of random variables. That is, independent random variables are uncorrelated, but not all uncorrelated random variables are independent (see Exercise 30.5).

Example 30.4 Suppose that we draw a hand of 3 cards from a standard deck (where each combination of cards is equally likely). Let us define random variables:

- H = number of hearts in the hand
- C = number of clubs in the hand

We wish to understand the correlation between H and C. (Before reading on, give this some thought. Do you expect larger values of H to coincide with larger values of C, or smaller values? Do you think that there is even a relationship at all?)

Let us calculate some values the pmf of H (which is the same as the pmf of C). Note that we can ignore $f_H(0)$, as this will be zero-ed out in the calculations of $\mathbb{E}[H]$ and $\text{Var}(H)$ anyway. We have,

$$f_H(1) = \frac{\binom{3}{1}\cdot 13\cdot 39\cdot 38}{52\cdot 51\cdot 50} = \frac{741}{1{,}700}, \qquad f_H(2) = \frac{\binom{3}{2}\cdot 13\cdot 12\cdot 39}{52\cdot 51\cdot 50} = \frac{234}{1{,}700}, \qquad f_H(3) = \frac{13\cdot 12\cdot 11}{52\cdot 51\cdot 50} = \frac{22}{1{,}700}.$$

Thus, we calculate

$$\begin{aligned}
\mathbb{E}[H] = \mathbb{E}[C] &= 1\cdot\tfrac{741}{1{,}700} + 2\cdot\tfrac{234}{1{,}700} + 3\cdot\tfrac{22}{1{,}700} = \tfrac{1{,}275}{1{,}700} = \tfrac{3}{4} \\
\mathbb{E}\left[H^2\right] = \mathbb{E}\left[C^2\right] &= 1\cdot\tfrac{741}{1{,}700} + 4\cdot\tfrac{234}{1{,}700} + 9\cdot\tfrac{22}{1{,}700} = \tfrac{1{,}875}{1{,}700} = \tfrac{75}{68} \\
\text{Var}(H) = \text{Var}(C) = \sigma(H)\sigma(C) &= \tfrac{75}{68} - \left(\tfrac{3}{4}\right)^2 = \tfrac{147}{272}
\end{aligned}$$

To calculate the covariance, we must consider joint probabilities of having a non-zero number of hearts and clubs. We have:

$$f_{H,C}(1,1) = \tfrac{3!\cdot 13\cdot 13\cdot 26}{52\cdot 51\cdot 50} = \tfrac{169}{850}, \qquad f_{H,C}(2,1) = f_{H,C}(1,2) = \tfrac{\binom{3}{1}\cdot 13\cdot 13\cdot 12}{52\cdot 51\cdot 50} = \tfrac{39}{850}.$$

Thus, we calculate

$$\begin{aligned}
\mathbb{E}[HC] &= 1\cdot f_{H,C}(1,1) + 2\cdot f_{H,C}(2,1) + 2\cdot f_{H,C}(1,2) = 1\cdot\tfrac{169}{850} + 2\cdot\tfrac{39}{850} + 2\cdot\tfrac{39}{850} = \tfrac{13}{34}\\
\mathrm{Cov}(H,C) &= \tfrac{13}{34} - \left(\tfrac{3}{4}\right)^2 = -\tfrac{49}{272}\\
\mathrm{corr}(H,C) &= \frac{-\frac{49}{272}}{\frac{147}{272}} = -\tfrac{1}{3}.
\end{aligned}$$

H and C are negatively correlated, and the strength of this correlation is moderate. Intuitively, a negative correlation makes sense because the more hearts that we selected, the less room there is in the hand for clubs.

Main Takeaways:

- The linearity of expectation is a very important property that can greatly simplify many calculations with random variables. It does *not* require independence.
- The expectation of a product of independent random variables is the product of their expectations. From this, we conclude that the variance of linear combinations of independent random variables is a linear combination of their variances.
- *Covariance* is a measurement of how the realized value of one random variable affects the realized value of another. The *correlation* of two random variables is a normalized version of their covariance.

Exercises

30.1 Use the linearity of expectation to argue that

$$\mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2.$$

This verifies that the two definitions that we have given for the variance are equivalent.

30.2 In this exercise, you'll compute the expectation of a random variable in three different ways. Suppose that 4 cards are drawn at random from a standard deck (with all combinations equally likely). Let R be the random variable representing the number of red cards that were drawn.

(a) Compute the probabilities that exactly $\{0,1,2,3,4\}$ red cards were drawn (i.e., the pmf of R). Use these to compute $\mathbb{E}[R]$ directly.

(b) Let $R_i = \mathbb{I}(i\text{'th drawn card is red})$ for each $i \in \{1,2,3,4\}$. Calculate $\mathbb{E}[R_i]$ for each i and apply linearity of expectation to compute $\mathbb{E}[R]$.

(c) Let B be the random variable representing the number of black cards that were drawn. Argue that $f_B = f_R$ and that $R + B = 4$. Then, use these facts and the linearity of expectation to compute $\mathbb{E}[R]$.

30.3 Suppose that a permutation σ of $[9]$ is selected uniformly at random. Use linearity to determine the expectation of the following random variables.

(Hint: Represent these random variables as sums of indicator random variables.)

(a) L = the number of digits to the left of 7 in σ

(b) B = the number of digits between 5 and 8 in σ

(c) F = the number of fixed points in σ (recall that a fixed point is a value $i \in [9]$ that appears in the i'th position of the permutation)

(d) G = the number of digits $i \in [9]$ that appear in a position $j > i$ in σ

(e) N = the number of consecutive runs of digits in σ that sum to 10 (for example, there are 3 such runs in $(9,7,3,2,5,4,1,8,6)$: $7+3$, $3+2+5$, and $5+4+1$.)

30.4 Use a proof by induction to generalize the linearity of expectation to any finite linear combination of random variables.

30.5 Let us define two independent random variables X and Y with $f_X(0) = f_X(1) = \frac{1}{2}$ and $f_Y(-1) = f_Y(1) = \frac{1}{2}$. Then, let $Z = XY$.

Argue that $\mathrm{Cov}(X,Z) = 0$ (so X and Z are uncorrelated) even though X and Z are *not* independent.

30.6 Repeat the calculations in Example 30.4 for a hand of 5 cards.

30.7 Suppose that $U_1, U_2 \sim \mathrm{Unif}\{0, 100\}$ are independent, and define random variables:

$$S = \min(U_1, U_2), \qquad L = \max(U_1, U_2).$$

(a) Find the pmfs f_S and f_L.

(b) Compute $\mathbb{E}[S]$ and $\mathbb{E}[L]$. (Once you compute one of these, it should be easy to calculate the other.)

(c) Calculate $\mathrm{Var}(S)$ and $\mathrm{Var}(L)$. (This should require two separate calculations.)

(d) Calculate $\mathbb{E}[SL]$ and use this to find $\mathrm{Cov}(S,L)$.

(Hint: Rather than reasoning about the joint distribution of S and L to compute $\mathbb{E}[SL]$, we can reason about the joint distribution of U_1 and U_2 (why?), which is much nicer to work with.)

(e) What is $\mathrm{corr}(S,L)$? Explain why the sign (positive/negative) of the correlation makes sense.

30.8 10 marbles are randomly selected and placed into a jar. Each marble is equally likely to be red, yellow, or blue, and these selections are made (mutually) independently. Define the random variables R, Y, and B, that count the number of red, yellow, and blue (respectively) marbles in the jar.

(a) What is the distribution of R?

(b) What are $\mathbb{E}[R]$, $\mathrm{Var}(R)$, and $\sigma(R)$?

By the symmetry of this experiment (red, blue, and yellow balls are treated exactly the same), the results from parts (a) and (b) also hold for random variables Y and B.

(c) Use the definition of expectation to compute $\mathbb{E}[RB]$. The following algebraic identities may be useful for these calculations:

For any $n, a \in \mathbb{N}$,

$$\sum_{k=0}^{n} \binom{n}{k} \frac{k}{a^k} = \frac{n(a+1)^{n-1}}{a^n}, \qquad \sum_{k=0}^{n} \binom{n}{k} \frac{k^2}{a^k} = \frac{n(a+n)(a+1)^{n-2}}{a^n}.$$

(d) Use your answer from part (c) to calculate $\mathrm{Cov}(R,B)$ and $\mathrm{corr}(R,B)$.

(e) Explanation of why the sign (positive/negative) of the correlation you computed in part (d) makes sense.

31. Probability Inequalities

The field of statistics focuses on drawing conclusions about a large population through observations or experiments on a much smaller sample. For example, medical statisticians at the FDA may wish to decide whether to approve a drug treatment based on a medical trial. For such approval to take place, a drug company must be able to establish a *statistically significant* connection between the administration of their drug and the improvement in health outcomes. When we talk about statistical significance, we mean the ability to confidently conclude that the outcomes were due to the drug treatment as opposed to the natural randomness in the sampled individuals (brought about by genetics, behaviors, etc.). To establish statistical significance, we need to be able to quantify the likelihood that an observation (that is, a particular realization of a random variable) deviates from its expectation. This is exactly the use case for many famous probability inequalities. We will introduce three such inequalities in this lecture.

31.1 Markov's Inequality

Perhaps the most straightforward of these inequalities is Markov's inequality, which is an immediate consequence of our definition of expectation.

Theorem 31.1 — Markov's Inequality. Suppose X is a *non-negative* random variable (that is, the support of X is a subset of $\mathbb{R}_{\geq 0}$). Then, for any $\alpha > 0$,

$$\Pr(X \geq \alpha) \leq \frac{\mathbb{E}[X]}{\alpha}.$$

Proof for Discrete Random Variables. Suppose that X is discrete with support R_X. By applying the definition of expectation but splitting the possible values $r \in R_X$ into two separate sums (based on whether they are $\geq \alpha$), we have,

$$\mathbb{E}[X] = \sum_{\substack{r \in R_X \\ r < \alpha}} r \cdot f_X(r) + \sum_{\substack{r \in R_X \\ r \geq \alpha}} r \cdot f_X(r) \geq 0 \sum_{\substack{r \in R_X \\ r < \alpha}} f_X(r) + \alpha \sum_{\substack{r \in R_X \\ r \geq \alpha}} f_X(r) = \alpha \cdot \Pr(X \geq \alpha).$$

Here, the inequality uses the fact that $f_X(r) \geq 0$ for all $r \in R_X$. Dividing both sides of these inequalities by α (noting that $\alpha > 0$, so the direction of the inequality is preserved) gives Markov's inequality. ■

Another way that you will often see Markov's Inequality stated is with a multiple of the expectation inside of the probability. Suppose we let $\alpha = c\mathbb{E}[X]$ for some $c \geq 1$. Then,

$$\Pr(X \geq c\mathbb{E}[X]) \leq \frac{\mathbb{E}[X]}{c\mathbb{E}[X]} = \frac{1}{c}.$$

Markov's inequality is useful in settings where we know the expectation of a random variable (or a good approximation to this expectation), but lack more fine-grained knowledge of its distribution.

Example 31.1 Suppose that the average GPA at a university is 2.75. At least how many students at the university have a GPA less than 3.75?

Let G be the random variable representing a student's GPA. Using Markov's Inequality, we find that

$$\Pr(G \geq 3.75) \leq \frac{\mathbb{E}[G]}{3.75} = \frac{2.75}{3.75} \approx 0.73.$$

Since at most 73 percent of students have a GPA at least 3.75, at least 27 percent must have a GPA below 3.75.

When we have more information about the distribution, however, the bounds returned by Markov's inequality can be very weak.

Example 31.2 Suppose that a fair coin is flipped 100 times, and the random variable H records the number of heads. $H \sim \text{Binom}(100, \frac{1}{2})$, so $\mathbb{E}[H] = 50$. We may use Markov's inequality to bound the probability of flipping 100 heads:

$$\Pr(H \geq 100) \leq \frac{\mathbb{E}[H]}{100} = \frac{50}{100} = \frac{1}{2}.$$

While this bound is correct, it is extremely loose. In fact,

$$\Pr(H \geq 100) = \Pr(H = 100) = \binom{100}{100} \cdot \left(\frac{1}{2}\right)^{100} \approx 7.9 \cdot 10^{-31}.$$

This looseness of Markov's inequality is a direct consequence of the expectation being just a small snapshot of the random variable. More sophisticated probability inequalities will utilize other information about a random variable, such as its variance, in order to establish a tighter bound. One such inequality is Chebyshev's inequality.

31.2 Chebyshev's Inequality

The expectation of a random variable gives us a rough approximation of the center of its distribution. Markov's inequality bounds the probability of an event based on its distance from this center. On the other hand, the variance of a random variable gives us a measurement of the spread, or width, or the distribution. How can we incorporate this information into a probability bound?

Intuitively, when a random variable has lower variance, we'd expect more samples from it to be very close to the expected value. That is, we should be more surprised when a sample is far away from this expected value; this is a rare event. However, when a random variable has a higher variance, it is more likely that a sample is far from the expected value. Therefore, the probability of being a fixed distance away from the mean should increase as the variance increases. This relationship is formalized by Chebyshev's inequality.

Theorem 31.2 — Chebyshev's Inequality. Suppose that X is a random variable. Then, for any $\beta > 0$,

$$\Pr\left(\left|X - \mathbb{E}[X]\right| \geq \beta\right) \leq \frac{\mathrm{Var}(X)}{\beta^2}.$$

There is a lot going on in this inequality. Before we see its proof, let's carefully unpack the statement. Inside of the probability, we see $|X - \mathbb{E}[X]|$. This is how far the value of X is from its expectation. We are concerned with the event that this distance is at least β. The probability is over the distribution of values that X can assume. The inequality says that we can bound this probability by $\frac{\mathrm{Var}(X)}{\beta^2}$. Does this fraction make sense? Well, from our intuition above, we are happy to see $\mathrm{Var}(X)$ in the numerator; a higher variance means we should be more likely to see outlying samples. Seeing β^2 in the denominator also seems reasonable. As β gets larger, we want our samples from X to be farther from the expectation, which should become less and less likely.

The proof of Chebyshev's inequality is a clever application of Markov's inequality.

Proof. Let the random variable $Y = (X - \mathbb{E}[X])^2$. Note that

$$\mathbb{E}[Y] = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathrm{Var}(X).$$

We have,

$$\Pr\left(\left|X - \mathbb{E}[X]\right| \geq \beta\right) = \Pr(Y \geq \beta^2) \leq \frac{\mathbb{E}[Y]}{\beta^2} = \frac{\mathrm{Var}(X)}{\beta^2},$$

where the inequality is an application of Markov's inequality. ■

Just as with Markov's inequality, we can re-express Chebyshev's inequality in a way that puts a constant on the right-hand side. If we let $\beta = c\sigma(X)$, where $\sigma(X)$ is the standard deviation of X, for some $c > 0$, we find that

$$\Pr\Big(\left|X - \mathbb{E}[X]\right| \geq c\sigma(X)\Big) \leq \frac{\mathrm{Var}(X)}{c^2\mathrm{Var}(X)} = \frac{1}{c^2}.$$

We can interpret this re-formulation as saying that the probability that X is c standard deviations away from its expectation is at most $\frac{1}{c^2}$.

As the first application of Chebyshev's inequality, let's revisit our coin example from earlier.

Example 31.3 A fair coin is flipped 100 times, and the random variable H records the number of heads. $H \sim \text{Binom}(100, \frac{1}{2})$, so $\mathbb{E}[H] = 50$ and $\text{Var}(H) = 25$. By Chebyshev's inequality,

$$\Pr(H \geq 100) = \frac{1}{2} \cdot \Pr\big(|H - 50| \geq 50\big) \leq \frac{1}{2} \cdot \frac{\text{Var}(H)}{(50)^2} = \frac{25}{5000} = \frac{1}{200}.$$

Here, the $\frac{1}{2}$ follows from the symmetry of the binomial distribution, since

$$\Pr\big(|H - 50| \geq 50\big) = \Pr(H \leq 0) + \Pr(H \geq 100) = 2 \cdot \Pr(H \geq 100).$$

This is a big improvement on our bound of $\frac{1}{2}$ from Markov's inequality, but is still many orders of magnitude off from the true probability.

As a second application, we will use Chebyshev's inequality to derive a central result in probability known as the Law of Large Numbers.

31.2.1 The Law of Large Numbers

When we introduced the expectation of a random variable, we motivated it as the "typical" value that the variable assumes. While in most cases, this is good intuition, it can sometimes lead us astray. For example, consider a high-stakes gamble where you flip a coin and receive 100 dollars if it comes up heads and lose 100 dollars if it comes up tails. We can evaluate the expected winnings of this wager as $100 \cdot \frac{1}{2} + (-100) \cdot \frac{1}{2} = 0$. However, 0 doesn't really represent a typical run of the game; the game always results in a transfer of 100 dollars. Instead, the better framing for the expectation in this case is as a "long-run average". If we were to repeatedly make this wager over and over and consider our overall winnings (or losses) averaged over the number of times we play, this average should be very close to the expected value. Here, since it is equally likely that we win or lose 100 dollars each time we play, we'd expect our winnings to roughly cancel our losses, giving an expected value of 0. This long-run tendency toward the expected value is formalized by the laws of large numbers.

Theorem 31.3 — The (Weak) Law of Large Numbers. Given any $n \in \mathbb{N}^+$, suppose that $X_1, \ldots, X_n$ are independent realizations of some random variable X. Then given any tolerance threshold $\varepsilon > 0$,

$$\lim_{n \to \infty} \Pr\left(\left|\frac{X_1 + \ldots + X_n}{n} - \mathbb{E}[X]\right| < \varepsilon\right) = 1.$$

The statement of this theorem can look rather daunting, so let's break it down. Here, the fraction $\frac{X_1 + \ldots + X_n}{n}$ represents the long-run sampled average of the random variable (in our example, the net winnings from our wagers divided by the number of wagers we made). We're concerned with the event that this long-run average is close to the expectation $\mathbb{E}[X]$, which happens when the absolute value of the difference between these two values is small. Here, the ε encodes the fact that no matter how close we want the average and the expectation to be, we can be almost certain that it will happen eventually. As we continually sample (take n larger and larger) the probability that the average and expectation are close together nears 1 (almost certainty).

We will prove this theorem in the special case where X has bounded variance; $\text{Var}(X) \in \mathbb{R}$. The proof is an application of Chebyshev's inequality.

Proof. Applying Chebyshev's inequality, we have

$$\Pr\left(\left|\frac{X_1+\ldots+X_n}{n}-\mathbb{E}[X]\right|\geq\varepsilon\right)\leq\frac{1}{\varepsilon^2}\cdot\mathrm{Var}\left(\frac{X_1+\ldots+X_n}{n}\right).$$

Now, since $X_1,\ldots,X_n$ are mutually independent, we can rewrite this variance as

$$\mathrm{Var}\left(\frac{X_1+\ldots+X_n}{n}\right)=\frac{1}{n^2}\cdot\mathrm{Var}(X_1)+\ldots+\frac{1}{n^2}\cdot\mathrm{Var}(X_n)=n\cdot\frac{1}{n^2}\mathrm{Var}(X)=\frac{\mathrm{Var}(X)}{n}.$$

Thus,

$$\Pr\left(\left|\frac{X_1+\ldots+X_n}{n}-\mathbb{E}[X]\right|\geq\varepsilon\right)\leq\frac{\mathrm{Var}(X)}{\varepsilon^2 n},$$

for each n. Plugging into the limit, we have

$$\begin{aligned}\lim_{n\to\infty}\Pr\left(\left|\frac{X_1+\ldots+X_n}{n}-\mathbb{E}[X]\right|<\varepsilon\right)&=1-\lim_{n\to\infty}\Pr\left(\left|\frac{X_1+\ldots+X_n}{n}-\mathbb{E}[X]\right|\geq\varepsilon\right)\\&\geq 1-\lim_{n\to\infty}\frac{\mathrm{Var}(X)}{\varepsilon^2 n}\\&=1.\end{aligned}$$

Here, the last equality uses our bounded variance assumption. Since this probability expression must be ≤ 1 for each n (by the definition of a probability function), so too must its limit, meaning the limit of the probabilities is equal to 1. ■

31.3 Hoeffding's Inequality

Chebyshev's inequality illustrated how we can use more fine-grained information about a random variable (its variance) to better estimate the probability of an outlying event. We can take this reasoning one step further with our third probability inequality, Hoeffding's Inequality. Hoeffding's Inequality deals with sums of independent random variables. That is, it reasons about a random variable $X=X_1+\ldots+X_n$, where $\{X_1,\ldots,X_n\}$ are mutually independent. Such a variable X is often referred to as a *random walk*, where we view each X_i as a "step", and the partial sums as tracing out a path on the real line. This decomposition is a very strong structural assumption on X. Note that we can use our linearity properties to decompose both the expectation and variance of X:

$$\begin{aligned}\mathbb{E}[X]&=\mathbb{E}[X_1]+\ldots+\mathbb{E}[X_n],\\\mathrm{Var}(X)&=\mathrm{Var}(X_1)+\ldots+\mathrm{Var}(X_n).\end{aligned}$$

Moreover, the independence of the parts give a nice "concentration" property for X toward its expectation. Intuitively, some of the X_i will take value less than their expectations, but others should take value greater than their expectations. These biases should roughly cancel out and result in X staying fairly close to its mean. This intuition is formalized by the inequality.

Theorem 31.4 — Hoeffding's Inequality. Suppose that $\{X_1, \ldots, X_n\}$ are mutually independent random variables. Further suppose that for each i, there are constants a_i, b_i such that $\Pr(a_i \le X_i \le b_i) = 1$. Let $X = X_1 + \ldots + X_n$. Then,

$$\Pr\Big(X - \mathbb{E}[X] \ge t\Big) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Here, the inequality looks similar to a one-sided version of Chebyshev's inequality; we are bounding the probability that X exceeds its expectation by more than t. If you're unfamiliar with this notation, $\exp(x)$ is another way to write e^x that we often use when the exponent is a complicated expression. Thus, the probability decays exponentially in t, extremely quickly.

We will not give a proof of Hoeffding's inequality. This is a technical proof involving some ideas from calculus, so falls beyond our scope. In the end, this proof boils down to an application of Markov's inequality for a carefully constructed random variable.

As a special case of Hoeffding's inequality, consider when each X_i is a Bernoulli random variable, so we may take each $a_i = 0$ and $b_i = 1$. Then, the inequality simplifies to,

$$\Pr\Big(X - \mathbb{E}[X] \ge t\Big) \le \exp\left(-\frac{2t^2}{n}\right).$$

We can use this to obtain a further improvement on our probability bound in our coin problem.

Example 31.4 A fair coin is flipped 100 times, and the random variable H records the number of heads. $H \sim \text{Binom}(100, \frac{1}{2})$ which is a sum of 100 independent $\text{Bern}(\frac{1}{2})$ random variables. By Hoeffding's inequality,

$$\Pr(H \ge 100) = \Pr\big(H - 50 \ge 50\big) \le \exp\left(-\frac{2 \cdot 50^2}{100}\right) = e^{-50} \approx 1.9 \cdot 10^{-22}.$$

This bound is a significant improvement of our earlier estimates, although it is still far from the actual probability.

Main Takeaways:

- Probability inequalities give us a way to obtain an upper bound on the probability that a random variable assumes a value far from its mean without needing to directly work with its distribution.
- There is a trade-off between the strength of the inequality and the distributional information it requires. *Markov's* and *Chebyshev's Inequalities* require only summary statistics but give relatively weak bounds. *Hoeffding's Inequality* requires a strong assumption of a decomposition of a random variable into mutually independent components but gives a much stronger bound.
- The *Law of Large Numbers* formalizes the fact that the expectation of a random variable is its long-run average value.

Exercises

31.1 Suppose that a fair dice is rolled 60 times.

(a) Find the (exact) probability that 6 is rolled at least 15 times. (This calculation will be quite tedious.)

Now, compute an upper bound on the probability that 6 is rolled at least 15 times using:

(b) Markov's Inequality

(c) Chebyshev's Inequality

(d) Hoeffding's Inequality

Evaluate the strength of each of these bounds.

31.2 Since Chebyshev's Inequality utilizes more information about a distribution (i.e. its variance), it is natural to assume that it will always provide a stronger bound. However, this is not the case. Consider a discrete random variable $D \sim 6 \cdot \text{Bern}(\frac{1}{3})$.

(a) What is $\Pr(D \geq 5)$?

(b) Use Markov's Inequality to bound $\Pr(D \geq 5)$.

(c) Use Chebyshev's Inequality to bound $\Pr(D \geq 5)$.

31.3 Suppose that a 20-sided dice is rolled repeatedly until a 7 is rolled. Let G be the random variable that represents the total number of rolls.

(a) Use Markov's Inequality to bound the probability that more than 10 rolls are needed. What is strange about this answer? Can you explain why this calculation gave this?

(b) Why can't we use Chebyshev's Inequality to bound the probability that more than 10 rolls are needed?

(c) Use Markov's Inequality to bound the probability that more than 25 rolls are needed.

(d) Use Chebyshev's Inequality to bound the probability that more than 25 rolls are needed. Give some intuition for why this bound is still relatively bad.

(e) Use Markov's Inequality to bound the probability that more than 50 rolls are needed.

(f) Use Chebyshev's Inequality to bound the probability that more than 50 rolls are needed.

(g) Why can't we use Hoeffding's Inequality to bound the probability that more than 50 rolls are needed?

31.4 Earlier in our probability module, we introduced the Union Bound, which said that given a finite collection of events $A_1, \ldots, A_n \subseteq \Omega$

$$\Pr\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \Pr(A_i).$$

Use Markov's Inequality to give an alternate proof of the Union Bound. (Hint: Introduce random variables $X_i = \mathbb{I}(A_i)$ and consider their sum X.)

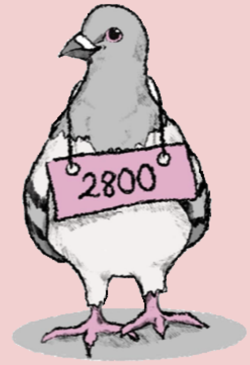
⋆ **31.5** Markov's and Chebyshev's Inequalities provide good probability estimates because they are, in a sense, *tight*. That is, there are some random variables for which the inequalities actually hold with equality. In this exercise, you'll construct some such random variables.

(a) Find a random variable X with support $R_X = \{0,5\}$ and a suitable choice of $\alpha > 0$ such that Markov's Inequality is tight:

$$\Pr(X \geq \alpha) = \frac{\mathbb{E}[X]}{\alpha}.$$

(b) Find a random variable Y with support $R_Y = \{0,3,6\}$ and a suitable choice of $\beta > 0$ such that Chebyshev's Inequality is tight:

$$\Pr(|Y - \mathbb{E}[Y]| \geq \beta) = \frac{\mathrm{Var}(Y)}{\beta^2}.$$



32. Continuous Random Variables

So far in our study of probability, we have restricted our attention to *discrete* probability spaces, which model randomized experiments that have a countable set of possible outcomes. In this lecture, we will extend many of these ideas to *continuous* probability spaces, in which Ω is an uncountable set.

32.1 Defining Continuous Random Variables

Suppose that we wish to sample a random number in the interval $[0,1]$. We don't want to limit the possibilities to a subset of the numbers but rather want to allow for the selection of *any* number in the interval. We can view the sampled number as a random variable U defined on a *continuous* probability space with sample space $\Omega = [0,1]$. The event space of this probability space includes all of the subsets of Ω that we can reasonably compute the probability of. In this case, we cannot take $\mathcal{F} = \mathcal{P}(\Omega)$, as trying to define a reasonable probability function with this domain will lead to a contradiction (see the lecture postscript for a discussion about this). Rather, we must restrict our attention to a collection of nice (more formally, *measurable*) subsets that we can compute with. This will not be an issue for us, because probably all of the subsets that you'd think of are measurable.

Now, how should we think about the distribution of the continuous random variable U? In discrete probability spaces, we described this using a probability *mass* function, which expressed the probability that the random variable assumed each value in its support. Doing this does not make sense for our variable U, which has $R_U = [0,1]$. Suppose that we assign a positive probability $\Pr(U = 0) = p > 0$. Then, since U is a uniform random variable, this should also be the probability that $U = 1$, and that $U = \frac{1}{2}$, and that $U = r$ for each $r \in [0,1]$. The events that U assumes each of these particular values are all disjoint. Thus, we have $\Pr(\Omega) = \sum_{r\in[0,1]} \Pr(U = r) = \sum_{r\in[0,1]} p$. However, the sum of uncountably many positive terms cannot equal 1, it must be infinite (see Exercise 32.2). Hence, the only possibility we are left with is to assign $\Pr(U = r) = 0$ for each $r \in [0,1]$.

Instead of thinking about the probability *mass* on each outcome in the sample space, we can conceptualize a probability *density*, which tells us how likely we are that the outcome lands in a particular region of the sample space. To calculate the probability that a continuous random variable lands in a particular subset (say, an interval) of its support, we need a way to aggregate all of the density in this subset. Integration gives us a natural way to do this and gives us a lens to formalize the distribution of a continuous random variable using a *probability density function*.

Definition 32.1 — probability density function (pdf).
Given a continuous random variable X on probability space $(\Omega, \mathcal{F}, \Pr)$, the *probability density function*, denoted f_X, is a function $f_X : \mathbb{R} \to \mathbb{R}_{\geq 0}$ defined such that for each $a \leq b \in \mathbb{R}$,

$$\Pr(a \leq X \leq b) \equiv \Pr\Big(\big\{\omega \in \Omega \mid a \leq X(\omega) \leq b\big\}\Big) = \int_a^b f_X(r)\, dr.$$

Note that we must have $\Pr(\Omega) = \Pr(-\infty < X < \infty) = 1$, which means that a probability density function f_X must satisfy the property that

$$\int_{-\infty}^{\infty} f_X(r)\, dr = 1.$$

The pdf may have value greater than 1 for some $r \in \mathbb{R}$, as long as this integral condition is satisfied; the values of the pdf are *probability densities* and not actual probabilities.

Our definition of the Cumulative Distribution Function remains the same for continuous random variables: $F_X : \mathbb{R} \to [0,1]$ with $F_X(s) = \Pr(X \leq s)$. Observe that

$$F_X(s) = \Pr(X \leq s) = \int_{-\infty}^{s} f_X(r)\, dr,$$

so F_X is the anti-derivative of f_X (or, in other words, the pdf f_X is the derivative of the CDF F_X wherever F_X is differentiable).

Example 32.1 Let's return our attention to the random variable U from above. Since U is a uniform random number from $[0,1]$, its probability density should be constant on its support (and 0 outside of its support). Thus, we should take

$$f_U(r) = \begin{cases} c & 0 \leq r \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

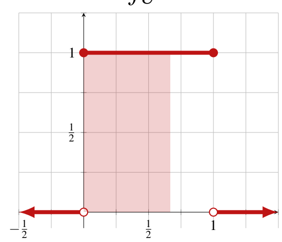
for some $c \in \mathbb{R}$. By the above property, we must have

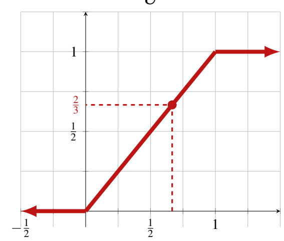
$$\int_{-\infty}^{\infty} f_U(r)\, dr = \int_0^1 c\, dr = \Big(cr\Big|_{r=0}^{r=1} = c = 1.$$

Taking the anti-derivative of f_U, we obtain the CDF

$$F_U(s) = \begin{cases} 0 & s \leq 0, \\ s & 0 \leq s \leq 1 \\ 1 & s \geq 1. \end{cases}$$

We visualize both of these functions at the top of the next page. Observe that $F_U(\frac{2}{3}) = \frac{2}{3}$, which corresponds to the area of the shaded region under f_U on the interval $[0, \frac{2}{3}]$.

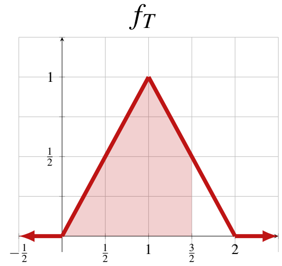


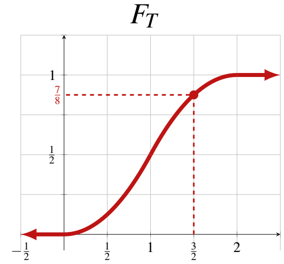


Example 32.2 Consider the random variable T with pdf and CDF,

$$f_T(r) = \begin{cases} r & 0 \le r \le 1, \\ 2-r & 1 \le r \le 2, \\ 0 & \text{otherwise,} \end{cases} \qquad F_T(s) = \begin{cases} 0 & s \le 0, \\ \frac{s^2}{2} & 0 \le s \le 1, \\ \frac{4s-s^2-2}{2} & 1 \le s \le 2, \\ 1 & s \ge 2. \end{cases}$$

We visualize these functions:





From its pdf, we see that T has support $R_T = [0,2]$, and it is more likely to assume a value closer to 1. One can verify that this pdf is well-defined by checking that this function is non-negative and the total area under it is 1. The CDF can be found by integrating f_T piecewise (and choosing the constant of integration appropriately to end up with a continuous function with range $[0,1]$). Observe that $F_T\left(\frac{3}{2}\right) = \frac{7}{8}$, since $\frac{7}{8}$ of the area under f_T sits to the left of $\frac{3}{2}$.

32.2 Continuous Distribution Families

Just as we saw for discrete random variables, there are some important families of continuous distributions. We mention two of these below. The next lecture focuses on the third, and undoubtedly the most important family of continuous distributions, *normal* distributions.

32.2.1 Uniform Continuous Random Variables

To generalize the distribution of the random variable U from above, we can imagine a random variable with a constant density on an interval $[a,b]$ for any $a < b \in \mathbb{R}$. We denote this distribution $\text{Unif}[a,b]$ (so $U \sim \text{Unif}[0,1]$). The pdf of a uniform continuous random variable should be some constant k on $[a,b]$ and otherwise 0. To ensure the total probability density is 1, we must have

$$\int_{-\infty}^{\infty} f_X(r)\,dr = \int_a^b k\,dr = k(b-a) = 1 \qquad \Longrightarrow \qquad k = \frac{1}{b-a}.$$

Definition 32.2 — Uniform Continuous Random Variable.

A random variable U has a *uniform* distribution with support $[a,b]$ for some $a<b\in\mathbb{R}$, which we denote $U\sim \text{Unif}[a,b]$, if its probability density function is

$$f_U(x)=\begin{cases}\frac{1}{b-a} & a\le x\le b,\\ 0 & \text{otherwise.}\end{cases}$$

For any $a\le c\le d\le b$ with $a<b$, we have

$$\Pr(c\le U\le d)=\int_c^d f_U(x)\,dx=\int_c^d \frac{1}{b-a}\,dx=\frac{d-c}{b-a}.$$

Example 32.3 Suppose that $U\sim\text{Unif}[-4,4]$. Let us compute $\Pr(|U|\ge 3)$. Note that

$$\begin{aligned}\Pr(|U|\ge 3)&=\Pr\Big((-4\le U\le -3)\cup(3\le U\le 4)\Big)=\Pr(-4\le U\le -3)+\Pr(3\le U\le 4)\\ &=\frac{-3-(-4)}{4-(-4)}+\frac{4-3}{4-(-4)}=\frac{2}{8}=\frac{1}{4}.\end{aligned}$$

Note that the second equality follows from the countable additivity property of probability, which also holds in uncountable probability spaces.

32.2.2 Exponential Random Variables

Exponential random variables are supported on the non-negative reals and have a decaying probability density. They are used to model random phenomena like radioactive decay.

Definition 32.3 — Exponential Random Variable.

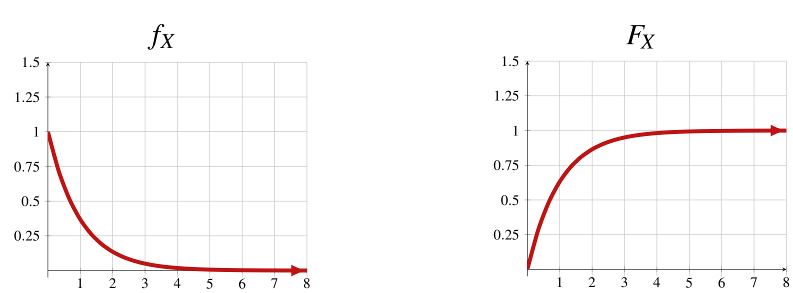
A random variable X has the *exponential* distribution with *rate* parameter $\lambda>0$, denoted $X\sim\text{Exp}(\lambda)$, if it has support $\mathbb{R}_{\ge 0}$ and probability density function

$$f_X(r)=\begin{cases}\lambda e^{-\lambda r} & r\ge 0,\\ 0 & r<0.\end{cases}$$

To verify that this is a valid probability density function, we check

$$\int_{-\infty}^{\infty} f_X(r)\,dr=\int_0^{\infty}\lambda e^{-\lambda r}\,dr=\int_0^{\infty} e^{-u}\,du=1,$$

where we've made the substitution $u=\lambda r$. We visualize the pdf and CDF for the Exp(1) distribution below.



32.3 Statistics of Continuous Random Variables

Just as we saw with discrete random variables, we can use *statistics* such as the expectation and variance to help summarize the distribution of a continuous random variable.

Computing the expectation of a continuous random variable is similar to the discrete case, but we must swap out the discrete components for their continuous counterparts. The summation is replaced with an integral, and the pmf is replaced with the pdf.

Definition 32.4 — Expectation of a Continuous Random Variable.
Suppose that X is a continuous random variable on probability space $(\Omega, \mathcal{F}, \Pr)$ with pdf f_X. Then, its expectation $\mathbb{E}[X]$ is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} r \cdot f_X(r)\, dr.$$

Example 32.4 — Uniform Random Variables. Suppose that $U \sim \text{Unif}[a,b]$. Then,

$$\mathbb{E}[U] = \int_a^b r \cdot f_X(r)\, dr = \tfrac{1}{b-a} \int_a^b r\, dr = \tfrac{1}{b-a} \cdot \tfrac{r^2}{2}\Big|_{r=a}^{r=b} = \tfrac{b^2-a^2}{2(b-a)} = \tfrac{b+a}{2}.$$

This result should be fairly intuitive. The expected value of a uniform random variable is at the midpoint of its support.

Example 32.5 — Exponential Random Variables. We again use the integral formula to compute the expectation of an exponential random variable. Suppose that $X \sim \text{Exp}(\lambda)$. Then,

$$\mathbb{E}[X] = \int_0^{\infty} \lambda r e^{-\lambda r}\, dr = -re^{-\lambda r}\Big|_{r=0}^{r=\infty} + \int_0^{\infty} e^{-\lambda r}\, dr = -re^{-\lambda r} - \tfrac{1}{\lambda} e^{-\lambda r}\Big|_{r=0}^{r=\infty} = \tfrac{1}{\lambda}.$$

Here, the second equality follows by integration by parts, taking $u = r$ and $dv = \lambda e^{-\lambda r}\, dr$. (Evaluating the infinite limit in the first term required us to handle an indeterminate form, the details of which are omitted here.)

32.3.1 Variance, Covariance, and Correlation

All of the familiar properties for expectations, such as linearity and the law of the unconscious statistician, extend to continuous random variables,

Example 32.6 — Uniform Random Variables. Suppose that $U \sim \text{Unif}[a,b]$. Then,

$$\mathbb{E}\left[U^2\right] = \frac{1}{b-a} \int_a^b r^2\, dr = \frac{1}{3(b-a)} r^3\Big|_{r=a}^{r=b} = \frac{b^3-a^3}{3(b-a)} = \frac{b^2+ab+a^2}{3}.$$

Thus,

$$\text{Var}(U) = \mathbb{E}\left[U^2\right] - \left(\mathbb{E}[U]\right)^2 = \frac{b^2+ab+a^2}{3} - \left(\frac{b+a}{2}\right)^2 = \frac{b^2-2ab+a^2}{12} = \frac{(b-a)^2}{12}.$$

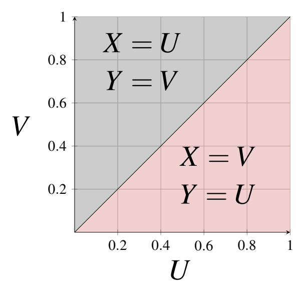
In Exercise 32.7, we provide some hints on how to calculate the variance of an exponential random variable. We can analogously consider joint distributions of continuous random variables and ask questions about independence, covariance, and correlation.

Example 32.7 Suppose that $U,V \sim \text{Unif}[0,1]$ are independent, and let $X = \min(U,V)$ and $Y = \max(U,V)$. How are X and Y correlated? (Before reading on, give this some thought. Do you expect larger values of X to coincide with larger values of Y, or smaller values? Do you think that there is even a relationship at all?)

To perform the covariance calculation, we record the following observations. First, since one of $\{X,Y\}$ equals U and the other equals V, we have

$$\mathbb{E}[XY] = \mathbb{E}[UV] = \mathbb{E}[U] \cdot \mathbb{E}[V] = \int_0^1 u\,du \cdot \int_0^1 v\,dv = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Here, the second equality uses the independence of U and V. To calculate the expectation of each of X,Y, we again need to consider the (joint) probability space of X,Y. We have the following picture:



Thus,

$$\mathbb{E}[X] = \int_0^1 \int_0^u v\,dv\,du + \int_0^1 \int_u^1 u\,dv\,du = \frac{1}{3}, \qquad \mathbb{E}[Y] = \int_0^1 \int_0^u u\,dv\,du + \int_0^1 \int_u^1 v\,dv\,du = \frac{2}{3}.$$

$$\mathbb{E}\left[X^2\right] = \int_0^1 \int_0^u v^2\,dv\,du + \int_0^1 \int_u^1 u^2\,dv\,du = \frac{1}{6}, \qquad \mathbb{E}\left[Y^2\right] = \int_0^1 \int_0^u u^2\,dv\,du + \int_0^1 \int_u^1 v^2\,dv\,du = \frac{1}{2}.$$

Finally, we have

$$\text{corr}(X,Y) = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]}{\sqrt{\left(\mathbb{E}[X^2] - \mathbb{E}[X]^2\right)\left(\mathbb{E}[Y^2] - \mathbb{E}[Y]^2\right)}} = \frac{\frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3}}{\sqrt{\left(\frac{1}{6} - \frac{1}{9}\right)\left(\frac{1}{2} - \frac{4}{9}\right)}} = \frac{1}{2}.$$

X and Y are positively correlated, and the strength of this correlation is moderate. We can provide the following intuition for why this correlation is positive. By definition $X \leq Y$. Therefore, if Y is small, so too must X be. Similarly, if X is large, so too must Y be.

Main Takeaways:

- *Continuous random variables* have uncountable supports. Rather than describing continuous distributions with a pmf, which assigns a probability to each point in the support, we describe them with a pdf.
- When we integrate the pdf over any interval, it gives the probability that the variable assumes a value in that interval.
- We can express the expectation of a continuous random variable as an integral. The remaining statistics of continuous random variables can be calculated analogously to discrete random variables.
- Two important families of continuous distributions are *Uniform Continuous* and *Exponential* distributions.

Exercises

32.1 Suppose that $U, V \sim \text{Unif}[0,1]$ are independent. You can visualize U, V as being the coordinates of a point chosen uniformly at random from the unit square. Calculate each of the following probabilities. It may help to draw pictures to visualize the events.

(a) $\Pr(U = \frac{1}{2})$ (b) $\Pr(U \leq \frac{2}{3})$ (c) $\Pr(\frac{1}{5} \leq V \leq \frac{4}{5})$

(d) $\Pr(U \geq \frac{2}{5} \cap V \leq \frac{2}{3})$ (e) $\Pr(U = V)$ (f) $\Pr(U \geq V)$

(g) $\Pr(U + V < 1)$ (h) $\Pr(U + V < \frac{3}{2})$ (i) $\Pr(U^2 + V^2 < 1)$

(j) $\Pr(UV = 0)$ (k) $\Pr(UV \geq \frac{1}{4})$ (l) $\Pr(|U - V| \leq \frac{1}{3})$

(m) $\Pr(U > \frac{1}{2} \mid U > V)$ (n) $\Pr(V \leq \frac{3}{4} \mid U = \frac{1}{4})$ (o) $\Pr(U + V \geq 1 \mid V \leq \frac{1}{2})$

32.2 Here, you'll work through the details of the proof that an uncountable sum of positive values is infinite.

(a) Argue that a countable union of finite sets is countable (i.e., describe a procedure to enumerate all elements of these sets).

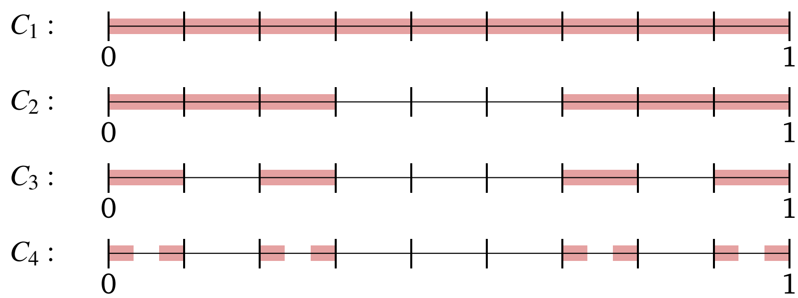
(b) We'll perform the proof by contradiction. We assume that there is an uncountable (multi)set $U \subseteq \mathbb{R}$ with each $u \in U$ strictly positive and with sum $\sum_{u \in U} u < \infty$. Now, for each $n \in \mathbb{N}$, define $A_n := \{u \in U : u > \frac{1}{n}\}$. Argue that $|A_n|$ is finite for all n.

(c) Use parts (a) and (b) to derive a contradiction.

32.3 Let $U \sim \text{Unif}[0,1]$. From the lecture, we know that $\Pr(U = x) = 0$ for all $x \in [0,1]$, so by countable additivity, the probability that U lands in any countable subset of $[0,1]$ is 0. In this exercise, you'll show that there are also uncountable sets that U lands in with probability 0.

The Cantor Set $C = \lim\limits_{n \to \infty} C_n$ of the following inductively defined sets.

- C_1 = the unit interval $[0,1]$.
- Construct C_{n+1} by removing the middle third from each contiguous interval of C_n.



(a) Argue that $\Pr(U \in C) = 0$ by calculating $\Pr(C_n)$ for each $n \in \mathbb{N}^+$, and then taking the limit.

(b) Argue that C is an uncountable set by establishing a bijection with B^∞, the set of infinite sequences of $\{0,1\}$.

32.4 Find $\mathbb{E}[T]$, $\mathbb{E}\left[T^2\right]$ and $\mathrm{Var}(T)$ for the triangular random variable T described in Example 32.2. It may help to break each of these calculations into multiple integrals.

32.5 Below, we give a few examples of pdfs of continuous random variables X. For each of them:

1. Verify that the given function is a valid pdf by checking that it is non-negative and integrates to 1 on $(-\infty,\infty)$.
2. Find the CDF F_X of X.
3. Compute $\mathbb{E}[X]$.
4. Compute“ $\mathbb{E}\left[X^2\right]$ and use this to find $\mathrm{Var}(X)$.

(a)

$$f_X(r) = \begin{cases} \frac{1}{3} & 0 < r \leq 1, \\ \frac{2}{3} & 1 < r \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(b)

$$f_X(r) = \begin{cases} r & 0 \leq r \leq \frac{1}{3}, \\ \frac{1}{3} & \frac{1}{3} \leq r \leq 1, \\ \frac{4}{3} - r & 1 \leq r \leq \frac{4}{3}, \\ 0 & \text{otherwise.} \end{cases}$$

(c)

$$f_X(r) = \begin{cases} \frac{1}{r^2} & r \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(d)

$$f_X(r) = \begin{cases} \frac{\sin(r)}{2} & 0 \leq r \leq \pi, \\ 0 & \text{otherwise.} \end{cases}$$

32.6 Let L be a continuous random variable supported on $[0,a]$ (for some $a > 0$) with linear density $f_L(x) = kx$. This models a random experiment where an outcome becomes more likely as it gets closer to a.

(a) Find the value of k in terms of a that makes f_L a well-defined pdf.

(b) What is the CDF F_L?

(c) Compute $\mathbb{E}[L]$. Does your answer seem reasonable?

(d) Compute $\mathrm{Var}(L)$. Does your answer seem reasonable?

◇ **32.7** Suppose that $X \sim \mathrm{Exp}(\lambda)$. Use integration by parts twice to calculate $\mathbb{E}\left[X^2\right]$. Then, use this to find a formula for $\mathrm{Var}(X)$ in terms of λ.

32.8 Suppose that $X \sim \mathrm{Unif}[-1,1]$ and $Y = X^2$.

(a) Argue that X and Y are not independent by exhibiting $r_1, r_2 \in \mathbb{R}$ for which $f_{X,Y}(r_1, r_2) \neq f_X(r_1) \cdot f_Y(r_2)$.

(b) Verify that $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$.

This exercise gives another example of two random variables (this time, continuous random variables) that are uncorrelated but not independent.

⋆ **32.9** Suppose that a random point (X,Y) is chosen uniformly from the *unit disk* $\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$.

(a) Find the joint probability density function $f_{X,Y}$.

(b) Find the marginal probability density function f_X with respect to X.

(c) Find the marginal probability density function f_Z with respect to the distance from the origin $Z = \sqrt{X^2 + Y^2}$.

32.10 Suppose that a 1-meter rod is cut at a uniform random position. Let L be the (continuous) random variable that represents the length (in meters) of the longer piece.

(a) Find the CDF and pdf of L. It will be easier to first compute the CDF by considering the value of L for cuts at various positions.

(b) Use your answers from part (a) to compute $\mathbb{E}[L]$ and $\mathrm{Var}(L)$.

(c) Repeat these calculations for the random variable S that represents the length of the shorter piece.

(d) If you know the value of $\mathbb{E}[L]$, how can you use this to calculate the value of $\mathbb{E}[S]$ without needing to know the distribution of S?

(e) You should have found that $\mathrm{Var}(L) = \mathrm{Var}(S)$. Provide some intuition for why this should be the case.

⋆ **32.11** Suppose that a 1-meter rod is cut at two (independent) uniform random points $U, V \sim \mathrm{Unif}[0,1]$ which separate it into three pieces with random lengths. Let us represent the lengths of the smallest, middle, and largest pieces with random variables S, M, and L (respectively).

(a) Draw a picture that visualizes the value of S in different regions of the unit square representing the values of U and V.

(b) On your picture from part (a), shade in the region that corresponds to the event $S \le x$ for some arbitrary $0 \le x \le \frac{1}{3}$. What is the area of the shaded region (in terms of x)? Use this to write the CDF F_S.

(c) Use your answer from part (b) to find the pdf f_S, $\mathbb{E}[S]$, and $\text{Var}(S)$.

(d) Repeat these calculations for random variables M and L.

◇ **32.12** Suppose that X and Y are two independent continuous random variables with pdfs f_X and f_Y (respectively). Let us define a third random variable $Z = X + Y$.

(a) Argue that the pdf f_Z of Z is given by the formula

$$f_Z(r) = \int_{-\infty}^{\infty} f_X(\tau) f_Y(r - \tau)\, d\tau.$$

This operation on the right side is known as the *convolution* of f_X and f_Y, and is often denoted $f_Z = f_X * f_Y$.

(b) Use the formula from part (a) to show that the triangular random variable T from Example 32.2 has $T \sim U + V$ for independent uniform random variables $U, V \sim \text{Unif}[0,1]$.

32.13 Suppose that $U_1, \ldots, U_{20}$ are 20 mutually independent samples from the Unif$[0,1]$ distribution, and let $U = \sum_{i=1}^{20} U_i$. Bound the probability that $U \ge 12$ using

(a) Markov's Inequality

(b) Chebyshev's Inequality

(c) Hoeffding's Inequality

32.14 The time (in years) that it takes for a radioactive Carbon-14 particle to decay is an exponential random variable $T \sim \text{Exp}(\frac{1}{5700})$.

(a) Calculate the (exact) probability that a Carbon-14 particle from a 20,000-year-old fossil has not yet decayed.

(b) Use Markov's Inequality to compute an upper bound on the probability from part (a).

(c) Use Chebyshev's Inequality to compute an upper bound on the probability from part (a).

32.4 Postscript: Non-Measurable Events

When we discussed sets at the beginning of the course, we worked mostly intuitively, staying away from the formal logical underpinning of the properties that we discussed. The most popular formalization comes from the *Zermelo-Fraenkel* axioms, a set of properties that describe what can constitute a set. While some of these axioms are rather straightforward (for example, one axiom simply states the existence of an empty set, others define set union, and the power set operation, etc.) others are more complicated or controversial. Undoubtedly the most controversial of the axioms is the so-called Axiom of Choice.

Definition 32.5 — Axiom of Choice.

Given any set X of non-empty sets, there exists a *choice function* $f\colon X \to \bigcup_{S\in X} S$ that maps each set $S \in X$ to an element in S. That is, $f(S) \in S$ for each $S \in X$.

As an example, if $X = \big\{\{1,2,3\},\{3,4\},\{5\}\big\}$, then $\bigcup_{S\in X} S = \{1,2,3,4,5\}$, and we could define

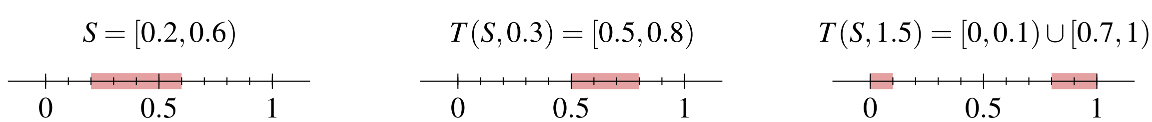
$$f\big(\{1,2,3\}\big) = 2, \qquad f\big(\{3,4\}\big) = 4, \qquad f\big(\{5\}\big) = 5.$$

We call f a choice function because it is choosing an element from each of the sets in X. Note that the axiom does not make explicit how these elements are chosen, but rather asserts only that there is a way to make all of the requisite choices.[1] We will use the Axiom of Choice in the following example to construct an event that cannot have a well-defined probability.

Example 32.8 Let us consider the random experiment where we choose a uniform random number from the interval $\Omega = [0,1)$. We'll attempt (dubiously) to take the event set $\mathcal{F} = \mathcal{P}(\Omega)$. Now, let's consider how to define a probability function Pr on $\mathcal{F}$. From the lecture, we know that this function should assign a probability to each event that is equal to its total "length", but which is independent of its position. This latter property is known as *translation invariance*, and we can formalize it as follows: Define a *translation* function $T\colon \mathcal{P}(\Omega) \times \mathbb{R} \to \mathcal{P}(\Omega)$ with

$$T(S,a) = \big\{x \in [0,1) \mid x = (s+a) - \lfloor s+a \rfloor \text{ for some } s \in S\big\}.$$

This function translates (slides over) the elements of a set S by a units, wrapping them back around to ensure that the points never go outside of the interval $[0,1)$. To illustrate this behavior, consider the following images.



Since the total length of the red intervals remains the same after translation, our probability function should satisfy

$$\Pr(S) = \Pr\big(T(S,a)\big)$$

for each $a \in \mathbb{R}$.

Now, let's construct a diabolical set D that cannot have this property. To do this, we'll define an equivalence relation $\approx$ on Ω such that

$$x \approx y \Leftrightarrow x - y \in \mathbb{Q}.$$

Here, all of the rational points in $[0,1)$ form one equivalence class since the difference of any two rational numbers is also rational. However, $\frac{\sqrt{2}}{2} \approx 0.707$ is in a different equivalence class since it is irrational. Similarly, $\frac{\pi}{4} \approx 0.785$ is in yet another equivalence class. Recall that the equivalence classes form a partition of the ground set, so every point in $[0,1)$ belongs to exactly one equivalence class.

[1] For this reason, the Axiom of Choice is rejected by *constructive* mathematicians, who believe that merely arguing existence is not enough. Rather, they believe that one must have an explicit formula or example to verify the truth of a claim of existence.

Now, let's use the Axiom of Choice to define a set D that includes exactly one member from each equivalence class in $\Omega/\approx$. For example, D may include $\frac{3}{7}, \frac{\sqrt{2}}{2}$, and $\frac{\pi}{4}$, but then can't include $\frac{1}{4}$ since $\frac{3}{7} - \frac{1}{4} \in \mathbb{Q}$. Now, let us consider the set

$$\mathcal{S} = \Big\{ S \mid S = T(D,q) \text{ for some } q \in \mathbb{Q} \cap [0,1) \Big\}.$$

Let us make the following two observations about the sets in $\mathcal{S}$:

1. The sets in $\mathcal{S}$ cover Ω.

Given any $x \in [0,1)$, the definition of D ensures that there is some $d \in D$ such that $x \approx d$. Thus, $x = d + q - \lfloor d + q \rfloor$ for some $q \in \mathbb{Q} \cap [0,1)$, meaning $x \in T(D,q) \in \mathcal{S}$.

2. The sets in $\mathcal{S}$ are pairwise disjoint.

Suppose that S_1 and S_2 are two distinct sets in $\mathcal{S}$. By definition, $S_1 = T(D,q_1)$ and $S_2 = T(D,q_2)$ for some $q_1 \neq q_2 \in \mathbb{Q} \cap [0,1)$. Note that there can be no $x \in S_1 \cap S_2$, as this would mean that $d_1 + q_1 = d_2 + q_2$ for some $d_1 \neq d_2 \in D$, but we could rearrange this to be $d_1 = d_2 + (q_2 - q_1)$, which would imply that $d_1 \approx d_2$, contradicting our definition of D. Hence, $S_1 \cap S_2 = \varnothing$, so the sets in $\mathcal{S}$ are pairwise disjoint.

We have,

$$1 = \Pr(\Omega) = \Pr\left(\bigcup_{S \in \mathcal{S}} S \right) = \sum_{S \in \mathcal{S}} \Pr(S) = \sum_{q \in \mathbb{Q} \cap [0,1)} \Pr\Big(T(D,q)\Big) = \sum_{q \in \mathbb{Q} \cap [0,1)} \Pr(D).$$

Here, the first equality is the total probability property, the next two equalities use our observations about $\mathcal{S}$ and the countable additivity property, the fourth equality applies the definition of $\mathcal{S}$, and the last equality uses the translation invariance property from above.

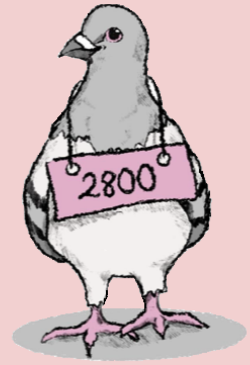
Now, consider the possible values of $\Pr(D)$. If $\Pr(D) = 0$, then a countable sum of 0s is $0 \neq 1$. On the other hand, if $\Pr(A) > 0$, then adding it to itself countably many times gives ∞. We can never obtain a sum of 1, meaning we cannot define a translation-invariant probability function Pr with domain $\mathcal{F} = \mathcal{P}(\Omega)$.

This example shows that we cannot include this devious event D in our event space while still respecting the desired properties of our probability function. Instead of altering these properties, we instead will exclude these devious events from our consideration. That is, we restrict our attention to a sub-collection of subsets of the sample space, the so-called *measurable* sets. The formal name for the continuous event spaces that we typically work with are *Borel sets*. The Borel sets include all intervals (open, closed, half-open/closed) and countable unions of them, among other sets. Rest assured that most of the events that you'd ever want to consider are Borel sets, so are measurable.

32.15 Consider the set B^∞ consisting of all infinite sequences of $\{0,1\}$.

(a) Let $\sim$ be the equivalence relation on B^∞ defined such that two sequences $b, b' \in B^\infty$ have $b \sim b'$ if and only if b and b' differ in only finitely many positions. Argue that $\sim$ is an equivalence relation.

(b) Following Example 32.8, we'll attempt to define a probability space on B^∞ that allows all events in its power set. Use $\sim$ to construct a devious event A and argue that its probability cannot be defined.



33. The Normal Distribution

In our final lecture, we will introduce the most important distribution in statistics, the *normal distribution*. The reason for their pervasiveness is their appearance in the Central Limit Theorem. We will introduce this theorem and explore some of its consequences. To conclude, we'll briefly discuss *Maximum Likelihood Estimation*, a technique to recover the unknown parameters of a random variable's distribution via sampling.

33.1 Normal Random Variables

Similar to the other families of continuous random variables that we have introduced, Unif(a,b) and Exp(λ), normal random variables realize a parameterized family of distributions, which we can describe by defining their support and probability density function.

Definition 33.1 — Normal Random Variable.

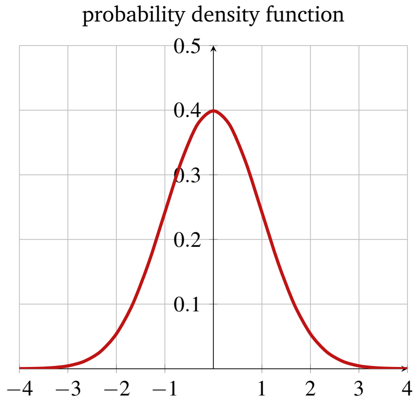
A *normal* random variable with *mean* parameter μ and *standard deviation* parameter σ, abbreviated $N(\mu,\sigma^2)$, is a continuous random variable N with support $\mathbb{R}$ and probability density function

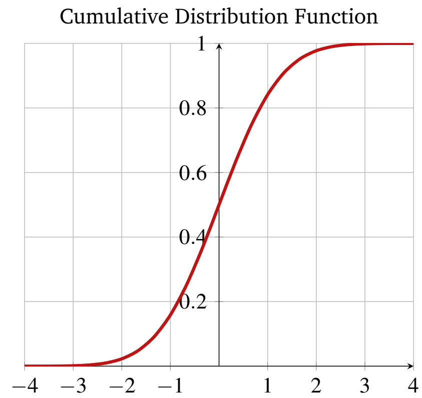
$$f_N(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right).$$

Recall that the notation $\exp(\cdot)$ is another way that we write $e^{(\cdot)}$ to more clearly display the exponent. The most popular settings of these parameters are $\mu = 0$ and $\sigma = 1$, giving the *standard normal distribution* $N(0,1)$ with density $f_N(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$. Interestingly, the Cumulative Distribution Function of a normal distribution $F_N(x)$ cannot be expressed in terms of the nice analytic functions we are used to (polynomials, exponentials, trig functions, etc.); this presents an interesting computational challenge.

It is a standard exercise in multivariable calculus to check that the area under this function is 1 (we give some helpful hints in the chapter exercises if you're interested). Similar calculations can verify that the expectation of $N(\mu,\sigma^2)$ is μ and its variance is σ^2 (so its standard deviation is σ).

We visualize the pdf and CDF for the standard normal distribution below.





You'll often hear the normal distribution be referred to as the "bell curve" because of the distinctive shape of its pdf. The mean of this distribution occurs at its maximum value. Its standard deviation is the horizontal distance between this maximum and either of its inflection points. In particular, a larger standard deviation results in a wider bell curve.

33.2 The Central Limit Theorem

Upon first seeing the definition of the normal distribution, it is natural to ask why it is significant. Its definition is rather complicated, and exact calculations with normal random variables are often difficult due to our inability to nicely integrate its density function. Nevertheless, the normal distribution is perhaps the most pervasive in all of probability and statistics. The main reason for this is its appearance in the *Central Limit Theorem*.

To motivate the Central Limit Theorem, we'll return to a setting that we briefly discussed in the previous lecture. Let $\mathcal{D}$ be any distribution (discrete or continuous) that has finite variance (this is true of all of the distributions that we have discussed). We'll let μ denote the expectation of this distribution and σ denote its standard deviation. Now, suppose that $X_1, X_2, \dots$ are a sequence of mutually independent samples from $\mathcal{D}$ (so each $X_i \sim \mathcal{D}$). We can imagine that we have some random object that conforms to the distribution $\mathcal{D}$, and we take many measurements of this object over a period of time.

Now, suppose that we consider the *prefix averages* of these measurements $A_1, A_2, \dots$ with each $A_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. The (Weak) Law of Large Numbers that we discussed in the last lecture tells us that $A_n \to \mu$ in probability, i.e., that the long run average tends toward the expectation. The Strong Law of Large Numbers gives a more significant convergence result: $A_n \to \mu$ along almost every possible realization of samples $X_1, X_2, \dots$.

Suppose that instead of concerning ourselves only with the long-term average, we care also about how far it strays from the expected value. To overcome the Law of Large Numbers, we will need to introduce a scaling factor to "exaggerate" this deviation. It turns out that scaling by $\sqrt{n}$ is appropriate. This is the setting of the Central Limit Theorem.

Theorem 33.1 — Central Limit Theorem. Suppose that $X_1, X_2, \ldots$ are independent and identically distributed (iid) random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Define the sequence of sample averages $A_1, A_2, \ldots$ with each $A_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then,

$$\sqrt{n}(A_n - \mu) \to N(0, \sigma^2).$$

The theorem tells us that as n grows large, these appropriately-scaled deviations tend toward a normal distribution. Truthfully, we are hiding a lot of details in the "$\to$" in this theorem statement; understanding different notions of convergence of random variables is a major topic of study in an advanced probability class. In this case, the convergence is "in distribution", meaning that for each $s \in \mathbb{R}$, the sequence of CDF values $F(s)$ of the random variables on the left side of this convergence statement tend toward $F_N(s)$.

An equivalent statement of the Central Limit Theorem is:

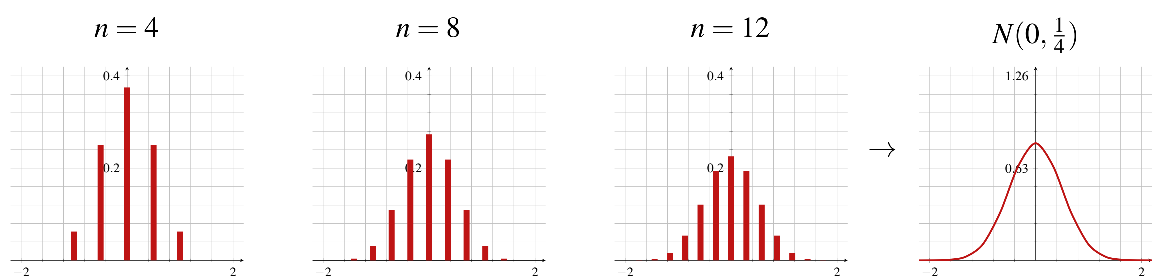
$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma} \to N(0, 1).$$

The fractions $\frac{X_i - \mu}{\sigma}$ are referred to as the *Z-scores* of the random variables X_i (Z is another common notation for the standard normal distribution). They describe the number of standard deviations above/below the mean that the sample falls.

33.2.1 The CLT and Binomial Random Variables

When we first introduced binomial random variables, we remarked that their probability mass function appeared to have a "bell-curve" shape. This suggests a link between binomial and normal random variables; since the normal random variable appears as a "smoothed out" version of a binomial distribution, it seems natural that it is their limiting distribution.

The following plots give a visual representation of this intuition. On the left, we visualize the pmfs of $\frac{1}{\sqrt{n}}\left(\mathrm{Binom}(n, \frac{1}{2}) - \frac{n}{2}\right)$ for a few values of n. The final plot is the pdf of $N(0, \frac{1}{4})$.



Note that there is a scaling of the y-axis by a factor of π in the last plot to account for the transition from measuring probability mass to measuring probability density.

We can formalize this intuition using the Central Limit Theorem.

Recall that we can interpret a binomial random variable $B \sim \mathrm{Binom}(n, p)$ as the sum of n mutually independent $\mathrm{Bern}(p)$ random variables. In the lens of the CLT, we take $\mathcal{D} = \mathrm{Bern}(\frac{1}{2})$, so that each $X_i \sim \mathrm{Bern}(\frac{1}{2})$. Note that $\mu = \mathbb{E}[X_i] = \frac{1}{2}$ and $\sigma^2 = \mathrm{Var}(X_i) = \frac{1}{4} \implies \sigma = \frac{1}{2}$. Then, note that each $A_n \sim \frac{1}{n} \cdot \mathrm{Binom}(n, \frac{1}{2})$, so by the CLT,

$$\sqrt{n}(A_n - \mu) \sim \frac{1}{\sqrt{n}}\left(\mathrm{Binom}(n, \tfrac{1}{2}) - \tfrac{n}{2}\right) \to N(0, \tfrac{1}{4}).$$

33.3 Statistical Inference

Another application of the Central Limit Theorem is as a tool for statistical inference. In this setting, we are given a large population of individuals who have characteristics drawn from some underlying distribution $\mathcal{D}$. We wish to gain insight about $\mathcal{D}$ (for example, understand its summary statistics such as its expectation, standard deviation, etc.). Often, our capacity to measure the individuals is limited by a resource budget (money, time, material). Thus, a task of statistical inference is to estimate properties of a population using only data from a much smaller random sample. It is this *random* sampling that allows us to leverage the Central Limit Theorem.

As a thought experiment, we can consider trying to estimate the average height of an American. It would be impractical to gather measurements of over 300 million individuals; in fact, even simply counting the population in the decennial census is a massive undertaking requiring years of planning. Instead, we'll select a smaller population (say 10,000 individuals) and use them as a proxy for the entire population. We cannot simply pick any 10,000 people to measure, as there are certainly dependencies built into the population. As a physical trait, there is a genetic component to height (people with tall parents are more likely to be tall). To have a representative sample, we'd like to select people from a wide cross-section of the population to reduce any bias from dependencies. One way to do this is with a uniform random sample of the population[1]. In such a sample, it is unlikely that individuals in the sample will be closely related, so we can treat them as independent realizations of $\mathcal{D}$.

Now, we can apply the Central Limit Theorem. We'll let $X_1, \ldots, X_n$ be the measurements (in our example, heights) of the individuals in our sample. By our assumption, these X_i are mutually independent, with each $X_i \sim \mathcal{D}$. By the CLT, for a large enough choice of n, the sample average $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ will be normally distributed with expected value equal to μ, the mean of $\mathcal{D}$. Thus, averaging heights of our randomly sampled individuals will almost always give a good estimate of the average height in the population (since a normal distribution *concentrates* its probability mass around its mean).

While everything said above holds for an arbitrary distribution (with finite variance), we can make a stronger claim when the underlying distribution $\mathcal{D}$ is actually normal.

Lemma 33.1 — Sampling Distribution of the Normal Distribution. Suppose that $X_1, \ldots, X_n$ are independent realizations of $N(\mu, \sigma^2)$. Then, their sample mean $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu, \frac{\sigma^2}{n})$.

The standard deviation of this sampling distribution is inversely proportional to $\sqrt{n}$. Thus, if we choose a larger sample from the population, our sample mean should provide a better estimate of the population mean. However, as n grows larger, we risk violating the independence condition of our sample. Therefore, the sample size must be carefully chosen, a topic that warrants further study.

[1] In practice, this is infeasible. Many people may choose not to participate, and willingness to participate itself may be dependent on the features we are trying to measure. There are many statistical techniques to deal with this.

33.3.1 Maximum Likelihood Estimation

Finally, we consider more carefully the setting where we know that the underlying distribution $\mathcal{D}$ is normal, but we are unsure of its parameters μ and σ. How can we use mutually independent samples $X_1, \ldots, X_n$ to estimate these parameters? In other words, what choice of estimate is the "best"? Since we are in a probabilistic setting, the notion of best is unclear; the quality of our estimate will depend on the values that we sampled $x_1, \ldots, x_n$. As our measure of the quality, we'll use the *likelihood*, which is another word for the joint probability density of the samples, now viewed as a function of parameters μ and σ.

Definition 33.2 — Likelihood Function.

The *likelihood* of realizing samples $x_1, \ldots, x_n$ from a $N(\mu, \sigma^2)$ distribution is the joint probability density $\mathcal{L}(\mu, \sigma^2) = f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)$ where each $X_i \sim N(\mu, \sigma^2)$ (mutually independently).

We would like the likelihood to be as large as possible, as this tells us that the probability that our samples actually came from the distribution with parameters μ and σ is large. The *Maximum Likelihood Estimate* is, as its name suggests, the maximizer (μ, σ) of $\mathcal{L}(\mu, \sigma)$.

We can use calculus to compute this maximizer. Using mutual independence to rewrite the joint pdf value as a product of its marginal pdf values, we find that

$$\mathcal{L}(\mu, \sigma^2) = \prod_{i=1}^{n} \tfrac{1}{\sigma\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}\left(\tfrac{x_i-\mu}{\sigma}\right)^2\right).$$

This function will be a pain to work with directly; however, there is a nice trick that will simplify the calculations. Note that the natural logarithm function $\ln(\cdot)$ is monotone increasing on $\mathbb{R}_{>0}$. Hence, $\mathcal{L}(\mu, \sigma)$ is maximized at the same point (μ, σ) as $\ln \mathcal{L}(\mu, \sigma)$. The latter function, the *log likelihood* is nicer to work with. By properties of logarithms, we have

$$\ln \mathcal{L}(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

To compute the maximum likelihood estimate of μ we should maximize $\ln \mathcal{L}(\mu, \sigma)$ with respect to μ, which we do by differentiating with respect to μ and setting this partial derivate equal to 0. We have

$$\frac{\partial \ln \mathcal{L}(\mu, \sigma^2)}{\partial \mu} = \frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(x_i - \mu) = 0 \qquad \Longrightarrow \qquad \mu = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Thus, the best estimator of the population mean μ (which we denote $\hat{\mu}$) is $\bar{x}$, the sample mean. While this is a consequence of our earlier reasoning with the CLT and verifies our intuition, it is beneficial to reach this conclusion from a different calculation.

We can similarly compute the maximum likelihood estimate of σ. We have

$$\frac{\partial \ln \mathcal{L}(\mu, \sigma^2)}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i - \mu)^2 \qquad \Longrightarrow \qquad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Since μ is an unknown parameter, we replace it in this estimator with our estimate $\hat{\mu}$, giving $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$, which is exactly the sample variance.

Main Takeaways:

- The *normal* distribution, $N(\mu, \sigma^2)$, is parameterized by its expectation μ and standard deviation σ, and has a density with a "bell-curve" shape.
- The Central Limit Theorem says that a suitably scaled average of independent samples from any distribution with finite variance will converge toward a normal distribution.
- Statistical inference is an area focused on estimating the properties of a distribution using relatively few random samples from it.
- When data is (approximately) normally distributed, we can use the sample mean and variance as maximum likelihood estimates of the true distribution mean and variance.

Exercises

◇ **33.1** Here, you'll verify the properties of the Normal distribution stated in the lecture.

(a) Prove that the *standard* normal ($N(0,1)$) density function

$$f_N(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

is a valid pdf.

(*Hint:* This is not a function that we can integrate directly. Instead, compute the square of this integral, applying Fubini's Theorem and a transformation to the polar coordinate system.)

(b) Suppose that $N \sim N(\mu, \sigma^2)$. Verify that $\mathbb{E}[N] = \mu$.

(c) Suppose that $N \sim N(\mu, \sigma^2)$. Verify that $\mathrm{Var}(N) = \sigma^2$.

33.2 Suppose that $N \sim N(\mu, \sigma^2)$.

(a) What is the distribution of aN where $a \in \mathbb{R}$?

(b) What is the distribution of $N + c$ where $c \in \mathbb{R}$?

⋆ **33.3** Given two continuous random variables X, Y with pdfs f_X and f_Y, let $Z = X + Y$. Recall (from an exercise in the previous lecture) that the pdf of f_Z is given by the convolution $f_Z = f_X * f_Y$ with

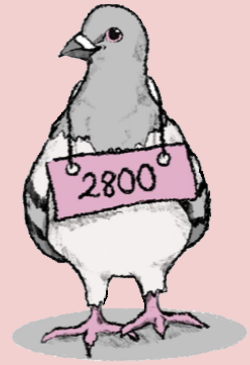
$$f_Z(r) = \int_{-\infty}^{\infty} f_X(\tau) f_Y(r - \tau)\, d\tau.$$

Suppose that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, and let $Z = X + Y$.

(a) Compute f_Z.

(b) What does this tell you about the distribution of Z?

VII Languages and Automata



34. Deterministic Automata

So far in this course, we have taken a whirlwind tour of the building blocks of mathematics. We've defined the main mathematical structures — sets, relations, and functions — and used formal logic as a means to develop different strategies for proving mathematical statements. Much of what we have covered has been very abstract, talking about properties of arbitrary sets or functions and introducing lots of definitions. While we will surely continue introducing new structures and new terminology, the latter modules in the course will be more concrete. We will apply the mathematical ideas that we have learned to reason about more concrete settings that are of interest to computer scientists. In this module, we will think about formalizing the behavior of computers as we study languages and automata.

34.1 Modeling Computation

As a student in a computer science class, you might have written and studied many computer programs. These programs likely had a lot of variety, being written in different languages and performing vastly different tasks. However, all of them likely conformed to the same basic paradigm. A program (which we will also call a *computation*) takes in some input from the user, whether this is in the form of some text, a file, or some interactions with an interface, carries out some internal calculations, and then returns to the user output. In other words, computer programs are functions from a set of possible inputs to a set of outputs. For our purposes, we will simplify down these inputs and outputs; we will only consider programs that take a single string as input and take one of two possible output actions, "accept" or "reject." Before we can go on, we'll formalize what we mean by an input *string*.

34.1.1 Strings

Strings are the analogs of words in natural language. We form a string by putting together (or *concatenating*) a sequence of characters. The set of characters that we can use to build strings is called an *alphabet*.

Definition 34.1 — Alphabet, Characters, Strings, Length.
An *alphabet*, often denoted by the letter Σ, is a *finite* set of *characters* (also called *letters* or *symbols*).

A *string* (also called a *word*) is a *finite* sequence (list) of characters. The *length* of a string w, denoted $|w|$, is the number of characters that it contains.

Example 34.1 — Alphabets.

- The *binary* alphabet consists of two characters $\{0,1\}$. This is the alphabet that is most natural for computers (in which each bit can encode two possible values), so will be the focus of many of our examples.
- The English alphabet consists of 52 characters $\Sigma = \{$a, b, ..., z, A, B, ..., Z$\}$.
- The digits $D = \{0,1,\ldots,9\}$ form an alphabet from which we can build numbers.
- The natural numbers $\mathbb{N} = \{0,1,\ldots\}$ are not an alphabet since they form an *infinite* set.

While, formally, the string "cat" in the English alphabet would be the sequence $(\mathrm{c},\mathrm{a},\mathrm{t}) \in \Sigma\times\Sigma\times\Sigma = \Sigma^3$, we will typically drop the brackets and commas and simply write "cat".

Example 34.2 — Strings.

- 010010, 01, 10010, 111 are all strings over the binary alphabet (or *binary strings*) with lengths 6, 2, 5, and 3 (respectively).
- cat, pigeon, Matt, cLxPpWma are all strings over the English alphabet.
- 1,800 is *not* a string over the digit alphabet since it includes the character ","$\notin D$.
- An infinite sequence of 1s is *not* a string because it does not have finite length.
- The *empty string*, which we represent with the symbol λ, is a sequence containing 0 characters. λ is a string over every alphabet.

We have a special notation to refer to the set of all strings over an alphabet. It utilizes an operator known as the *Kleene Star* that we will study more in a couple of lectures.

Definition 34.2 — Set of all Strings, Σ^*.
Given an alphabet Σ, Σ^* refers to the set of all strings over Σ. Formally,

$$\Sigma^* = \{\lambda\} \cup \Sigma \cup (\Sigma\times\Sigma) \cup (\Sigma\times\Sigma\times\Sigma)\cup\ldots = \bigcup_{n\in\mathbb{N}} \Sigma^n.$$

Example 34.3 The set $\{0,1\}^*$ consists of all binary strings $\{\lambda,0,1,00,01,10,11,000,001,010,011,100,101,110,111,\ldots\}$.

We will study computations of the form $\Sigma^* \to \{\text{Accept},\text{Reject}\}$ for some alphabet Σ.

34.2 Deterministic Finite Automata

For the rest of this lecture, we will introduce a very simple model of computation called a *Deterministic Finite Automaton* (DFA) which will process a string over some alphabet Σ and decide whether to accept or reject that string. A DFA processes its input string one character at a time from left to right. The DFA has a set of internal *states* that it uses during its computation. Whenever it processes (or *reads*) a character, it chooses its next internal state based entirely on the current state and the character that it reads. Based upon which state it is in after it finishes reading the input string, it either accepts or rejects the input.

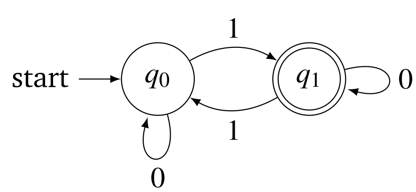
We can give the following formal definition of a DFA.

Definition 34.3 — DFA.
A DFA M is a 5-tuple $M = (\Sigma, Q, s, F, \delta)$ where:

- Σ is an alphabet
- Q is the *finite* set of machine states
- $s \in Q$ is the *starting* or *initial* state
- $F \subseteq Q$ is the set of *final* or *accepting* states
- $\delta : (Q \times \Sigma) \to Q$ is a (deterministic) transition function

We can give a more palatable description of a DFA by drawing a directed graph. The nodes in this graph are the machine states (Q). We draw a double circle around a state when it is a final state (F). The initial state (s) is labeled with a "start" arrow. Out of each state q, there is an arrow labeled with each character $\sigma \in \Sigma$ that points to state $\delta(q, \sigma)$.

Example 34.4 Consider the following DFA:



This machine has binary alphabet $\Sigma = \{0, 1\}$, which we determine by looking at the labels on the arrows leaving any of its states. It has state set $Q = \{q_0, q_1\}$, initial state $s = q_0$, and final state set $F = \{q_1\}$. Its transition function is given by $\delta(q_0, 1) = \delta(q_1, 0) = q_1$ and $\delta(q_0, 0) = \delta(q_1, 1) = q_0$.

To determine the behavior of a DFA, we start in the initial state. Then, each time we read a character from the input string, we follow the arrow leaving our current state that is labeled with that character. For example, on input "01011" we begin in state q_0, loop around the 0 arrow to remain in q_0, follow the 1 arrow to q_1, loop around the 0 arrow to remain in q_1, follow the 1 arrow to q_0, and follow the 1 arrow to q_1. Since we ended in a final state, the machine accepts the input "01011". If we ended in a state that was not final (e.g. on input "1001") the machine would reject that string.

Notice that the fact that δ is a well-defined function ensures that there will always be exactly one arrow labeled with each character leaving each state. Thus, the "arrow tracing" procedure will (deterministically) map every string to one of the machine states, meaning the DFA defines a function $\Sigma^* \to \{\text{accept}, \text{reject}\}$.

34.2.1 The Language of a DFA

Just as the English language includes only some of the strings that can be formed from its alphabet (cat, pigeon, Matt, but not cLxPpWma), a formal *language* is a subset of strings.

Definition 34.4 — Language.
A *language* L over an alphabet Σ is a subset of its strings $L \subseteq \Sigma^*$.

There is natural choice for a language that is defined by a DFA M which includes all of the strings that M accepts.

Definition 34.5 — Language of a DFA, Recognize.
Given a DFA M, the language of M, denoted $\mathcal{L}(M)$ is the set of all strings that it accepts. We say that M *recognizes* $\mathcal{L}(M)$.

Let's return to the DFA from Example 34.4. What language does this machine recognize? To answer this question, let's observe what this machine does. Whenever it reads a 0 from the input, it stays in the same state. Whenever it reads a 1, it moves to the other state. Thus, the states are somehow keeping track of the number of 1s that we have seen. When we are in state q_0, we know that we have read an even number of 1s, and when we are in state q_1, we have read an odd number of 1s. Since q_1 is the only accepting state, this machine accepts strings with an odd number of 1s.

34.3 Constructing a DFA

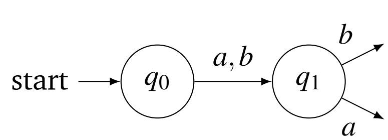
To conclude this lecture, we'll look at an example of how to construct a DFA that recognizes a language.

Example 34.5 We'll construct a DFA that recognizes the language

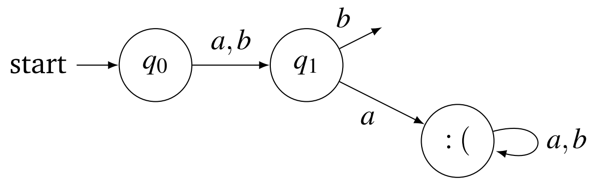
$$L = \{w \in \{a,b\}^* \mid \text{the second letter of } w \text{ is } b \text{ and } w \text{ ends with } a\}.$$

All of the "computation" in a DFA happens when we transition between states; we need to use the states to keep track of any information about what we have read so far in the string. Thus, a good way to start designing a DFA is to decide which states we will need.

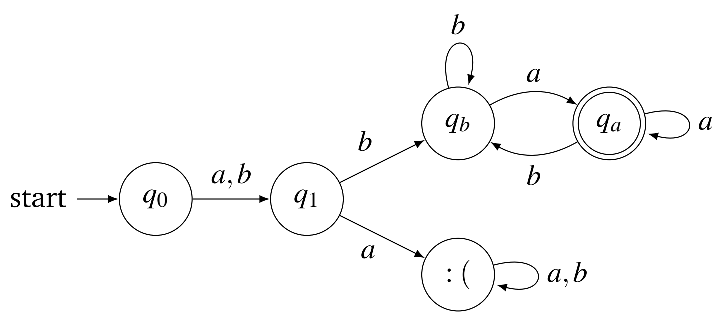
In this example, we will need to check something about the second character, meaning we'll need to know when we are reading the second character. To do this, we can have the initial state (q_0) represent "we've read 0 characters", which always transitions to another state (q_1) representing "we've read 1 character".



Here, we use the label a,b to mean that we should follow this arrow from q_0 if we read either an a or a b. Now, once we are in state q_1, we will need to do something different based on which character we see. If we see an a, then we know that we should not accept this string (no matter what characters come later). We can transition to a new state (called $:($, because it is sad) that does not accept. Once we are in state $:($, we can continue to loop there forever.



If we are in state q_1 and we read a b, then we should not transition to $:($ since it is possible that we accept the input. At this point, the only remaining thing that we must check is whether the input ends with an a or a b. To do this, we can have two states keeping track of the last character that was seen (q_a and q_b), since we just read b, we should transition from q_1 to q_b. Then, the a arrows leaving q_a and q_b should both point to q_a, and the b arrows leaving q_a and q_b should both point to q_b. q_a will be a final state since if we end up there, we know that the last character we read was an a. Our DFA, which we call M, is:



As an initial check, we can confirm that each state has one outgoing a arrow and one outgoing b arrow. Thus, we have drawn a valid DFA. Furthermore, by the reasoning that we used to construct M, we conclude that $\mathcal{L}(M) = L$. Trace through the computation of M on a few sample inputs to verify that this is true.

Main Takeaways:

- Languages are sets of strings, which are finite sequences of characters from a finite alphabet.
- A DFA is a simple computational model that can recognize a language.
- DFAs process their inputs one character at a time and perform their computation by transitioning between states. The state the machine is in when it finishes reading the string determines whether it accepts or rejects the input.
- When we construct or reason about a DFA, we should reason about the meaning of being in each state.

Exercises

34.1 Write out all strings that belong to each of the following languages.

(a) $L_1 = \{w \in \{a,b,c\}^* \mid |w| \leq 2\}$

(b) $L_2 = \{w \in \{d,e\}^* \mid |w| = 5, |w|_d = 2\}$

Here, the notation $|w|_d$ refers to the number of occurrences of character d in string w.

(c) $L_3 = \{w \in \{0,1\}^* \mid |w| \leq 3, |w|_0 \leq |w|_1\}$

(d) $L_4 = \{w \in \{0,1\}^* \mid |w| \leq 4, \text{all 0s in } w \text{ appear before all 1s in } w\}$

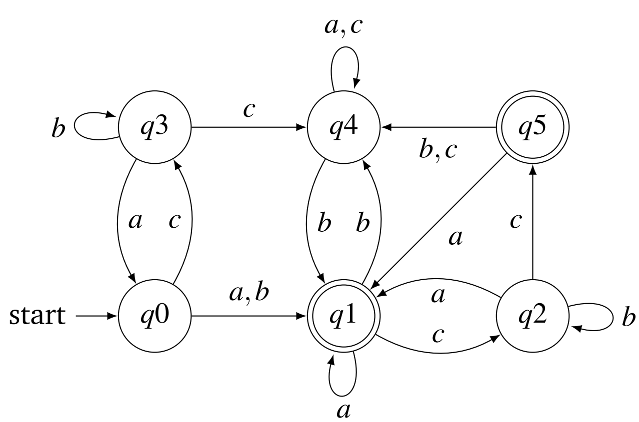
(e) $L_5 = L_3 \cup L_4$

(f) $L_6 = L_3 \cap L_4$

(g) $L_7 = \{w \in \{2,3\}^* \mid |w| \leq 5, w \text{ is a palindrome}\}$

A *palindrome* is a string that reads the same forward and backward, such as "racecar".

34.2 Determine whether the following DFA accepts or rejects each of the following strings.



(a) λ

(b) *acc*

(c) *abc*

(d) *cbbcb*

(e) *abcabc*

(f) *aabbcc*

(g) *cabacbbac*

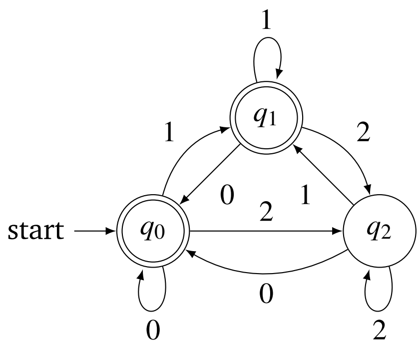
(h) *cbccbcccbccccb*

(i) *cabacab*

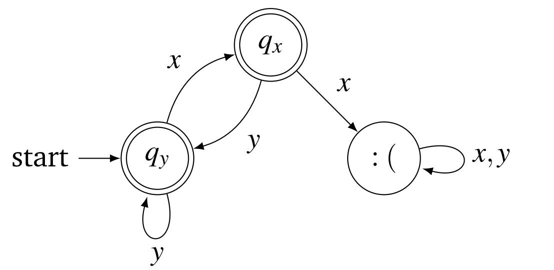
34.3 For each of the following DFAs:

- Give its formal description. That is, write out its components $(\Sigma, Q, s, F, \delta)$.
- Describe the language that the machine recognizes.

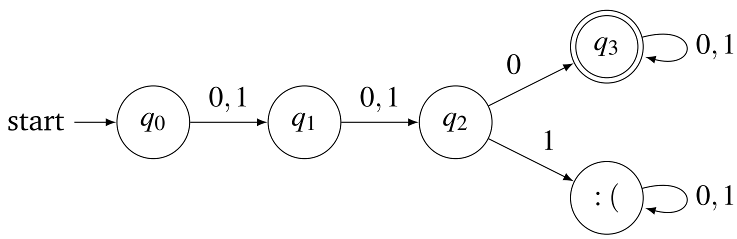
(a)



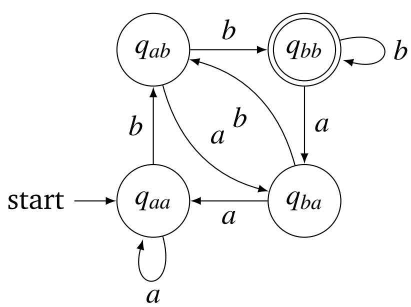
(b)



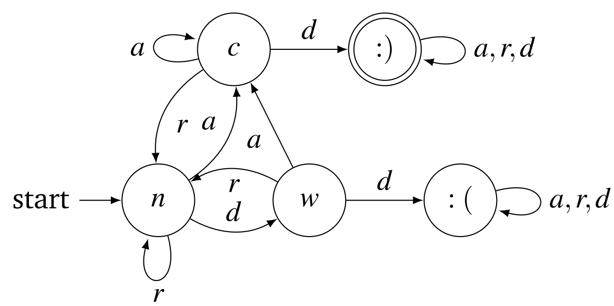
(c)



(d)



(e)



34.4 Construct a DFA that recognizes each of the following languages.

(a) $L_1 = \{w \in \{0,1\}^* \mid w \text{ contains a } 1\}$

(b) $L_2 = \{w \in \{0,1\}^* \mid w \text{ does not contain more than two 0s}\}$

(c) $L_3 = \{w \in \{a,b,c\}^* \mid w \text{ starts with } c\}$

(d) $L_4 = \{w \in \{a,b,c\}^* \mid w \text{ ends with } b\}$

(e) $L_5 = L_3 \cap L_4$

(f) $L_6 = \{w \in \{x,y,z\}^* \mid$ every x in w is immediately followed by a $z\}$

(g) $L_7 = \{w \in \{0,1\}^* \mid$ the second-last character in w is $1\}$

(h) $L_8 = \{w \in \{a,b,c\}^* \mid w$ does not contain two consecutive copies of the same character$\}$

⋄ **34.5** In this exercise, you'll reason about certain languages of binary numbers. Recall that the digit in the i'th position (from the right, starting from 0) of a number's binary representation indicates whether that number includes 2^i in its binary expansion. For example, $100101 = 1\cdot 2^5 + 0\cdot 2^4 + 0\cdot 2^3 + 1\cdot 2^2 + 0\cdot 2^1 + 1\cdot 2^0 = 32+4+1 = 37$.

(a) Design a DFA that accepts binary numbers that are even. In this and the following parts, your automaton should not accept the empty string.

(b) Design a DFA that accepts binary numbers that are multiples of 4.

(c) Design a DFA that accepts binary numbers that are multiples of 3. (Hint: The DFA that we have in mind has 3 states, one for each residue modulo 3.)

(d) Design a DFA that accepts binary numbers that are multiples of 5.

34.6 In this exercise, you'll revisit some ideas from our discussion of set cardinality.

(a) Given any finite alphabet Σ, is the set Σ^* countable or uncountable? Explain your answer.

(b) Given any finite alphabet Σ, is the set of all languages over Σ countable or uncountable? Explain your answer.

34.7 Consider a DFA $M = (\Sigma, Q, s, F, \delta)$.

(a) Describe how we can view the outgoing transitions from each state as describing a function $\Sigma \to Q$.

(b) Expand on your answer to part a, describing how we can recast the transition function $\delta\colon (Q\times\Sigma) \to Q$ as a map $\delta'\colon Q \to Q^\Sigma$. Recall that the notation Q^Σ refers to the set of all functions $\Sigma \to Q$.

Switching between these two perspectives — a function with multiple arguments and a function whose outputs are also functions — is referred to as *currying/uncurrying* and plays an important role in functional programming.

34.8 Given a language $L \subseteq \Sigma^*$, its *complement* $\overline{L} = \Sigma^* \setminus L$ (recall that languages are sets of strings, so we can perform set operations on them). Suppose that M is a DFA with $\mathcal{L}(M) = L$. Describe how to modify M so that it recognizes $\overline{L}$.

34.9 Given strings $v, w \in \Sigma^*$ we say that v is a *substring* of w if w contains the letters in v in consecutive positions (in order). For example, 010 and 1101 are both substrings of 11010, but 111 is not.

Construct DFAs that recognize each of the following languages, whose descriptions involve substrings.

(a) $L_1 = \{w \in \{0,1\}^* \mid 00 \text{ is a substring of } w\}$

(b) $L_2 = \{w \in \{0,1\}^* \mid 000 \text{ is a substring of } w\}$

(c) $L_3 = \{w \in \{0,1\}^* \mid 010 \text{ is a substring of } w\}$

(d) $L_3 = \{w \in \{0,1\}^* \mid \text{ both 00 and 11 are substrings of } w\}$

⋆ **34.10** Argue that any *finite language* (that is, any language that consists of finitely many strings) can be recognized by a DFA. To do this, describe a procedure to build such a DFA for any finite language.

34.4 Postscript: Structural Induction on Automata

In this lecture, we studied two DFAs and reasoned about the languages that they recognize. The arguments that we gave were somewhat “informal”; they were based on intuition that we built by reasoning about how computations on different strings would flow through the states. In order to give a more formal proof, we can use a flavor of an inductive argument which is often called *structural induction*.

In a structural induction proof, we give a formal description of the set of strings that the machine processes to each of the states. Then, we argue that these descriptions are all accurate by carrying out induction on the length of the strings. This is analogous to the steps of a traditional induction proof. The following outlines how such proofs typically go:

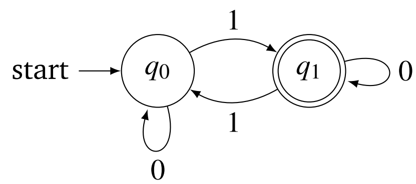
Base Case: Argue that the empty string fits the description assigned to the start state.

Inductive Hypothesis: Given any state q, suppose that the machine has just processed a string w that meets the description assigned to state q.

Inductive Step: For each character $\sigma \in \Sigma$, argue that the string $w\sigma$ (this is notation for the string w with the character σ appended at the end) meets the description of state $\delta(q, \sigma)$.

The inductive step is essentially ensuring that the transition function δ preserves the state descriptions, which it does by separately reasoning about each arrow in the automaton. Once we have done this, we know that our descriptions are correct for any string; here we are using the fact that strings have finite length. Thus, the language of the machine is exactly the union of the descriptions of its final states. Let’s put this proof technique into practice for the two example automata from the lecture.

First, let us recall the DFA from Example 34.4:



We will argue by structural induction that

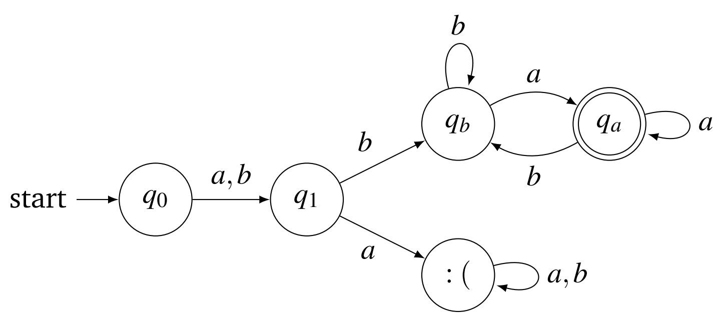
- This DFA processes string w to $q_0 \Leftrightarrow w$ contains an even number of 1s.
- This DFA processes string w to $q_1 \Leftrightarrow w$ contains an odd number of 1s.

For the base case, we note that λ contains an even number of 1s, so fits the description of q_0. For the inductive step, we note that:

- If the DFA processes string w to q_0, then w contains an even number of 1s. Then, the string $w0$ also contains an even number of 1s, and the DFA processes $w0$ to state $\delta(q_0, 0) = q_0$.
- If the DFA processes string w to q_0, then w contains an even number of 1s. Then, the string $w1$ contains an odd number of 1s, and the DFA processes $w1$ to state $\delta(q_0, 1) = q_1$.
- If the DFA processes string w to q_1, then w contains an odd number of 1s. Then, the string $w0$ also contains an odd number of 1s, and the DFA processes $w0$ to state $\delta(q_1, 0) = q_1$.
- If the DFA processes string w to q_1, then w contains an odd number of 1s. Then, the string $w1$ contains an even number of 1s, and the DFA processes $w1$ to state $\delta(q_1, 1) = q_0$.

In every case, the transition function preserves our state descriptions. By structural induction, we conclude that the language of this machine consists of those strings that fit the description of q_1, which are exactly the strings with an odd number of 1s. ■

Next, let us recall the DFA from Example 34.5:



We will argue by structural induction that:

- M processes string w to $q_0 \Leftrightarrow w = \lambda$.
- M processes string w to $q_1 \Leftrightarrow w$ contains 1 character.
- M processes string w to : ($\Leftrightarrow$ the second character of w is a.

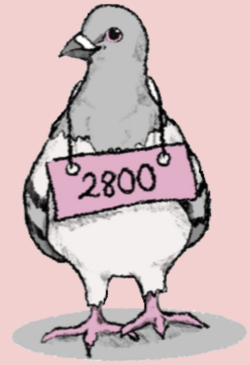
- M processes string w to $q_b \Leftrightarrow$ the second character of w is b and w ends with b.
- M processes string w to $q_a \Leftrightarrow$ the second character of w is b and w ends with a.

The base case of this induction is trivial, since M processes λ to q_0. For the inductive step, we note that:

- If the DFA processes string w to q_0, then $w = \lambda$. Then, the strings $wa = \lambda a = a$ and $wb = \lambda b = b$ both contain 1 character and are processed to state $\delta(q_0, a) = \delta(q_0, b) = q_1$.
- If the DFA processes string w to q_1, then w contains one character. Then, the string wa has a as its second character and is processed to state $\delta(q_1, a) =: ($.
- If the DFA processes string w to $:($, then the second character of w is a. Then, the strings wa and wb also both have a as their second character and are processed to state $\delta(:(, a) = \delta(:(, b)) =: ($.
- If the DFA processes string w to q_1, then w contains one character. Then, the string wb has b as its second character and ends with b and is processed to state $\delta(q_1, b) = q_b$.
- If the DFA processes string w to states q_a or q_b, then w has b as its second character. Then, the string wb also has b as its second character and ends with b and is processed to state $\delta(q_a, b) = \delta(q_b, b) = q_b$.
- If the DFA processes string w to states q_a or q_b, then w has b as its second character. Then, the string wa also has b as its second character and ends with a and is processed to state $\delta(q_a, a) = \delta(q_b, a) = q_a$.

In every case, the transition function preserves our state descriptions. By structural induction, we conclude that $\mathcal{L}(M)$ consists of those strings that fit the description of q_a, which are exactly the strings with b as their second character that end with a.

For more practice with structural induction, use it to reason formally about the languages of the machines described in the lecture exercises.



35. The Pumping Lemma

In the previous lecture, we introduced DFAs as a simple computational model for recognizing a language. We posed the question

Which languages can be recognized by a DFA?

which will occupy us for the next few lectures. Today, we will make some progress toward an answer. We will introduce and prove the *Pumping Lemma*, which describes a property of any language recognized by a DFA. We can use the Pumping Lemma as a tool for finding languages that a DFA *cannot* recognize.

35.1 Substrings

In order to state and understand the Pumping Lemma, we will need to understand substrings, concatenation, and string decomposition. All of these concepts relate to breaking down strings into smaller pieces and recombining these pieces to form new strings.

Definition 35.1 — Substring.
A string v is a *substring* of string w if the characters in v appear in consecutive positions in w.

Example 35.1 — Substrings.

- The strings “disc” and “crete” are both substrings of the string “discrete”.
- The string “01001” is a substring of “01001001”. In fact, this substring appears two times in the longer string.
- The string $v =$“abba” is *not* a substring of $w =$“abaaba”. Although the letters in v appear in order in w, they are not in consecutive positions.

Definition 35.2 — Concatenation, Decompositions.

Given two strings v and w, the *concatenation* of v and w, which we write as vw, is the string that is formed by first writing all characters (in order) in v and then appending all characters (in order) in w.

When we can write a string w as the concatenation of two strings $w = uv$, we say that uv is a *decomposition* of w.

To avoid confusion about which symbols are characters and which represent strings, we will adopt the convention that the later letters in the alphabet (u, v, w, x, y, z) stand for strings, and numbers and earlier letters (a, b, c, d, e) are characters.

Example 35.2 We'll consider strings $u = 101$, $v = 11$, $w = 0$, and $x = \lambda$. Then,

- $uv = 10111$, $vu = 11101$. This shows that concatenation is *not* a *commutative* operation.
- $ux = xu = 101$. Concatenating any string with the empty string gives the same string.
- The notation w^i for some $i \in \mathbb{N}$ refers to the string formed by concatenating together i copies of w. So $w^4 = 0000$, and $u^0 = \lambda$ (in fact, $z^0 = \lambda$ for any string z).
- $wv^3 = 0111111$, $(wv)^3 = 011011011$.

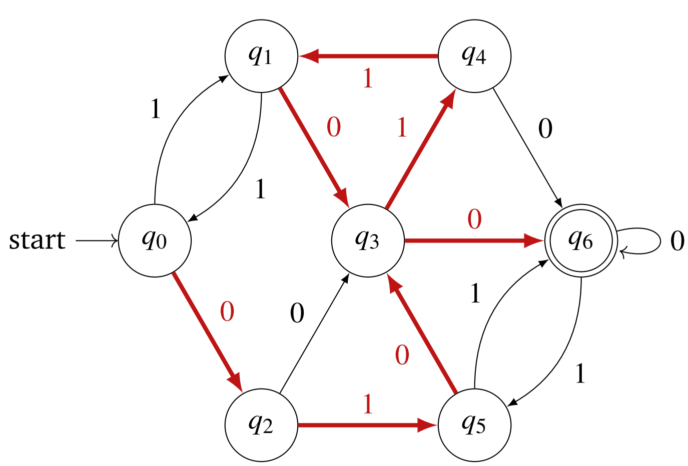
35.2 The Pumping Lemma

Lemma 35.1 — The Pumping Lemma.

Let L be a language that is recognized by a DFA with k states, and let $w \in L$ be a string with $|w| \geq k$. Then, there is a decomposition $w = xyz$ with $|xy| \leq k$, $|y| > 0$, and $xy^iz \in L$ for all $i \in \mathbb{N}$.

The proof of the Pumping Lemma relies on tools that we introduced earlier in the semester.

Proof. Since $w \in L$, we know that tracing along the path of arcs labeled with the characters in w takes us from the initial state to an accepting state in the DFA. As an example, the 7-state DFA shown below accepts the string 0101100, as shown by the highlighted arcs.



Consider the first k arcs along this path (our assumption $|w| \geq k$ ensures that the path is always at least this long). The path starts at the initial state vertex. Then, for each of the k arcs, it will transition to another (perhaps the same) state. This means that we will make $k+1$ visits to states along this portion of the path. However, we assumed that the DFA includes only k states. Thus, by the Pigeonhole Principle, we must have visited one of the states q multiple times. In our example, we visited the state $q = q_3$ twice along the red path.

Let's think about the arcs that we traveled along between the first and second times that we visited q. These arcs must form a cycle (in our example, the cycle $(q_3, q_4), (q_4, q_1), (q_1, q_3)$). Now, we can break the full path that our computation traced into three pieces:

1. The path before we reach the vertex q for the first time.
2. The cycle that returns us to q for the second time.
3. The remaining path after we visit q the second time.

We will let x refer to the characters that label the arcs in the first piece, y refer to the characters from the arcs in the second piece, and z refer to the characters from the arcs in the third piece. In our example

$$x = 010, \qquad y = 110, \qquad z = 0$$

Notice that x, y, and z are all substrings of w with $w = xyz$. In addition, note that

- $|xy| \leq k$, since we located the cycle within the first k arcs in the path.
- $|y| > 0$, since the arcs in the cycle connect two *distinct* visits to state q.

Finally, we note that the path formed by following the arcs in the first piece, traveling around the cycle any number (0 or more) of times, and then following the arcs in the third piece always connects the initial state (q_0) to a final state (q_6). Thus, that $xy^iz \in L$ for all $i \in \mathbb{N}$. ■

Using the Pumping Lemma can be challenging at first because it is a rather complicated proposition that involves multiple layers of quantifiers. Written symbolically, it says that if a language L is recognized by a DFA, then

$$\exists_{k \in \mathbb{N}^+}, \ \forall_{\substack{w \in L \\ |w| \geq k}}, \ \exists_{\substack{x,y,z \\ w = xyz \\ |xy| \leq k \\ |y| > 0}}, \ \forall_{i \in \mathbb{N}}. \quad xy^iz \in L.$$

The contrapositive form of the Pumping Lemma will be more useful to us, as it gives us a way to establish that a language *cannot* be recognized by *any* DFA. Using DeMorgan's Laws, we can write this contrapositive: If

$$\forall_{k \in \mathbb{N}^+}, \ \exists_{\substack{w \in L \\ |w| \geq k}}, \ \forall_{\substack{x,y,z \\ w = xyz \\ |xy| \leq k \\ |y| > 0}}, \ \exists_{i \in \mathbb{N}}. \quad xy^iz \notin L,$$

then L cannot be recognized by a DFA.

35.3 Proofs using the Pumping Lemma

To conclude the lecture, we will see some examples of how to use the Pumping Lemma to show that a language cannot be recognized by a DFA. The structure of these proofs are all very similar and follow from strategies for proving quantified statements that we discussed at the beginning of the semester. The high-level skeleton for these arguments is:

1. Consider an *arbitrary* variable k that represents a potential number of states in a DFA that could recognize L.
2. Choose a string $w \in L$ to reason about. This choice should depend on k, as we must verify that $|w| \geq k$. You might need to choose w very cleverly to make the proof go through.
3. Consider an *arbitrary* decomposition of $w = xyz$ with $|xy| \leq k$ and $|y| > 0$.
4. Use the latter two properties of this decomposition and your choice of k to find an $i \in \mathbb{N}$ for which $xy^iz \notin L$.

This argument shows that there cannot be a DFA with k states that recognizes L, and since our choice of k was arbitrary, this means that there cannot be any such DFA.

Example 35.3 We will argue that the language $L = \{w \in \{0,1\}^* \mid w = 0^n 1^n \text{ for some } n \in \mathbb{N}\}$ is not recognized by any DFA.

Proof. Given any $k \in \mathbb{N}^+$, we consider the string $w = 0^k 1^k \in L$. Note that $|w| = 2k \geq k$. Now, consider any decomposition $w = xyz$ with $|xy| \leq k$ and $|y| > 0$. The first k characters of w are all 0s, meaning substrings x and y consist entirely of 0s, so $x = 0^n$ for some $n \in \mathbb{N}$, $y = 0^m$ for some $m \in \mathbb{N}^+$ and $z = 0^{k-n-m}1^k$. But then, the string $xy^2z = 0^{k+m}1^k$, and $k+m \neq k$ since $m > 0$. Thus, $xy^2z \notin L$. By the contrapositive of the Pumping Lemma, we conclude that L cannot be recognized by a DFA with k states. Since k was chosen arbitrarily, no DFA can recognize L. ■

Before we look at a second example, let's stop to think about what makes this language impossible for a DFA to recognize. To check if a string belongs to L, we would need to know how many 0s we had read at the time when we read the first 1 (so that we could check to make sure we see the same number of 1s). The only mechanism that a DFA has to keep track of what it has seen is its states. We need a different state for each possible number of 0s. Since there can be any $n \in \mathbb{N}$ number of $0s$, our machine would need infinitely many states. Since a DFA is limited to finitely many states, we have uncovered the impossibility.

Example 35.4 We will argue that the language $L = \{w \in \{0\}^* \mid |w| = n^2 \text{ for some } n \in \mathbb{N}\}$ is not recognized by any DFA.

Proof. Given any $k \in \mathbb{N}^+$, we consider the string $w = 0^{k^2} \in L$. Note that $|w| = k^2 \geq k$. Now, consider any decomposition $w = xyz$ with $|xy| \leq k$ and $|y| > 0$. We must have $y = 0^j$ for some $0 < j \leq k$ so that the string $xy^2z = 0^{k^2+j}$. Note that $k^2 < k^2 + j \leq k^2 + k < k^2 + 2k + 1 = (k+1)^2$. Since $k^2 + j$ sits between two adjacent perfect squares, it is not a perfect square. Thus, $xy^2z \notin L$. By the contrapositive of the Pumping Lemma, we conclude that L cannot be recognized by a DFA with k states. Since k was chosen arbitrarily, no DFA can recognize L. ■

Main Takeaways:

- Strings can be broken up into smaller pieces called *substrings* or *concatenated* to form larger strings.
- The Pumping Lemma tells us that in any language that is recognized by a DFA, sufficiently long strings must have repeating pieces.
- To prove the Pumping Lemma we can examine the path that a string takes through the graph representation of a DFA and apply the Pigeonhole Principle.
- The contrapositive of the Pumping Lemma is useful for proving that a language cannot be recognized by a DFA.

Exercises

35.1 Argue that each of the following languages cannot be recognized by a DFA.

(a) $L_1 = \{w \in \{0,1\}^* \mid w = 0^n 1^{n+1} \text{ for some } n \in \mathbb{N}\}$

(b) $L_2 = \{w \in \{0,1\}^* \mid w = 0^n 1^m \text{ for some } n,m \in \mathbb{N} \text{ with } n < m\}$

(c) $L_3 = \{w \in \{0,1\}^* \mid w = 0^n 1^m \text{ for some } n,m \in \mathbb{N} \text{ with } n > m\}$

(d) $L_4 = \{w \in \{0\}^* \mid |w| = n^3 \text{ for some } n \in \mathbb{N}\}$

(e) $L_5 = \{w \in \{0,1\}^* \mid |w|_0 = |w|_1\}$.
Recall that $|w|_0$ refers to the number of occurrences of 0 in w (and similar for $|w|_1$).

(f) $L_6 = \{w \in \{0,1\}^* \mid w \text{ is a palindrome}\}$.
Recall that a *palindrome* is a string that reads the same forward and backward.

(g) $L_7 = \{w \in \{0,1\}^* \mid w = vv \text{ for some } v \in \{0,1\}^*\}$.

◇ (h) $L_8 = \{w \in \{0\}^* \mid |w| \text{ is prime}\}$

35.2 For each of the following languages, draw a DFA with the indicated number of states that recognizes the language. Then, for the provided string w, find a decomposition $w = xyz$ that satisfies the conditions of the Pumping Lemma.

(a) $L_1 = \{w \in \{0,1\}^* \mid w \text{ begins with } 0\}$, 3 states, $w = 0100101101$

(b) $L_2 = \{w \in \{0,1\}^* \mid w \text{ ends with } 11\}$, 3 states, $w = 0011011$

(c) $L_3 = \{w \in \{0,1\}^* \mid w \text{ contains the substring } 101\}$, 4 states, $w = 1010001$

(d) $L_4 = \{w \in \{0,1\}^* \mid \text{ every 0 in } w \text{ is followed by two 1s }\}$, 4 states, $w = 110111011$

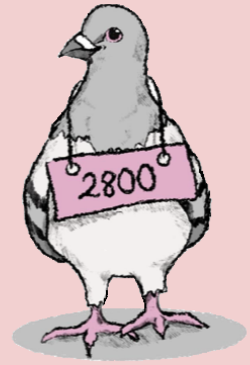
35.3 The Pumping Lemma tells us that if a language is recognized by a DFA, then it is *pumpable* (the adjective we use to describe the condition $xy^iz \in L$ for all $i \in \mathbb{N}$, since we are "pumping up" the string with more copies of y). We saw that its contrapositive, if a language is not pumpable then it is not recognized by a DFA, is useful to argue that a language cannot be recognized by a DFA. However, this method is not foolproof because the converse of the Pumping Lemma is not true; there are languages that are *pumpable* but still not recognized by a DFA.

Consider the language

$$L = \{w \in \{0,1\}^* \mid w = 1^m 0^n 1^n \text{ with } n \in \mathbb{N}, m \in \mathbb{N}^+\} \cup \{w \in \{0,1\}^* \mid w = 0^m 1^n \text{ with } m, n \in \mathbb{N}\}.$$

Argue that for any $k \in \mathbb{N}^+$ and any $w \in L$ with $|w| \geq k$, we can find a decomposition $w = xyz$ that satisfies the conditions of the Pumping Lemma.

Nevertheless, L cannot be recognized by a DFA for the same reason that we saw in the lecture (we'd need infinitely many states to count the number of 0s). Later in the module, we'll see an alternate tool for showing that a language cannot be recognized by a DFA, the *Myhill-Nerode Theorem*, that does not have this issue.



36. The Myhill-Nerode Theorem

In the previous lecture, we saw how to construct a DFA that recognizes a particular language. This begs the question, is it possible to build a DFA to recognize *any* language? The, perhaps surprising, answer to this question is "no;" there are some languages which cannot be recognized by DFAs. In today's lecture, we will discuss the Myhill-Nerode Theorem, which is a cool result for many reasons. First, it provides a necessary and sufficient ($\Leftrightarrow$) condition for a language to be recognized by a DFA, so it gives a technique for proving that a language cannot be recognized. Second, the theorem uses an equivalence relation on strings, so it provides opportunity for us to review these concepts from our relations module. Finally, this theorem has a Cornell connection; one of its discoverers, Anil Nerode, is Cornell's longest-tenured professor, working in our math department since 1959.

36.1 An Equivalence Relation on Strings

The central object in the Myhill-Nerode Theorem is an equivalence relation on strings known as the *Nerode congruence* relation. To define this relation, we must first define an operation on strings known as *concatenation*;

Definition 36.1 — Concatenation.

Given two strings over the same alphabet $u, v \in \Sigma^*$, their *concatenation*, uv is the string formed by appending the sequence of characters making up v to the end of the sequence of characters making up u.

Example 36.1

- Over the English alphabet, the concatenation of the words $u =$"race" and $v =$"car" is $uv =$"racecar". Concatenation is not commutative since $vu =$"carrace"
- Over the binary alphabet, let us define strings $u = 010$ and $v = 1101$. Then $uv = 0101101$, $vu = 1101010$, and $uu = 010010$.
- Given any string $w \in \Sigma^*$, the concatenation $w\lambda = \lambda w = w$

Definition 36.2 — Nerode congruence.
Given any language L over a finite alphabet Σ, the *Nerode congruence* relation $\sim_L$ is defined such that for any strings $u,v \in \Sigma^*$, $u \sim_L v$ if and only if for all $w \in \Sigma^*$ either both $uw, vw \in L$ or $uw, vw \notin L$.

Example 36.2 Let L be the language of binary strings that contain at least one "0".

- $0100 \sim_L 001$. Since both $u = 0100$ and $v = 001$ contain a "0", both uw and vw will belong to L no matter which string $w \in \{0,1\}^*$ we choose to concatenate to them.
- $111 \sim_L 11$. Neither $u = 111$ nor $v = 11$ contain a "0". If we choose a string $w \in \{0,1\}^*$ that does not contain a "0", then both $uw, vw \notin L$. On the other hand, if we choose a string w that contains a "0", then both $uw, vw \in L$.
- $10 \not\sim_L 11$. If we (for example) concatenate the string $w = 11$ to both of these strings, we'd find that $uw = 1011 \in L$, but $vw = 1111 \notin L$.

Lemma 36.1 For any language L, $\sim_L$ is an equivalence relation.

Proof. We must verify the three properties of an equivalence relation. In this case, all of the properties follow immediately from the definition

Reflexivity: Given any string $u \in \Sigma^*$, $u \sim_L u$ because uw either deterministically belongs to L or does not for each $w \in \Sigma^*$.

Symmetry: This follows from the symmetry in the definition. Suppose that $u \sim_L v$ for some $u,v \in \Sigma^*$. Then, for any $w \in \Sigma^*$, $\big((uw \in L) \wedge (vw \in L)\big) \vee \big((uw \notin L) \wedge (vw \notin L)\big)$ is true. By symmetry of $\wedge$, this means that $\big((vw \in L) \wedge (uw \in L)\big) \vee \big((vw \notin L) \wedge (uw \notin L)\big)$ is also true. As this is true for arbitrary $w \in \Sigma^*$, we conclude by definition that $v \sim_L u$.

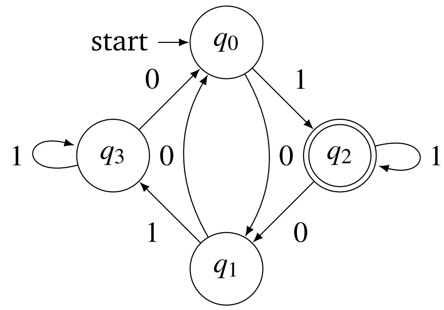
Transitivity: Suppose that $u \sim_L v$ and $v \sim_L x$ for $u,v,x \in \Sigma^*$. Then, given any $w \in \Sigma^*$ there are two possibilities. If $uw \in L$, then $vw \in L$ (since $u \sim_L v$), so $xw \in L$ (since $v \sim_L x$). Otherwise, if $uw \notin L$, then $vw \notin L$ (since $u \sim_L v$), so $xw \notin L$ (since $v \sim_L x$). Thus, either both $uw, xw \in L$ or $uw, xw \notin L$. As this is true for arbitrary $w \in \Sigma^*$, we conclude by definition that $u \sim_L x$. ■

As we learned, before, the Nerode congruence relation divides up the set of all strings in Σ^* into a set of equivalence classes (called the *quotient* of the relation). The Myhill-Nerode theorem establishes a connection between this quotient and the DFAs that recognize a language.

36.2 DFAs and Nerode Congruence

Suppose that a language L is recognized by a DFA $M = (\Sigma, Q, s, F, \delta)$ that recognizes L. When M processes a string $u \in \Sigma^*$, its transition function describes how we jump from state to state as we read each character. At the end of the computation (that is, once we have processed all of the characters), we will be sitting in one of the states, which we'll denote by $q(u) \in Q$. In this way, M divides the set of strings Σ^* based on which state they are processed to. Recall that since M is *deterministic*, it maps every string in Σ^* to exactly one state. How does this division relate to the equivalence classes of $\sim_L$?

Suppose that $u, v \in \Sigma^*$ are both processed from the initial state s to the same state; in our new notation, $q(u) = q(v)$. As an example, you can check that the strings $u = 011$ and $v = 10001$ are both processed to state q_3.



u follows the path

$$q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_3 \xrightarrow{1} q_3,$$

and v follows the path

$$q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{0} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_3.$$

Since M's decision to accept or reject a string is based only on which state it is processed to, this means that either both u and v are accepted (so $u, v \in L$) or both u and v are rejected (so $u, v \notin L$). In our example, since q_3 is not a final state, $u, v \notin L$. However, we can expand this idea further. Suppose that we append a common string $w \in \Sigma^*$ to both u and v. The machine M was in state $q(u) = q(v)$ after processing u or v. If from here it continues to process the characters of w, it will follow a deterministic path from state $q(u) = q(v)$ processing the characters in w until it reaches the state $q(uw) = q(vw)$. In our example, if $w = 01$, then the string uw will follow the path

$$\underbrace{q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_3 \xrightarrow{1} q_3}_{\text{process } u}\underbrace{\xrightarrow{0} q_0 \xrightarrow{1} q_2}_{\text{process } w}$$

and vw will follow the path

$$\underbrace{q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{0} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_3}_{\text{process } v}\underbrace{\xrightarrow{0} q_0 \xrightarrow{1} q_2}_{\text{process } w}.$$

In this case, both uw and vw are processed to state q_2 and accepted, meaning $uw, vw \in L$. No matter which string w we append, the fact that M is deterministic will always ensure that the second (red) part of the path will be identical and $q(uw) = q(vw)$. Said differently, the only "memory" that a DFA has is its current state; once two strings are processed to the same state, they appear identical to the DFA, so will be processed exactly the same from that point onward. This condition should look familiar. We have argued that if $q(u) = q(v)$, then $q(uw) = q(vw)$ for *any* string $w \in \Sigma^*$, meaning either both $uw, vw \in L$ or $uw, vw \notin L$. This exactly tells us that $u \sim_L v$. We'll record this observation, which will serve as the basis for the Myhill-Nerode Theorem.

> If $u, v \in \Sigma^*$ are processed to the same state by a DFA that recognizes L, then $u \sim_L v$.

36.3 The Myhill-Nerode Theorem

We have just observed that for any language L recognized by a DFA $M = (\Sigma, Q, s, F, \delta)$, the number of equivalence classes of $\sim_L$ is at most $|Q|$. In particular, this tells us that the quotient of Σ^* under $\sim_L$ must be finite (since a DFA, by definition, has a finite number of states). In fact, the converse of this is also true, as we will show below. This gives an equivalent characterization of languages recognized by DFAs, which is known as the Myhill-Nerode Theorem.

Theorem 36.1 — Myhill-Nerode Theorem.

A language L over a finite alphabet Σ can be recognized by a DFA if and only if the quotient $\Sigma^*/\sim_L$ is finite.

Proof. We have already established the forward implication, since $|\Sigma^*/\sim_L| \leq |Q| \in \mathbb{N}$. For the reverse implication, suppose that L is a language such that $\Sigma^*/\sim_L$ is finite. We will describe how to construct a DFA $M = (\Sigma, Q, s, F, \delta)$ that recognizes L.

Σ: The alphabet is the same as that of language L.

Q: The (finite) set of states will correspond to the equivalence classes in $Q = \Sigma^*/\sim_L$.

s: The initial state will be the equivalence class of the empty strings $[\lambda]_{\sim_L}$.

F: By taking $w = \lambda$ in the definition of $u \sim_L v$, we see that two strings from the same equivalence class either both belong to L or both do not, so L consists of a union of $\sim_L$ equivalence classes. The set of final states includes the equivalence classes in this union:

$$F = \{[v]_{\sim_L} \in \Sigma^*/\sim_L \mid v \in L\}.$$

δ: We can describe the transition function for this DFA as

$$\delta([u]_{\sim_L}, \sigma) = [u\sigma]_{\sim_L}$$

for any $u \in \Sigma^*$ and $\sigma \in \Sigma$. That is, on reading the character σ, if we are in the equivalence class with representative u, we transition to the equivalence class that includes $u\sigma$.

To argue that this transition function is well-defined (i.e. does not depend on the choice of representative), consider any $u \sim_L v \in \Sigma^*$. Given any string $w \in \Sigma^*$, the concatenation σw is also in Σ^*. By definition of $\sim_L$, $u\sigma w \in L \Leftrightarrow v\sigma w \in L$. Since w was chosen arbitrarily, we conclude that $u\sigma \sim_L v\sigma$, meaning $[u\sigma] = [v\sigma]$.

It remains to argue that our DFA accepts exactly the language L. For this, note that our definition of transition function δ ensures that upon processing any string $u \in L$, M will transition from initial state $[\lambda]_{\sim_L}$ to state $[u]_{\sim_L}$. Thus,

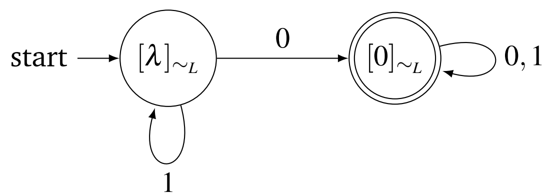
$$\begin{aligned}
u \in L &\Leftrightarrow [u]_{\sim_L} \in F && \text{(definition of } F\text{)}\\
&\Leftrightarrow M \text{ processes } u \text{ from } [\lambda]_{\sim_L} \text{ to an accepting state} && \text{(definition of } \delta\text{)}\\
&\Leftrightarrow u \in \mathcal{L}(M). && \text{(definition of } s\text{)}
\end{aligned}$$

■

Example 36.3 The language L from Example 36.2 has a quotient $\{0,1\}^*/\sim_L$ with two equivalence classes:

- $[0]_{\sim_L}$ includes all strings that contain at least one 0, $[0]_{\sim_L} = \{u \in \{0,1\}^* \mid |u|_0 \geq 1\}$.
- $[\lambda]_{\sim_L}$ includes all strings that consist only of 1s, $[\lambda]_{\sim_L} = \{u \in \{0,1\}^* \mid |u|_0 = 0\}$.

The DFA for this language is given by



36.4 A Language that cannot be Recognized by a DFA

The Myhill-Nerode Theorem gives us a way to argue that a language cannot be recognized by a DFA. Namely, if we can show that the quotient $\Sigma^*/\sim_L$ is infinite, L cannot be recognized by a DFA; such a DFA would need infinitely many states to keep track of the infinitely many equivalence classes, but DFAs can only have finitely many states. To argue that this quotient is infinite, we can exhibit some infinite collection of strings, no pair of which are equivalent under $\sim_L$.

Example 36.4 We will argue that the language $L = \{w \in \{0,1\}^* \mid w = 0^n 1^n \text{ for some } n \in \mathbb{N}\}$ is not recognized by any DFA.

Proof. Consider the infinite set of strings $U = \{0\}^* \subseteq \{0,1\}^*$ (that is U contains strings consisting of any number of 0s). Suppose that u and v are two distinct strings from U. Then, $u = 0^i$ and $v = 0^j$ with $i \neq j$. Letting $w = 1^i$, we see that $uw = 0^i 1^i \in L$, but $vw = 0^j 1^i \notin L$. Thus, $u \not\sim_L v$. As this is true for an arbitrary pair of distinct strings $u \neq v \in U$, we conclude that $\{0,1\}^*/\sim_L$ is infinite, so the Myhill-Nerode Theorem tells us that L cannot be recognized by a DFA. ■

Main Takeaways:

- *Nerode congruence* describes an equivalence relation on a set of strings.
- Two strings u, v processed by a DFA M to the same state have $u \sim_{\mathcal{L}(M)} v$.
- The Myhill-Nerode Theorem says that a language can be recognized by a DFA if and only if the quotient of the Nerode congruence relation for that language is finite.
- To prove a language cannot be recognized by a DFA, find an *infinite* set of strings U, no pair of which are Nerode congruent.

Exercises

36.1 Below, we ask you to consider various languages L. For each language, decide whether the given pair of strings $u, v \in \Sigma^*$ are Nerode congruent. If they are, explain why. If they are not, give an example of a string w that distinguishes them. Here, the notation $|u|_\sigma$ refers to the number of occurrences of the character σ in the string u.

(a) $L = \{u \in \{0,1\}^* \mid \quad |u|_1 = 3\}$

(i) 01101, 100101 (ii) 100, 10
(iii) 11011, 0111110 (iv) 0010, 1010

(b) $L = \{u \in \{0,1\}^* \mid u \text{ ends with } 101\}$

(i) λ, 00 (ii) 1101, 0101
(iii) 001, 100 (iv) 110, 100
(v) 01010, 10010 (vi) λ, 101

(c) $L = \{u \in \{0,1\}^* \mid \quad |u|_0 > |u|_1\}$

(i) 010, 101 (ii) 100, 010
(iii) 111000, 0101 (iv) 11010, 0101101

36.2 For each of the following languages L, describe the equivalence classes of the Nerode congruence relation $\sim_L$. Here, the notation $|u|_\sigma$ refers to the number of occurrences of the character σ in the string u.

(a) $L = \{u \in \{0,1\}^* \mid u \text{ ends with } 0\}$

(b) $L = \{u \in \{0,1\}^* \mid u \text{ starts with } 110\}$

(c) $L = \{u \in \{0,1\}^* \mid u \text{ does not contain the substring “00”}\}$

(d) $L = \{u \in \{0,1\}^* \mid \quad |u|_0 = 7\}$

(e) $L = \{u \in \{0,1\}^* \mid \quad |u|_0 = |u|_1\}$

(f) $L = \{u \in \{0,1\}^* \mid u \text{ is a palindrome}\}$

36.3 Use the Myhill-Nerode Theorem to argue that the following languages cannot be recognized by a DFA.

(a) $L_1 = \{w \in \{0,1\}^* \mid w = 0^n 1^{n+1} \text{ for some } n \in \mathbb{N}\}$

(b) $L_2 = \{w \in \{0,1\}^* \mid w = 0^n 1^m \text{ with n < m}\}$

(c) $L_3 = \{w \in \{0,1\}^* \mid |w|_0 = |w|_1\}$

(d) $L_4 = \{w \in \{0,1\}^* \mid w \text{ is a palindrome}\}$

(e) $L_5 = \{w \in \{0,1\}^* \mid w = vv \text{ for some } v \in \{0,1\}^*\}$.

⋆ **36.4** Here, we consider the language $L = \{w \in \{0\}^* \mid |w| = n^2 \text{ for some } n \in \mathbb{N}\}$.

(a) Prove the following algebraic fact:

Suppose that $i, j \in \mathbb{N}$ with $i < j$. Then, $(i+j)^2 + (j-i)$ is *not* a perfect square.

(Hint: What is the next larger perfect square after $(i+j)^2$?)

(b) Use the Myhill-Nerode Theorem and the fact from part (a) to argue that L is not regular.

36.5 We say that a language is *regular* if it can be recognized by a DFA. (We will see this terminology more when we study *regular expressions* in a couple of lectures.)

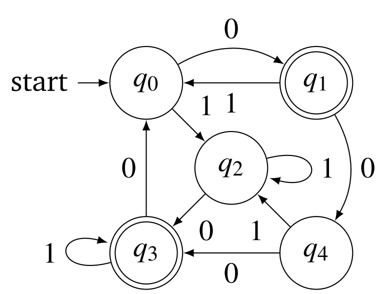
(a) Give an example (with proof) that the union of a regular language L_1 with a non-regular language L_2 can be regular.

(b) Give an example (with proof) that the union of a regular language L_1 with a non-regular language L_2 can be non-regular.

36.6 In our proof of the Myhill-Nerode Theorem, we used the fact that if two strings u, v are processed by a DFA M to the same state, then they are Nerode congruent with respect to $\mathcal{L}(M)$. However, the converse of this implication is not true. Give an example of a L and a DFA M such that $\mathcal{L}(M) = L$, M processes strings u and v to different states, but $u \sim_L v$.

⋆ **36.7** From Exercise 36.6, we see that an equivalence class in $\Sigma^*/\sim_L$ is not necessarily composed of the set of strings processed to one state in a DFA. Rather, it can consist of the union of strings processed to multiple DFA states.

(a) Consider the following DFA, M, that recognizes language L.



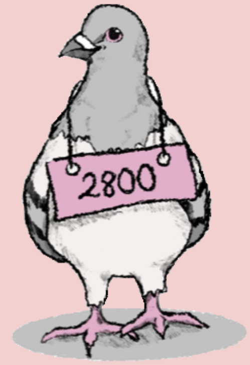
Although M has 5 states, we claim that $\Sigma^*/\sim_L$ has only 4 equivalence classes. Which two states correspond to the same equivalence class? How do you know?

(b) Since $\Sigma^*/\sim_L$ from part (a) has only 4 equivalence classes, we can use the procedure from the proof of Theorem 36.1 to construct a 4-state DFA that recognizes L. Draw the resulting DFA.

(c) Explain why no DFA with fewer than 4 states can recognize L. (Hint: Perform a proof by contradiction. Use the definition of Nerode congruence to locate a string that is mishandled.)

More generally, any DFA recognizing L must have at least $|\Sigma^*/\sim_L|$ states.

In light of part (c), we say that the DFA from part (b) is *minimal* for language L since it has the fewest possible number of states. Given any DFA, we can always "merge" the states corresponding to Nerode congruent strings to obtain a minimal DFA. However, for larger automata, figuring out which states to merge can become more demanding. Nevertheless, *Hopcroft's Algorithm* gives an efficient procedure based on properties of partitions to find the minimal DFA. John Hopcroft, the creator of this algorithm was a Cornell CS professor.



37. Non-Deterministic Automata

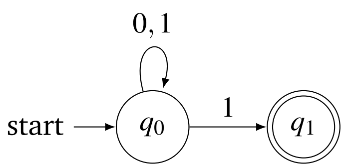
In the previous lecture, we saw that there are some languages that cannot be recognized by a DFA. In today's lecture, we will introduce a new computational model, non-deterministic finite automata, that generalize DFAs, and consider which languages they can recognize.

37.1 Non-Deterministic Finite Automata

One of the main properties of a DFA is that it has *exactly one* outgoing arrow from each state labeled with each character in the alphabet. This is enforced by the fact that the transition function $\delta\colon (Q \times \Sigma) \to Q$ is well defined and ensures that no matter which state we are in, reading the next character will move the machine to exactly one state.

What happens if we relax this requirement? In this case, we can end up with something like the following automaton with alphabet $\Sigma = \{0, 1\}$:

Example 37.1



Here, the state q_0 has two outgoing arrows that are both labeled with 1, one pointing to q_1 and the other looping back to q_0. It also has no outgoing arrows from the state q_1. In order to work with these new automata, which we will call *Non-Deterministic Finite Automata* (NFAs), we will first need to settle two things:

1. **Definition:** How can we formally describe an NFA using sets and functions?
2. **Semantics:** How do we determine whether a string w is accepted by an NFA?

37.1.1 Definition

Most of the components of an NFA are the same as for a DFA; both types of automata have an alphabet Σ, a state set Q, a starting state s, and accepting states F. Their difference comes from their transition functions δ, which prescribes a rule for what to do when we read a character σ while sitting in a current state q.

The DFA's transition function tells us that we should always follow the σ arrow leaving q to a new state q'. The codomain of this transition function is Q, since q' will be one of the states. In an NFA, we can have multiple (or no) arrows labeled with σ leaving state q. To account for these possibilities, we'll define δ to map the pair (q,σ) to the *set* of all states q' that are reached from q by σ arrows:

$$\delta(q,\sigma) = \{q' \in Q \mid \text{ there is an arrow from } q \text{ to } q' \text{ labeled with } \sigma\}.$$

In this definition, $\delta(q,\sigma)$ is a subset of Q, so δ is a function $(Q\times\Sigma) \to \mathcal{P}(Q)$. Overall, we end up with the following definition.

Definition 37.1 — NFA.
An NFA N is a 5-tuple $N = (\Sigma, Q, s, F, \delta)$ where:

- Σ is an alphabet
- Q is the *finite* set of machine states
- $s \in Q$ is the *starting* or *initial* state
- $F \subseteq Q$ is the set of *final* or *accepting* states
- $\delta\colon (Q\times\Sigma) \to \mathcal{P}(Q)$ is a (non-deterministic) transition function

In the NFA from Example 37.1, $\Sigma = \{0,1\}$, $Q = \{q_0,q_1\}$, $s = q_0$, $F = \{q_1\}$, and δ is given by $\delta(q_0,0) = \{q_0\}$, $\delta(q_0,1) = \{q_0,q_1\}$, and $\delta(q_1,0) = \delta(q_1,1) = \varnothing$.

37.1.2 Semantics

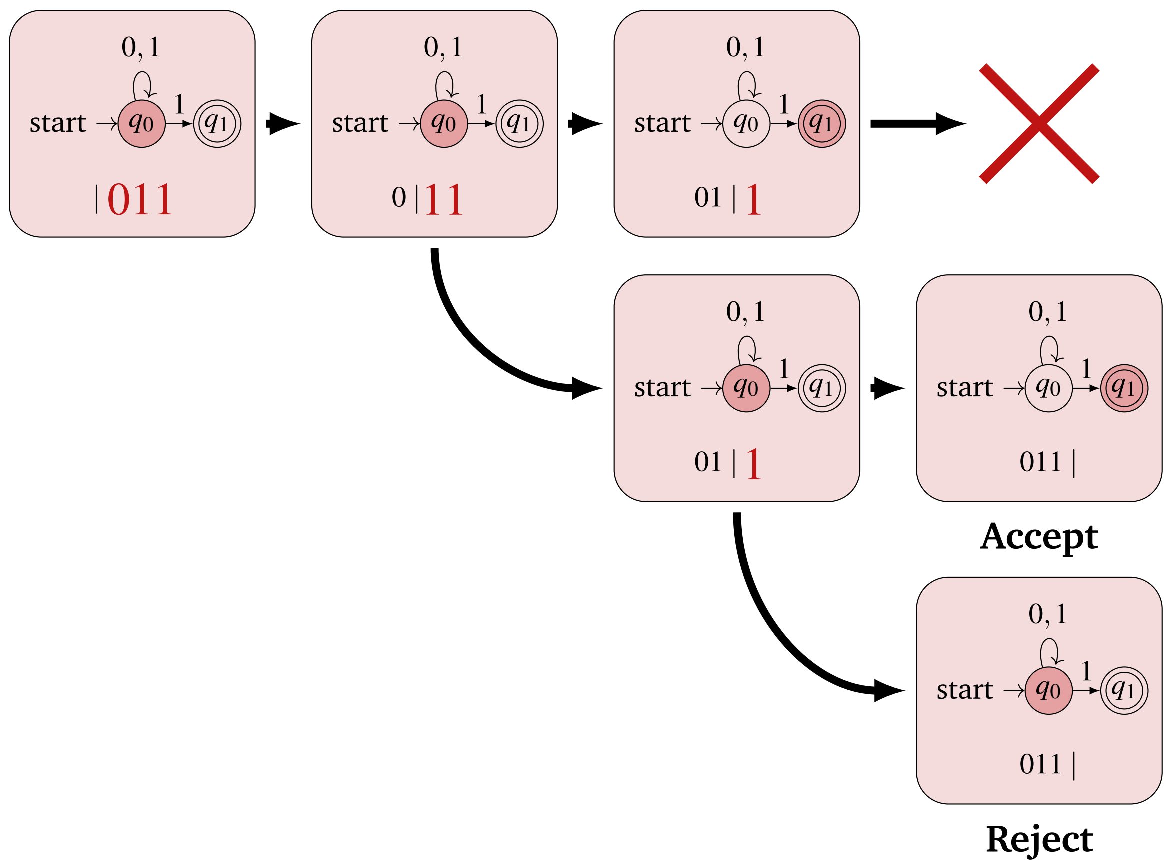
We can summarize the behavior of NFAs in one sentence as follows:

> *An NFA N accepts a string w if there is at least one path from s to an accepting state that is labeled with the characters in w.*

Since NFAs can have multiple arrows with the same label, it's possible that we'd need to consider multiple different computation paths. As long as one of these paths ends at an accepting state, we should accept the string.

Perhaps the best way to conceptualize how an NFA does its computation is to imagine that it has the ability to clone itself. If it is in state q, reads character σ from the input, and sees multiple outgoing σ arrows, it clones itself and has each copy follow a different one of these arrows. If there are no σ arrows leaving q, that clone is destroyed. If one of the clones survives to the end of the string and is sitting in a final state, then the string is accepted. Otherwise, if all of the clones are destroyed or they all sit in non-final states at the end of the string, then the string is rejected. We trace through some computations on the NFA from Example 37.1 on the next page.

We reason about the string 011.



We start with a single automaton in state q_0 that hasn't processed any of the string. When we process 0, we follow the single outgoing 0 arrow from state q_0 back to q_0. When we process the first character 1, there are two outgoing arrows, so we must split to two machine configurations. The first configuration (shown in the top row) follows the arrow to transition to state q_1. The second configuration (shown in the middle row) follows the looping arrow to remain in state q_0. From each of these configurations, the second character 1 is processed. There is no outgoing 1 arrow from state q_1 so the first configuration is culled. In the second configuration, we again follow the two 1 arrows from state q_0 to simultaneously transition to states q_0 and q_1. At this point, we have processed the entire input string. Since the configuration in the middle line has ended in the accepting state q_1, our NFA accepts string 011. This accepting configuration corresponds to the path $q_0 \to q_0 \to q_0 \to q_1$ in the NFA.

By similar reasoning, we see that any string that ends with 1 will be accepted by this NFA. The NFA can follow the looping transition for all but the last 1 and then use the final 1 to transition to state q_1. In addition, no string that ends with 0 will be accepted by this NFA, as there is no incoming arrow to the final state q_1 that is labeled with 0. Hence, the language recognized by this NFA is exactly the set of binary strings that end with 1.

The word "non-deterministic" in NFA describes how the machine simultaneously explores all possible paths in parallel; it does not decide (or determine) a single state to be in at any time.

37.2 Languages of NFAs and the Subset Construction

Just as for DFAs, we define the language of an NFA N, $\mathcal{L}(N)$, to be the set of all strings that are accepted by N. As we just saw, the language of the NFA from Example 37.1 is all strings ending with 1. It is relatively easy to draw a DFA that also recognizes this language (try it!). This motivates us to ask about the relationship between the set of languages recognized by DFAs versus those recognized by NFAs.

For part of this answer, note that DFAs are essentially a special case of NFAs (where δ maps each (q, σ) pair to a 1-element set). This means that every language that is recognized by a DFA can be recognized by an NFA. Perhaps more surprisingly, the other inclusion is also true. Every language that is recognized by an NFA can also be recognized by a DFA! This means that the addition of non-determinism does not make automata any more expressive as a computational model. This is the content of the following theorem.

Theorem 37.1 Let L be a language over a finite alphabet Σ. Then, L can be recognized by a DFA if and only if L can be recognized by an NFA.

Proof. We have already reasoned about the forward direction, so it remains to show the reverse direction. To prove this, we must take an NFA $N = (\Sigma_N, Q_N, s_N, F_N, \delta_N)$ that recognizes L and transform this to a DFA $M = (\Sigma_M, Q_M, s_M, F_M, \delta_M)$ that also recognizes L. We describe each component of M below and give an example of this construction (often called the *Subset Construction*) after the proof.

Alphabet: We use the same alphabet: $\Sigma_M = \Sigma_N$

State Set: The states in M represent subsets of states in our NFA: $Q_M = \mathcal{P}(Q_N)$

Start State: The start state of M contains only the start state of N: $s_M = \{s_N\}$

Accepting States: Sets in $\mathcal{P}(Q_N)$ with at least one accepting state from N are accepting states in M: $F_M = \{S \subseteq Q_N \mid S \cap F_N \neq \varnothing\}$

Transition Function: Given a state $S \in Q_M$, and a character $\sigma \in \Sigma$, $\delta_M(S, \sigma)$ includes all states that are reachable in N from a state $q \in S$ by following an arc labeled with σ:

$$\delta_M(S, \sigma) = \bigcup_{q \in S} \delta_N(q, \sigma)$$

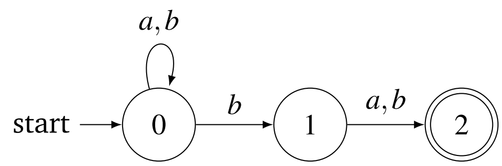
This construction ensures that after M has processed a string w, it will be in the state S corresponding to the set of all states that N would (simultaneously) be in after processing w (this should be argued more carefully to make this proof completely rigorous, but doing so would require more notation than we wish to introduce). Thus, M accepts exactly those strings that have a path in N to an accepting state, which are exactly those strings accepted by N. ■

The following example will show the subset construction in action. In fact, it will actually show a slightly "streamlined" version of the construction that often produces a DFA with fewer than $2^{|Q_N|}$ states that recognizes the same language.

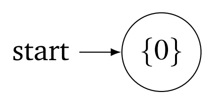
Example 37.2 We will consider the language

$$L = \{w \in \{a,b\}^* \mid \text{the second-to-last character of } w \text{ is } b\}.$$

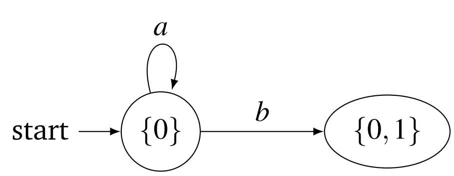
L can be recognized by the following NFA, N where the loop on state 0 processes the beginning of the string, the transition to state 1 ensures that the second-to-last character is a b, and the transition to state 2 ensures we end at an accepting state.



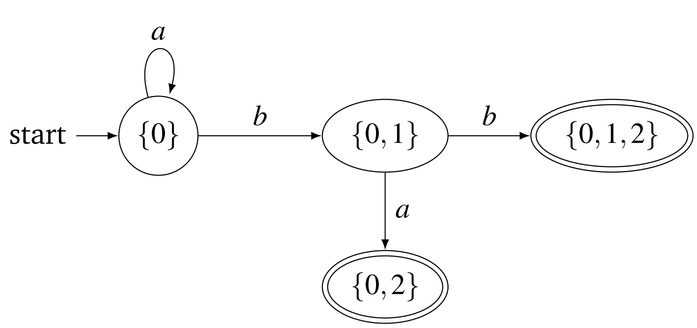
Let's use the subset construction to build a DFA that recognizes L. We start with the initial state, which represents N being in only state 0 (which it is before it reads any characters).



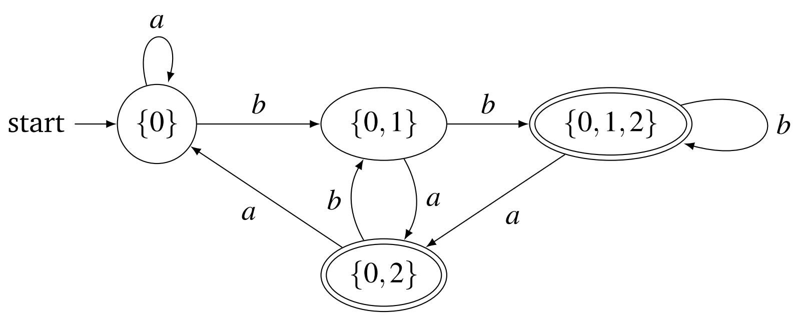
To be a DFA, M must have an arrow leaving state $\{0\}$ for each character in $\{a,b\}$. When N reads an a while in state 0, it loops back to state 0. Thus, the a arrow from state $\{0\}$ in M will loop back to state $\{0\}$. When N reads a b while in state 0, it non-deterministically transitions to both states 0 and 1. Thus, the b arrow from state $\{0\}$ in M will point to a new state $\{0,1\}$.



Next, we must draw the outgoing arrows from the new state $\{0,1\}$ that we introduced in the previous step (and continue to repeat this until all necessary arrows have been drawn). From state 0, we can reach state 0 by following an a arrow in N. From state 1, we can reach state 2 along an a arrow in N. Hence, the a arrow out of $\{0,1\}$ in M points to a new state $\{0,2\}$. This new state should be accepting since 2 is an accepting state in N. Following a b arrow from either states 0 or 1 in N can take us to any of states $0,1,2$, so the b arrow from $\{0,1\}$ points to another new accepting state $\{0,1,2\}$ in M.



Using similar reasoning, we can add the transitions out of states $\{0,2\}$ and $\{0,1,2\}$. Doing this does not result in any additional states, so our final DFA M is given by the diagram:



The subset construction gives us a way to transform any NFA into an equivalent (i.e., recognizing the same language) DFA. However, since the states in the DFA represent subsets of states in the NFA, it is possible that the DFA is exponentially larger. Thus, although NFAs are no more expressive than DFAs, they can often have structures that are smaller and simpler to reason about.

Main Takeaways:

- NFAs are a generalization of DFAs that allow for any number of outgoing transitions from a state labeled with each character.
- An NFA accepts a string as long as there is at least one path of arcs labeled with those characters that ends in an accepting state.
- The *Subset Construction* can be used to argue that a language is recognized by a DFA if and only if it is recognized by an NFA.
- Oftentimes, a DFA recognizing a language will be significantly larger than an NFA recognizing the same language.

Exercises

37.1 Determine whether each of the following statements is true or false. Explain your answer with either a short justification or a counterexample.

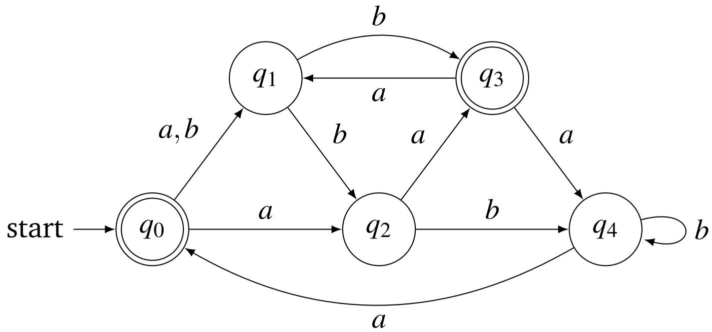
(a) If an NFA N has $F = \varnothing$ (i.e. no states are accepting), then $\mathcal{L}(N) = \varnothing$.

(b) If an NFA N has $F = Q$ (i.e. all states are accepting), then $\mathcal{L}(N) = \Sigma^*$.

(c) Swapping the accepting and non-accepting states in an NFA N (that is, replacing F with $Q \setminus F$) causes it to recognize the complement language $\Sigma^* \setminus \mathcal{L}(N)$.

(d) Two different NFAs can recognize the same language.

37.2 Determine whether each of the following strings is accepted or rejected by the following NFA. For each string that is accepted, write out a path to an accepting state. Do any of the strings have multiple such paths?



(a) *ab* (b) *aab* (c) *bbaba*

(d) *aaaa* (e) *aaaaa* (f) *ababab*

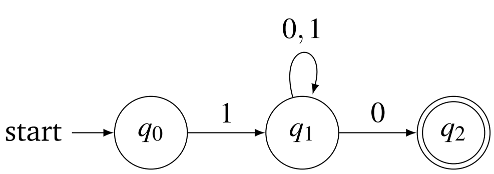
(g) *abbba* (h) *bbaabb* (i) *aaabbbaaa*

37.3 For each of the following NFAs:

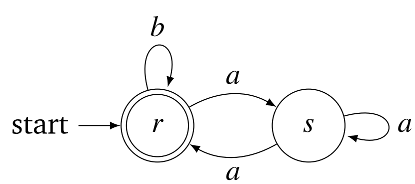
- Give its formal description. That is, write out its components $(\Sigma, Q, s, F, \delta)$.
- Describe the language that the machine recognizes.

You may assume that there is at least one arc in the machine labeled with each character in the alphabet.

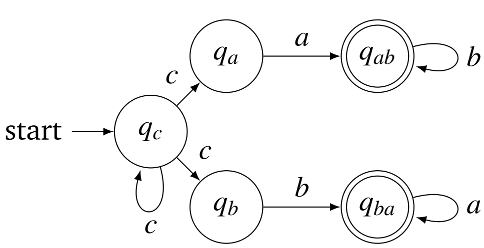
(a)



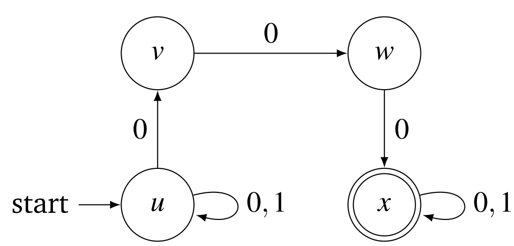
(b)



(c)



(d)



37.4 Draw an NFA that recognizes each of the following languages.

(a) $L_1 = \{w \in \{0,1\}^* \mid |w|_0 \leq 2\}$.

Recall that $|w|_0$ refers to the number of occurrences of character 0 in string w.

(b) $L_2 = \{w \in \{0,1\}^* \mid \ w \text{ begins with 3 0s}\}$

(c) $L_3 = \{w \in \{0,1\}^* \mid \ w \text{ ends with 3 0s}\}$

(d) $L_4 = L_2 \cup L_3$

(e) $L_5 = L_2 \cap L_3$ (Be careful!)

(f) $L_6 = L_2 \setminus L_3$

(g) $L_7 = L_2 \triangle L_3$ (Recall that $\triangle$ denotes the *symmetric difference* operation on sets.)

(h) $L_8 = \{w \in \{a,b\}^* \mid \text{every } a \text{ in } w \text{ is immediately followed by a } b\}$

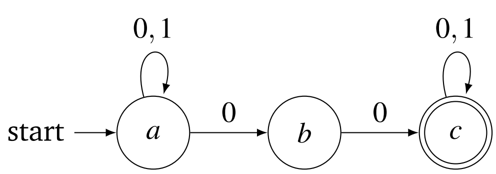
(i) $L_9 = \{w \in \{a,b,c\}^* \mid w \text{ contains the substring } baba\}$

(j) $L_{10} = \{w \in \{a,b\}^* \mid w \text{ does not contain the substring } aab\}$

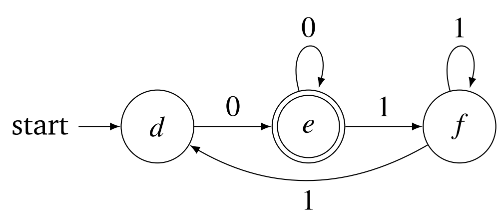
37.5 Let's revisit the DFA that was constructed in Example 37.2. Describe the sets of strings that are processed to each state in this DFA. These descriptions should provide a more intuitive understanding of how this DFA behaves, and suggest how one would construct the DFA without converting from an NFA.

37.6 Use the subset construction to draw a DFA that recognizes the same language as each of the following NFAs.

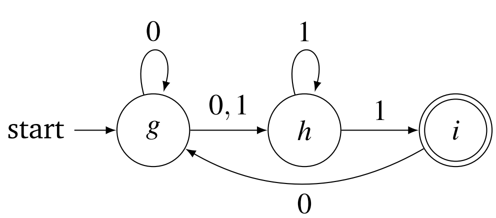
(a)



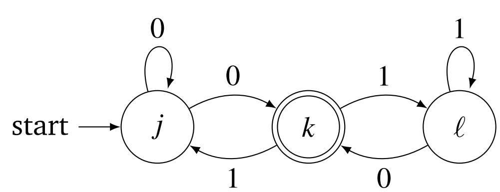
(b)



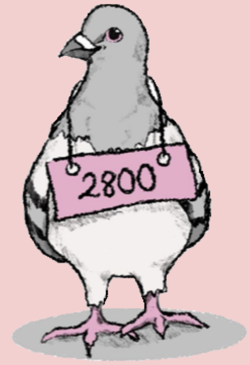
(c)



(d)



⋆ **37.7** (From Dexter Kozen) Draw an NFA over alphabet $\Sigma = \{a\}$ that rejects some string, but the length of the shortest rejected string is greater than $|Q|$.



38. Regular Expressions

So far we have thought about languages from a computational perspective, they are sets of strings that can (or cannot) be recognized by an automaton. Today, we will consider an alternate view of languages. We'll think about languages as mathematical objects in their own right. Similar to what we did for logical expressions at the beginning of the semester, we'll define a set of operations on languages that we can use to build up more "complicated" languages from simpler ones. This will give rise to a new language class, *regular languages*, and a new syntactic tool *regular expressions* for describing them. At the end of the lecture (and all of the next lecture) we will bridge these two perspectives as we explore the connection between regular expressions, DFAs, and NFAs.

38.1 Three Operations on Languages

Recall that a language L is simply a set of strings over some finite alphabet Σ. Therefore, all of the set operations that we are familiar with can be applied to languages. One of these operations, set union, will be useful for our definition of regular languages.

Definition 38.1 — Union of Languages.
Given languages $L_1, L_2 \subseteq \Sigma^*$, their *union* $L_1 \cup L_2$ contains all strings belonging to either language.

The other two operations that we will use are not ordinary set operations; rather, they are unique to languages. The first of these is the *concatenation* of two languages. We have already seen the concatenation of strings, which takes one string and appends it to the end of another. When we concatenate languages, we form a set of all possible ways to concatenate one string taken from each language.

Definition 38.2 — Concatenation of Languages.
Given languages $L_1, L_2 \subseteq \Sigma^*$, their *concatenation*

$$L_1 \cdot L_2 = \{vw \in \Sigma^* \mid v \in L_1, w \in L_2\}.$$

That is $L_1 \cdot L_2$ contains all strings that we can form by concatenating a string $v \in L_1$ with a string $w \in L_2$.

Example 38.1 — Concatenation of Languages.

- If $L_1 = \{0, 00\}$ and $L_2 = \{1, 11\}$, then $L_1 \cdot L_2 = \{01, 011, 001, 0011\}$.
- If $L_1 = \{01, 101, \lambda\}$ and $L_2 = \{10, 001\}$, then $L_1 \cdot L_2 = \{0110, 01001, 10110, 101001, 10, 001\}$.
- If $L_1 = \{0, 01\}$ and $L_2 = \{0, 10\}$, then $L_1 \cdot L_2 = \{00, 010, 0110\}$. Study this example carefully.
- Given any language $L \subseteq \Sigma^*$, $L \cdot \varnothing = \varnothing \cdot L = \varnothing$; there are no strings in $\varnothing$ that we can concatenate with a string from L.

The final operation, called the *Kleene star* is a unary operation on a language. We have seen similar notation when defining the set of all strings Σ^*. The Kleene star of a language L consists of all strings that we can form by concatenating together any 0 or more strings from L.

Definition 38.3 — Kleene Star of a Language.
Given a language $L \subseteq \Sigma^*$, we recursively define L^n for all $n \in \mathbb{N}$ with $L^0 = \{\lambda\}$ and $L^{k+1} = L^k \cdot L$ for all $k \in \mathbb{N}$. Then, the *Kleene star* of L is given by

$$L^* = \bigcup_{n \in \mathbb{N}} L^n.$$

Example 38.2 — Kleene Star of Languages.

- If $L = \{0\}$, then $L^* = \{\lambda, 0, 00, 000, \ldots\}$.
- Let's consider $L = \{0, 1\}$. We have $L^0 = \{\lambda\}$, $L^1 = L = \{0, 1\}$, $L^2 = L \cdot L = \{00, 01, 10, 11\}$. Continuing this, we'll find that every string $w \in \{0, 1\}^*$ belongs to some L^n (in fact, to $L^{|w|}$), so $L^* = \{0, 1\}^*$.
- If $L = \{00, 101\}$, then $L^* = \{\lambda, 00, 101, 0000, 00101, 10100, 101101, 000000, \ldots\}$.
- $\{\lambda\}^* = \varnothing^* = \{\lambda\}$.

38.2 Regular Expressions

Next, we will define a new syntactical object (i.e. a system of notation) called a *regular expression* that we can use to describe a language. The definition that we will give will be inductive. We'll start by defining the regular expressions for very simple languages and then give rules for how to combine these expressions to describe more complicated languages.

Throughout our discussion, we'll assume that we are working over a fixed alphabet Σ. To make things easier, we'll also assume that Σ does not contain any of the characters $\{+, ^*, (\,,)\,, \varnothing, \lambda\}$. Then, a *regular expression* (or *regex*) is simply a string (sequence of characters) over the alphabet $\Sigma \cup \{+, ^*, (\,,)\,, \varnothing, \lambda\}$. The power of regular expressions is that they provide us a way to give a concise representation of a language, as we describe below.

Definition 38.4 — Semantics of Regular Expressions.

Each regular expression R over alphabet Σ describes a language $\mathcal{L}(R) \subseteq \Sigma^*$. We can describe $\mathcal{L}(R)$ inductively as follows.

Base Cases:

1. "$\varnothing$" represents the empty language. $\mathcal{L}(\varnothing) = \varnothing$.
2. "λ" represents the language with only the empty string. $\mathcal{L}(\lambda) = \{\lambda\}$.
3. For each $\sigma \in \Sigma$, "σ" represents the one-string language $\{\sigma\}$. $\mathcal{L}(\sigma) = \{\sigma\}$.

Recursive Cases: Given regular expressions R_1 and R_2,

4. "$R_1 + R_2$" represents the union of $\mathcal{L}(R_1)$ and $\mathcal{L}(R_2)$. $\mathcal{L}(R_1 + R_2) = \mathcal{L}(R_1) \cup \mathcal{L}(R_2)$.
5. "R_1R_2" represents the concatenation of $\mathcal{L}(R_1)$ and $\mathcal{L}(R_2)$. $\mathcal{L}(R_1R_2) = \mathcal{L}(R_1) \cdot \mathcal{L}(R_2)$.
6. "${R_1}^*$" represents the Kleene star of $\mathcal{L}(R_1)$. $\mathcal{L}({R_1}^*) = \mathcal{L}(R_1)^*$.

Just as with ordinary arithmetic expressions, we use parentheses in regular expressions to ensure that the operations are performed in the correct order. E.g., we might write $(0+1)^*$, which represents the language of all bit strings. There is also an order of operations for regular expressions. Parentheses always take precedence, of course. Then, the Kleene star operation takes precedence over the concatenation operation which takes precedence over the union operation. For example, the regular expression $1 + 10^*$ is equivalent to the regular expression $1 + (1(0^*))$.

Example 38.3 — Regular Expressions. We consider the following regular expressions over alphabet $\Sigma = \{0, 1\}$.

- $\mathcal{L}(0 + 10^*) = \mathcal{L}(0) \cup \mathcal{L}(10^*) = \{0\} \cup \mathcal{L}(1) \cdot \mathcal{L}(0^*) = \{0\} \cup \{1\} \cdot \mathcal{L}(0)^* = \{0\} \cup \{1\} \cdot \{0\}^*$.

  This set consists of the string "0" and all strings of the form "1" followed by any number of 0s.
- $\mathcal{L}(0 + (10)^*) = \mathcal{L}(0) \cup \mathcal{L}((10)^*) = \{0\} \cup \mathcal{L}(10)^* = \{0\} \cup (\{1\} \cdot \{0\})^* = \{0\} \cup \{10\}^*$.

  This set consists of the string "0" and strings containing any number of copies of "10".
- $\mathcal{L}((0+10)^*) = (\mathcal{L}(0+10))^* = \{0, 10\}^* = \{\lambda, 0, 10, 00, 010, 100, 1010, \ldots\}$.
- $\mathcal{L}((0+1)0^*) = \{0, 1\} \cdot \{0\}^* = \{0, 1, 00, 10, 000, 100, 0000, 1000, \ldots\}$.
- $(0+1)^*$ represents all binary strings.
- $(0+1)^*0$ represents all binary strings that end with 0.
- $0^*10^*10^*$ represents all binary strings with exactly two 1s.

Any language that can be represented by a regular expression is called a *regular language*.

Definition 38.5 — Regular Language.

A *regular language* is any language that is represented by a regular expression. In other words, regular languages are those that can be constructed from the simplest languages $\varnothing, \{\lambda\}, \{\sigma\}$ using *finitely many* operations from Section 38.1.

As the main result of this module, which will be our focus for the rest of this lecture as well as all of the next lecture, we will draw a connection between regular languages and finite automata.

Automata Recognize Every Regular Language

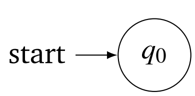
We will argue the following lemma.

Lemma 38.1 Let R be a regular expression. Then there is an NFA N such that $\mathcal{L}(R) = \mathcal{L}(N)$.

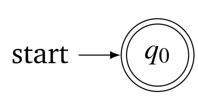
In other words, every regular language is recognized by an NFA. Since we already know that NFAs and DFAs recognize the same languages, this also tells us that every regular language is recognized by a DFA. Notice that this lemma universally quantifies the following predicate over regular expressions: $P(R) =$ "R represents a language that is recognized by a DFA." Since regular expressions are defined inductively, we can use an inductive argument (often called *structural induction*) to prove the lemma. We will show that the lemma holds for the base case regular expressions (1-3); we can construct NFAs that recognize these regular languages. Then, we will reason about the recursive cases. Given NFAs for that recognize the languages $\mathcal{L}(R)$ and $\mathcal{L}(S)$, we will show how to use these to construct NFAs that recognize $\mathcal{L}(R) \cup \mathcal{L}(S)$, $\mathcal{L}(R) \cdot \mathcal{L}(S)$ and $\mathcal{L}(R)^*$. This proof is a modification of a construction originally described by Ken Thompson (the creator of UNIX).

Proof. We separately consider the six cases from Definition 38.4. We can easily draw NFAs that recognize the languages of the three base cases.

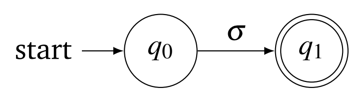
<u>Case 1:</u> $\mathcal{L}(\varnothing) = \varnothing$



<u>Case 2:</u> $\mathcal{L}(\lambda) = \{\lambda\}$



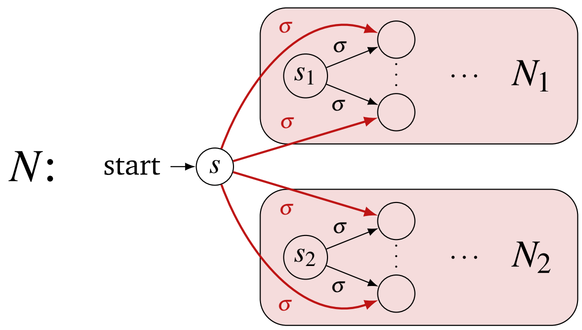
<u>Case 3:</u> $\mathcal{L}(\sigma) = \{\sigma\}$



Next, we consider the recursive cases. Any regular expression that does not belong to one of the base cases must have the form $R = R_1 + R_2$, $R = R_1 R_2$, or $R = R_1{}^*$ for some "smaller" regular expressions R_1, R_2. As our inductive hypothesis, we assume that we have NFAs $N_1 = (\Sigma, Q_1, s_1, F_1, \delta_1)$ that represents $\mathcal{L}(R_1)$ and $N_2 = (\Sigma, Q_2, s_2, F_2, \delta_2)$ that represents $\mathcal{L}(R_2)$. We'll use these as building blocks to describe an NFA N that represents R.

<u>Case 4:</u> $R = R_1 + R_2$

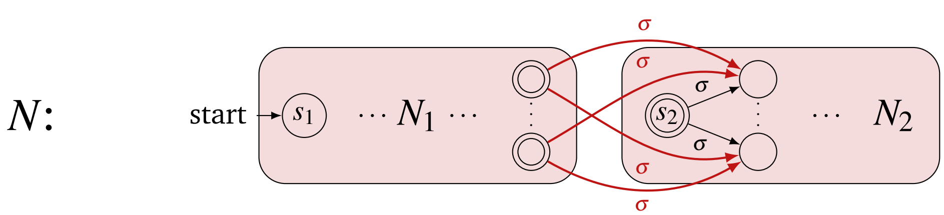
In this case, N must represent every string that is accepted by either N_1 or N_2. To do this, we'll introduce a new starting state s that acts as a "shared starting state" for both N_1 and N_2. For each $\sigma \in \Sigma$, s has an outgoing σ arrow pointing to each state in $\delta_1(s_1, \sigma) \cup \delta_2(s_2, \sigma)$. If either s_1 or s_2 is final, s should be final as well. Pictorially,

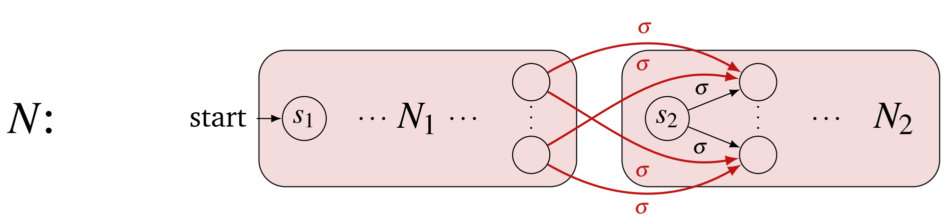


Any string w beginning with σ that is accepted by N_1 must first follow a σ-arrow from s_1 to some state $q \in Q_1$. Our new construction ensures that there is a σ-arrow from initial state s to q. By following this arrow and then the rest of the accepting path in N_1, we locate an accepting path in N, so N also accepts w. The same reasoning holds for strings accepted by N_2. Hence, $\mathcal{L}(N) = \mathcal{L}(N_1) \cup \mathcal{L}(N_2) = \mathcal{L}(R_1) \cup \mathcal{L}(R_2) = \mathcal{L}(R_1 + R_2) = \mathcal{L}(R)$.

<u>Case 5:</u> $R = R_1 R_2$

In this case, N must represent strings $w = uv$ with $u \in \mathcal{L}(R_1)$ and $v \in \mathcal{L}(R_2)$. To do this, we want to link the machines together so that the computation first goes through N_1 and then goes through N_2. For each $\sigma \in \Sigma$, we draw a σ-arrow from each final state in N_1 (representing the end of a string from $\mathcal{L}(R_1)$) to each state in $\delta_2(s_2, \sigma)$ (representing the first character in a string from $\mathcal{L}(R_2)$). If $\lambda \notin \mathcal{L}(R_2)$ we must change all states in F_1 to be non-final. Pictorially,

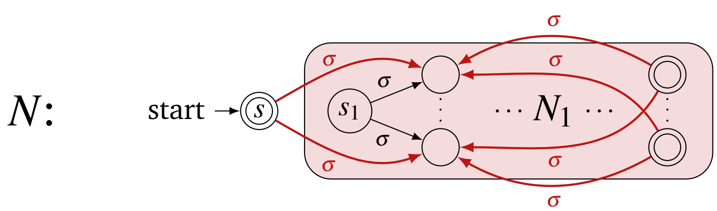




To reach a final state, a computation on string w must first transition from initial state s_1 to a final state in F_1. If $\lambda \in \mathcal{L}(R_2)$, then the computation can end here. Otherwise, it follows along a red arc and completes a path in N_2 to a final state in F_2. We can split the overall path into two pieces: the piece contained in N_1 and the piece consisting of the red arc and the computation in N_2. These correspond to strings $u \in N_1$ and $v \in N_2$ (respectively), with $w = uv$, so $w \in \mathcal{L}(N_1) \cdot \mathcal{L}(N_2)$. Thus, $\mathcal{L}(N) \subseteq \mathcal{L}(N_1) \cdot \mathcal{L}(N_2) = \mathcal{L}(R_1) \cdot \mathcal{L}(R_2) = \mathcal{L}(R_1 R_2) = \mathcal{L}(R)$. Arguing the reverse inclusion is similar and left as an exercise (essentially, given any string in the concatenation, decompose it and use the decomposition to describe a path to a final state). Thus, $\mathcal{L}(N) = \mathcal{L}(R)$.

<u>Case 6:</u> $R = {R_1}^*$

In this case, N must represent strings that can be decomposed into any number of substrings that are each represented by R_1. In this machine, we must both ensure that it recognizes λ and that whenever it finishes processing a string in R_1, it has a way to loop back and process another. We'll introduce a new initial state s that is also final. Then, for each $\sigma \in \Sigma$, we draw a σ-arrow from s and from each final state in N_1 to each state in $\delta_1(s_1, \sigma)$. Pictorially,



Similar to the previous case, we can decompose any string w accepted by N into substrings, where the computation of each substring begins by following a red arc (or the first arc in the case of the first substring) and ends at a final state. Each of these substrings corresponds to a path from s_1 to a final state, so belongs to $\mathcal{L}(N_1)$. Hence, $w \in \mathcal{L}(R_1)^* = \mathcal{L}({R_1}^*) = \mathcal{L}(R)$, so $\mathcal{L}(N) \subseteq \mathcal{L}(R)$. Again the reverse direction is argued similarly and left as an exercise. Thus, $\mathcal{L}(N) = \mathcal{L}(R)$.

By repeatedly applying these "wiring procedures" to larger machines, we can construct an NFA that recognizes any regular language. ■

We will see examples of this construction at the beginning of the next lecture.

Main Takeaways:

- *Regular expressions* describe languages that can be built using three language operations: *union*, *concatenation*, and the *Kleene star*.
- The semantics of regular expressions are defined inductively. We reason about regular expressions inductively by separately reasoning about the six cases in the definition.
- *Regular languages* are *represented* by regular expressions.
- We can use a wiring procedure called *Thompson's Construction* to show that every regular language can be recognized by an NFA (or DFA).

Exercises

38.1 Let L, K, and J be languages over a finite alphabet Σ. Determine whether each of the following statements about operations on languages is true or false. If it is true, justify why. If it is false, provide a counterexample.

(a) If $L \neq \varnothing$, then L^* is an infinite set.

(b) $|LK| = |L| \cdot |K|$.

(c) $LK = KL$.

In other words, concatenation is a *commutative* operation on languages.

(d) $(LK)J = L(KJ)$.

In other words, concatenation is an *associative* operation on languages.

(e) $L(K+J) = LK + LJ$.

In other words, concatenation *distributes over* union.

(f) $(L+K)^* = L^* + K^*$.

(g) $(L^*)^* = L^*$.

In other words, the Kleene star is an *idempotent* operation.

(h) $L^* = LL^*$.

38.2 Describe the language (over the specified alphabet) that is represented by each of the following regular expressions. (Rather than "translating" each regular expression into English, try to find a simple property that characterizes each language.)

(a) $R = a^*b^*c^*d^*, \qquad \Sigma = \{a,b,c,d\}$

(b) $R = (0+1)^*0(0+1)^*, \qquad \Sigma = \{0,1\}$

(c) $R = (a+b)^* + (a+c)^* + (b+c)^*, \qquad \Sigma = \{a,b,c\}$

(d) $R = (00+1)^*, \qquad \Sigma = \{0,1\}$

(e) $R = (0+10^*1)^*, \qquad \Sigma = \{0,1\}$

(f) $R = (x+\lambda)(yx)^*(y+\lambda), \qquad \Sigma = \{x,y\}$

(g) $R = (0+\lambda)(1+101+1001)^*(0+\lambda), \qquad \Sigma = \{0,1\}$

(h) $R = (01+001+(10^*01)^*)^*\varnothing(1+\lambda)(0^*(10^*)^*1+010^*), \qquad \Sigma = \{0,1\}$

38.3 Write a regular expression that represents each of the following languages.

(a) $L_1 = \{w \in \{0,1\}^* \mid w \text{ starts with 0 and ends with 1}\}$

(b) $L_2 = \{w \in \{a,b,c\}^* \mid w \text{ contains at least two } c\text{s}\}$

(c) $L_3 = \{w \in \{0,1\}^* \mid w \text{ contains the substring } 01001\}$

(d) $L_4 = \{w \in \{0,1\}^* \mid \text{the number of 0s in } w \text{ is a multiple of } 3\}$

(e) $L_5 = \{w \in \{d,e,f\}^* \mid w \text{ contains at least 3 occurrences of (at least) one letter}\}$

(f) $L_6 = \{w \in \{0,1\}^* \mid \text{every 0 in } w \text{ is next to another 0}\}$

⋆ (g) $L_7 = \{w \in \{g,h,i\}^* \mid w \text{ does not contain consecutive repeating characters}\}$

⋆ (h) $L_8 = \{w \in \{j,k,\ell\}^* \mid w \text{ does not contain the substring } \ell jk\}$

38.4 We say that two regular expressions R_1 and R_2 are *equivalent* if $\mathcal{L}(R_1) = \mathcal{L}(R_2)$. Determine whether each of the following pairs of regular expressions is equivalent or not. If they are, describe the language that they represent. If they are not, give an example of a string that is represented by only one of them.

(a) $R_1 = 0(10)^*1, \qquad R_2 = 01(01)^*$

(b) $R_1 = (0+01)^*, \qquad R_2 = (0+10)^*$

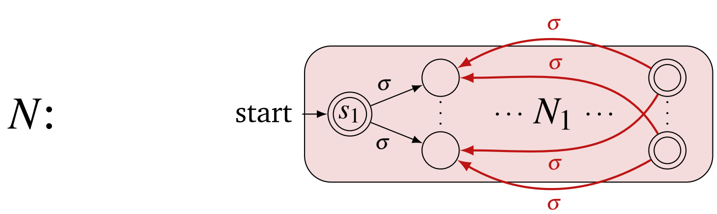
(c) $R_1 = (0^*1)^*0^*, \qquad R_2 = (0+1)^*$

(d) $R_1 = (0+1)^*01(0+1)^*, \qquad R_2 = 1^*0(0+1)^*10^*$

(e) $R_1 = (100+00)^*, \qquad R_2 = (10+\lambda)0(010+00)^*0$

⋆ **38.5** Argue that the set of all regular languages over a finite alphabet Σ is countable. (Hint: Use the characterization of regular expressions as strings over another alphabet.)

38.6 A keen student questions the handling of the Kleene star operation in the proof of Lemma 38.1 (Case 6). They suggest an alternate, simpler construction where a new state s is not added, but rather the previous initial state s_1 is made final. Pictorially,



Describe an example that shows that this alternate construction will not work. Your example should include a regular expression R, an NFA N_1 that recognizes this language, and a string that shows that $\mathcal{L}(R^*)$ is different from the language accepted by the NFA N that is output by the construction.

38.7 Given any string $w \in \Sigma^*$, its *reverse* which we denote w^R is the string formed by reversing the order of its characters. For example, $10110^R = 01101$. We can similarly define the reverse of a language L over Σ by

$$L^R = \{w \in \Sigma^* \mid w = v^R \text{ for some } v \in L\}.$$

Use structural induction on regular expressions to argue that the reverse of any regular language is also regular. In the argument, you should consider an arbitrary regular expression R and describe how to construct a new regular expression S such that $\mathcal{L}(S) = \big(\mathcal{L}(R)\big)^R$. You may use (without proof) the fact that for any strings $v, w \in \Sigma^*$, $(vw)^R = w^R v^R$, but you should prove any other facts that you use.

39. Kleene's Theorem

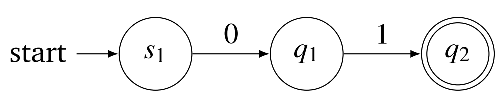
At the end of the previous lecture, we introduced a variant of Thompson's construction. This procedure can be used to construct an NFA that recognizes the language represented by any regular expression. Today, we'll practice using this construction. We'll also continue to study the link between regular expressions and finite automata by considering the converse question: can the language of any finite automaton be represented with a regular expression?

39.1 Using Thompson's Construction

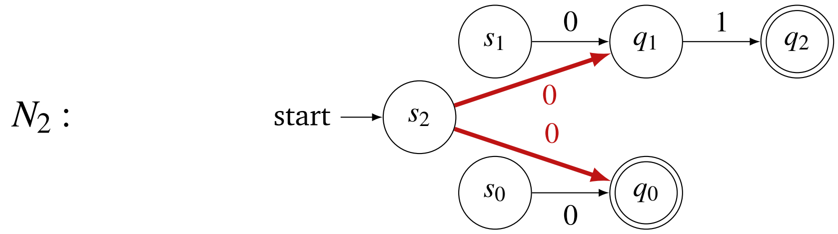
The procedure from last lecture separately considered the six cases in the inductive description of regular expressions. We described three simple automata to capture the base cases $\varnothing, \lambda$, and σ (for any $\sigma \in \Sigma$). For the recursive cases, we described how we can add additional wiring to mimic the behavior of the regular language operations union, concatenation, and the Kleene star. We give two examples of using this construction below. The labels that we assign the states remain consistent throughout the construction so you can keep track of how the wiring process is working.

Example 39.1 Consider the regular expression $(01+0)(1+00)$ over the alphabet $\Sigma = \{0,1\}$. We will build an NFA that represents this regular expression by working from inside out (that is, in the order that the language operations are applied). First, we can recognize "01" by concatenating the base case automata that recognize "0" and "1". This gives the automaton:

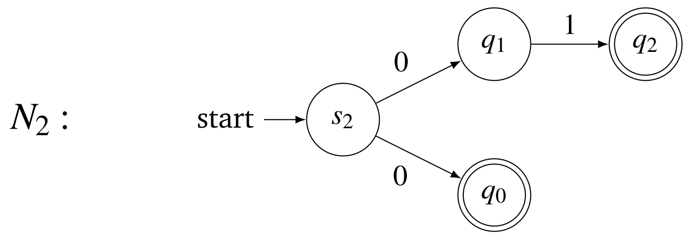
N_1 :



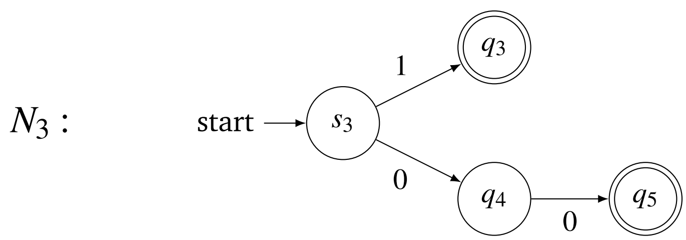
Now, to recognize "$01+0$", we must apply the union case to wire together N_1 and the base case automaton that recognizes "0". We get,



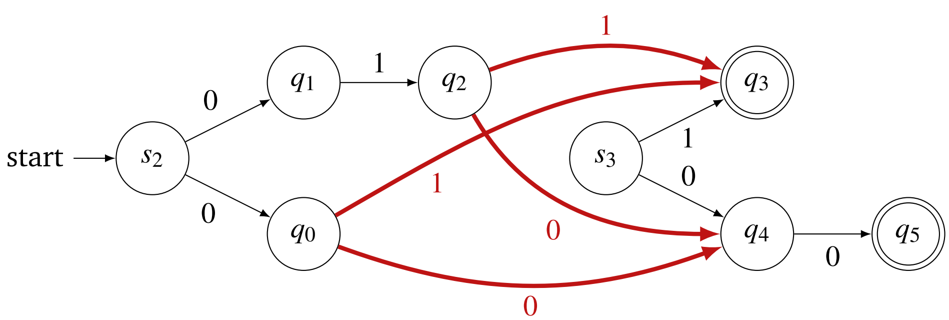
Since s_0 and s_1 have no incoming arcs, they are *unreachable* states in N_2. Thus, we will frequently remove them to obtain a simpler automaton:



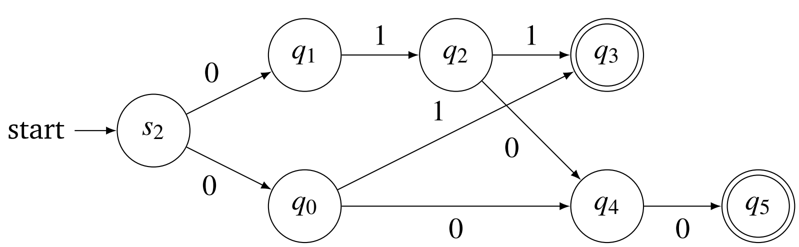
We can similarly construct an NFA N_3 that recognizes "$1+00$".



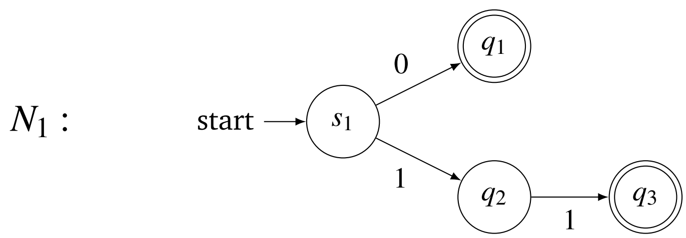
We use the concatenation construction to join N_2 and N_3:



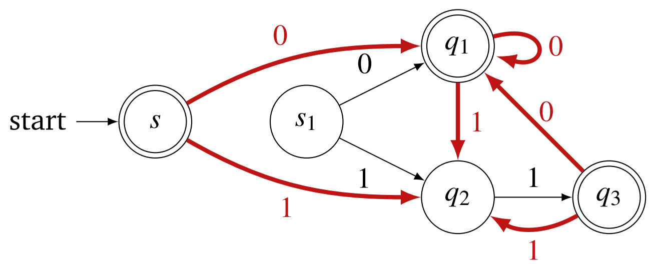
and then (optionally) remove the unreachable state s_3 to obtain our final automaton:



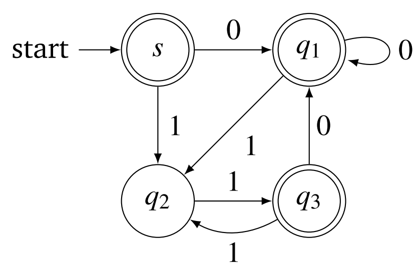
Example 39.2 To demonstrate how to work with the Kleene star, we'll build an NFA that recognizes $(0+11)^*$. Using the same ideas as the previous example, we can recognize "$0+11$" with the NFA:



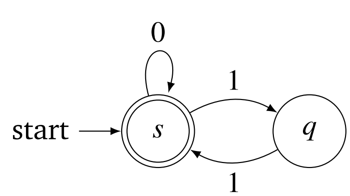
As described by the construction, we add a new initial state s and transitions out of the final states of N_1 to capture the behavior of the Kleene star. We end up with the automaton:



Removing the unreachable state s_1 and rearranging, we are left with:



This example illustrates something important about this construction; although it is guaranteed to produce an automaton that recognizes the language given by the regular expression, it does not guarantee to find the simplest machine (i.e., the machine with the smallest number of states) that does this. Note that the following NFA also recognizes this language:



39.2 Kleene's Theorem

We have shown that every regular language can be recognized by an NFA (so also a DFA using the subset construction from two lectures ago). One might wish to ask the opposite question: Is every language that is recognized by an NFA regular? The answer to this question turns out to be yes, and this celebrated result is referred to as Kleene's Theorem.

Theorem 39.1 — Kleene's Theorem. A language L is regular if and only if it is recognized by an NFA (or, equivalently, a DFA).

We have already proven the forward direction of this theorem. For the rest of today's lecture, we will focus on proving the reverse direction. That is, we will give a procedure to write a regular expression that represents the language of a given NFA. At a high level, this procedure will iteratively remove the states of the automaton. As it does this, it will need to replace the character labels on its transition arrows with more complicated regular expressions. Once we are left with a machine with at most 2 states, we will give an explicit regular expression for its language.

Step 1: Merge Final States

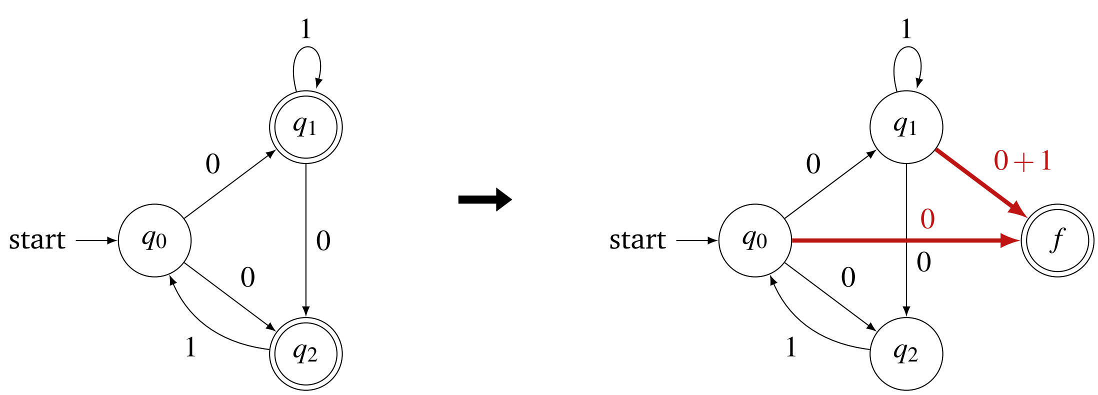
The rest of the procedure will require the following property of our NFA:

> Besides the starting state, N can contain at most one other final state f, which must have no outgoing arcs.

To get our machine in this form, we do the following:

1. Introduce a new state f to Q.
2. For each state $q \in F \setminus \{s\}$, each character $\sigma \in \Sigma$, and each incoming transition to q labeled with σ, let q' be the origin of this transition (i.e. $q \in \delta(q', \sigma)$). Add a σ-transition from q' to f.
3. Make all states in $F \setminus \{s\}$ non-final, and make state f final.

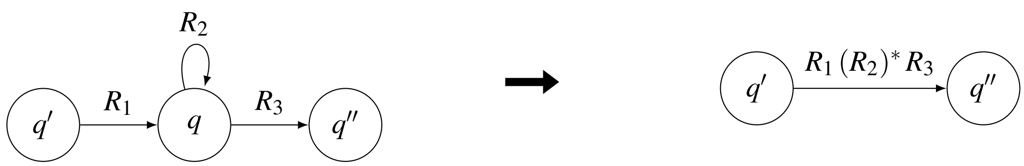
As an example, the machine on the left would be transformed into the machine on the right.



Step 2: Eliminate Intermediary States

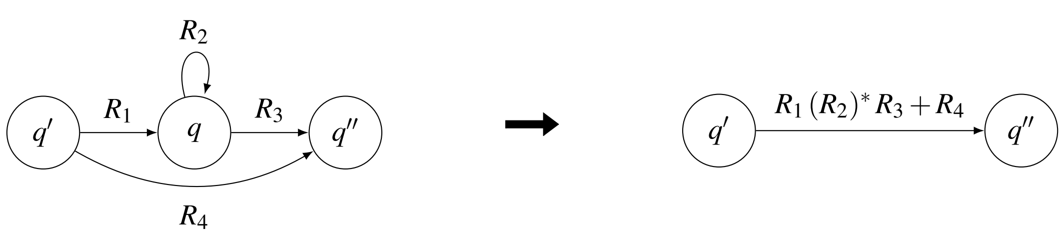
Now, we will eliminate all of the states that are neither the starting state nor the final state one at a time. If we simply delete the states, we would be losing information about the language of the machine; some paths to an accepting state may disappear. Instead, we'll retain this information by adding it to the transition labels. Now, we'll allow each label to be a regular expression rather than a single character. We'll call these machines with regular expression transition labels *Generalized NFAs* (GNFAs). We follow an arc in a GNFA to process any substring that is represented by its regular expression label. We leave it as an exercise (Exercise 39.2) to give a formal definition of this class of automata.

Suppose that we are currently eliminating state q. Then, we need the transition labels in the GNFA to encode the computation paths that could have gone through q. Each such path must enter q from another state q', perhaps loop in state q for a while, and then exit along a transition to state q'', as shown in the following figure.

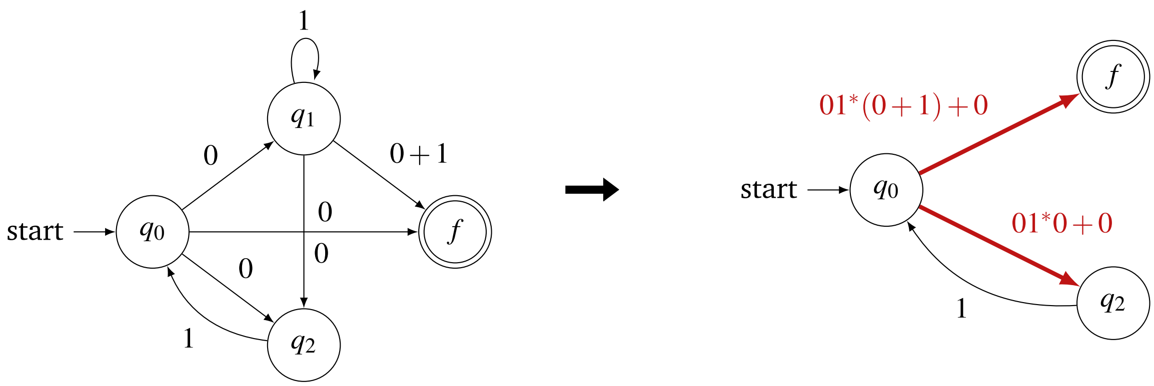


R_1, R_2, and R_3 are the regular expressions that label these arcs. If our input string is processed along this path, then it will contain some string represented by R_1 concatenated to any number of strings represented by R_2 and then concatenated to a string represented by R_3. Thus, we can account for this path by adding an arrow with regular expression "$R_1\,(R_2)^*R_3$" from q' to q'', as shown above. There are some important things to note with this step of the procedure:

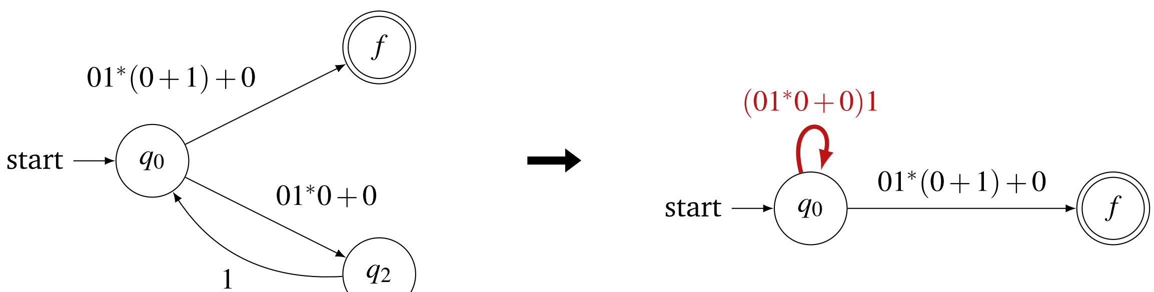
- If there is no self-loop on state q, we leave this part out of the regular expression and instead label the transition with $R_1 R_3$.
- This substitution needs to be made for *every* (q', q'') pair before q can be deleted. This includes the possibilities where $q' = q''$, which can introduce a self-loop on this state.
- It's possible that this procedure will add an arrow between two states that already have an arrow (i.e., there could already be an arrow labeled R_4 from q' to q''). In this case, we can "merge" the two arrows by taking the union of their regular expressions (here, giving "$R_1\,(R_2)^*R_3 + R_4$"). See the following figure.



Let's carry out this step on our example machine from earlier. First, we can eliminate the state q_1, which gives,



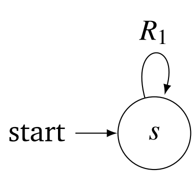
Next, we can eliminate state q_2, which leaves



Step 3: Write the Final Regular Expression

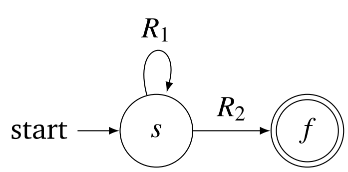
After eliminating all of the intermediary transitions, we are left with a machine that has either one (just the starting state s) or two (s and f) states. The following cases enumerate all possible configurations and how to write the final regular expression R. These rules can be constructed by reasoning about which paths can reach a final state in this machine. Any missing arrows can be added and labeled with the regular expression λ

Case 1:



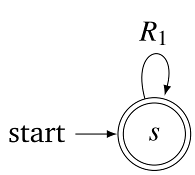
$R = \varnothing$

Case 2:



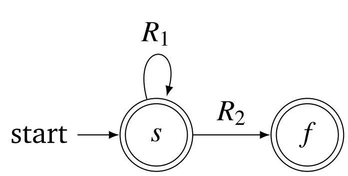
$R = (R_1)^* R_2$

Case 3:



$R = (R_1)^*$

Case 4:



$R = (R_1)^*(\lambda + R_2)$

In our example, we end up with the regular expression $((01^*0+0)1)^*(01^*(0+1)+0)$.

We have given a complete (albeit intricate) description of a procedure to convert an NFA into a regular expression that represents the same language. This completes the proof of Kleene's Theorem. Notice again that there is no guarantee that the regular expression that is produced is the simplest one possible. In fact, most of the regular expressions that it produces will be long and incomprehensible.

Main Takeaways:

- *Unreachable* states in an automaton have no incoming transitions and can be removed without changing the language of the machine.
- *Kleene's Theorem* gives an equivalence between regular languages and languages recognized by NFAs/DFAs.
- A *Generalized NFA* (GNFA) has regular expressions labeling its transitions.
- To construct a regular expression for an NFA's language, first merge the final states, then introduce new regular expression transitions that allow you to delete intermediary states one at a time, and then write the final regular expression once there are at most two states left.

Exercises

39.1 For each of the following regular expressions R, draw an NFA N such that $\mathcal{L}(N) = \mathcal{L}(R)$.

As further practice, find a DFA that also recognizes the same language, either using the subset construction to convert the NFA, or coming up with one from scratch.

(a) $R = 101 + 01$

(b) $R = (ab+ba)(aa+bb)$

(c) $R = (01)^*$

(d) $R = (xyx)^* + (zyz)^*$

(e) $R = (01+110)^*$

(f) $R = (cd^*c+c)^*$

39.2 Give a formal definition of a Generalized Non-deterministic Finite Automaton. Similar to our definitions of other classes of automata, you should list out a tuple of components of this definition and then describe each of them. In your description of the transition function, be sure to list its domain, codomain, and describe how the mapping is defined (it will likely look very different from the other transition functions we've seen).

⋆ **39.3** In this exercise, you'll think about the relationship between the languages that are recognized by NFAs and by GNFAs.

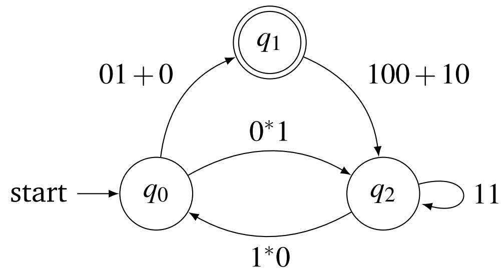
(a) The name *Generalized* NFA suggests that every NFA can be viewed as a GNFA. Explain why this is true.

(b) Given any GNFA, describe how to construct an NFA that recognizes the same language.

(Hint: The arcs of a GNFA are labeled with regular expressions. What procedure do we have to convert regular expressions to automata?

Together, parts (a) and (b) show that GNFAs are just as powerful as NFAs. Both classes recognize the same set of languages.

39.4 Consider the following GNFA over alphabet $\Sigma = \{0, 1\}$:



Determine whether this automaton accepts each of the following strings. For each that it does, describe its decomposition into substrings such that each substring corresponds to a transition that processes this string to the accepting state.

(a) $w_1 = 01100$

(b) $w_2 = 1111$

(c) $w_3 = 111100$

(d) $w_4 = 001001$

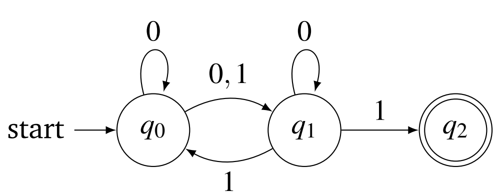
(e) $w_5 = 0001001011$

(f) $w_6 = 1101100110001$

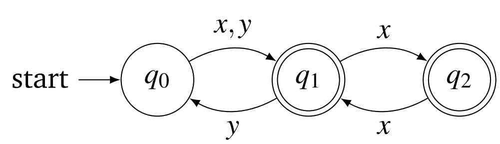
Can you find a regular expression that describes the language of this machine?

39.5 Find a regular expression that represents the language of each of the following automata.

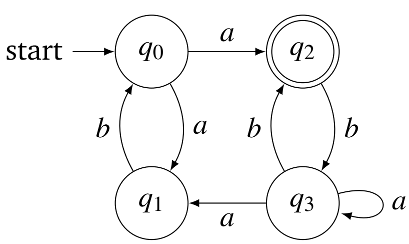
(a)



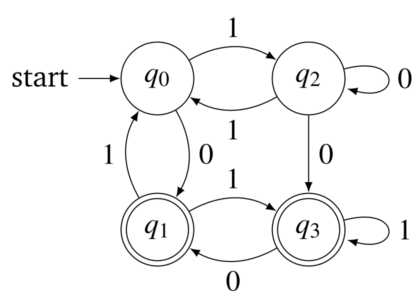
(b)



(c)



(d)



39.6 In the third step of the procedure to convert an NFA to a regular expression, we claimed that there were only four possible forms for the GNFA that results after completing steps 1 and 2 (technically eight possibilities if you allow for the inclusion or exclusion of the self-loop on state s). Explain why these are the only possibilities. In particular, how do we know that there will not be a self-loop on state f or a transition from f to s?

One nice thing about Kleene's Theorem is that it gives us two very different frameworks for reasoning about regular languages: the inductive structure of regular expressions and the more procedural framework of automata. When we wish to prove properties of regular languages, it is sometimes more useful to consider one of these frameworks over the other. We explore this phenomenon in this Exercises 39.7-39.9.

39.7 In Exercise 21.7, you were asked to argue that the set of regular languages is closed under the reversal operation. This follows from a nice inductive argument on regular expressions.

Suppose instead that we wish to argue this using NFAs. Given an NFA that recognizes a language L, describe how to construct an NFA that recognizes L^R.

(Hint: Reversing the direction of the arrows is not enough. You'll need to use some other tricks from this lecture to take care of the starting/final states.)

39.8 Exercise 19.8 uses DFAs to show that the complement of any regular language is also regular. This was a very simple argument. It is not so simple to take the complement of an NFA; the best approach is to use the subset construction to find an equivalent DFA and then take the complement of that.

It is also non-obvious how to take the complement of a regular expression (that is, find a regular expression that represents the complement language). Converting Regular Expression $\to$ NFA $\to$ DFA $\to$ Complement DFA $\to$ Complement Regular Expression would be extremely tedious for even the simplest of regular expressions. Nevertheless, find the complement of the following regular expressions over $\Sigma = \{0,1\}$ by reasoning about the languages that they represent.

(a) $R_1 = (0+1)^*0$

(b) $R_2 = (01^*0+1)^*$

(c) $R_3 = (00+1)^*$

(d) $R_4 = (0+1)^*100(0+1)^*$

39.9 Argue that the set of regular languages is closed under the following set operations (that is, for each operation $\star$, argue that if L_1, L_2 are both regular, then so is $L_1 \star L_2$).

(a) Set union $(\cup)$

(b) Set intersection $(\cap)$

(Hint: This follows from part (a) and Exercise 19.8. For another nice proof, given DFAs M_1 and M_2, construct DFA that recognizes $\mathcal{L}(M_1 \cap M_2)$ which has as its states $Q_1 \times Q_2$.)

(c) Set difference ($\setminus$)

(Hint: Same as part (b).)

39.10 Describe a procedure that determines whether two regular expressions R_1 and R_2 are equivalent (i.e., $\mathcal{L}(R_1) = \mathcal{L}(R_2)$).

(Hint: Use Exercise 39.9(c).)